Community Medicine AIIMS 2017

Question 1:

Which of the following wastes are disposed in the bag shown below?

- **a)** Urine bags
- **b)** Anatomical waste
- **c)** Soiled waste
- **d)** Sharp waste

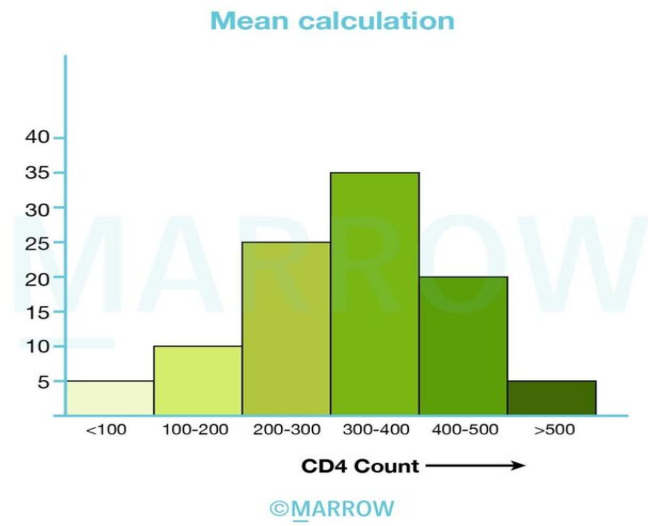
Question 2:

A researcher discovered a new drug which is effective in chronic hypertensives with a p value of <0.10. which of the following is true regarding the same

- **a)** Not more than 10% of the people benefitted by the drug could be due to chance
- **b)** 90% of patients will be benefitted by giving the drug
- **c)** The test is 90% reproducible
- **d)** 90% of test results could have occurred by chance

Question 3:

Below is the graph of CD4 count in HIV patients. From the graph below identify the mean CD 4 count



a) 300-400

b) 200-300

c) Below 250

d) Above 350

Question 4:

Which of the following is true about triage?

a) Yellow – least priority

b) Red – mortality

c) Green – ambulatory

d) Blue – ambulatory

Question 5:

A randomized trial comparing the efficacy of two drugs showed a difference between the two with a p value of <0.005. In reality, both drugs do not differ at all. This is an example of:

a) Alpha error

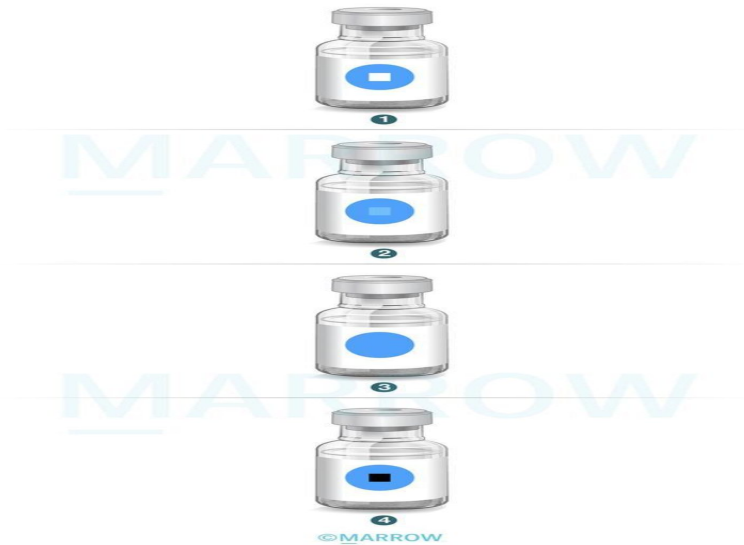
b) Beta error

c) $1 - \alpha$

d) 1 – β

Question 6:

Which of the following vaccine vials can be used based on the vaccine vial monitor (VVM) color changes shown?



a) Only vial 1 can be used

b) Vials 1 and 2 can be used

c) Vials 1, 2, and 3 can be used

d) All vials can be used

Question 7:

Under the Mid-Day Meal (MDM) scheme, supplementary nutrition provided should cover at least _______

a) 1/3 of the total protein requirement + ½ of total energy requirement

b) 2/3 of the total protein requirement + ½ of total energy requirement

c) ½ of the total protein requirement + 1/3 of total energy requirement

d) ½ of the total protein requirement + 2/3 of total energy requirement

Question 8:

In a subcentre area with a crude birth rate (CBR) of 20, what would be the expected number of antenatal care (ANC) registrations by the village health nurse (VHN) in a year?

a) 60

b) 80

c) 100

d) 110

Question 9:

As per the syndromic approach to sexually transmitted infections (STIs) and reproductive tract infections (RTIs), what is the treatment of choice of gonococcal and non-gonococcal urethritis?

a) Azithromycin + Metronidazole

b) Cefixime + Azithromycin

c) Azithromycin + Doxycycline

d) Doxycycline + Cefixime

Question 10:

Annual new case detection rate (ANCDR) of leprosy in India, as of 31st March 2016 was ________.

a) 0.66/10,000 population

b) 0.66/1 lakh population

c) 9.7/10,000 population

d) 9.7/1 lakh population

Question 11:

The recommended calorie intake for a pregnant woman in the second trimester, who is a sedentary worker would increase by _______

a) 600 kcal/day

b) 520 kcal/day

c) 450 kcal/day

d) 350 kcal/day

Question 12:

Match the following declarations to their significance.

Declaration	Significance
I. Geneva	A. Guidelines for physicians concerning torture and other cruel, inhuman or degrading treatment or punishment in relation to detainment or imprisonment
II. Oslo	B. Set of ethical principles directed to the medical community for human experimentation.
III. Tokyo	C. Physician's dedication to the humanitarian goals of medicine
IV. Helsinki	D. Statement on therapeutic abortion

a) I – A, II – B, III – C, IV – D

b) I - C, II - D, III - A, IV - B

c) I – D, II – C, III – A, IV - B

d) I – B, II – A, III – D, IV - C

Question 13:

Chronic liver disease is classified using the Child-Pugh score as follows: Class A: 5-6 , Class B: 7-9, Class C: 10-15. What type of data is this scoring system?

a) Nominal

b) Ordinal

c) Qualitative

d) Metric

Question 14:

An 18 month old child is brought with no history of vaccination. From the following, which vaccines would be appropriate to give this child according to the National Immunization

Schedule (NIS), India?

a) DPT -1, OPV-1 only

b) BCG, DPT-1, OPV-1

c) BCG, DPT-1, OPV-1, Measles – 1, Vit. A -1

d) DPT-1, OPV-1, Measles – 1, Vit. A – 1

Question 15:

P-value is the probability of which of the following?

a) Not rejecting a null hypothesis when it is true

b) Rejecting a null hypothesis when it is true

c) Not rejecting a null hypothesis when it is false

d) Rejecting a null hypothesis when it is false

Question 16:

While testing the deleterious health hazards on a group of people after giving a new drug and a placebo, the following results were obtained as tabulated. Based on these results, which of the following is true?

a) Absolute risk reduction (ARR) is 1% and relative risk reduction (RRR) is 0.1

b) Absolute risk reduction (ARR) is 10% and relative risk reduction (RRR) is 0.9

c) Absolute risk reduction (ARR) is 1% and relative risk reduction (RRR) is 10

d) Absolute risk reduction (ARR) is 10% and relative risk reduction (RRR) is 9

Question 17:

A new drug combination of chemotherapy and immunotherapy for metastatic melanoma was prescribed by a radiotherapist. It prolongs the survival. Which of the following is true in this situation?

a) Incidence increases and prevalence reduces

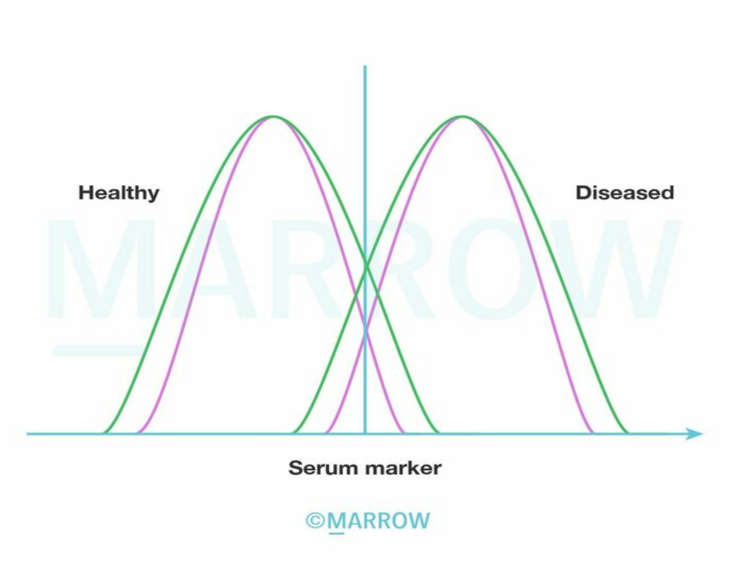
b) Incidence reduces and prevalence remains the same

c) Incidence remains the same and prevalence increases

d) Incidence reduces and prevalence increases

Question 18:

Observe the following curves, which of the following is a true statement about sensitivity and specificity when the curve changes from green to pink?



a) Both sensitivity and specificity increase

b) Both sensitivity and specificity decrease

c) Sensitivity increases while specificity decreases

d) Sensitivity decreases while specificity increases

Question 19:

A study was conducted to compare the incidence of postpartum depression in mothers as compared to the sex of the child. For assessment, Edinburg Depression scaled score (EDSS) was used. The study was divided into mothers with male babies as one group to be compared with mothers with female babies. Which of the following statistical test will be used to determine the association?

a) Students t test

b) Paired t test

c) ANOVA

d) Chi square test

Question 20:

A one-and-half-year-old child was brought to the hospital with a history of breathlessness, fever, cough, and cold. On examination, respiratory rate was 46/min, and there were no chest retractions. Which is the most appropriate management?

a) Give IV antibiotics and observe

b) Give IV antibiotics and refer to higher centre

c) Oral antibiotics, warn of danger signs, and send home

d) No need of treatment only home remedies

Question 21:

The following are the disease frequencies in a population. Based on the number of cases which of the following is the correct match?

a) Pandemic, endemic, sporadic

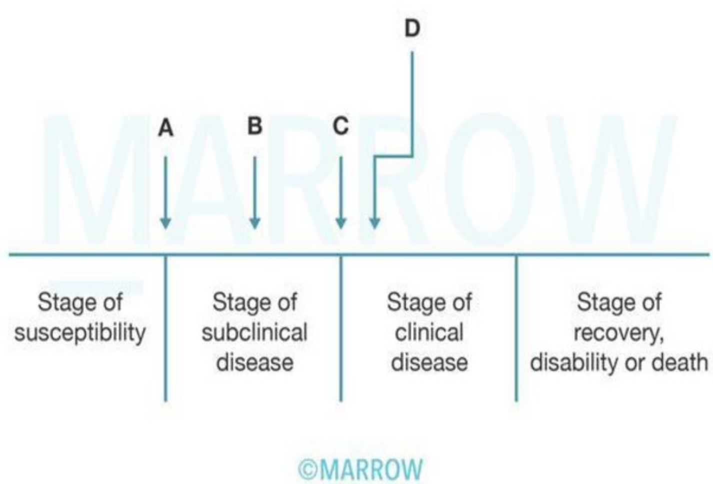
b) Sporadic, endemic, epidemic

c) Endemic, epidemic, sporadic

d) Endemic, sporadic, epidemic

Question 22:

Below is a diagram depicting the natural history of the disease. Which point marks the onset of symptoms?



a) Point A

b) Point B

c) Point C

d) Point D

Question 23:

A researcher finds out a sensitive test to be positive for a person. For diagnosis, which of the following will hold true?

a) It confirms that the person does not have the disease

b) It suggests that if the disease is rare, the person surely has the disease

c) It confirms that the person has the disease if the disease is common

d) It is an expected finding in case the disease is prevalent

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	c
5	a
6	b
7	c
8	a
9	b
10	d
11	d
12	b
13	b
14	d
15	b
16	a
17	c
18	a

19	d
20	c
21	c
22	c
23	c

Detailed Explanations

Solution to Question 1:

The above picture shows a red coloured bag which is used to dispose contaminated wastes (recyclable) such as tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes (without needles) and gloves.

Option B: Anatomical wastes go into yellow non-chlorinated plastic bags.

Option C: Soiled wastes go into yellow coloured non-chlorinated plastic bags.

Option D: Sharp wastes go into white puncture proof, leak proof, tamper proof containers.

Solution to Question 2:

P value <0.10 indicates that not more than 10% of people benefitted by the drug could be due to chance.

It can also be interpreted as

• If the test is repeated 100 times, the difference will be statistically non significant only 10 times and difference will be statistically significant 90 times.

• Probability of false positive conclusion that drug is beneficial which in reality is not, is 10%.

It arises in case of statistical studies to accept or reject a null hypothesis.

P value is compared with a predetermined value α; P value < α means the test is statistically significant and null hypothesis is rejected.

Solution to Question 3:

The given data should be converted into numerical table. Small approximations can be made while converting into numerical form.

Since intervals are given, midpoint of each interval has to be calculated.

It should then be multiplied with frequency of that particular interval.

Mean is given by the formula: Mean = Sum of all observations / number of observations.

Mean = 34500/100 = 345.

Between 300 -400 is the most suitable answer.

CD4 Count	Number of people
<100	5
100 - 200	10
200 - 300	25
300 - 400	35
400 - 500	20
>500	5

CD4 Count	Mid point	Number of people (n)	Mid point × n
0 - 100	50	5	250
100 - 200	150	10	1500
200 - 300	250	25	6250
300 - 400	350	35	12250
400 - 500	450	20	9000
500 - 1600	1050	5	5250
TOTAL		100	34500

Solution to Question 4:

In triage, green signifies low priority or ambulatory patients and those with minor illnesses.

An internationally accepted four-color coded system, with red having the highest priority is used commonly here.

Solution to Question 5:

• In this scenario, let’s take the null hypothesis as the two drugs being the same, which the questions states, is true in reality.

• The alternative hypothesis will then be that both drugs are different.

• The questions states that p value was found to be < 0.005, which is statistically significant and so the null hypothesis is rejected.

• But we already know that the null hypothesis is true and we are forced to accept the alternative hypothesis even though it is false.

• This is a type I error or alpha error.

• When an alternative hypothesis is accepted even when it is false and the null hypothesis is true, it is a type I error or false positive error.

Solution to Question 6:

From the above image, only vials 1 and 2 can be used as the inner squares of the VVMs are lighter than the outer circles.

Vial 3 has crossed the discard point as the inner square and outer circle are the same color and thus cannot be used. Vial 4 cannot be used as the VVM shows a darker inner square compared to the outer circle.

Solution to Question 7:

Under the Mid-Day Meal (MDM) scheme, supplementary nutrition provided must cover at least ½ of the total protein requirement and 1/3 of total energy requirement to children in schools.

The MDM scheme is also called School Lunch Programme and has been operational since 1961 in India. Currently, it functions under the broad coverage of the Minimum Needs Programme.

Some other principles of the MDM are:

• The meal provided under the MDM should only be a supplement and not a substitute to home diets.

• The cost of the meals should be reasonably low.

• The meal should be easy to cook in schools.

• Locally available foods must be used, for as far as possible.

• The menu should be regularly changed to avoid monotony.

Solution to Question 8:

• According to the Indian Public Health Standards (IPHS) laid down by the Government of India, a subcentre is to serve a population of 5000 in plains.

• In this scenario, the CBR is 20.

• Probable number of pregnancies (Y) = Birth rate x population of the area / 1000 = 20 x 5000/1000 = 100.

• But, we add a correction factor of 10% to adjust for pregnancy wastage in the form of abortions or stillbirths.

• Thus, total number of expected pregnancies= Y + 10% of Y= 110

• As all the pregnancies are not registered, and at any given time an ANM should have atleast half the expected pregnancies registered with her, the number of ANC registrations is taken as 50% of the expected pregnancies.

• Therefore, Total ANC registrations = total number of expected pregnancies/2 = 55

• Hence, the closest answer is 60.

Solution to Question 9:

As per the syndromic approach to STIs/RTIs, the treatment of choice of for gonococcal and non-gonoccocal urethritis is azithromycin + cefixime.

Solution to Question 10:

For the year 2015 – 2016, the annual new case detection rate (ANCDR) of leprosy in India was 9.7/100,000 population.

Solution to Question 11:

The recommended calorie intake for a pregnant woman who is a sedentary worker would increase by ~ 350 kcal/day.

Solution to Question 12:

Declaration	Significance
Geneva	Physician's dedication to the humanitarian goals of medicine.
Oslo	Statement on therapeutic abortion.

Declaration	Significance
Tokyo	Guidelines for physicians concerning torture and other cruel, inhuman or degrading treatment or punishment in relation to detainment or imprisonment.
Helsinki	Set of ethical principles directed to the medical community for human experimentation.

Solution to Question 13:

The Child-Pugh score is an ordinal scale of data as:

There is a rank order (Class A having a better prognosis than classes B and C) but no regular interval between each class. The final division of prognosis is based on class and not the numerical values arrived at by the summation of a set of variables.

Other options:

Option A: The Child-Pugh score cannot be a nominal scale of data as there is a rank order with class A having better survival statistics than B or C. Nominal data do not have a rank order.

Option C: This is also not merely a qualitative scale as there are scores given to each variable to arrive at the final class.

Option D: The data used in the Child-Pugh score cannot be merely metric as the final division of prognosis is based on class and not the numerical values. Also, there is no definite interval between the classes.

Solution to Question 14:

- An 18 month old, completely un-immunized child should receive first doses of DPT, OPV, measles vaccine and vitamin A, according to the National Immunization Schedule (NIS), India.
- BCG can be given only up to one year of age.

Solution to Question 15:

p-value is the probability of rejecting a null hypothesis when it is true.

The p-value is the criterion or significance level (α), below which an outcome can be said to be statistically significant. From this, we can also deduce that the p-value is the probability that a null hypothesis will be rejected even when it is true, thereby making a type I error.

Solution to Question 16:

In the given scenario, absolute risk reduction (ARR) is 1% and relative risk reduction (RRR) is 0.1.

Let us reconstruct the design table in the following way:

Relative risk (RR) is given as a/(a+b) ÷ c/(c+d).

RR = 1603/15225 ÷ 1804/15225 = 0.106/0.118 = 0.89 ≈ 0.90.

Relative risk reduction (RRR) = 1 – Relative risk. In this scenario, RRR = 1 – 0.9 = 0.1 or 10%.

Incidence of a disease is the absolute risk of contracting it.

Absolute risk rate of the group using new drug = a/(a+b) = 1603/15225 = 0.106 or 10.6% ≈ 11%.

Absolute risk rate of the group using placebo = c/(c+d) = 1804/15225 = 0.118 or 11.8% ≈ 12%.

Absolute risk reduction (ARR) is the difference in the absolute risk rates between the placebo and the drug. In this scenario, ARR = 12% - 11% = 1%.

	Health hazard (+)	Health hazard (-)	Total
New drug	1603 (a)	13622 (b)	15225 (a +b)
Place bo	1804 (c)	13421 (d)	15225 (c+ d)

Solution to Question 17:

By the new drug combination, the survival of the patient has increased, hence incidence remains the same while prevalence increases.

Incidence: Number of new cases in a defined population in a specified period of time.

Prevalence: Number of old and new cases at a given point of time over a period of time.

Options A, B, D: The treatment only prolongs the disease, it has nothing to do with the occurrence of cases thereby nothing to do with incidence.

Solution to Question 18:

Both sensitivity and specificity are higher in the test represented in pink (Test B) compared to the test represented in green (Test A).

Sensitivity is the proportion of all people with the disease who test positive. It is the probability of the test being positive when the disease is present.

Sensitivity = TP / (TP + FN) or Sensitivity = 1 – false-negative rate.

Specificity is the proportion of all people without the disease who test negative. It is the probability of the test being negative when the disease is absent.

Specificity= TN / (TN + FP) or specificity= 1 – false-positive rate.

Test B has an increased number of true positives and true negatives and decreased number of false positives and negatives, compared to Test A.

Hence leading to increased sensitivity and specificity.

Solution to Question 19:

The Chi-square test is used to compare qualitative data and compares 2 groups with independent measures. The variables used here are yes/no signifying the presence/absence of postpartum depression. Hence it is the most appropriate test here.

Option A: Student's t-test is for quantitative data. Here the data is qualitative, hence it can't be used.

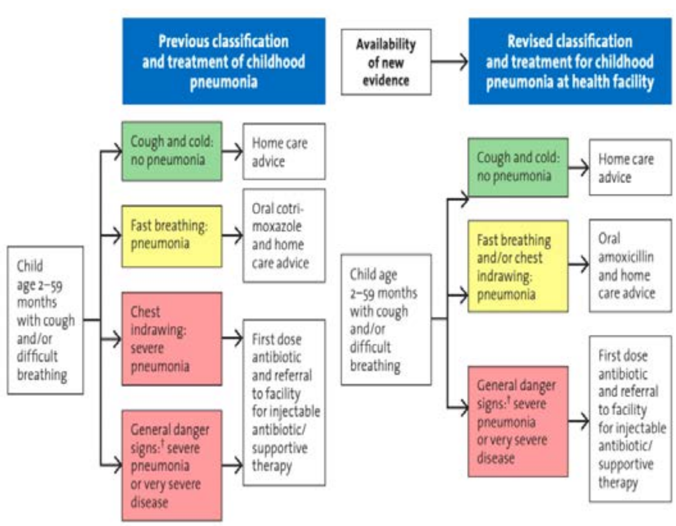
Option B: Paired t-test compares paired data or pre-intervention and post-intervention data of the same group. This doesn't apply here as the groups are not paired.

Option C: ANOVA is used to compare more than 2 groups.

Solution to Question 20:

From the given clinical scenario, it can be easily inferred that the child has pneumonia. The treatment for pneumonia for a child between 2 months and 5 years of age is oral amoxicillin and home care advice.

The following image gives the IMNCI classification for cough/difficulty in breathing in children.



† Not able to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition.

Solution to Question 21:

Disease 1- the cases have been almost the same so it has to be endemic.

Disease 2- there is an alarming increase in the number of cases. It has to be an epidemic.

Disease 3- the number of cases has decreased without any explanation so it has to be sporadic.

Epidemic refers to the occurrence of a disease in excess of expected frequency.

A pandemic is an epidemic affecting a large population over a large geographical area crossing international boundaries.

An endemic disease is a regularly occurring disease in a population in expected frequency without any importation from outside.

Sporadic diseases present as scattered cases of infection without relation temporally/spatially.

Solution to Question 22:

Point C represents a point where signs and symptoms appear.

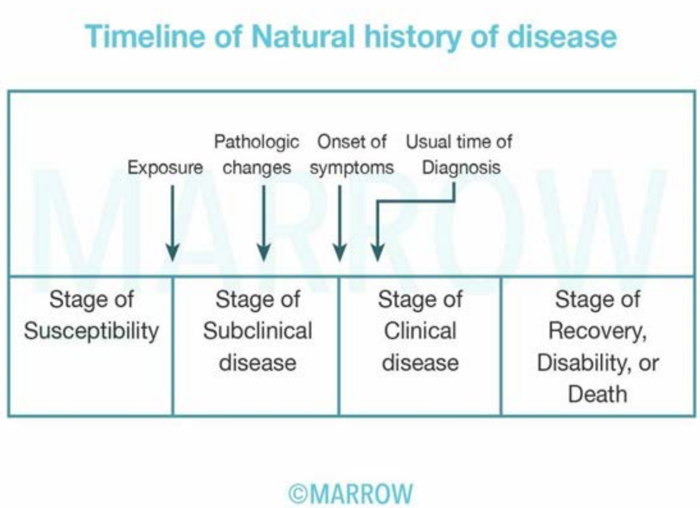
Various points marked in the image represent the following:

Point A: Entry of infectious agent into the host.

Point B: Pathological and physiological changes occur in the stage of the subclinical phase. These changes are induced by the causative agent but are not sufficient enough to cause clinical signs and symptoms.

Point C: Marks onset of clinical signs and symptoms.

Point D: Usual time of diagnosis at which clinical signs and symptoms are well evident.



Solution to Question 23:

A sensitive test, if it is positive it confirms that the person has the disease if the disease is common.

The sensitivity of a screening test is the ability of the test to identify correctly those who have the disease. In other words, Sensitivity is the ability of a test to detect true positives (TP) out of total disease (TP+FN).

Sensitivity = TP/ TP+FN

Community Medicine AIIMS 2018

Question 1:

A patient from a North-Eastern state was diagnosed to have an infection with P. falciparum malaria. What is the most appropriate treatment for this patient?

a) Artemether plus lumefantrine

b) Sulfadoxine plus pyrimethamine

c) Choroquine

d) Mefloquine

Question 2:

Choose the correct order of steps of disaster management.

a) Disaster impact – Mitigation – Rehabilitation – Response

b) Disaster impact – Response– Rehabilitation – Mitigation

c) Rehabilitation – Response – Disaster Impact – Mitigation

d) Response – Disaster impact – Mitigation – Rehabilitation

Question 3:

Chi square test is used to test the significance of association between:

a) Age and serum cholesterol level

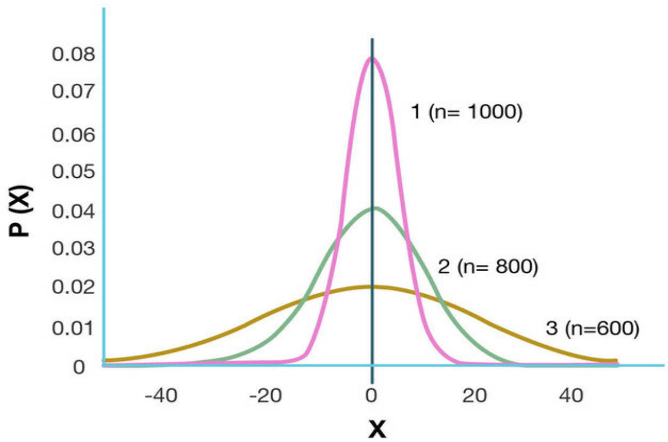
b) Heart rate and body mass index

c) Sex and type of cancer

d) Weight of a patient before and after a treatment

Question 4:

Sample sizes of samples 1, 2 and 3 are 1000, 800, 600 respectively. The order of margin of error will be ___________.



a) 3 >2>1

b) 3>1>2

c) 1>2>3

d) 1=2=3

Question 5:

Which of the following is a heat-sensitive indicator of vaccine viability?

a) e-VIN

b) Freeze indicator

c) Vaccine vial monitor

d) Fridge indicator

Question 6:

Which of the following is set up for health planning at the village level?

a) Panchayat Health Committee

b) Village Health Planning and Management Committee

c) Rogi Kalyan Samiti

d) Village Health Sanitation and Nutrition Committee

Question 7:

You have gone to a subcenter as part of an audit. How many infants should be registered in a year with a health worker working there?

a) 50

b) 100

c) 150

d) 200

Question 8:

As per the national immunization schedule, which vaccines are indicated at the first visit for immunization of a 13-month-old baby who has not been previously immunized?

a) BCG, OPV 1st dose, Pentavalent 1st dose, Rota virus 1st dose

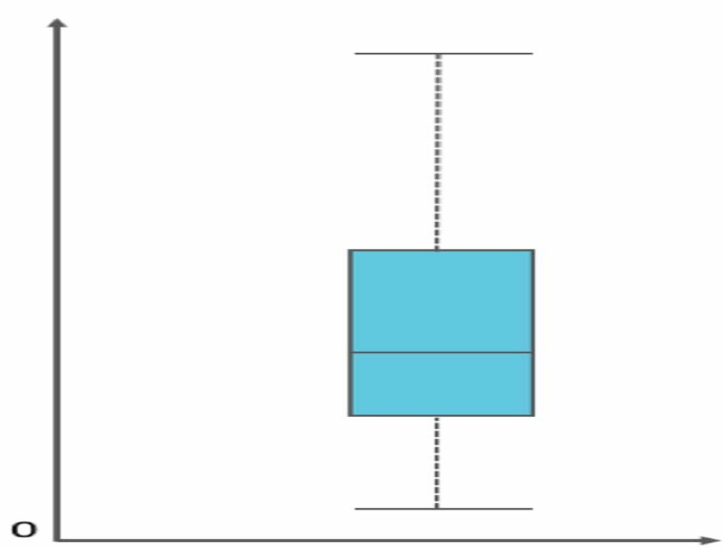
b) OPV 1st dose, Pentavalent 1st dose, Measles 1st dose

c) BCG, OPV 1st dose, Measles 1st dose

d) OPV 1st dose, DPT 1st dose, Measles 1st dose

Question 9:

Regarding the given box-whisker plot, the true statement is:



a) Mean > Median > Mode

b) Mode > Median > Mean

c) Mode = Mean = Median

d) Peaked symmetrical distribution

Question 10:

A randomised control trial is being conducted in patients with direct inguinal hernia, to compare the outcomes of a new surgery 'A' and the gold standard surgery 'B'. The p value of this trial is found to be 0.04. What can we conclude from this?

a) Type II error is small and we can accept the findings of the study

b) The probability of false negative conclusion that operation A is better than B is 4%

c) The power of the study to detect the difference between operation A and B is 96%

d) The probability of a false positive conclusion that operation A is better than B is 4%

Question 11:

A large number of tests were done in series. Which of the following will be true?

a) Sensitivity and specificity will increase

b) Sensitivity will decrease, specificity will increase

c) Sensitivity and specificity will decrease

d) Sensitivity will increase, specificity will decrease

Question 12:

Blood bag is to be disposed in:

a) Red bin

b) Yellow bin

c) Blue bin

d) Black bin

Question 13:

Phase 1 of a clinical trial is done to test:

a) Safety

b) Efficacy

c) Dose range

d) Pharmacoeconomics

Question 14:

A child was brought to you with complaints of fever, malaise and a rash. You diagnose the child with measles. What is the recommended period of isolation for this disease?

a) From onset of rash to 3 days after rash

b) 2 weeks before rash to 1 week after rash

c) From onset of rash to 5 days after rash

d) From catarrhal stage to 5 days after rash

Question 15:

BCG vaccine provides protection against which of the following?

a) Pulmonary TB

b) CNS and disseminated TB

c) Pulmonary and CNS TB

d) Pulmonary and skeletal TB

Question 16:

In a community, there are usually 40-50 reported cases of dengue every week. This week, there were 48 reported cases. This would classify as:

a) Epidemic

b) Endemic

c) Sporadic

d) Outbreak

Question 17:

Which of the following vaccines has not been introduced to the immunization programme in India?

a) Adult Japanese encephalitis

b) Influenza

c) Measles-Rubella

d) Rotavirus

Question 18:

The life cycle of which of the following is depicted in the image below?

a) Japanese encephalitis

b) Zika virus

c) Rabies

d) Nipah virus

Question 19:

While collecting blood for samples, you have spilled some on the floor. What would be the immediate next step?

a) Call the infection control team of the hospital

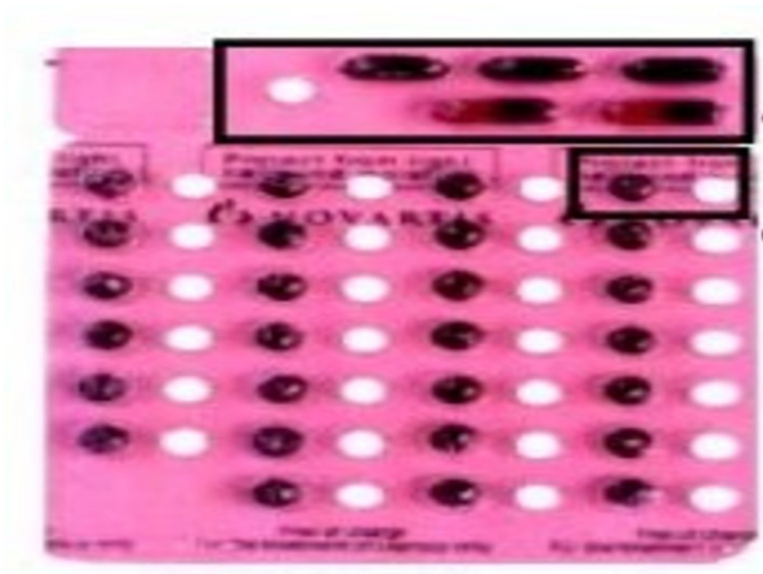
b) Mop it

c) Cover it with 1% sodium hypochlorite

d) Clean with absorbable material

Question 20:

The following drug packet is used in:



a) TB

b) Leprosy

c) Vaginal discharge

d) Gonorrhea

Question 21:

The symbol shown in the image depicts:

a) Cytotoxic waste

b) Biomedical waste

c) Radiation hazard

d) Biohazard

Question 22:

The Mid Day Meal Programme aims to supplement:

a) 1/2 protein and 1/3 calorie requirements

b) 1/3 protein and 1/3 calorie requirements

c) 1/2 protein and 1/2 calorie requirements

d) 1/3 protein and 1/2 calorie requirements

Question 23:

What test is used to check for freeze damage to vaccines?

a) Shake test

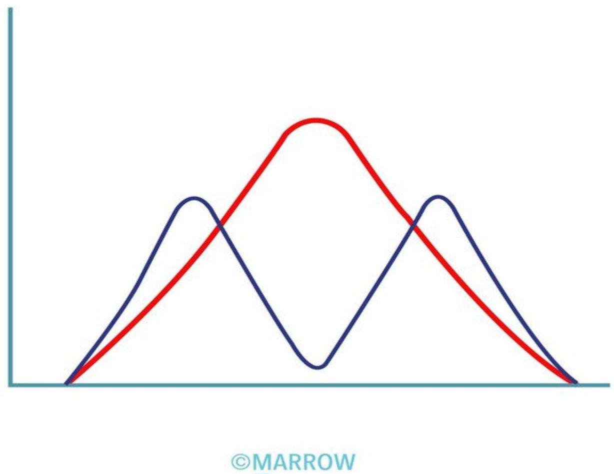
b) Schick test

c) Habel test

d) Tine test

Question 24:

Study A(red) and Study B(blue) done in a given population yields two different distribution curves as shown below. Comparing both the curves, which of the following will be true?



a) Mean = mode = median

b) Mean = median, not equal to mode

c) Mean, median, mode not equal

d) Mode = median, not equal to mean

Question 25:

A clinical trial was done between two groups to measure their blood pressure after treatment. Which of the following tests would be appropriate to use?

a) Paired t test

b) Unpaired t test

c) ANOVA

d) Chi-square test

Question 26:

Maternal mortality ratio (MMR) for a country is best given as __________

a) Maternal deaths per 100 live births

b) Maternal deaths per 10,000 live births

c) Maternal deaths per 100,000 live births

d) Maternal deaths per 1,000 live births

Question 27:

SPIKES protocol is for:

a) Triage

b) Consent for study participation

c) Hospital infection control

d) Breaking bad news

Question 28:

According to the WHO Mental Health Gap Action Programme, which of the following is not included in the Mental Health Care Act, 2017?

a) Screening of family members

b) Mobilization of resources

c) Social support

d) Interpersonal therapy

Question 29:

Which of the following is the appropriate schedule for the treatment of multi-drug resistant tuberculosis?

a) 6 drugs for 4-6 months followed by 4 drugs for 12 months

b) 7 drugs for 4-6 months followed by 4 drugs for 5 months

c) 4 drugs for 4 months followed by 6 drugs for 10-12 months

d) 5 drugs for 6 months followed by 4 drugs for 12 months

Question 30:

The POCSO Act is related to which of the following?

a) Women empowerment

b) Human trafficking

c) Juvenile delinquency

d) Child sexual abuse

Question 31:

Droplet precaution and isolation is to be done for all of the following conditions except:

a) Pertussis

b) Diptheria

c) Localized varicella zoster

d) Mycoplasma pneumonia

Question 32:

A patient presents with fever for 1 week, with abdominal distension, loss of appetite, and lymphadenopathy. There is a poor response to antibiotics and antimalarials. The Widal test is negative but rk39 is positive. Which drug can be used in the treatment of this patient?

a) Liposomal Amphotericin B

b) Fluconazole

c) Linezolid

d) Bedaquiline

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	a
5	c
6	d
7	c

8	d
9	a
10	d
11	b
12	b
13	a
14	d
15	b
16	b
17	b
18	d
19	d
20	b
21	c
22	a
23	a
24	b
25	b
26	c
27	d
28	a
29	b
30	d
31	c
32	a

Detailed Explanations

Solution to Question 1:

The most appropriate treatment for Plasmodium falciparum infection in North-Eastern states is artemether (20 mg) plus lumefantrine (120 mg).

As per National Vector Borne Disease Control Program (NVBDCP):

• Plasmodium malariae: treat as Plasmodium falciparum

• Plasmodium ovale: treat as Plasmodium vivax

• Severe and complicated malaria

-Initial parenteral treatment x 24-48 hr: quinine/artemether/artesunate/arteether

-Oral treatment after 48 hrs-after parenteral quinine: quinine + doxycyclinex 7 days or quinine + clindamycin x 7 days (pregnancy and chidren<8yrs)

-After parenteral artemisinin derivates: in NE states: ACT -AL X 3 days+primaquine (day2), in other states: ACT-SPX 3 days+primaquine (day 2)

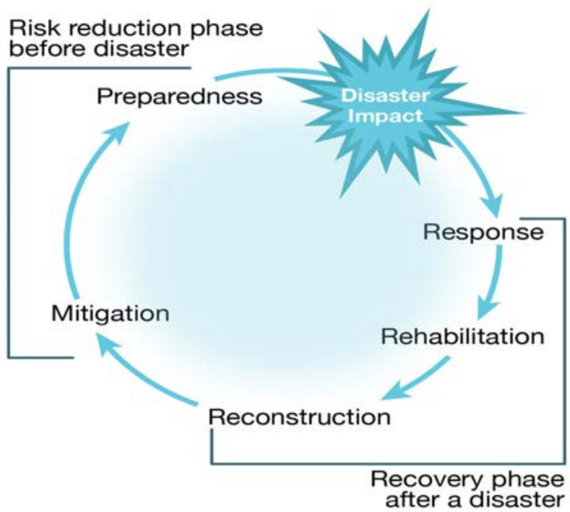
• Chemoprophylaxis

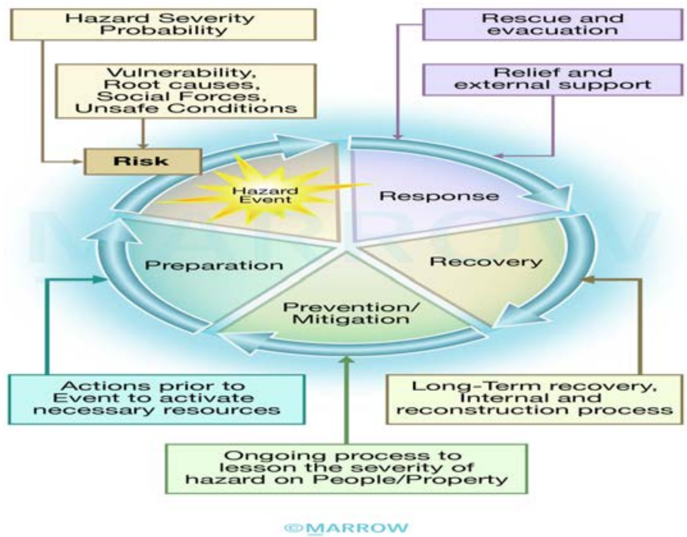
Short-term, less than or equal to 6 weeks: Doxycycline OD (start one day before travel and continue for 4 weeks after return)

Long-term, more than 6 weeks: mefloquine weekly (start 2 weeks before travel and 4 weeks after return).

Solution to Question 2:

The correct sequence of steps is disaster impact – response – rehabilitation – mitigation.





Solution to Question 3:

Chi square test is a nonparametric test that assesses the significance of association between two qualitative variables.

Among the given options, sex and type of cancer is the only pair of qualitative variables. Hence the answer.

Solution to Question 4:

Curve 1 has the least margin of error, followed by curve 2, and then curve 3. So the order is 3>2>1.

The image shows three normal distribution curves of different sample sizes, having the same mean but different standard deviations.

As the sample size is decreasing, the peak decreases and the curve becomes more flat, implying a wider confidence of interval for the given mean.

The width of the base of the curve is directly proportional to the margin of error.

Solution to Question 5:

Vaccine vial monitors (VVM) are the only heat-sensitive indicator of vaccine viability that are routinely used in vaccines throughout the supply chain.

VVM records cumulative heat exposure through a gradual change in color. If the color of the inner square is the same or is darker than the outer circle, the vaccine has been exposed to too much temperature and should be discarded.

Solution to Question 6:

Village Health Sanitation and Nutrition Committee (VHSNC) has been formed to take collective actions on issues related to health and its social determinants at the village level (local level community action).

The committee is formed at the revenue village level and it should act as a sub-committee of the gram panchayat. It should have a minimum of 15 members and ASHA residing in the village shall be the member secretary and convener of the committee.

Solution to Question 7:

150 infants should be registered in a subcentre per year.

Annual infant population = (total population) x (% infants in population or expected birth rate)

The percentage of infants in the population, or the expected birth rate, should be obtained from local data. If a specific local percentage is not available, it is estimated to be 3% of the total.

Total population in a subcenter is 5000.

Here, annual infant population = 5000 × 3/100 = 150

Solution to Question 8:

DPT-1, OPV-1 and Measles 1st dose will be given to this child.

BCG can only be given upto 1 years of age, according to the National Immunisation Schedule. BCG will not be given to this child, if the National Immunisation Schedule is followed.

Please note that BCG can be given upto 5 years of age, as part of the catch-up immunization schedule, which is put forth by the Indian Academy of Paediatrics and that is not followed in government setups.

OPV can be given till 5 years of age.

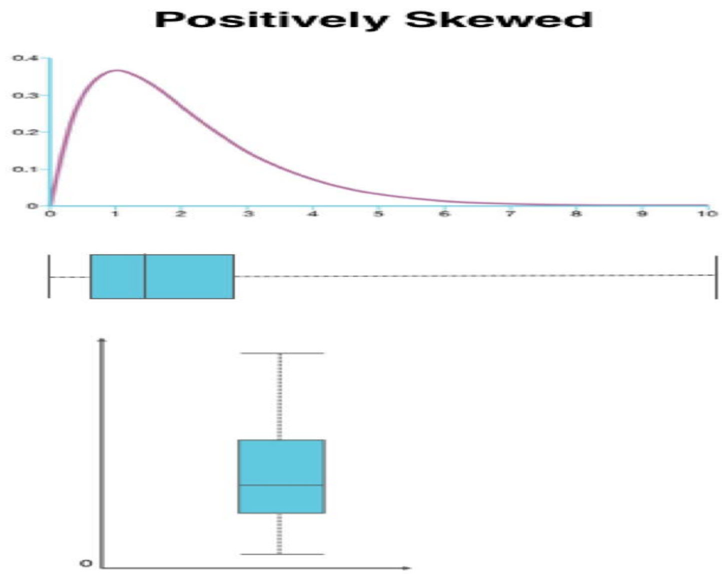
Measles can be given till 5 years of age.

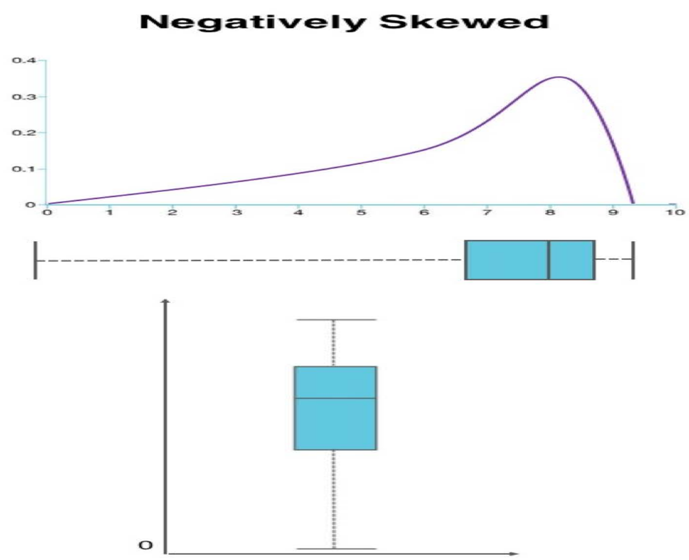
DPT can be given till 7 years of age (with booster doses).

Solution to Question 9:

The given box-whisker plot represents a positively (or right) skewed distribution. In a positively skewed distribution, the mean > median > mode.

To further understand and correlate the fact, have a look at the following images:





Solution to Question 10:

Since the p-value of the study is found to be 0.04, we can infer that the probability of a false positive conclusion that the operation A is better than B is 4%.

In this scenario, the p value is 0.04 which makes it, by convention, statistically significant. We can reject the null hypothesis and accept the alternative hypothesis that operation A is better than operation B.

Solution to Question 11:

When a large number of tests are done in series, net sensitivity is decreased and net specificity is increased.

Screening tests are done in series when the result of the first test determines whether to run the second test or not. e.g., HIV screening is done using consecutive ELISA.

Screening tests are done in parallel when the second test is independently performed regardless of what the first test result is. Both the tests are done at the same time. For e.g., In LFT, multiple tests are done together like SGOT, SGPT, total bilirubin, alkaline phosphatase, etc.

Following are the trends that are observed while evaluating screening tests:

	Tests in Series	Tests in parallel
Combined sensitivity	Decreases	Increases
Combined specificity	Increases	Decreases
Combined PPV	Increases	Decreases
Combined NPV	Decreases	Increases

Solution to Question 12:

Blood bag should be disposed off in the yellow bin.

Solution to Question 13:

The phase I of clinical trial is done to test Safety.

Solution to Question 14:

The isolation period for measles is from the catarrhal (prodromal) stage to 5 days after the onset of the rash.

The period of communicability is approximately 4 days before and 5 days after the appearance of the rash. Isolation of the patient for a week from the onset of rash adequately covers the period of communicability.

Solution to Question 15:

BCG vaccine provides protection against CNS and disseminated TB.

It does not prevent primary infection. More importantly, it does not prevent reactivation of latent pulmonary infection, the principal source of bacillary spread in the community. The impact of BCG vaccination on transmission of mycobacterium tuberculosis is therefore limited.

Solution to Question 16:

The given scenario is an example of an endemic disease.

Endemic diseases are those that have a constant presence and /or usual prevalence of a disease or infectious agent in a population within a geographic area.

Epidemic refers to an increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area.

Outbreak carries the same definition of an epidemic, but is often used for a more limited geographic area.

Sporadic diseases occur in a haphazard and irregular manner from time to time.

Solution to Question 17:

Influenza vaccine (Flu shot) has not been introduced to the National Immunization Programme in India.

The Influenza vaccine provides protection against viral flu caused by the influenza virus.

Note: Bacterial Hemophilus influenzae type-b (Hib) vaccine is different from the viral influenza (Flu) vaccine. HiB is a part of the pentavalent vaccine.

Solution to Question 18:

The given image depicts the life cycle of the Nipah virus.

Nipah virus is an RNA virus that belongs to the family Paramyxoviridae, genus Henipavirus. It is a zoonotic virus that shows both animal-to-human (bats, pigs) and human-to-human transmission. The natural hosts are fruit bats of the Pteropodidae family.

The clinical manifestations include asymptomatic subclinical infection, acute respiratory infection, and fatal encephalitis. There is no treatment or vaccine available for either people or animals. The primary treatment for humans is supportive care.

Nipah virus is on the WHO list of Blueprint priority diseases.

Solution to Question 19:

The immediate next step in this scenario is to clean the spilled blood with absorbable material.

This absorbable material is then discarded in the yellow bin. Following this, sodium hypochlorite solution is used to decontaminate the surface. The concentrations of sodium hypochlorite used are:

• < 10 mL blood spill - 1:100 dilution to decontaminate non-porous surfaces

• > 10 mL blood spill or culture spill - 1:10 dilution for the first application, followed by 1:100 dilution

Solution to Question 20:

The given image shows the adult blister packet used in the treatment of leprosy.

Solution to Question 21:

The black propellers on a yellow background is the logo for radiation hazard.

This symbol is used to let everyone be aware of the hazard in an area where there is a source of radiation. This indicates the need for precautionary measures to avoid contamination.

The other logos representing cytotoxic waste and biohazard are given in the image below.

Solution to Question 22:

Under the Mid-Day Meal (MDM) scheme, supplementary nutrition provided must cover at least 1/2 of total daily protein requirement and 1/3rd of total daily energy requirement of children in schools.

The MDM scheme is also called School Lunch Programme and has been operational since 1961 in India. Currently, it functions under the broad coverage of the Minimum Needs Programme.

Some other principles of the MDM are:

• The meal provided under the MDM should only be a supplement and not a substitute for home diets.

• The cost of meals should be reasonably low.

• The meal should be easy to cook in schools.

• Locally available foods must be used, for as far as possible.

• The menu should be regularly changed to avoid monotony.

Note: In September 2021, the Mid-Day Meal Scheme was renamed PM POSHAN or Pradhan Mantri Poshan Shakti Nirman.

Solution to Question 23:

The freeze damage to vaccines is detected by Shake test.

• A frozen vaccine is used as a control vial and the vial suspected to be damaged by freezing is taken as the test vial.

• Both are shaken after thawing and the rate of sedimentation is compared.

• If the rate of sedimentation in test vial is slower than the control vial, then it is usable and has not been damaged.

• Some of the most widely used vaccines are freeze-sensitive. Examples: Diphtheria, Tetanus, Pertussis, Liquid Haemophilus influenza type b (Hib), Hepatitis B.

Solution to Question 24:

On comparing the 2 graphs, we can say mean=median for both study A and B but the mode is different.

Study A is a unimodal symmetrical distribution, so mean=median=mode.

Study B is a bimodal distribution. Here, mean=median, but the curve has two highest points, therefore it has 2 modes.

Solution to Question 25:

In the given scenario, two groups are being compared independent of each other.
Hence, students t test/ unpaired t test should be used.

T tests are used if 2 groups are being compared. If the same group is compared before and after, paired T test is used. Otherwise, unpaired T test is used.

Solution to Question 26:

Maternal mortality ratio is the number of maternal deaths in a given period per 100,000 live births during the same period.

Solution to Question 27:

SPIKES protocol is for breaking bad news.

Each letter represents a phase in the six-step sequence.

• S - setting

• P - perception

• I - invitation or information

• K - knowledge

• E - empathy

• S - summarizing or strategizing

Key components of the SPIKES strategy include:

• Demonstrating empathy

• Acknowledging and validating the patient's feelings

• Exploring the patient's understanding and acceptance of the bad news

• Providing information about possible interventions

Solution to Question 28:

According to the WHO Mental Health Gap Action Programme (mhGAP), screening of family members is not included in the Mental Health Care Act, 2017.

The Mental Healthcare Act of 2017 has provisions that include:

• Rights of persons with mental illness to access health care, live without discrimination and enjoy confidentiality.

• Advance directives to allow mentally ill people to make decisions about their treatment

• Procedures for admission, treatment (behavioral and interpersonal therapy, medications), and discharge

• Decriminalizing suicide and prohibiting electroconvulsive therapy unless anesthesia is used

These are achieved through the mobilization of resources to required areas, and by providing social support.

Solution to Question 29:

The appropriate schedule for the treatment of multi-drug resistant tuberculosis (MDR-TB) is 7 drugs for 4-6 months followed by 4 drugs for 5 months.

Note: Pearl 1720 has been updated according to NTEP 2021 guidelines.

Solution to Question 30:

The POCSO (Protection of Children from Sexual Offences) Act is related to child sexual abuse.

POCSO Act defines a child as any person below eighteen years of age. It defines different forms of sexual abuse, including penetrative and non-penetrative assault, as well as sexual harassment and pornography.

The Act deems a sexual assault to be "aggravated" under certain circumstances, such as when the abused child is mentally ill or when the abuse is committed by a person in a position of trust or authority vis-à-vis the child, like a family member, police officer, teacher, or doctor.

People who traffic children for sexual purposes are also punishable under the provisions relating to abetment in the Act. The Act prescribes stringent punishment graded as per the gravity of the offence, with a maximum term of rigorous imprisonment for life and a fine.

Solution to Question 31:

Droplet precaution and isolation is not done for localized varicella zoster, as it is only transmitted through direct contact.

Varicella-zoster localized in the patient with an intact immune system with a lesion that can be contained/covered do not require droplet precaution. Disseminated varicella in any patient requires airborne and contact precautions.

Diphtheria, pertussis, and mycoplasma are transmitted by droplets and hence require droplet precaution and isolation.

Although patients are not contagious starting 48 hours after antibiotic therapy, droplet precautions are typically maintained until the completion of the antibiotic course and are culture-negative on two cultures taken 24 hours apart. Droplet precautions include the use of masks, gowns, gloves and protective eye gear when providing direct patient care.

Solution to Question 32:

This clinical scenario is suggestive of visceral leishmaniasis. This patient should be managed with liposomal amphotericin B.

This condition causes fever with rigor and chills associated with splenomegaly and hepatomegaly. Though lymphadenopathy is common in most endemic regions, it is rare in India.

It is diagnosed by ELISA, indirect immunofluorescent antibody test (IFAT), and rk39 which detects antibodies to Leishmania donovani.

The treatment is as follows:

• In most endemic regions:

• Drug of choice - pentavalent antimonial (sodium stibogluconate or meglumine antimoniate)

• If it fails or is unavailable - amphotericin B

• In India and Mediterranian countries, the drug of choice is liposomal AmB.

• Other drugs - paromomycin and miltefosine

Option B: Fluconazole is an antifungal and is not useful in visceral leishmaniasis

Option C: Linezolid is used in the treatment of MRSA and related infections.

Option D: Bedaquiline is a newer anti-TB drug used in MDR-TB.

Community Medicine AIIMS 2019 (May & Nov)

Question 1:

Scope of modern concept of family planning services include all of the following except:

a) Screening for cervical cancer

b) Providing services for unmarried mothers

c) Screening for HIV infection

d) Providing adoption services

Question 2:

Sandfly does not transmit:

a) Kala azar

b) Oriental sore

c) Oraya fever

d) Trench fever

Question 3:

The seasonal variations of a disease were studied in children, in a group from June to July and in another group of similar characteristics from August to September. The test of significance that should be used to compare the data is:

a) Chi square

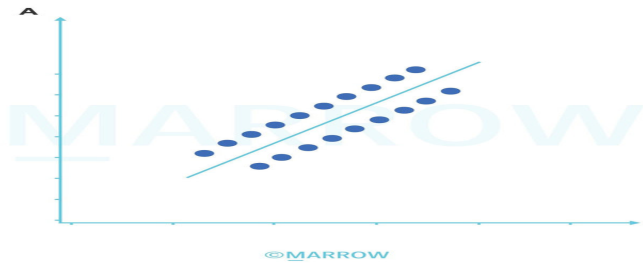
b) Paired t test

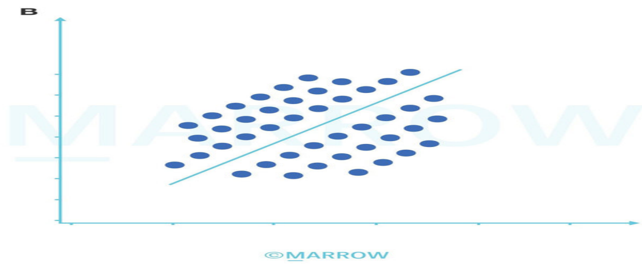
c) Wilcoxan test

d) ANOVA

Question 4:

The bone marrow density along with the gestational age is plotted from two different studies. Which of the following is true for the relation between bone marrow density and gestational age?





a) Strength of correlation is same in A and B

b) Strength of the correlation is more in A than in B

c) Strength of the variation is more in B than A

d) Variation in variables is more with A than B

Question 5:

A study was conducted to assess the effect of an intervention for certain disease. The p values for two similar studies A and B with 95% confidence limits, was found to be 0.02 and 0.001 respectively using interventions "X" and "Y" with a control group. Based on preliminary analysis, the result may be interpreted as:

a) If A, B and C are correct

b) If A and C are correct

c) If B and D are correct

d) If all (A, B, C and D) are correct

Question 6:

In a study, group A received placebo while group B received a new drug. In group A, 36 deaths were reported out of a sample of 120. In group B, 26 deaths were reported out of a sample size of 130. How many patients should be treated to avert 1 death?

a) 100

b) 10

c) 250

d) 160

Question 7:

Regarding ASHA, all are true except:

a) Malaria slide preparation

b) Ensure immunization as per schedule

c) Accompany pregnant women to hospital

d) Spread awareness about contraception

Question 8:

A study was conducted on two different samples for comparing variation between DBP and Vitamin D. The mean diastolic blood pressure is 110 with standard deviation 11 and the mean vitamin D level is 18 with standard deviation 3. Which of the following statement is true regarding the coefficient of variation (CV)?

a) CV of DBP is more than Vitamin D

b) CV of Vitamin D is more than DBP

c) Both DBP and Vitamin D have equal variability

d) It cannot be ascertained from information provided

Question 9:

Which is the study design of choice for finding out circadian variation of fat content in expressed breast milk of mother of preterm babies?

a) Prospective Cohort

b) Ambispective Cohort

c) Case control

d) Cross sectional

Question 10:

Mass vaccination is most effective in which of■ the following outbreaks?

a) Cholera

b) Typhoid

c) Polio

d) Measles

Question 11:

Program for improvement of quality of labor room comes under:

a) LaQshya

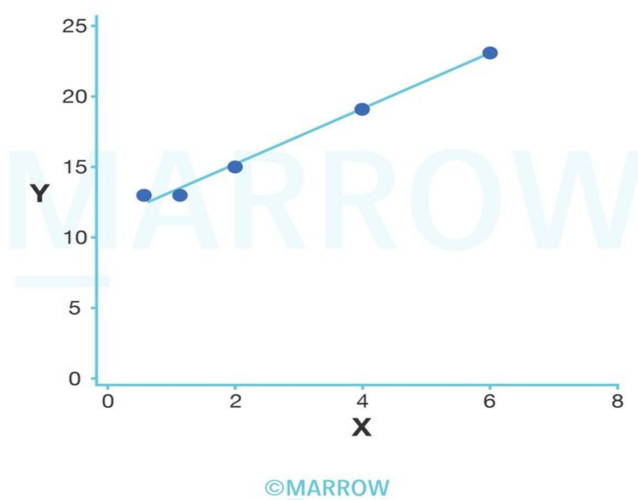
b) Ayushman Bharat

c) Janani Shishu Suraksha Karyakram

d) Pradhan Mantri Surakshit Matritva Abhiyan

Question 12:

Look at the image below and select the linear regression equation corresponding to the graph.



a) y=10-x

b) x=10+y

c) y=11+2x

d) x=11+2y

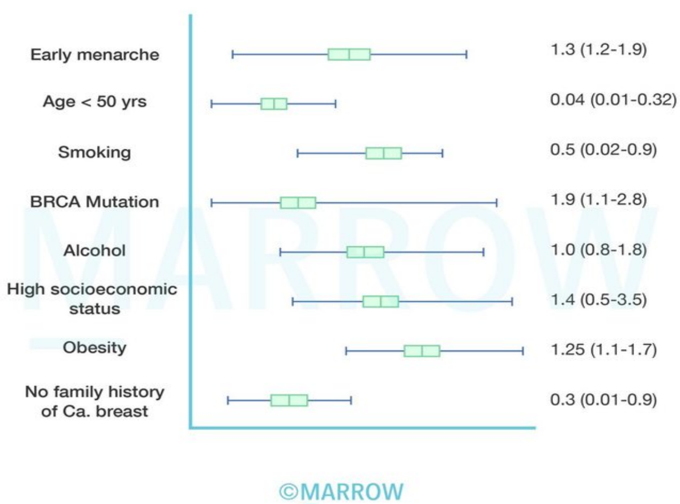
Question 13:

All of the following can be done to remove confounders except:

- **a)** Randomization
- **b)** Blinding
- **c)** Restriction
- **d)** Matching

Question 14:

A study was done to assess the risk factors for breast cancer in 100 females from urban cities. A statistical group and analysis are shown below. The values indicate the odds ratio and the 95% confidence interval. How many risk factors may be ascertained from the statistical analysis of the research?



- **a)** 3
- **b)** 4
- **c)** 5
- **d)** 6

Question 15:

In a town of 20,000 populations, total 456 births were there in a year out of which 56 were dead born. The total death were 247 out of which 56 deaths were within 28 days of life and another 34 had dies after 28 days and before completing the first birthday. Calculate the infant mortality rate of the area?

a) 197

b) 392

c) 225

d) 344

Question 16:

A nurse sustained a needle stick injury while sampling the blood of a patient who is HIV-positive. Which of the following is incorrect?

a) Washing hands with soap and water

b) Administration of tenofovir monotherapy

c) Measure baseline viral markers of health care worker immediately

d) Measure viral markers after 6 weeks

Question 17:

The renal feed of a patient suffering from end-stage renal disease (stage 5) should ideally be:

a) Low volume, high calories

b) High volume, low calories

c) High volume, high calories

d) Low volume, low calories

Question 18:

Which of the following is best method to survey the injection practices of health workers?

a) Direct observation with a structured checklist

b) Closed circuit camera surveillance

c) In-depth interview with the patient

d) In-depth interview with the healthcare provider

Question 19:

A polysaccharide vaccine against Streptococcus pneumoniae has 23 serotypes. In which of the following patients is this vaccine most beneficial?

a) A child < 2 years

b) A patient with sickle cell disease

c) A patient with cystic fibrosis

d) A patient with recurrent otitis media

Question 20:

A 5-year-old boy had received a full course rabies vaccine taken in December 2018. His pet dog which has been fully vaccinated now (November 2019) bit him on the toe. Bleeding was present at the site of the bite. What would be the line of treatment?

a) No treatment

b) 2 doses of vaccine on Day 0 & 3

c) 5 doses of vaccine with Immunoglobulin

d) 5 doses of vaccine

Question 21:

Mass drug administration is not done for which of the following conditions?

a) Worm infestation

b) Vitamin A deficiency

c) Scabies

d) Filariasis

Question 22:

Primary level of prevention for specific protection includes all of the following except?

a) Meningitis prophylaxis in exposed contacts/relatives

b) Hand washing

c) Vaccination

d) Adding iron to wheat flour

Question 23:

The treatment of paucibacillary leprosy consists of MDT (which comprises rifampicin, dapsone, and clofazimine). The latest recommended regimen is:

a) 2 drug therapy for 6 months

b) 3 drug therapy for 12 months

c) 2 drug therapy for 12 months

d) 3 drug therapy for 6 months

Question 24:

The pathogen transmitted by the vector is ?

a) Kyasanur forest disease virus

b) Orientia tsutsugamushi

c) Chandipura virus

d) Anaplasma phagocytophilia

Question 25:

The numerator of perinatal mortality rate includes ___________

a) Post-neonatal period death

b) Early neonatal death of fetus > 1000 g

c) Stillbirth of fetus < 500 g

d) Abortion after 16 weeks

Question 26:

What is the cornerstone of treatment for a patient of dengue with warning signs?

a) IV Fluids

b) Platelets

c) Cryoglobulins

d) FFP

Question 27:

When can a patient with dengue hemorrhagic shock be discharged?

a) No episode of fever for more than 24 hours with paracetamol

b) After the return of appetite

c) Urine output more than 200ml

d) After 24hrs from recovery of shock

Answer Key

Question No.	Correct Option
1	c
2	d
3	a
4	b
5	b
6	b
7	a
8	b
9	a
10	c
11	a

12	c
13	b
14	a
15	c
16	b
17	a
18	a
19	b
20	b
21	b
22	b
23	d
24	b
25	b
26	a
27	b

Detailed Explanations

Solution to Question 1:

Screening for HIV infection does not come under the purview of family planning.

The modern concepts of family planning services include:

• Proper spacing and limitation of birth

• Advice on sterility

• Education for parenthood

• Sex education

• Screening for pathological conditions related to the reproductive system (e.g. cervical cancer)

• Genetic counseling

• Premarital consultation and examination

• Pregnancy tests

• Marriage counseling

• Preparation of couples for the first child

• Providing services to unmarried mothers

• Teaching home economics and nutrition

• Providing adoption services

Solution to Question 2:

Trench fever is not transmitted by sandflies. It is transmitted by lice and caused by Bartonella quintana.

Solution to Question 3:

Chi-square test is used to test the significance of the difference between two proportions.

• Analysis of seasonal or cyclic variation of a disease provides the data, which is expressed as proportion of children affected by the disease in various months. Though the data seems to be expressed in numbers, it is actually a qualitative or nominal variable as we assess the proportion of children who are affected by the disease and those who are not affected by the disease.

• The chi-square test can be used to compare two or more groups.

Solution to Question 4:

The strength of the correlation is more in graph A than in graph B.

Graphs A and B show a positive correlation between bone marrow density and gestational age.

In graph A: The data points in the scattergram are closer to the line of best fit, indicating a strong correlation.

In graph B: The data points are scattered farther away from the line of best fit, indicating a weak correlation.

Therefore, the strength of the correlation is more in graph A than B.

Solution to Question 5:

Statements A and C are correct.

The p-value of study A with intervention X at 95% confidence interval is 0.02.

The p-value of study B with intervention Y at 95% confidence interval is 0.001

The probability (p) values of both the studies are less than 0.05 (level of significance - α(alpha)). Therefore, both studies A, and B are significant, thus, rejecting the null hypothesis.

However, study B has a lower p-value compared to study A, which suggests that the study B with intervention Y is more significant and precise than study A with intervention X.

Solution to Question 6:

10 patients need to be treated with the new drug to avert 1 death.

The number needed to treat (NNT) is defined as the number of individuals to be treated with the new drug to prevent 1 death (or to save 1 life).

NNT = 100/ARR

Where, ARR is the absolute risk reduction, which is the difference in the absolute risk rates between the placebo group and drug group. The absolute risk is usually expressed in percentage.

ARR = (No. of deaths/Sample population) x 100

Calculation:

Step 1: Calculation of absolute risk rate

The absolute risk rate in group A (placebo)

= (36/120)x100

= 30%

The absolute risk rate in group B (new drug)

= (26/130)x100

= 20%

Step 2: Calculation of absolute risk reduction (ARR)

ARR = Absolute risk rate in group A - Absolute risk rate in group B

= 30 - 20

= 10%

Step 3: Calculation of number needed to treat

NNT = 100/ARR

= 100/10

= 10

Therefore, 10 individuals need to be treated with the new drug to save 1 life.

Solution to Question 7:

Preparation of malaria slides is the responsibility of health worker male (HWM), not ASHA.

Roles and responsibilities of ASHA:

A. Health education: The ASHA provides information and awareness to the community regarding nutrition, basic sanitation and hygienic practices, healthy living and working conditions, and timely utilization of health and family welfare services.

B. Maternal and child health:

• Facilitate utilization of health services such as immunization, antenatal and postnatal checkup and supplementary nutrition, sanitation and other services provided by the government.

• Provide counseling to women on birth preparedness, safe delivery, breastfeeding and complementary feeding, immunization, contraception, and prevention of common infections.

• Escort/accompany pregnant women and children requiring treatment or admission to the nearest health center.

• Upon graded training, provide newborn care and management of common childhood illnesses.

C. Primary care:

• Provide primary care for minor ailments like diarrhea and fever and provide first aid for injuries.

• Provide DOTS under RNTCP.

• Depot holder of essential provisions such as oral rehydration therapy, iron and folic acid tablets, chloroquine, disposable delivery kits, oral pills, condoms, etc.

D. Sanitation:

• Work with village health and sanitation committee to create a comprehensive village health plan.

• Promote the construction of household toilets under total sanitation campaign

E. Notification: ASHA will inform births, deaths, and any disease outbreaks to the sub-center or PHC.

Solution to Question 8:

The coefficient of variation of vitamin D is more than that of diastolic blood pressure.

The coefficient of variation (CV) is given by the formula:

CV = 100(s/x■)

where,

CV = Coefficient of variation

s = standard deviation

x■ = mean

Substituting the data given in the question, in the above formula,

The CV of diastolic blood pressure is:

CV(DBP)= 100(11/110)

= 100(1/10)

= 10%

The CV of vitamin D is

CV(Vitamin D)= 100(3/18)

= 100(1/6)

= 16.67%

Therefore, the CV of vitamin D is more than the CV of DBP.

Solution to Question 9:

The study of choice is a prospective study.

In the above scenario, the study group would consist of the mothers of preterm babies, while the comparison group would consist of the mothers of term babies. To study the circadian variation of the fat content in the expressed breast milk, the samples of breast milk would be taken at specific time intervals every day and followed up for several days and would be analyzed. This is a prospective study.

Other options:

Option b: An ambispective cohort is the combination of both retrospective and prospective cohort studies.

Option c: In a case-control study, the study begins from the outcome and goes retrospectively to find out the exposure among the cases and controls.

Option d: Cross-sectional study is done to identify the prevalence of a disease.

Solution to Question 10:

Mass immunization is effective in case of an outbreak of poliomyelitis.

The WHO has laid down standard operating procedures to be followed for responding to a poliovirus event or outbreak.

In the case of detection of an event or outbreak, the WHO recommends the following step-wise approach:

• Notification of the disease to international health regulations (IHR) focal point at the WHO regional centre.

• Detailed investigation and risk assessment.

• Enhanced surveillance to detect ongoing person-person transmission.

• Vaccination response to reach every child.

The vaccination response includes:

• A rapid response (RR) vaccination campaign: This campaign is initiated with 14 days of the laboratory sequencing result and targets the immediate area of the virus isolation to rapidly stop further transmission. The target population is about 2 lakh to 5 lakh children.

• Supplementary immunization activities (SIA): Two high-quality large-scale SIA campaigns (SIA1and SIA2) are carried out, to ensure that >90% of children are vaccinated. These campaigns should be completed within 8 weeks of laboratory sequencing result. The target population in the large scale rounds is about 2 million children.

• Mop-up round or additional SIAs: If the monitoring suggests that children have been missed in certain areas, then a mop-up round of immunization is done.

The target age group for immunization under SIAs is children less than five years of age. If there is evidence of virus circulation among older age groups, then the target age group can be expanded to 10 or 15 years or even to the whole population depending on the local context.

Other options:

Options a: In case of a cholera outbreak, the mainstay of response includes health education on the role of hygiene, rehydration therapy, and sanitation measures.

Option b: The mainstay of typhoid control in case of an outbreak includes health education, water quality and sanitation improvement, and training of health professionals in diagnosis and treatment. Although WHO recommends immunization as a complementary method to the above measures, there is insufficient data to ascertain the effectiveness of vaccination in case of a typhoid outbreak.

Option d: In case of measles outbreak, the control measures include isolation for seven days after the onset of rash and immunization of the contacts within 2 days of exposure.

Solution to Question 11:

The Laqshya initiative launched recently by the government of India refers to initiative for labor room quality improvement.

There is enough evidence to show that improving the quality of care in labor rooms, especially on the day of birth, is central to maternal and neonatal survival.

• Laqshya is expected to improve the quality of care that is being provided to the pregnant mother in the labor room and maternity operation theatres, thereby preventing the undesirable adverse outcomes associated with childbirth.

• This initiative will be implemented in government medical colleges (MCs) besides district hospitals (DHs), and high delivery load sub-district hospitals (SDHs) and community health centers (CHCs).

• The initiative plans to conduct quality certification of labor rooms and also incentivize facilities achieving the targets outlined.

• The goal of this initiative is to reduce preventable maternal and new-born mortality, morbidity, and stillbirths associated with the care around delivery in the labor room and maternity OT and ensure respectful maternity care.

Solution to Question 12:

Among the given options, the equation $y=11+2x$ corresponds to the linear regression equation of the form $y=\alpha + \beta x$.

When 2 variables are correlated, such that the dependent variable is represented on the y-axis and the independent variable on the x-axis, then the value of y can be obtained by using the linear regression equation $y = \alpha + \beta x$, where:

y - dependent variable

α - intercept

β - slope of the line

x - independent variable

In the equation $y=11+2x$, the intercept(α) = 11, while the slope of the line (β) = 2.

Option A: $y=10-x$, represents an intercept of 10, and a slope of -1 indicating a negative correlation. However, the given image shows a positive correlation.

Options B and D: They do not fit the linear regression equation of the form $y = \alpha + \beta x$.

Solution to Question 13:

Blinding does not eliminate confounding factors. Blinding is done to eliminate the bias (such as subject bias, observer bias, or investigator bias).

A confounding factor is defined as the one associated with both exposure and disease and is distributed unequally in study and control groups.

• For example, a prospective study is being conducted to know the effect of alcohol on the development of lung cancer.

• However, individuals who drink alcohol are most likely to smoke. So, smoking is associated with the exposure. Also, smoking is an independent causative factor for lung cancer. Thus, smoking is a confounding factor.

• In the above example, if the study is conducted without eliminating the confounding factor, then the result may falsely suggest that alcohol is a causative factor for lung cancer.

Techniques to eliminate confounding factors include

• Randomization

• Matching

• Stratified randomization

• Restriction

Randomization is the random allocation of the participants
in experimental studies (Ex.randomized control trial) into the study group and control group.

Matching, which is usually done in analytical studies (for example, case–control and cohort studies), is the process where the participants in two groups are matched based on factors such as age, gender, race, smoking status, etc. to eliminate the potential confounders.

Stratified randomization, which is also known as prestratification or blocking, is the process that combines both randomization and matching techniques. In this method, the participants are first grouped into subgroups which internally match with potential confounding factors such as age, race, gender, etc. An equal number of participants form each subgroup are then allocated to the study and control groups.

Restriction: This method restricts the study to only those participants with particular characteristics. For example, a study can be restricted to participants of only one gender, race, age group, geographical area, disease severity, etc., thus eliminating the potential confounders.

Solution to Question 14:

Among the factors shown in the above statistical group, the odds ratio of early menarche, BRCA mutation, and obesity show positive association and are statistically significant. Hence these 3 factors can be ascertained as risk factors.

High socioeconomic status is not a risk factor. Even though there is a positive association, it is not statistically significant. Any association that has a 95% confidence interval range that includes the value of 1.0 is considered to be non-significant. The value of 1.0 in the range means that the two factors may co-exist together but one is not causing the other to happen.

Odds ratio (OR) is the key parameter in case-control studies to assess the strength of association between the exposure and outcome. The interpretation of the odds ratio can be done as below:

• OR > 1 indicates a positive association. If both upper and lower limits of the 95% confidence limits are also >1, it suggests that the odds ratio is significant and the factor can be ascertained as a risk factor.

• OR = 1indicates no association. Irrespective of the odds ratio, if the lower limit of 95% CI is <1 and the upper limit of 95% CI >1, the OR is non-significant and cannot be ascertained either as a protective or a risk factor.

• OR < 1 indicates a negative association. If both upper and lower limits of the 95% confidence limits are also <1, it suggests that the odds ratio is significant and the factor can be ascertained as a protective factor.

Interpretation of the individual factors given in the question:

Factor	Odds ratio(95%confidence limits)	Interpretation
Early menarche	1.3(1.2-1.9)	Positive association, significant,risk factor
Age <50 years	0.04 (0.01-0.32)	Negative association, significant, protective factor
Smoking	0.5 (0.02-0.9)	Negative association, significant, protective factor
BRCA mutation	1.9 (1.1-2.8)	Positive association, significant,risk factor
Alcohol	1.0 (0.8-1.8)	No association, non-significant
High socioeconomic status	1.4 (0.5-3.5)	Positive association, non-significant
Obesity	1.24 (1.1-1.7)	Positive association, significant,risk factor

Factor	Odds ratio(95%confidence limits)	Interpretation
No family history of Ca. breast	0.3 (0.01-0.9)	Negative association, significant, protective factor

Solution to Question 15:

The infant mortality rate is 225 per 1000 live births.

Infant mortality rate (IMR) is defined as the ratio of infant deaths registered in a given year to the total number of live births in the same year. IMR is usually expressed as a rate per 1000 live births. Infants include those under the age of 1 year.

Calculation:

• IMR = (Number of deaths of children under 1 year of age in a year x 1000)/(Number of live births in the same year).

• In the given question, Out of total 456 births, 56 were dead born. Therefore, the number of live births= 456-56 = 400.

• In the same year, it is given that the number of deaths before 28 days is 56, and the number of deaths after 28 days but before 1 year. Therefore, the total number of deaths of children under the age of 1 year is 56 + 34 = 90

• IMR = (90x1000)/400 = 900/4 = 225

• Hence, the IMR is 225 per 1000 live births.

Solution to Question 16:

Triple therapy (tenofovir 300mg + lamivudine 300mg + dolutegravir 50mg) is recommended for HIV post-exposure prophylaxis (PEP), not monotherapy.

Steps to be undertaken after a needlestick injury are:

• Wash the wound and skin with water and soap. Do not scrub and avoid antiseptics.

• Assess exposed individual:

• Check baseline viral markers

• Monitor for acute seroconversion illness within 3-6 weeks after exposure

• If testing is positive, refer to ART center

• Assess exposure source and determine the risk of transmission

• Counseling and support

• Initiate PEP at the earliest - preferably < 2 hours and ideally < 72 hours

• 28-day prescription with drug information

• HIV test at 3 months after exposure

Solution to Question 17:

The renal feed of a patient suffering from end-stage renal disease should ideally be low volume, high calories.

Dietary factors may have an effect on the progression of kidney disease and its complications. A low-volume feed is preferred in order to avoid:

• Sodium and volume overload

• Hyperkalemia

• Hyperphosphatemia

• Accumulation of toxic metabolites of protein degradation

A high-calorie feed is preferred in order to combat the patient's risk of malnutrition and wasting seen in ESRD.

Solution to Question 18:

Direct observation with a structured checklist is the best method to survey the injection practices of health workers.

Checklist of practices for assessment or evaluation include:-

• In order not to harm the patient, each procedure should be administered with a new sterile single-use device, using the right medication, vaccine or fluid for infusion.

• In order not to expose the provider to any avoidable risk, in general, any needles used during a procedure should be placed in a puncture-proof closed container immediately after use without recapping. In addition, providers of these procedures should be fully vaccinated against Hepatitis-B.

• In order that any waste produced during the performance of a procedure does not become a hazard for other people, used sharps waste and infectious non-sharps waste should be safely managed.

Solution to Question 19:

The polysaccharide vaccine against Streptococcus pneumoniae (PPSV-23) is most beneficial for patients with sickle cell disease.

S. pneumoniae is an encapsulated organism that is detected by the spleen and is removed. However, in sickle cell disease, the sickled cells block the vessels in the spleen which results in splenic infarction. This hyposplenism results in an increased risk of pneumococcal infection. Hence, pneumococcal vaccine (PPSV-23) is given to these patients as a prophylactic measure to reduce this risk.

Solution to Question 20:

This boy should receive 2 doses of vaccine on days 0 & 3 as he has received a prior course of post-exposure prophylaxis (PEP) > 3 months ago.

Management of a new dog bite in a previously immunized individual with either pre-exposure prophylaxis (PrEP) or post-exposure prophylaxis (PEP) is as follows:

• Category I: Wound treatment only

• Category II/ III:

• Wound treatment with soap and running water for 15 min and apply antiseptic

• Receive Anti-rabies vaccine (ARV) 2 doses ID (0.1 ml, 1 site)/ 2 doses IM (1 VIAL, 1 site) on days 0 and 3.

*Note: In developing countries like India, the vaccination status of animals alone, doesn't play a significant role in determining the initiation of prophylaxis in humans.

Solution to Question 21:

Mass drug administration is not done in case of vitamin A deficiency.

Mass Drug Administration (MDA) is the administration of a drug to an entire concerned population irrespective of age and gender. For example, antimalarials are given as MDA in malaria-endemic island communities and in low-endemic non-island settings approaching elimination, where there is minimal risk of re-introduction of infection, good access to treatment, and implementation of vector control and surveillance.

Vitamin A prophylaxis is given under National Immunization Schedule (NIS) only to children up to 5 years. It is given as mass prophylaxis and not as a drug for treatment. Vitamin A as a treatment is given only to a child with features of vitamin A deficiency and such doses are not instituted to everyone in the community. Therefore, it is not considered under MDA.

On the other hand, MDA is done in scabies (ivermectin), worm infestation (albendazole), and filariasis (DEC + albendazole).

Solution to Question 22:

Handwashing is not a specific protection measure under primary prevention. All other options are protection measures against specific diseases.

Solution to Question 23:

The latest recommendation is 3 drug therapy for 6 months to treat paucibacillary (PB) leprosy by WHO. The regimen consists of rifampicin, dapsone, and clofazimine.

Meanwhile, multibacillary (MB) leprosy is treated using the same 3 drugs for a duration of 12 months. While the NLEP guidelines earlier mentioned a 2-drug MDT for PB leprosy, the National Strategic Plan and Roadmap for Leprosy (2023-2027) now recommends a uniform MDT of 3 drugs for both PB and MB leprosy.

Note: The central government has approved a new treatment regimen for paucibacillary (PB) leprosy. The Ministry of Health and Family Welfare has introduced a 3-drug regimen for PB leprosy instead of a 2-drug regimen. This revised drug regimen will be applicable in India from April 1, 2025.

Solution to Question 24:

The given image is of a Trombiculid mite and it transmits Orientia tsutsugamushi.

Orientia tsutsugamushi is an obligate, intracellular bacterium. It causes scrub typhus disease in humans.

Hard ticks are covered by a chitinous shield on their dorsal surface which is called scutum. They have a head or capitulum present at the anterior end, which is visible from above. Hard ticks transmit certain diseases like tick typhus, viral encephalitis, hemorrhagic fever, etc.

Solution to Question 25:

Early neonatal death of fetus > 1000 g is included in the numerator of perinatal mortality rate .

Perinatal mortality rate is described as late foetal and early neonatal deaths weighing over 1000 grams at birth expressed as a ratio per 1000 live births weighing over 1000 gram at birth.

$$\text{Perinatal mortality rate} = \frac{\text{Late foetal and early neonatal deaths weighing over 1000g at birth}}{\text{Total live births weighing over 1000g at birth}} \times 1000$$

Solution to Question 26:

The primary modality of treatment in patients with dengue with warning signs is IV fluids.

The main pathology in dengue is plasma leakage resulting in intravascular volume depletion and thereby shock. Warning signs in dengue include:

• Abdominal pain or tenderness

• Persistent vomiting

• Clinical fluid accumulation (ascites, pleural effusion)

• Mucosal bleeding

• Lethargy or restlessness

• Hepatomegaly >2 cm

• Increase in hematocrit concurrent with a rapid decrease in platelet count

Inpatient management is advised to all patients of dengue with warning signs. The management of dengue includes:

• IV crystalloids - mainstay of treatment

• Platelet transfusion - severe thrombocytopenia (<10,000/mm3) and active bleeding

FFP and cryoglobulins are not the preferred for treatment in dengue.

Solution to Question 27:

A patient with dengue hemorrhagic shock can be discharged after the return of appetite.

Criteria for the discharge of patients recovering from dengue:

• Absence of fever for at least 24 hours without the use of antipyretic drugs

• Platelet count >50,000/cu.mm

• Return of appetite

• Adequate urine output

• Visible clinical improvement

• Minimum of 2-3 days after recovery from shock

• No respiratory distress from pleural effusion or ascites

Community Medicine AIIMS 2020 (May and Nov INI-CET)

Question 1:

A resident doctor gets pricked with a needle while attending an HIV positive patient. The regimen to be given is?

a) Zidovudine + Lamivudine

b) Zidovudine + Lamivudine + Nevirapine

c) Zidovudine + Lamivudine + Indinavir

d) Zidovudine + Nevirapine

Question 2:

Airborne transmission precaution is necessary for all of the following except:

a) Tuberculosis

b) Herpes Zoster

c) Measles

d) Chicken Pox

Question 3:

The maternal mortality ratio is defined as?

a) Maternal Deaths per 10,000 live births

b) Maternal Deaths per 100,000 live births

c) Maternal Deaths per 10,000 women in reproductive age group

d) Maternal Deaths per 100,000 women in reproductive age group

Question 4:

What is the dosage of iron and folic acid for adult women under the Anemia Mukth Bharath initiative?

a) 60 mg elemental iron + 500 mcg folic acid daily

b) 60 mg elemental iron + 500 mcg folic acid weekly

c) 100 mg elemental iron + 500 mcg folic acid weekly

d) 100 mg elemental iron + 500 mcg folic acid daily

Question 5:

Which of the following is true regarding nikshay poshan yojana?

a) Rs 500 is given per week

b) Rs 500 is given per month

c) Mid day meals are given

d) Dietary advice is given

Question 6:

Diphtheria vaccine is a:

a) Polysaccharide vaccine

b) Toxoid

c) Killed vaccine

d) Live attenuated vaccine

Question 7:

Under RBSK screening which one of the following diseases is not covered?

a) Congenital cataract

b) Congenital glaucoma

c) ROP

d) Vitamin A deficiency

Question 8:

The serum level of tetanus antitoxin that gives clinical protection is?

a) >0.1 IU/ml

b) >0.01 IU/ml

c) >0.001 IU/ml

d) >0.0001 IU/ml

Question 9:

Which of the following is true about the reference woman used in the calculation of the recommended daily allowance (RDA)?

a) Weight 55 kgs

b) Height 1.73 m

c) Works 10 hours/day

d) Age of 18-25 years

Question 10:

Which of the following methods is not effective against preventing the spread of COVID 19?

a) Face mask

b) 1% glutaraldehyde

c) 70% alcohol

d) Hand washing

Question 11:

According to biomedical waste management, used pacemakers should be disposed of in which of the following colored bag?

a) Yellow

b) White

c) Black

d) Blue

Question 12:

In which of the following bags are latex gloves disposed of after use?

a) Yellow

b) Red

c) Blue

d) Transparent

Question 13:

Which of the following is a non-parametric test?

a) Student t test

b) Friedman test

c) ANOVA

d) Pearson

Question 14:

Glasgow coma scale is a type of?

a) Nominal scale data

b) Ordinal scale data

c) Alphanumerical data

d) Interval scale data

Question 15:

A new drug combination of chemotherapy and immunotherapy for metastatic melanoma was prescribed by a radiotherapist. It prolongs the survival. Which of the following is true in this situation?

a) Incidence increases and prevalence reduces

b) Incidence reduces and prevalence remains the same

c) Incidence remains the same and prevalence increases

d) Incidence reduces and prevalence increases

Question 16:

The HIV sentinel surveillance includes all of the following populations except

---

a) Single male migrants

b) Truck drivers

c) STD clinic attendees

d) Antenatal woman

Question 17:

You are a medical officer at a PHC. Which of the following would you advise for mass prophylaxis/ treatment for anemia?

a) $FeSO_4$ + Vit C + Folate + Vit B12

b) Iron (III) hydroxide polymaltose

c) Ferrous fumarate

d) Inj. iron sucrose

Question 18:

Which of the following quantitative methods is the best to study previous research studies?

a) Cohort study

b) Systematic review

c) Meta-analysis

d) RCT

Question 19:

Discrete probability distribution that expresses probability of a given number of events in a particular time frame is known as:

a) Binomial distribution

b) Gaussian distribution

c) Poissons distribution

d) Normal distribution

Question 20:

The data which best explains the difference between the highest and the lowest values is:

a) Variance

b) Coefficient of variation

c) Range

d) Interquartile range

Question 21:

Which of the following is a marker of the heat stability of vaccines?

a) VMV

b) VVM

c) MVV

d) VMM

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	b
5	b
6	b
7	b
8	b
9	a
10	b
11	d
12	b
13	b
14	b
15	c
16	c
17	a
18	c

19	c
20	c
21	b

Detailed Explanations

Solution to Question 1:

Amongst the given options, a three-drug combination of Zidovudine+Lamivudine+Indinavir is preferred.

Three drug combination is preferred as per the US Public health service guidelines, WHO guidelines, and MOHFW 2020 guidelines.

A PEP regimen for HIV should have Lamivudine + Tenofovir + Lopinavir/ritonavir to be taken for 4 weeks after exposure. It should ideally be started within 2 hours and definitely within 72 hours. Here as the Lopinavir/ritonavir is not part of the options, one can choose, the indinavir-based combination as the best of the available options.

Nevirapine is not preferred for use in PEP. Previous NACO 2018 guidelines mentioned the use of 2 drug combinations but it is no longer preferred.

Exposures that may warrant occupational PEP include:

• Parenteral or mucous membrane exposure (splashes to the eye, nose, or oral cavity)

• The following bodily fluids may pose a risk of HIV infection: blood, blood-stained saliva, breast-milk, genital secretions, and cerebrospinal, amniotic, rectal, peritoneal, synovial, pericardial, or pleural fluids.

Exposures that do not require PEP include:

• When the exposed individual is already HIV-positive

• When the source is established to be HIV-negative

• Exposure to bodily fluids that do not pose a significant risk: tears, non-blood-stained saliva, urine, and sweat.

To ensure no transmission has occurred, follow-up with HIV antibody testing should be done for at least 6 months post-exposure (e.g. at baseline, 6 weeks, 3 months, and 6 months).

Note: This is an AIIMS 2020 recall question. It may not reflect the latest guidelines. Kindly refer to the attached pearl for the same.

Solution to Question 2:

Measles, Tuberculosis, and chickenpox require airborne transmission precaution as does the disseminated herpes zoster.

The patient should be made to wear a mask for source control. A fit-tested NIOSH-approved N95 or higher level respirator should be used by the healthcare personnel whilst care delivery.

CDC recommends the display of signs similar to the one below in room doors of patients with the above conditions:

Solution to Question 3:

Maternal mortality ratio is defined as the number of maternal deaths per 100,000 live births during a given time period.

Maternal mortality rate is the number of maternal deaths per 100,000 women of reproductive age in a given time period.

Solution to Question 4:

According to the Anemia Mukt Bharat initiative, adult women need to take 60 mg elemental iron + 500 mcg folic acid tablets once a week. These tablets are sugar-coated, and red in color.

Solution to Question 5:

Under nikshay poshan yojana, a financial incentive of Rs.500 per month is provided for nutritional support to each notified TB patient for the duration of anti-TB treatment.

For pediatric TB patients who do not have bank accounts, the money shall be deposited in parent's/guardian's accounts whose details are to be sought and entered in nikshay against the patient's records.

The processing of payment is on the 7th of every month.

Solution to Question 6:

Diptheria vaccine is a toxoid.

Toxoids: Certain organisms like diphtheria and tetanus bacilli produce exotoxins. The toxins produced by these organisms are detoxicated and used in the preparation of vaccines.
The antibodies produced to neutralize the toxic moiety produced during infection, rather than acting upon the organisms. In general, toxoid preparations are highly efficacious and safe immunizing agents.

Solution to Question 7:

All of the above conditions are included in child health screening & early intervention services under RBSK (Rashtriya Bal Swasthya Karyakram) except congenital glaucoma.

The RBSK aims at the universal screening of the 4 'D's i.e Defects at birth, Deficiencies, Diseases, Developmental delays including disabilities that would lead to early identification, timely management, and ultimately a reduction in morbidity, mortality, and lifelong disability in children.

Health conditions selected for the child (0-18 years) health screening and early intervention services are:

Defects:

• Neural tube defect

• Down's syndrome

• Cleft lip and palate

• Talipes

• Developmental dysplasia of the hip

• Congenital cataract, congenital deafness, congenital heart disease

• Retinopathy of prematurity

Deficiencies:

• Anemia especially severe anemia

• Vitamin A deficiency

• Vitamin D deficiency

• Severe acute malnutrition

• Goitre

Diseases:

• Scabies, fungal infections, and eczema

• Otitis media

• Rheumatic heart disease

• Reactive airway disease

• Dental caries

• Convulsive disorders

Developmental delays and disabilities:

• Vision, hearing, and neuro-motor impairment

• Motor, cognitive and language delay

• Autism

• Learning disorder

• Attention deficit hyperactivity disorder

Solution to Question 8:

The serum level of tetanus antitoxin that gives clinical protection is 0.01 IU/ml.

Tetanus is best prevented by tetanus toxoid. It provides active immunization by stimulating the production of protective antitoxin. Vaccination should be aimed at maintaining the serum protective level of antitoxin (approximately 0.01 IU/ml) throughout life and covering the entire community.

Solution to Question 9:

The reference woman used in the calculation of recommended daily allowance (RDA) weighs about 55 Kgs and is aged between 19-39 years, nonpregnant, non-lactating with a height of 1.61 meters, and a BMI of 21.2, is free from disease and physically fit for active work.

On each working day, she must be engaged in 8 hours of occupation/work, which usually involves moderate activity. When not at work she spends 8 hours in bed, 4-6 hours in sitting and moving about, 2 hours in walking, and in active recreation or household duties.

Solution to Question 10:

1% glutaraldehyde is not effective against preventing the spread of COVID 19.

Effective everyday practices that prevent the spread include actions like keeping a social distance (at least 6 feet) from others, washing your hands with soap and water for 20 seconds or using a hand sanitizer with at least 70% alcohol, and wearing a face mask when you have to go out in public.

Solution to Question 11:

According to biomedical waste management, used pacemakers should be disposed of in a blue colored bag. All the metallic body implants and glasswares are discarded in a blue colored bag.

Solution to Question 12:

Latex gloves are disposed of in red-colored bag after use.

Solution to Question 13:

Friedman test is a non-parametric test. It is used to test for differences between groups when the dependent variable being measured is ordinal.

Other options:

Student t-test, analysis of variation (ANOVA) test, and pearson correlation are all based on quantitative data and hence grouped under parametric tests.

Solution to Question 14:

Glasgow coma scale is a type of ordinal data as the GCS score can be graded as mild, moderate, and severe based on the order of severity.

Solution to Question 15:

By the new drug combination, the survival of the patient has increased, hence incidence remains the same while prevalence increases.

Incidence: Number of new cases in a defined population in a specified period of time.

Prevalence: Number of old and new cases at a given point of time over a period of time.

Options A, B, D: The treatment only prolongs the disease, it has nothing to do with the occurrence of cases thereby nothing to do with incidence.

Solution to Question 16:

Sentinel surveillance of HIV is usually not done for STD clinic attendees.

Sentinel surveillance is used to augment the notification systems used in the surveillance of diseases.

Till 2012-13, STD sites were also included in the HIV sentinel surveillance but from 2014-15, these sites were entirely discontinued, since the STD data had significant biases.

The study population for HIV sentinel surveillance includes:

• Female sex worker: Women who are engaged in consensual sex for money or payment in kind, as a means of livelihood in the last six months

• Men who had anal or oral sex with a male partner in the last month.

• Injection drug users: Men and women who use addictive substances or drugs for recreational or non-medical reasons, through injections, at least once in the last three months.

• Hijra/Transgender: Person whose identity does not conform unambiguously to conventional notions of male or female gender roles, but combines or moves between these.

• Single male migrants: Single male, living at a place other than the place of usual residence without his spouse or family, for the purposes of work and visiting his home town at least once a year.

• Long-distance truckers: Truckers who travel more than 800 km one way between source and destination.

• A pregnant woman of age 15-49 years and attending the antenatal clinic for the first time.

Solution to Question 17:

Among the given options, $FeSO_4$ + vitamin C + Folate + vitamin B12 may be considered for mass prophylaxis against anemia.

Vit C enhances the absorption of iron from the stomach by acting as a reducing substance.

Subclinical folate deficiency is seen in up to 30% of North Indian pregnant women. Moreover, it may be masked by widespread iron deficiency. It is, for this reason, iron and folate supplementation is carried out in national programmes.

Clinical vitamin B12 deficiency is not commonly encountered, but subclinical cases might be seen in up to 30% of adults and children due to widespread restriction to consumption of vegetarian products only. Hence supplementing vitamin B12 is a good idea provided the cost factor is not an issue.

Option B: Iron (III) hydroxide polymaltose was marketed as being superior to iron sulfate in terms of efficacy, but it lacked any data to back up the evidence.

Option C: Ferrous fumarate may be considered for the treatment of iron deficiency anemia with similar efficacy as that of ferrous sulfate.

Option D: Injection iron sucrose or any parenteral form is not used as a first-line option. It is only indicated if the oral form is not tolerated.

Solution to Question 18:

The quantitative method which is the best to study previous research studies of a selected topic/research question is meta-analysis.

A systematic review is a process of collecting, reviewing, and presenting information from earlier research to answer a question. It is a qualitative technique. It may or may not involve a meta-analysis.

Solution to Question 19:

Poissons distribution is the probability distribution that describes the chances of an event in a particular time frame.

For example, if you plot the number of patients visiting the OPD of your hospital for a week on a chart, you can conclude that on a particular day(say Monday) there are more number of patients. And if the pattern of patients is similar you can estimate the number of patients in OPD next week or so.

It is usually done for rare events and is depends on $\mu = \lambda t$ (where μ defines the number of events in happening in time t)

Other options

Gaussian distribution is another name for Normal distribution

Binomial distribution examples consist of those events which can only have two possible results given as yes or no / success or failure. It is usually done for finite number of events unlike Poissons distribution.

Solution to Question 20:

Range gives the difference between the highest and lowest values in a given set of data.

In a dataset of marks of students in a subject, if the highest score is 100 and the lowest is 51, the range would be 49.

Option A: Variance (var or SD^2) is the mean of the squares of all the deviation scores of the distribution. It is used to measure how far data is spread out.

Option B: Coefficient of variation (CV) is used to compare the variability. For example, whether the pulse varies more in old or young and growth varies more in boys or girls. It is calculated by expressing SD as a percentage of the mean, i.e., CV = SD/mean x 100. Since it is expressed as a percentage it doesn't have any units.

Option D: Quartiles divide the distribution into 4 equal groups. The difference between mid values of each quartile gives the interquartile range.

Solution to Question 21:

Vaccine Vial Monitors (VVM) are heat-sensitive indicators that are routinely used to monitor vaccine viability as the vaccine vial moves across the supply chain.

VVM records cumulative heat exposure through a gradual change in color. If the color of the inner square is the same or is darker than the outer circle, the vaccine has been exposed to too much temperature and should be discarded.

Community Medicine NEET 2018

Question 1:

The Chadah committee recommended all of the following except:

- **a)** PHC at block level
- **b)** One PHC for 50,000 population
- **c)** One basic health worker per 10,000 population
- **d)** Responsibility of PHCs in malaria eradication

Question 2:

Which of the following is the most peripheral centre under the Revised National Tuberculosis Control Programme organization structure?

- **a)** District TB centre
- **b)** Intermediate Reference Laboratory
- **c)** Tuberculosis Unit
- **d)** Designated Microscopy Centre

Question 3:

Which of the following is true about Factories Act?

- **a)** Children under 14 years of age must not be employed in factories
- **b)** A child of 16 years can be employed only between 6 AM and 8 PM
- **c)** The maximum working hours for an adult is 72 hours per week
- **d)** A child of 16 years can work till 5 hours per day

Question 4:

The dose of iodine for lactating women in India according to the recommended dietary allowance is:

- **a)** 200 µg/day
- **b)** 220 µg/day

c) 250 µg/day

d) 280 µg/day

Question 5:

Which of the following is true for non-parametric tests?

a) ANOVA is an example of a non parametric test

b) It is used for skewed distributions

c) It involves the assumption that the data has a normal distribution

d) It cannot be used for small sample sizes

Question 6:

Which article of the Indian Constitution confers Right to Life to the citizens of India?

a) Article 11

b) Article 21

c) Article 23

d) Article 25

Question 7:

According to the WHO, what percentage of immunization coverage is needed for eradication of measles?

a) 68%

b) 72%

c) 84%

d) 96%

Question 8:

In a well-designed cancer study, similar remission rates were obtained for a new treatment (33.2%) and the usual treatment (32.2%), with the p-value being 0.04. Which of the following statements is true regarding the study?

a) Both the treatments are equally effective

b) Neither of the treatments is effective

c) The new treatment is more effective than the usual treatment

d) The information given is not adequate to compare the efficacy of the treatments

Question 9:

A 9-month-old baby with diarrhea is feeding well and is thirsty. Skin pinch when done returns in 2 seconds. What is the treatment?

a) 200-400 ml of ORS to be given in the first four hours.

b) 400-700 ml of ORS to be given in the first four hours.

c) 700-900 ml of ORS to be given in the first 4 hours.

d) In addition to the usual fluid intake, 100-200 ml should be given after each loose stool.

Question 10:

What value of BMI is considered as 'lethal' in men?

a) < 12

b) < 13

c) < 14

d) < 15

Question 11:

Which of the following is not used in the assessment of body fat?

a) Quetelet's index

b) Total body potassium

c) Total body water

d) Breslow index

Question 12:

What is the ratio of the incidence of the disease in the exposed to that of the non-exposed?

a) Attributable risk

b) Odds ratio

c) Relative risk

d) Population attributable risk

Question 13:

In order to reach the demographic goal of NRR=1, the couple protection rate should exceed?

a) 50%

b) 40%

c) 70%

d) 60%

Question 14:

Which is the new online software launched under the national tuberculosis elimination program (NTEP) by the government to monitor TB patients?

a) NIKSHAY

b) NISCHAY

c) E-DOTS

d) NIKUSTH

Question 15:

Cytotoxic and expired drugs are disposed by:

a) Dumping

b) Autoclave

c) Incineration

d) Chemical disinfection

Question 16:

In a screening test for DM out of 1000 population, 90 were positive. However, a gold standard test for DM was done of which 100 were positive. Calculate the sensitivity of the screening test?

a) 90/100

b) 100/110

c) (90-10)/1000

d) 90/1000

Question 17:

The incidence of a disease is 4 per 1000 of the population with an average duration of 2 years. What is the prevalence?

a) 8/100

b) 2/1000

c) 4/1000

d) 8/1000

Question 18:

The probability that infection will occur among susceptible persons following contact with an infectious person within the incubation period is known as the_______.

a) Secondary attack rate

b) Case fatality rate

c) Primary attack rate

d) Tertiary attack rate

Question 19:

Which of the following does not cause hardness of water?

a) Calcium carbonate

b) Calcium sulphate

c) Calcium bicarbonate

d) Magnesium bicarbonate

Question 20:

Which of the following is not a method of direct transmission of communicable diseases?

a) Vertical transmission

b) Contact with soil

c) Droplet nuclei

d) Droplet transmission

Question 21:

What should be the minimum interval between the administration of two live vaccines?

a) 2 weeks

b) 4 weeks

c) 8 weeks

d) 12 weeks

Question 22:

Kala-Azar is endemic in all these areas except:

a) West Bengal

b) UP

c) Bihar

d) Assam

Question 23:

What is the unit of observation in ecological studies?

a) Population

b) Patient

c) Healthy inividuals

d) Case

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	d
5	b
6	b
7	d
8	c
9	b
10	b
11	d
12	c
13	d
14	a
15	c
16	a
17	d
18	a
19	a
20	c
21	b
22	d
23	a

Detailed Explanations

Solution to Question 1:

The recommendation of one PHC for a 50,000 population was given by Kartar Singh Committee.

The recommendations given by the Chadah Committee of 1963:

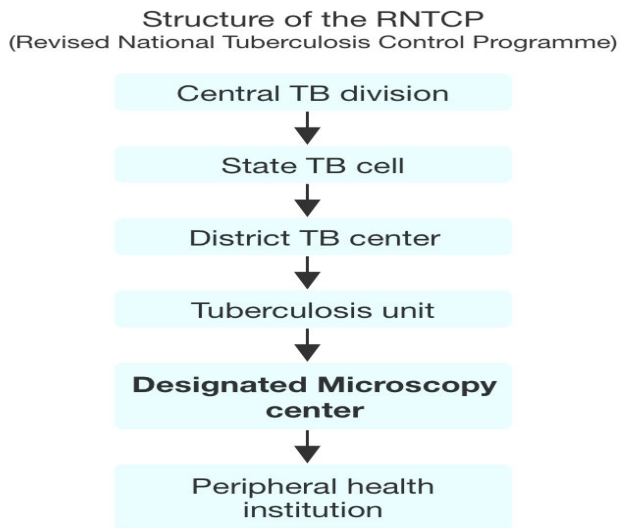
• PHC at the block level

• One basic health worker per 10,000 population

• Responsibility of PHCs in malaria eradication

Solution to Question 2:

Among the options given, Designated Microscopy Centre is the most peripheral centre.



Solution to Question 3:

Factories act prohibits children of <14 years from being employed.

Children between 15–18 years can be employed only in certain industries and work only between 6 AM–7 PM. They should not be made to work for more than four and a half hours.

The maximum working hours for an adult industrial worker should be not more than 48 hours per week, not exceeding 9 hours a day; if overtime is paid, then it should be not more than 60 hrs per week.

Solution to Question 4:

According to the National Institute of Nutrition (NIN) 2020 guidelines, the recommended dietary allowance (RDA) for dietary iodine in lactating women in India is 280 µg/day.

NIN guidelines on recommended dietary allowance for iodine per day:

Note: According to the WHO guidelines, the RDA during both pregnancy and lactation is 250 mcg.

Age	Male	Female
Birth to 6 months	100 mcg	100 mcg
7–12 months	130 mcg	130 mcg

Age	Male	Female
1–9 years	90-120 mcg	90-120 mcg
10–12 years	150 mcg	150 mcg
14–18 years	150 mcg	150 mcg
> 19 years	150 mcg	150 mcg
Pregnancy	—	250 mcg
Lactation	—	280 mcg

Solution to Question 5:

Non-parametric tests are tests that do not assume that the data is normally distributed. Thus, they are used for skewed (asymmetric) distributions {skewed either to the lesser side (left-sided skewed) or towards the greater side (right-sided skewed)}.

The chi-square test is an example of a non-parametric test that is used to test hypotheses in relation to nominal scale data or categorical data.

ANOVA is an example of a parametric test.

Solution to Question 6:

The Right to Life is given in Article 21 of the Indian Constitution.

Option A: Article 11 defines the right of the parliament to regulate the right of citizenship by law.

Option C: Article 23 defines the prohibition of traffic in human beings and forced labor.

Option D: Article 25 defines the freedom of conscience and free profession, practice, and propagation of religion.

Solution to Question 7:

According to the WHO, eradication of measles requires achieving an immunization coverage of at least 96% of children under 1 year of age.

Solution to Question 8:

As the p-value is 0.04, we can conclude that the new treatment is more effective than the usual treatment.

Even though both treatments have similar rates of remission, a valid statistical test is required to determine the difference in their effectiveness. Statistically, p-value < 0.05 implies that the alternate hypothesis is true.

In this scenario, we can consider:

• Null hypothesis: Both the treatments are equally effective (in other terms, there is no difference between the effect of the two treatments).

• Alternative hypothesis: New treatment is more effective than the usual treatment.

As the p-value is 0.04, which is statistically significant (<0.05), the null hypothesis is rejected and the alternative hypothesis is accepted. Hence the new treatment is more effective than the usual treatment.

Solution to Question 9:

According to the IMNCI classification, the baby has some dehydration and comes under Plan B. The amount of ORS given is 75ml/kg body weight.

Here, the weight of the child is not specified. Hence, 400-700 ml of ORS at home is to be given in the first four hours according to the age of the child. The below table shows the amount of ORS to be given according to the age of the child.

In addition to this, 50-100 ml extra fluid is given after each loose stool. After four hours, the baby is reassessed and the appropriate plan is selected again for the continuation of the treatment.

The four rules of home treatment are:

• Give extra fluid

• Give zinc supplements (1 tablet per day for 14 days, dissolved in a small amount of ORS or breast milk or clean water)

• Continue breastfeeding

• Counsel the mother to return if there is blood in stools and if the child is feeding poorly

Age	under 4 months	4-11 months	1-2 years	2-4 years	5-14 years	15 years and over
Weight(kg)	under 5	5-7.9	8-10.9	11-15.9	16-29.9	30 or over
ORS solution (ml)	200-400	400-600	600-800	800-1200	1200-2200	2200-4000

SIGNS	CLASSIFY AS	IDENTIFY TREATMENT (Urgent pre-referral treatments are in bold print.)
Two of the following signs: ● Lethargic or unconscious ● Sunken eyes ● Not able to drink or drinking poorly ● Skin pinch goes back very slowly	SEVERE DEHYDRATION	➤ If child has no other severe classification: — Give fluid for severe dehydration (Plan C). OR **If child also has another severe classification:** **— Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way.** **Advise the mother to continue breastfeeding** ➤ **If child is 2 years or older and there is cholera in your area, give antibiotic for cholera.**
Two of the following signs: ● Restless, irritable ● Sunken eyes ● Drinks eagerly, thirsty ● Skin pinch goes back slowly	SOME DEHYDRATION	➤ Give fluid and food for some dehydration (Plan B). ➤ **If child also has a severe classification:** **— Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way.** **Advise the mother to continue breastfeeding** ➤ Advise mother when to return immediately. ➤ Follow-up in 5 days if not improving.
Not enough signs to classify as some or severe dehydration.	NO DEHYDRATION	➤ Give fluid and food to treat diarrhoea at home (Plan A). ➤ Advise mother when to return immediately. ➤ Follow-up in 5 days if not improving.

Solution to Question 10:

A body mass index (BMI) of 13 or less is considered lethal in men.

However, females can survive even at a lower BMI up to 11.

Solution to Question 11:

Breslow's staging is used for the staging of malignant melanoma, not for the assessment of body fat.

Solution to Question 12:

The ratio of incidence of disease or death among exposed to that among the non-exposed is called relative risk (RR).

Solution to Question 13:

In order to reach the demographic goal of NRR=1, the couple protection rate (CPR) should exceed 60%. This coincides with every couple having 2 children.

Solution to Question 14:

Nikshay is the new online software launched under the national tuberculosis elimination program for monitoring TB patients.

Nikusth is the new online software for monitoring leprosy patients.

Nischay is a pregnancy test kit.

Solution to Question 15:

Cytotoxic and expired drugs are collected in a yellow bag and then disposed of by incineration.

Expired cytotoxic drugs are to be returned to the manufacturer and subjected to incineration, encapsulation, or plasma pyrolysis. Other discarded medicines can be disposed by incineration.

Solution to Question 16:

The screening test only identified 90 people as positive for diabetes, while the gold standard test identified 100 people. This means that out of the 100 confirmed, 90 have been identified as truly positive by the screening test and 10 are falsely negative. Therefore,

• True positive = 90

• False negative = 100-90= 10

• Sensitivity = TP/(TP+FN) =90/(90+10) = 90/100 = 90%

Solution to Question 17:

The prevelance in the given situation is 8/1000 population.

Prevalence is given by the formula :

Prevalance = Incidence x Mean Duration

Prevalence = 4/1000 x 2 = 8/1000 population

Solution to Question 18:

The probability that infection will occur among susceptible persons following contact with an infectious person within the incubation period is called secondary attack rate.

It is calculated by the following formula:

Secondary attack rate (SAR) = (Number of exposed or susceptible persons developing the disease within the incubation period x 100)/(Total number of close or exposed contacts that are susceptible)

Primary cases are usually excluded from the numerator and denominator.

Secondary attack rate measures the communicability of the disease.

Solution to Question 19:

Calcium carbonate does not cause hardness of water. Hardness of water is caused by bicarbonates and sulphates of calcium and magnesium.

There are two types of hardness of water:

• Temporary (carbonate) hardness is due to bicarbonates of calcium and magnesium.

• Permanent (non-carbonate) hardness is due to sulphates, chlorides and nitrates.

Solution to Question 20:

Airborne transmission by droplet nuclei is not a method of direct transmission of diseases.

It is a method of indirect transmission of diseases.

Droplet nuclei are tiny particles of size 1-5 microns range that may be formed by either due to evaporation of droplets coughed or sneezed into the air or are formed from aerosols.

These droplet nuclei remain airborne for long periods of time and can transmit diseases like tuberculosis, influenza, chickenpox, measles, Q fever and other respiratory infections.

Droplet transmission, vertical transmission, and contact with soil are methods of direct transmission of diseases.

Differences between droplet and droplet nuclei are summarised below:

DROPLET NUCLEI	DROPLET
Indirect transmission	Direct transmission
It is the residue from evapora ted droplets	It is produced while coughing, sneezing, talking
<5microns in diameter	≥5 microns in diameter
No need for close contact for transmission	Requires closeness and direct contact for transmission
Diseases transmitted: chicke npox, measles etc.	Diseases transmitted: diphth eria, meningococcal meningit is etc.

Solution to Question 21:

The minimum interval between the administration of two live vaccines is 4 weeks.

This is because decreasing the interval between two doses may interfere with the antibody response and protection. Thus, live vaccines can either be given on the same day according to the schedule or 4 weeks apart.

Note: In Park 27th ed. the minimum interval between two live vaccines is given as 3 weeks. However, several other standard sources including CDC and UNDP suggested an interval of 4 weeks. When both are given in options, it is recommended to mark 4 weeks as the correct answer.

Solution to Question 22:

Kala-Azar is not endemic in Assam. It is endemic in the following states in 54 districts:

• Uttar Pradesh

• West Bengal

• Bihar

• Jharkhand

Solution to Question 23:

In ecological studies, the unit of observation is the population or community.

Type of study	Unit
Ecological/ Correlational	Population
Cross-sectional/ Prevalence	Individual
Case-control/ Case reference	Individual
Cohort/ Follow up	Individual
Randomised Controlled Tria ls	Patients
Field Trial	Healthy peo ple
Community trial	Community

Community Medicine NEET 2019

Question 1:

A 4-year-old male child developed sore throat and difficulty in swallowing and mild fever. On examination, the clinical picture shown below is seen. Which of the following sentences is false regarding the vaccine that could have prevented this condition?

a) Its administered on the anterolateral aspect of thigh

b) A prior adverse reaction with temperature above 37 degrees is a contraindication

c) Cellular pertussis component in the vaccine is not recommended after 7-years of age

d) The diphtheria toxoid dose in adults is much lesser than the pediatric dose

Question 2:

Which of the following parameters would you use to check the efficiency of the surveillance system for malaria under the National Vector Borne Disease Control Programme?

a) Annual Parasite Index

b) Annual Blood Examination Rate

c) Slide positivity rate

d) Slide falciparum rate

Question 3:

Which of the following is false regarding confounding factor?

a) It can be reduced by matching

b) It is associated individually with both cause and effect

c) It is distributed equally in both study and control groups

d) It is associated with the exposure of the study

Question 4:

Which of the following is the best level of prevention of breast cancer?

a) Specific protection

b) Early diagnosis and treatment

c) Disability limitation

d) Rehabilitation

Question 5:

What is the dosage of vitamin A given for a 2-year-old baby with keratomalacia?

a) 2,00,000 IU immediately, followed by same dose after 24 hours

b) 1,00,000 IU immediately, followed by the same dose after 1 week

c) 1,00,000 IU immediately, followed by the same dose after 24 hours and after a week

d) 2,00,000 IU immediately, followed by the same dose 24 hours later and after 2 weeks

Question 6:

In a normal curve, what is the percentage of distribution of one standard deviation?

a) 68%

b) 34%

c) 99%

d) 95%

Question 7:

According to IMNCI, a 6-month-old child is said to have pneumonia if the child has fast breathing with a respiratory rate more than:

a) 40

b) 60

c) 30

d) 50

Question 8:

Which of the following is part of the concurrent list of the Indian Constitution?

a) Adulteration of food

b) Fishing and fisheries beyond territorial waters

c) Regulating labor in mines

d) Public health and sanitation

Question 9:

The principal unit of administration in India is ______

a) Village

b) Centre

c) District

d) State

Question 10:

Cross product ratio can be arrived from:

a) Ecological study

b) Cohort study

c) Cross sectional study

d) Case control study

Question 11:

In controlled tipping, the amount of land required for the trench method to be practiced for a population of 10,000 is:

a) 4 acres

b) 5 acres

c) 2 acres

d) 1 acre

Question 12:

The test of choice to measure the difference in means between two groups in a study, with one group being a control is:

a) Chi-square test

b) Z test

c) Unpaired T test

d) Paired T test

Question 13:

The test of significance done for two or more proportions:

a) Chi- square test

b) Student's test

c) Z test

d) ANOVA test

Question 14:

If a patient tests positive for a disease, which of the following represents the probability of her actually having the disease?

a) Sensitivity

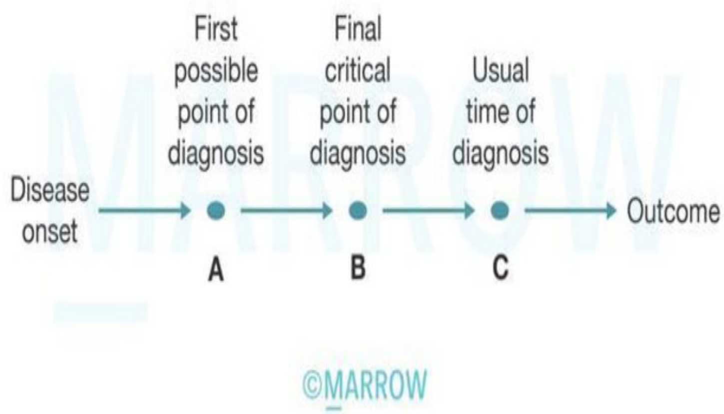
b) Specificity

c) Positive predictive value

d) Negative predictive value

Question 15:

In the following natural history of disease flowchart, point A to C is called as?



a) Screening time

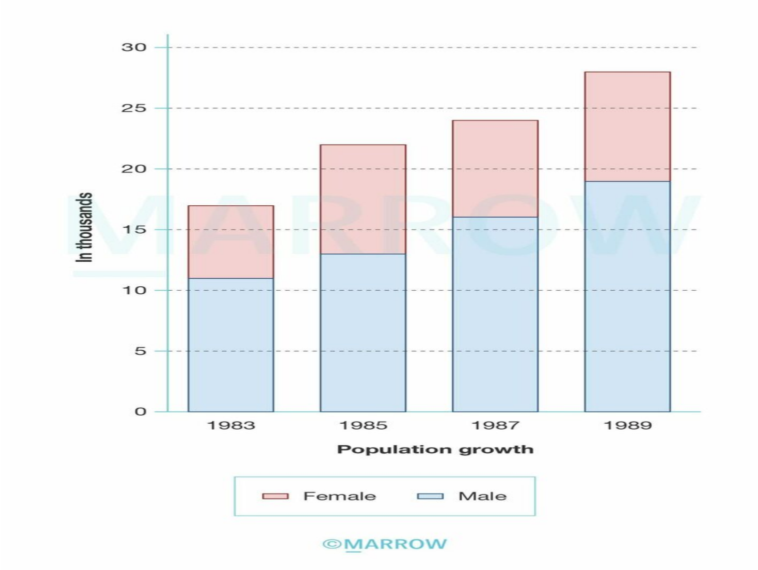
b) Lead time

c) Lag time

d) Generation time

Question 16:

The following diagram charts the population change over years from 1983 to 1989 in a given place. What is the statistical diagram used here?



- **a)** Histogram
- **b)** Line diagram
- **c)** Component bar graph
- **d)** Multiple bar graphs

Question 17:

Triage is done during disaster management to ensure treatment of casualties is done properly. What is true about triage done at the disaster site?

- **a)** First come first treated basis
- **b)** Green labels are for those who need to be transported on priority
- **c)** Moribund patients receive the lowest priority
- **d)** Most commonly used system is three color code system

Question 18:

The risk of genetic diseases in consanguineous marriage between first cousins is?

- **a)** 1-2%
- **b)** 4-8%
- **c)** 8-10%
- **d)** 12-14%

Question 19:

A paradoxical carrier is defined as:

a) A person who sheds pathogens during incubation period

b) A carrier who acquired pathogen from another carrier

c) A person who acquired pathogen from patient

d) A patient who became a carrier

Question 20:

Which of the following vaccine is contraindicated in pregnancy?

a) Hepatitis A

b) Hepatitis B

c) Rabies

d) Varicella

Question 21:

Which of the following is not a personal protective equipment?

a) Goggles

b) Gloves

c) Face shield

d) Lab coat

Question 22:

What is the best way to plot the changes in incidence of a disease in a given area over time?

a) Line graph

b) Histogram

c) Ogive

d) Tree diagram

Question 23:

Which of the following is false about Rubella infection?

a) It is a type of droplet infection

b) Vertical transmission is possible

c) Infection in early pregnancy causes a milder form of disease in the fetus

d) Fetus affected in late pregnancy may have only deafness

Question 24:

A resident doctor sustained a needlestick injury while sampling blood of a patient who is HIV positive. A decision is taken to offer him post-exposure prophylaxis. Which one of the following would be the best recommendation?

a) Zidovudine + Lamivudine for 4 weeks

b) Lamivudine + Tenofovir + Efavirenz for 4 weeks

c) Lamivudine + Tenofovir + Dolutegravir for 4 weeks

d) Zidovudine + Lamivudine + Nevirapine for 4 weeks

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	b
5	d
6	a
7	d
8	a
9	c
10	d
11	d
12	c
13	a
14	c

15	b
16	c
17	c
18	b
19	b
20	d
21	d
22	a
23	c
24	c

Detailed Explanations

Solution to Question 1:

The image shows a dirty white pseudomembrane classic of diphtheria. A prior adverse reaction with a temperature above 37 is not a contraindication.

Precautionary measures are taken when the temperature is above 40 degrees, but it is not considered a contraindication for subsequent doses of DPwT.

Anaphylaxis is a contraindication for subsequent vaccination with diphtheria, pertussis, or tetanus vaccine. Encephalopathy following vaccination is a contraindication for pertussis vaccine. DT is given for successive doses.

Events such as inconsolable cry, fever more than 40.5°C, seizures with or without fever are considered as precautions, and not contraindications for DTwP.

The cellular component of pertussis is not recommended after 7 years.

• Unvaccinated child 5-7 years: Give DPT 1, 2 and 3 at 1-month intervals. Administer booster dose of DPT at a minimum of 6 months

after administering DPT-3 up to the age of 7 years.

• Unvaccinated child above 7 years: Administer Tdap (Tetanus and diphtheria toxoids and acellular pertussis) at 0, and Td (Tetanus, Diphtheria vaccine) at 1 and 6 months, respectively.

The adult dose of diphtheria toxoid is almost 12 times lesser in comparison with the pediatric dosage.

Note: IAP guidelines state that "temperature more than >40.5°C is a precaution and not contraindication". Park's 27th Ed states that "temperature above 40°C is a contraindication to DPT". This topic is controversial, however, IAP guidelines are commonly followed by practitioners.

Solution to Question 2:

The Annual Blood Examination Rate (ABER) is used to check the efficiency of the surveillance system for malaria. It takes into account the number of slides examined for the population under surveillance.

Solution to Question 3:

A confounding factor is a factor that is associated with both exposure and diseases and is distributed unequally in study and control groups. It can be reduced by matching.

For example, if the study is that a sedentary lifestyle can lead to coronary artery diseases, then age can act as a confounding factor as it is individually linked to both sedentary lifestyle and coronary artery diseases.

Solution to Question 4:

Early diagnosis and treatment is the best level of prevention of breast cancer. It is a secondary level of prevention.

The earlier a disease is diagnosed and treated, the better is the prognosis. The occurrence of secondary cases or any long term disability can be prevented. This is of particular importance in chronic diseases.

Early diagnosis and treatment is not as effective and economical as primary prevention. However, it may be critically important in reducing the high morbidity and mortality in diseases such as essential hypertension, cancer cervix and breast cancer.

Solution to Question 5:

The recommended vitamin A dosage for a 2-year-old child with keratomalacia is 2,00,000 IU given immediately, followed by the same age-specific dose 24 hours later and after 2 weeks.

Solution to Question 6:

Approximately 68% of a normal distribution falls within ±1 standard deviations of the mean. This is as per the empirical rule of a normal distribution.

Approximately 95% of the distribution falls within ±2 standard deviations of the mean.

Approximately 99.7% of the distribution falls within ±3 standard deviations of the mean.

Solution to Question 7:

Three key clinical signs are used to assess a sick child with cough or difficult breathing:

• Respiratory rate (RR): distinguishes children who have pneumonia from those who do not. Children are classified as pneumonia if:

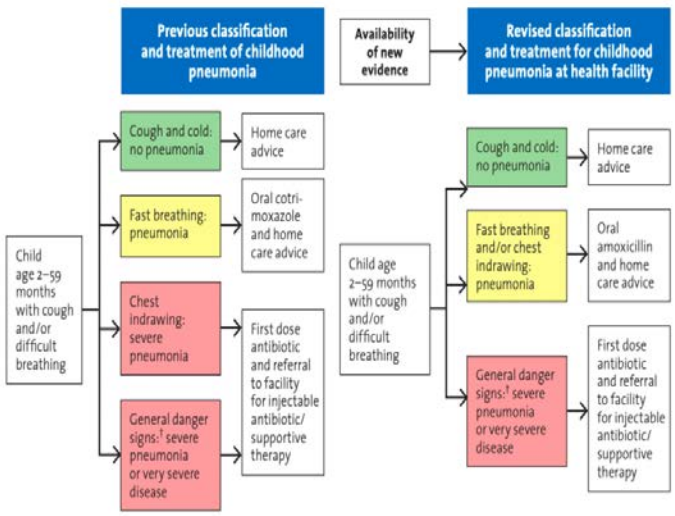
• > 50/min RR for infants 2 months to 12 months

• > 40/min RR for children 12 months to 5 years of age

• Lower chest wall indrawing indicates severe pneumonia

• Stridor indicates those with severe pneumonia requiring hospital admission.

The following image gives the IMNCI classification for cough/difficulty in breathing in children.



† Not able to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition.

Signs of severe pneumonia are :

• Not able to drink

• Persistent vomiting

• Convulsions

• Lethargic or unconscious

• Stridor in a calm child

• Severe malnutrition

Solution to Question 8:

Adulteration of food is included under the concurrent list.

The legislative section of the seventh schedule is divided into three lists:

• Union list - a list of 100 items on which the parliament has exclusive power to legislate

• State list - a list of 61 items on which the state government has exclusive power to legislate

• Concurrent list - a list of 52 items on which both central and state governments have the power to legislate

Solution to Question 9:

• The principal unit of administration in India is the district under a Collector.

• Under districts are the sub-divisions, tahsils or talukas, municipalities or corporations, villages, panchayats, etc.

• As of the 2019 census, there are 725 districts in India.

Solution to Question 10:

Cross product ratio (Odds ratio) can be obtained from the case-control study.

Solution to Question 11:

In controlled tipping, one acre of land is required by the trench method to be practiced for a population of 10,000.

It is also known as a sanitary landfill. It is the most satisfactory method of refuse disposal where suitable land is available. In this method, waste material is placed in a trench or other prepared area. Then, it is adequately compacted and covered with earth at the end of the day.

Methods used in controlled tipping are:

• Trench method: Done in places where level ground is available

• Ramp method: Well suited where the terrain is moderately sloping

• Area method: Used for filling land depressions, disused quarries, and clay pits

Solution to Question 12:

The test of choice to measure the difference in means between two groups in a study, with one group being a control is unpaired T test.

Solution to Question 13:

Chi-Square(χ2) test is a test used for proportions and associations.

It is a non-parametric (non-numeric) test and it is used to test hypotheses in relation to nominal scale data or categorical data.

Solution to Question 14:

Positive predictive value implies the probability of a person with a positive test, actually having the disease.

It is the ability of a test to detect true positive(TP) out of total positive(TP+FP) and it is directly proportional to the prevalence of a disease.

PPV= TP/TP+FP

Solution to Question 15:

Lead time is the period between the first possible time of detection (point A) and the usual time of diagnosis (point C).

Screening time is the time difference between the first point of diagnosis and the critical point of diagnosis.

Diagnosis of a disease before the critical point has a good outcome and reduces the severity by reversing the disease process. Diagnosis beyond the critical point treatment would be unsuccessful and/or permanent damage would have occurred.

Screening must be done in conditions where there is a considerable time lag between the onset of disease and the usual time of diagnosis. That is conditions with a long lead time.

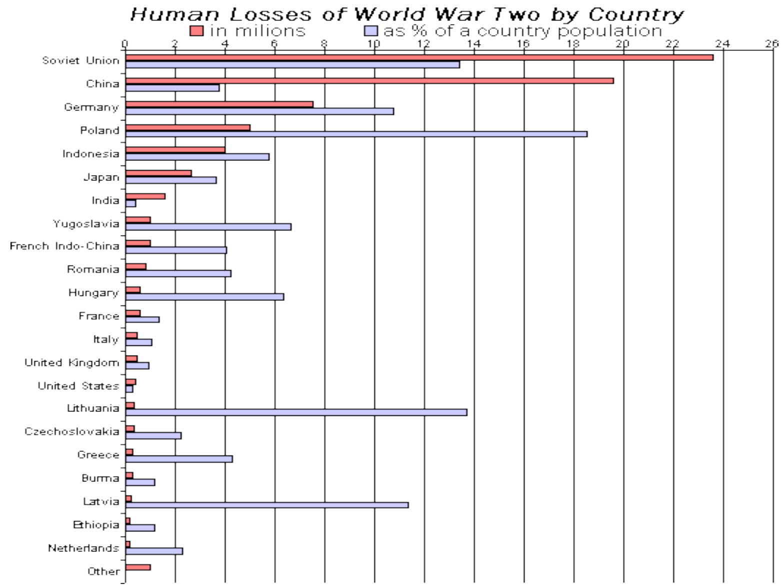
Generation time is defined as the interval of time between the onset of infection by a host and the maximal infectivity of that host.

Solution to Question 16:

The given image shows a component bar graph. It is used to represent data in which the total magnitude is divided into different subsets or components.

A bar chart or bar graph is used mainly for nominal or ordinal scale data. They can be simple, multiple or component bar graphs.

The image given below shows a multiple bar graph showing the human losses of world war II by country. In a multiple bar chart, 2 or more sets of data are shown simultaneously.



A bar chart can be distinguished from a histogram because the rectangles are separated by the spaces. This indicates that the data forms discrete categories.

In a histogram, the rectangles are not separated by spaces as the data represented forms a continuous set instead of discrete groups.

Solution to Question 17:

Moribund patients receive the lowest priority.

This is because despite paying quite a lot of attention, the benefit is still questionable.

Solution to Question 18:

The risk of genetic diseases in consanguineous marriage between first cousins is 4-8%.

The risk of genetic diseases in non-consanguineous marriage is 2-4%.

Solution to Question 19:

A paradoxical carrier is a carrier that acquired a pathogen from another carrier.

Incubatory carriers are those who shed the infectious agents during the incubation period of the disease.

Convalescent carriers are those who continue to shed the disease agent during the period of convalescence.

Healthy carriers are those who had a subclinical infection and developed a carrier state without suffering from overt disease.

Solution to Question 20:

Varicella vaccine is contraindicated in pregnant women as it is a live vaccine.

The live attenuated varicella virus vaccine is currently recommended for children between 12 to 18 months of age who have not had chickenpox.

Solution to Question 21:

A lab coat is not a personal protective equipment.

Personal Protective Equipment (PPE) are:

• Gloves – protect hands.

• Gowns/ aprons – protect skin and clothing.

• Masks protect the mouth/nose and respirators protect the respiratory tract from airborne infectious agents.

• Goggles – protect eyes.

• Face shields – protect face, mouth, nose, and eyes.

Solution to Question 22:

Line graphs or line diagrams are the best way to plot the changes in the incidence of a disease in a given area over time.

Solution to Question 23:

Earlier the rubella infection occurs in pregnancy, greater is the damage to the foetus.

The major determinant of the extent of fetal infection in congenital rubella syndrome is the gestational age at which fetal transmission occurs.

Rubella infection in 1st trimester can cause:

• Abortions

• Still birth

• Skin lesions - blueberry muffin rash

• Congenital rubella syndrome

Solution to Question 24:

Triple therapy with tenofovir 300mg, lamivudine 300mg, and dolutegravir 50mg for 4 weeks is recommended for HIV post-exposure prophylaxis (PEP).

Steps to be undertaken after a needlestick injury are:

- Wash the wound and skin with water and soap. Do not scrub and avoid antiseptics.
- Assess exposed individual:
- Check baseline viral markers
- Monitor for acute seroconversion illness within 3-6 weeks after exposure
- If testing is positive, refer to ART center
- Assess exposure source and determine the risk of transmission
- Counseling and support
- Initiate PEP at the earliest - preferably < 2 hours and ideally < 72 hours
- 28-day prescription with drug information
- HIV test at 3 months after exposure

Community Medicine NEET 2020

Question 1:

Sharp waste is disposed in:

- **a)** Yellow bag
- **b)** Red bag
- **c)** Blue container
- **d)** White container

Question 2:

According to the Factories Act 1948, what is the maximum work hours per week including overtime work hours?

- **a)** 48 hours
- **b)** 72 hours
- **c)** 60 hours
- **d)** 66 hours

Question 3:

Which of the following was launched to ensure complete immunization of children?

- **a)** Mission Dhanush
- **b)** Mission Indradhanush
- **c)** Mission Indira Yojana
- **d)** Mission Suraksha Yojana

Question 4:

Which of the following is an example of prospective screening?

- **a)** Cervical Pap smear in a 40 year old patient
- **b)** Neonatal screening of a new-born baby for hypothyroidism

c) Screening of immigrants to a country

d) Urine for sugar screening in a 40 year old man

Question 5:

Monetary benefit is measured in which of the following analyses?

a) Cost effective analysis

b) Cost benefit analysis

c) Network analysis

d) Input Output analysis

Question 6:

A 24-month-old child weighing 11 kg was brought to the pediatric OPD. He had a respiratory rate of 38/min. On examination, chest indrawing was observed. What is the next step of management?

a) Oral amoxicillin for five days

b) Urgent referral to the tertiary care centre immediately

c) Administer IV antibiotics and ask the patient to be brought after 24 hours

d) Start antipyretics urgently

Question 7:

An employee diagnosed with TB gets extended sickness benefit for:

a) 1 year

b) 2 years

c) 3 years

d) 4 years

Question 8:

Chemical waste is disposed in _____ bags.

a) Yellow

b) White

c) Red

d) Green

Question 9:

Which of the following techniques is based on behavioral sciences?

a) Decision making

b) Systems analysis

c) Network analysis

d) Management by objective

Question 10:

A study was done to assess nutritional status among young children in rural and urban areas. 30 children in rural and 20 children in urban areas were malnourished. Which of the following tests of significance is the most appropriate in this case?

a) Chi Square test

b) Paired T test

c) Standard error of the mean

d) ANOVA

Question 11:

As per NCEP- ATP III, which of the following is not a criterion for metabolic syndrome?

a) Hypertriglyceridemia

b) High LDL

c) Central obesity

d) Hypertension

Question 12:

In a population study, the value of the mean is 200. If the standard deviation is 20, 68% of the population will lie between?

a) 180- 220

b) 160- 240

c) 170- 230

d) 190- 210

Question 13:

Which of the following is disposed in the yellow bag?

a) Blood bag

b) Gloves

c) Sharps

d) Urine bag

Question 14:

Calculating one variable using another variable is done by

a) Coefficient of correlation

b) Coefficient of Regression

c) Coefficient of variation

d) Coefficient of determination

Question 15:

The best indicators for routine monitoring of air pollution are:

a) Sulfur dioxide, smoke, and suspended particles

b) Sulfur dioxide, lead, and particulate matter

c) Sulfur dioxide and carbon monoxide

d) Carbon monoxide and hydrogen sulfide

Question 16:

A person wants to study about the relationship between smoking and lung cancer, so he collects data about people diseased with lung cancer from government hospitals and the number of cigarette packets sold at the same time period. Which type of study is this?

a) Cross sectional

b) Ecological

c) Experimental

d) Quasi-experimental

Question 17:

Project MONICA is related to:

a) Cardiovascular diseases

b) Cervical Cancer

c) Breast Cancer

d) Road Traffic Accidents

Question 18:

Which of the following compounds is predominantly responsible for the disinfecting property of bleaching powder?

a) Hypochlorite ion

b) Hydrochloric acid

c) Hypochlorous acid

d) Chloride ion

Question 19:

The difference between the incidence of disease among exposed and incidence among non exposed is given by which of the following variables?

a) Relative risk

b) Attributable risk

c) Population Attributable risk

d) Odds Ratio

Question 20:

In Sustainable Development Goal 3, the target 3.1 aims to reduce the Maternal Mortality Ratio by 2030 to ____ (per 1 lakh live births):

a) <70

b) <100

c) <50

d) <130

Question 21:

A patient comes with a ■history of clean-cut injury which is not lacerated. He says that he received his last dose of tetanus vaccination about 10 years ago. What should be given next?

a) Full course of tetanus vaccination

b) Single dose tetanus toxoid

c) Tetanus toxoid + immunoglobulin

d) No vaccination needed

Question 22:

The variation in a data set is compared with that of another data set using:

a) Variance

b) Coefficient of variation

c) Standard error of mean

d) Standard deviation

Question 23:

Pandemic that occurred in 2009 was due to which influenza virus?

a) H1N1

b) H5N1

c) H5N7

d) H3N2

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	c
5	b
6	a
7	b
8	a
9	d
10	a
11	b
12	a
13	a
14	b
15	a
16	b
17	a
18	c
19	b
20	a
21	b
22	b
23	a

Detailed Explanations

Solution to Question 1:

Sharp waste is disposed of in the puncture and leak-proof white containers.

Solution to Question 2:

Maximum work hours per week including overtime work hours is 60 hours.

Solution to Question 3:

Mission Indradhanush was launched by the Ministry of Health and Family Welfare in 2014 to fully immunize children (>90%) who are either partially immunized or completely unimmunized by 2020. It was later modified in 2017 to the Intensified Mission Indradhanush, to achieve the same goal by December 2018.

Intensified Mission Indradhanush includes 12 vaccine-preventable diseases (updated from the 7 diseases that Mission Indradhanush focussed on). 4 rounds of immunization are proposed to be held for 7 days each month (from the 7th- 13th of each month).

The diseases included in Mission Indradhanush are:

• TB

• Polio

• Hep B

• Diphtheria

• Pertussis

• Tetanus

• Measles

The diseases included recently in Intensified Mission Indradhanush are:

• Rotavirus diarrhoea

• Rubella (as MR vaccine)

• H. influenzae B diseases (as part of the pentavalent vaccine)

• Pneumococcal meningitis

• Japanese Encephalitis

The Intensified Mission Indradhanush 2.0:

The government of India introduced IMI 2.0 to accelerate the vaccine drive and extend the coverage to children and pregnant women in identified districts and blocks from December 2019-March 2020.

Note: Currently, the government has completed Intensified Mission Indradhanush 5.0:

Intensified Mission Indradhanush (IMI 5.0), the flagship routine immunization campaign of the Union Ministry of Health and Family Welfare concluded all 3 rounds in October 2023. This is the first campaign to be conducted across all the districts in the country and includes children up to 5 years of age (Previous campaigns included children up to 2 years of age). It also had a special focus on improvement of Measles and Rubella vaccination coverage.

Solution to Question 4:

Screening of immigrants to a country is an example of prospective screening.

Screening of disease is done to detect unrecognized disease or defect, in apparently healthy individuals, by means of rapidly applied tests, examinations, or other procedures.

The difference between prospective and prescriptive screening is given below:

Note: Most common neonatal disease to be screened for is neonatal hypothyroidism.

	Prospective screening	Prescriptive screening
Definition	People screened for others be nefit	People screened solely for the ir own benefit
Essential purpose	Disease control	Case detection
Request for screenin g	Specific request from the aut hority	Request from physician
Example	Immigrantsfor infectious dise ases like tuberculosis syphilis, yellow fever, etc;HIV screeni ngduring antenatal visits, sex workers.Screening ofblood d onors.	Neonatalscreening for hypot hyroidism.Pap smear for dete ction of cervicalcancer.Testin g of urine sugar for diabetes mellitus.

Solution to Question 5:

The monetary benefit is measured in a cost-benefit analysis.

In cost-benefit analysis, the economic benefits of any program are compared with the cost of the program. The benefits are expressed in monetary terms to determine whether a given programme is economically sound and to select the best out of several alternate programmes.

The main drawback with this technique is that the benefits in the health field, as a result of a particular programme, cannot always be expressed in monetary terms. We generally express the benefit in terms of births or deaths prevented, or illness avoided or overcome.

In cost-effective analysis, the benefit is expressed in terms of results achieved. (Eg: number of lives saved or number of days free from disease).

Input-output analysis: how much input is needed to produce a unit amount of output. In the health field, input refers to all health service activities which consume resources (manpower, money, materials, and time); and output refers to useful outcomes such as cases treated, lives saved, or inoculations performed.

Network analysis: a network is a graphic plan of all events and activities to be completed in order to reach an end objective.

Management Methods in Health Programmes:

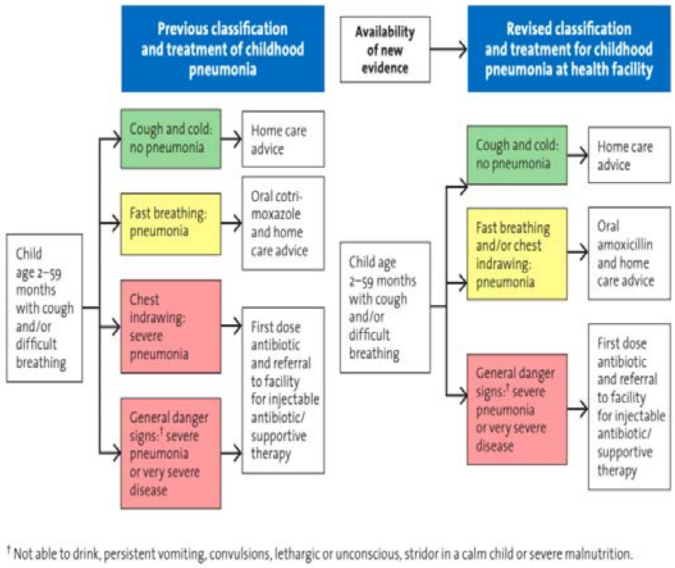
Methods based on behavioral sciences	Quantitative methods
Organisational designPersonnel managementCommunicationInformation systemsManagement by objectives (MBO)	Cost–Benefit analysisCost-Effective analysisCost–AccountingInput–Output analysisModelSystems analysisNetwork analysis- Includes Programme Evaluation and Review Technique (PERT) and Critical Path Method (CPM)Planning–Programming–Budgeting System (PPBS)Work samplingDecision making

Solution to Question 6:

From the given clinical picture, it appears that the child has pneumonia as diagnosed from the presence of chest indrawing. The next best step of management is to give oral amoxicillin for five days and send the child home.

This is based on the revised WHO classification and treatment of childhood pneumonia at health facilities. Pneumonia with general danger signs like inability to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition needs antibiotics with urgent referral to higher centre.

The following image distinguishes the previous and revised classification and treatment of the childhood pneumonia.



Solution to Question 7:

Extended sickness benefit is given for a period of 2 years.

Solution to Question 8:

Chemical waste is disposed of in yellow-colored containers or non-chlorinated plastic bags.

Solution to Question 9:

Management by objective is based on behavioral sciences.

Management methods used in health programmes are either quantitative or based on behavioral sciences.

Methods based on behavioral sciences include:

- Organizational design
- Personnel management
- Communication
- Information systems
- Management by objectives (MBO)

Quantitative methods include:

- Cost-benefit analysis
- Cost-effective analysis

• Cost–accounting

• Input-output analysis

• Model

• Systems analysis

• Network analysis - includes programme evaluation and review technique (PERT) and critical path method (CPM)

• Planning–programming–budgeting system (PPBS)

• Work sampling

• Decision making

Solution to Question 10:

The most appropriate test of significance, in this case, is the chi-square test. It is used to compare qualitative data between 2 groups with independent measures.

Option B: Paired t-test compares paired data or pre-intervention and post-intervention data of the same group. This doesn't apply here as the groups are not paired.

Option C: The standard error of the mean is calculated for quantitative data, with a measure of central tendency as mean.

Option D: ANOVA is used to compare more than 2 groups.

Solution to Question 11:

As per NCEP- ATP III, high LDL is not included in the definition criteria of metabolic syndrome.

Metabolic syndrome is present if three or more of the following five criteria are met, which is given by NCEP-ATP III (National Cholesterol Education Program and Adult Treatment Panel III). It includes:

• Central obesity: waist circumference >102 cm (M), >88 cm (F)

• Hypertriglyceridemia: triglyceride level ≥ 150 mg/dL

• Low HDL cholesterol: <40 mg/dL and <50 mg/dL for men and women, respectively.

• Hypertension: blood pressure ≥130 mmHg systolic or ≥ 85 mmHg diastolic.

• Fasting plasma glucose level: ≥100 mg/dL or previously diagnosed type 2 diabetes.

Solution to Question 12:

68% of the population will lie between 180- 220.

In a normal distribution curve, 68% of the values lie within the mean +/- 1 standard deviation and the mean=median=mode. Since the mean = 200 and the standard deviation = 20:

68% of population will lie between Mean +/- 1SD, that is, 200 +/- 20 = 180-220

95% of population lies between Mean+/-2SD, 200+/- 2*20 = 160- 240

99% of population lies between Mean+/-3SD, 200+/- 3*20 = 140- 260

Solution to Question 13:

Blood bag is disposed in the yellow bag.

Option B: Gloves are disposed in the red bag.

Option C: Sharps are disposed in the white container.

Option D: Urine bags are disposed in the red bag.

Solution to Question 14:

Calculating one variable using another variable is done using coefficient of regression.

Solution to Question 15:

The best indicators for routine monitoring of air pollution are sulfur dioxide, smoke, and suspended particles.

Solution to Question 16:

The type of study described is an ecological study as the unit of study is the population. The number of cigarette packets sold at the same time is data about the population.

Option A: The unit of study in a cross-sectional study is an individual.

Option C: There is no experiment done here. Hence, it is not experimental study.

Option D: Quasi-experiments are nonrandomized, pre-post intervention studies done when it is not feasible or ethical to conduct a randomized controlled trial. The unit of study is therefore patients or individuals and not population.

Solution to Question 17:

Project MONICA is related to cardiovascular diseases.

It stands for multinational monitoring of trends and determinants in cardiovascular diseases. It was done to study the reasons for the changes in the trends of cardiovascular diseases in different countries.

Solution to Question 18:

The disinfectant property of bleaching powder is predominantly due to the action of hypochlorous acid. It is 70-80 times more effective than hypochlorite ions.

Bleaching powder, also known as chlorinated lime ($CaOCl_2$), has 33 per cent of chlorine in it. When chlorine is added to water, there is the formation of hydrochloric and hypochlorous acids. The hydrochloric acid is neutralized by the alkalinity of the water. The hypochlorous acid ionizes to form hydrogen ions and hypochlorite ions, as follows:

$H_2O + Cl_2 \rightarrow HCl + HOCl$

$HOCl \rightarrow H + OCl$

Solution to Question 19:

The difference between the incidence of disease among exposed and incidence among non-exposed is given by attributable risk.

Attributable risk is often expressed as a percentage.

$$\text{Attributable risk (AR)} = \frac{\begin{array}{c}\text{Incidence of}\\ \text{disease rate}\\ \text{among } \textbf{exposed}\end{array} - \begin{array}{c}\text{Incidence of}\\ \text{disease rate}\\ \text{among } \textbf{non-exposed}\end{array}}{\text{Incidence of disease rate among } \textbf{exposed}} \times 100$$

Solution to Question 20:

In Sustainable Development Goal 3, the target 3.1 aims to reduce the Maternal Mortality Ratio by 2030 to <70 per 1 lakh live births.

There are 17 global goals and 169 targets set for 2030 as part of the official Sustainable Development Goals of the Agenda 2030.

Solution to Question 21:

A single dose of tetanus toxoid is given as the wound is clean and the last dose was received more than 10 years ago.

Solution to Question 22:

The variation in one data set is compared with that of another data set using the coefficient of variation (CV). It measures the relative variability by using the mean and standard deviation.

Solution to Question 23:

Recent influenza pandemic occurred in 2009 due to the emergence of a novel H1N1 influenza A virus.

Past influenza pandemics:

- 1968 pandemic due to H3N2 virus
- 1957-58 pandemic due to H2N2 virus
- 1918 pandemic due to H1N1 virus

Community Medicine INI-CET 2021

Question 1:

Select the correct option regarding Mobile Medical Units (MMUs).

a) Only 1

b) 1 and 2

c) 1, 2, and 3

d) 1, 2, 3, and 4

Question 2:

Which of the following statements is true?

a) In a population with low prevalence of the disease, positive result with a highly sensitive test means true positive

b) Sensitivity depends on prevalence of the disease

c) Positive predictive value depends on prevalence of disease

d) Specificity depends on prevalence of the disease

Question 3:

You are starting services for hypertension in your PHC. 50 patients who required antihypertensive treatment were transferred from another center. 40 of them were on amlodipine (5mg PO) and 10 were on lisinopril (10 mg PO) as they had contraindications to the use of amlodipine. The drugs are supplied at the PHC on a monthly basis and you have to place an order for their medications. What is the number of tablets that you will order and the reorder factor?

a) 1000, rf= 3

b) 1200, rf=2

c) 1400, rf= 3

d) 1600, rf= 2

Question 4:

Match the following STI kit colors and their use:

a) A-4, B-3, C-2, D-1

b) A-4, B-1, C-3, D-2

c) A-1, B-4, C-3, D-2

d) A-3, B-2, C-4, D-1

Question 5:

Compared to a pregnant female, a lactating female would require a higher level of nutrient supplementation for which of the following?

a) Vitamin A

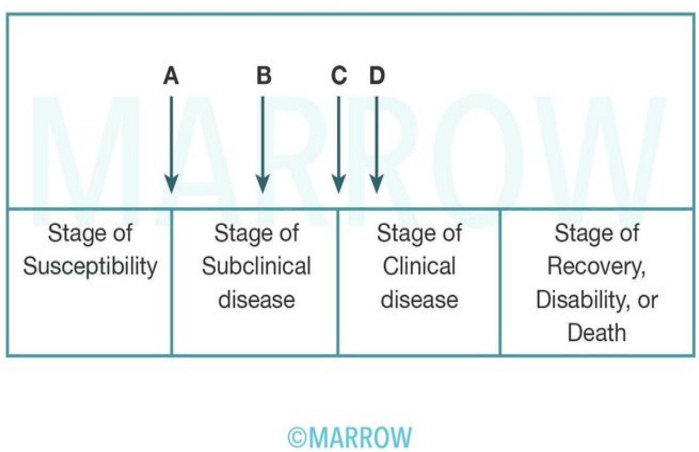
b) Calcium

c) Iron

d) Folic acid

Question 6:

What does the point labeled C denote?



a) Onset of symptoms

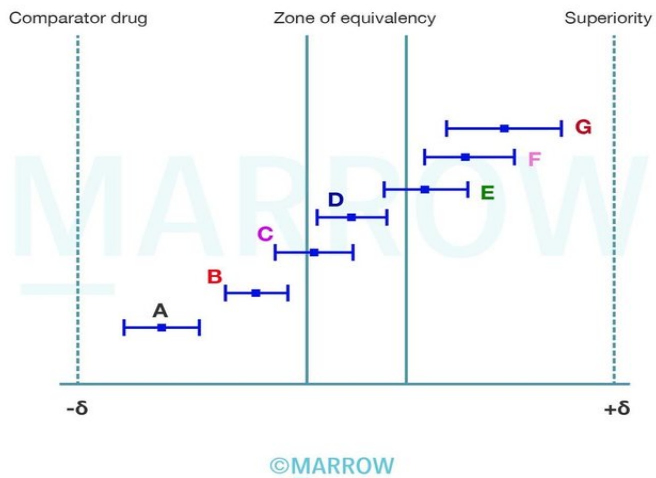
b) Onset of pathological changes

c) Usual time of diagnosis

d) Exposure to etiologic agent

Question 7:

Certain new vaccines are being studied in the market and their efficacies are compared and presented in the image as shown below. Which of the newer vaccines can be used with the same efficacy as compared to the previous vaccine?



a) D only

b) D and E

c) C, D and E

d) A, B, C and D

Question 8:

As per the verbal autopsy report 2010-2013, arrange the most common cause of death among 1-4-year-old children in descending order:

a) 2 > 1 > 3 > 4

b) 4 > 1 > 2 > 3

c) 1 > 2 > 4 > 3

d) 2 > 1 > 4 > 3

Question 9:

The most common mode of absorption of inorganic lead in industries leading to lead poisoning is:

a) Absorption through skin

b) Inhaled lead dust

c) Ingestion of contaminated food and water

d) Contaminated hands

Question 10:

Pentavalent vaccine includes which of the following?

a) Diphtheria, Pertussis, Tetanus , HBV, Injectable Polio

b) Diphtheria, Pertussis, Typhoid, HBV, Hib

c) Diphtheria, Pertussis, Tetanus, Haemophilus influenza B, Pneumococcal vaccine

d) Diphtheria, Tetanus, HBV, Haemophilus influenza B, Pertussis

Question 11:

A nurse keeps the bins as shown in the image in the hospital ward. Which of the following items would go into the black bin?

©MARROW

a) Soiled linen bedsheet

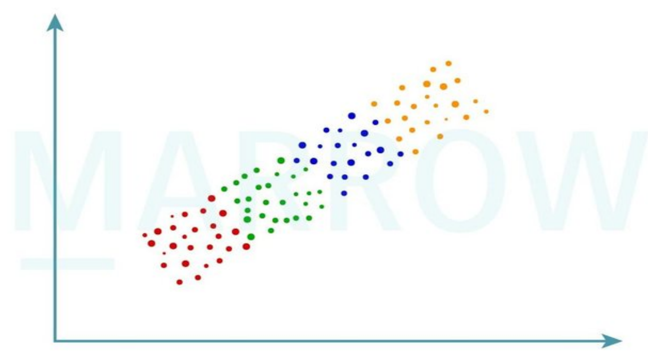
b) Glove paper cover

c) Gloves

d) Contaminated gloves

Question 12:

A homogenous sample of 4 groups was taken as shown below. The height and weight of each group had a correlation coefficient of 0.6. What will be the total coefficient of correlation of the whole group?



a) Equal to 0.6

b) Less than 0.6

c) More than 0.6

d) Skewed coefficient of 0.6

Question 13:

The graphical representation for flow cytometry analysis is done by which of the following?

a) Bar diagram, dot plot

b) Line diagram, dot plot

c) Histogram, dot plot

d) Pie chart, dot plot

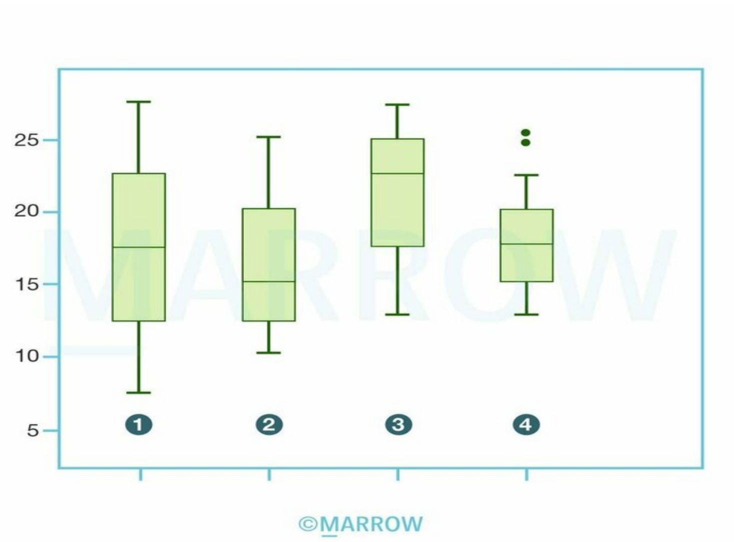
Question 14:

Administration of which of the following vaccines is associated with the risk of intussusception?

- **a)** Rotavirus vaccine
- **b)** Oral Polio vaccine
- **c)** Adenovirus vaccine
- **d)** Measles vaccine

Question 15:

Interpret the following graph.



- **a)** Normal, negatively skewed, positively skewed, skewed with outliers
- **b)** Normal, positively skewed, negatively skewed, normal with outliers
- **c)** Normal, negatively skewed, positively skewed, normal with outliers
- **d)** Skewed with outliers, positively skewed, negatively skewed, normal

Question 16:

The standardized age mortality ratio is an indicator for which of the following?

- **a)** Death rate of a population adjusted to a standard age distribution
- **b)** Death rate of a population adjusted to a standard sex distribution
- **c)** Removal of bias due to different age group population in different countries
- **d)** To eliminate age structure of different countries

Question 17:

According to WHO, which of the following are components of the Human Development Index (HDI)?

- **a)** Life expectancy at birth, literacy and standard of living
- **b)** Life expectancy at 1 year, literacy and standard of living
- **c)** Infant mortality rate, literacy and standard of living
- **d)** Life expectancy at 1 year, literacy and infant mortality rate

Question 18:

The external validity of a screening test is related to?

- **a)** Stability
- **b)** Objectivity
- **c)** Generalizability
- **d)** Repeatability

Question 19:

All the following tests are related to pasteurization of milk except

- **a)** Phosphatase test
- **b)** Iodine test
- **c)** Standard plate count
- **d)** Methylene blue reduction test

Question 20:

Over the past few years, there is an increase in the incidence of non-communicable diseases. What is this type of trend called?

- **a)** Seasonal
- **b)** Cyclical
- **c)** Periodic

d) Secular

Question 21:

Which of the following is not a part of case-control studies?

a) Follow up

b) Matching

c) Strength of association

d) Selection of study subjects

Question 22:

Which of the following is the best study design to demonstrate the relationship between cause and effect?

a) Case control study

b) Cohort study

c) Cross-sectional study

d) Clinical trial

Question 23:

Hepatitis B subunit vaccine contains which of the following antigen?

a) HbeAg

b) HbsAg

c) HbcAg

d) HBV DNA

Question 24:

Arrange the following causes of neonatal mortality from maximum to the least (according to the verbal autopsy report 2010-2013 data).

a) 1>2>4>3

b) 1>4>2>3

c) 4>1>2>3

d) 1>2>3>4

Question 25:

Which of the following is not involved in meta-analysis?

a) Selection

b) Analysis

c) Randomization

d) Abstraction

Question 26:

A new drug has been introduced into the market which was found to decrease mortality but it does not cure the disease. Which of the following is a true statement regarding prevalence and incidence?

a) Decrease in incidence

b) Decrease in prevalence

c) Increase in incidence

d) Increase in prevalance

Answer Key

Question No.	Correct Option
1	c
2	c
3	d
4	b
5	b
6	a
7	c
8	d
9	b

10	d
11	b
12	a
13	c
14	a
15	b
16	a
17	a
18	c
19	b
20	d
21	a
22	b
23	b
24	a
25	c
26	d

Detailed Explanations

Solution to Question 1:

Statements 1, 2, and 3 are correct. MMUs can be run by either Government or an external agency, but the drugs and medical supplies are to be provided always by Government only.

Mobile Medical Units (MMU) is a major initiative under NRHM to provide a range of health care services for people living in remote, inaccessible, un-served, and underserved areas in both rural as well as urban areas, with the aim of taking healthcare services to the doorsteps of such people.

There are 3 models of operationalizing MMUs:

• Government operated MMU

• Operation of MMU on outsourcing basis by an external agency but capital expenditure, drugs, and supplies provided by Government.

• Outsourcing by an external agency which provides both capital expenditure and operational expenditure. However, drugs and supplies are to be provided by Government.

MMUs provide primary care services for common diseases (communicable and non-communicable), RCH services, screening tests, and referral to higher facilities and specialists as required. Clinical services are provided by a Medical Officer and team, and mobilization and outreach services are provided by ANMs and ASHA.

Solution to Question 2:

The positive predictive value (PPV) of a screening test depends on the prevalence of the disease. The positive predictive value (PPV) of a test indicates the probability that a patient with a positive test result has actually contracted the disease i.e., the likelihood that a person with a positive test result truly has the disease.

If the prevalence of a disease in a population is high, the more accurate will be the positive predictive value. Therefore, in a population with a high prevalence of the disease, a positive result with a highly sensitive test means true positive. Example: Hypothyroidism is more prevalent in hilly areas than in coastal areas. If someone from a hilly area tests positive he is more likely to have the disease than someone who tests positive in a coastal area.

Likewise, if the prevalence decreases, the PPV will also become low. So in a population with a low prevalence of the disease, positive result with a highly sensitive test need not always mean true positive as it can produce an unacceptably high proportion of false positive results.

Predictive value is affected the most by prevalence. However, both sensitivity and specificity can change the predictive value.

Solution to Question 3:

The Quantity to Order (QO) or the reorder quantity of medicines to be placed at each order interval is calculated using the reorder formula as given below:

QO = CA x (LT + PP) = CA x Rf

where QO is the quantity to order, CA is average consumption in a month, LT is lead time, PP is procurement period and Rf is the reorder factor. Reorder factor (Rf) is the sum of lead time (LT) and procurement period (PP). Lead time (LT) is the time between the initiation of a purchase order and receipt at the warehouse from the supplier. The procurement period (PP) covers the time until the next regular order will be placed.

In the given question, lead time (LT) and procurement period (PP) are 1 month each.

Average consumption (CA) in a month = (40 tablets of amlodipine + 10 tablets of lisinopril) x 30 days = 1500 tablets

Reorder factor (Rf) = LT + PP = 1 + 1 = 2

Since these patients got transferred from another center there is no safety stock for them at present at the facility. So in order to prevent stockouts for them, an extra number of tablets must be ordered, which makes 1600 tablets with a reorder factor (Rf) of 2 the correct option.

Factors determining the reorder formula are:

• Average consumption or demand of the drug (CA) in a month

• Supplier Lead time (LT): is the time between initiation of a purchase order and receipt at the warehouse from the supplier.

• Safety stock (SS): is the stock that should always be on hand to prevent stockouts. The safety stock level and reorder level will be the same if consumption and lead times are predictable and stable.

• Procurement period (PP): it covers the time until the next regular order will be placed. In a scheduled system this is in multiples of one month and in a perpetual system, it is counted in days or weeks. Note: Quantity ordered (QO) + Safety Stock (SS) must cover the time until the next order is received, which is given by the Reorder factor (Rf) = Procurement Period (PP) + Lead Time (LT).

• Reorder level: is the quantity of remaining stock that should trigger a reorder of the item. In a basic purchasing formula, reorder level is calculated by multiplying the average lead time by the average consumption of the drug during the lead time.

• Maximum stock level: is the same as the target stock level, which is required to satisfy demand until the next order is received.

• Stock position: is the sum of stock on hand (working and safety stock) and stock on order, minus any stock back-ordered to clients.

Solution to Question 4:

The correct match is A-4, B-1, C-3, D-2

The National Aids Control Organization (NACO) has provided color-coded kits for the syndromic management of sexually transmitted diseases:

Kit	Syndrome/s
Kit 1(Grey)	Urethral dischargeCervical dischargeAno-rectal dischargePainful scrotal swelling
Kit 2 (Green)	Vaginal discharge/ Vaginitis
Kit 3 (White)	Non-herpetic genital ulcers
Kit 4(Blue)	Non-herpetic genital ulcers if allergic to penicillin
Kit 5(Red)	Herpetic genital ulcers
Kit 6 (Yellow)	Lower abdominal pain
Kit 7(Black)	Inguinal bubo
Kit 8(Brown)	Anorectal discharge syndrome

Solution to Question 5:

According to the Recommended daily allowance (RDA) guidelines set by the National Institute of Nutrition (NIN), the requirement of calcium intake is increased in a lactating woman (1200mg/d) when compared to that of a pregnant woman (1000mg/d). Calcium intake of 1200mg/d is advised in postmenopausal women as well.

The RDA of vitamin A is increased by 50mcg in lactation as compared to pregnancy. But the increased vitamin A demand is met from the diet and it is not given as a supplement nutrient. Vitamin A supplements are given only when there is vitamin A deficiency.

The RDA of iron and folic acid is decreased during lactation compared to pregnancy.

The salient features of updated RDA guidelines, 2020 by NIN are:

Variable	Non pregnant female	Pregnancy	Lactation			
1st TM	2nd TM	3rd TM	0-6 m	6-12 m		
Body weight	55 kg	55 kg + 10 kg in entire pregnancy				
Energy (Kcal/d)SedentaryModerateHeavy	166021302720	+350	+600	+520		
Protein (g/d)	45.7	45.7	+9.5g	+22g	+16.9 g	+13.2 g
Carbs (g/d)	130	175g	200g			
Iron (mg/d)	29 mg/d	27 mg/d	23 mg/d			
Iodine (mcg/d)	150	250	280			
Calcium (mg/d)	1000	1000	1200			
Vit A (mcg/d)	840	900	950			
Folate (mcg/d)	220	570	330			

Solution to Question 6:

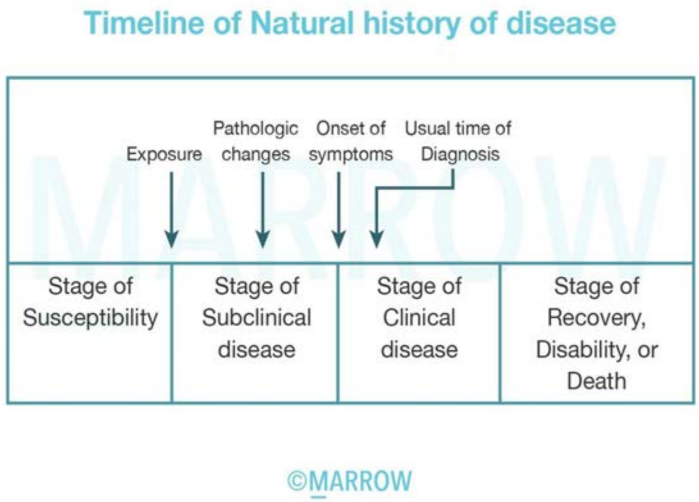
The given diagram shows the natural history of disease in an individual over time, in which the point labeled C denotes the onset of symptoms. The natural history of disease refers to the progression of a disease in an individual over time, in the absence of treatment.

Exposure (point A): the disease process begins with appropriate exposure to or accumulation of factors sufficient for the disease process to begin in a susceptible host. For example in infectious disease, the exposure is a microorganism. For lung cancer, it is any factor that initiates the process, such as asbestos fibers or components in tobacco smoke.

Pathological changes (point B): after the disease process has been triggered, pathological changes occur without the individual being aware of it. This stage of subclinical disease, extending from the time of exposure to onset of symptoms, is usually called the incubation period for infectious diseases, and the latency period for chronic diseases. During this stage, the disease remains asymptomatic.

The onset of clinical symptoms (point C): it is the beginning of the clinical stage of the disease.

The usual time of diagnosis (point D): the onset of symptoms is followed by a short interval (varies from disease to disease) after which the patient presents to a healthcare setup where a diagnosis of the disease is made. Based on treatment and prognosis, the patient can progress to either recovery, disability, or death.



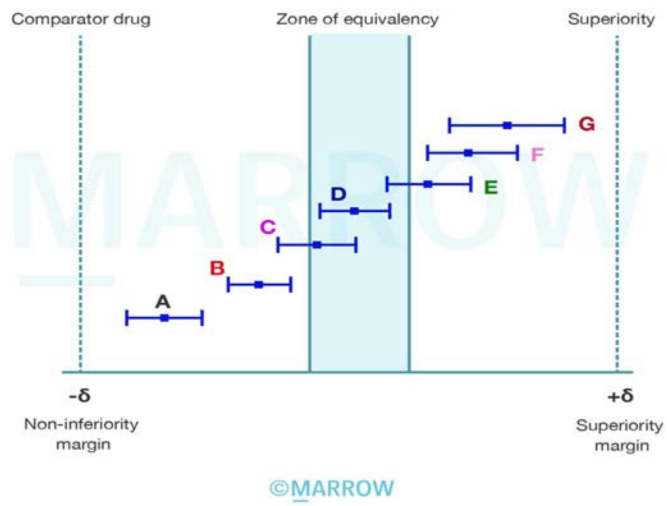
Solution to Question 7:

Vaccines C, D, and E can be used with the same efficacy as compared to the previous vaccine, as their confidence interval lies within the zone of equivalence.

In a classical randomized controlled trial (RCT), the aim is to evaluate a difference between two drugs by proving the superiority of the new drug over the standard drug. However, an equivalence study is carried out with the aim of showing there are no significant differences in the efficacy between the new drug and the standard drug.

To define 'no significant differences' the limit of equivalence (δ) must be defined first. δ must be set as superiority margin (+δ) and non-inferiority margin (-δ), so that they are clinically appropriate and the differences can be tolerated as clinically irrelevant. To conclude equivalence the confidence interval (CI), i.e., the whisker (tail) of the box-whisker plot must lie within the zone of equivalence.

Therefore in the given question, vaccines C, D, and E have CI in the zone of equivalence and thus have the same efficacy as the previous vaccine. Vaccines F and G lie outside the zone of equivalence but well within +δ, making them superior to the previous vaccine. Vaccines A and B lie outside the zone of equivalence but well within -δ, making them non-equivalent and at the same time non-inferior to the previous vaccine. If the CI of any vaccine includes or crosses the -δ limit, they will be considered inferior to the standard vaccine.



Note: A superiority trial is done to show that the new drug is superior to the standard drug. A non-inferiority trial is done to show that the new drug is not worse than the standard drug. This is done for newer interventions or drugs that may be less costly, less invasive, and have fewer side effects.

Solution to Question 8:

According to the verbal autopsy report of 2010-2013, the most common cause of death among 1-4 year old children in descending order is Pneumonia > Diarrhea > Injuries > Malaria.

According to the Verbal autopsy report 2010-2013, the top 5 causes of death among children aged 1-4 years are:

- Pneumonia
- Diarrheal diseases
- Injuries
- Other non-communicable diseases
- Malaria

1-4 year mortality rate also known as the child death rate is the number of deaths of children aged 1-4 years per 1000 children in the same age group in a given year. It thus excludes infant mortality. It is given by the formula,

1-4 year mortality rate (child death rate) = (no. of deaths of children aged 1-4 years during a year x 1000) / Total no. of children aged 1-4 years at the middle of the year.

It is a more refined indicator of the social situation than the infant mortality rate as it reflects adverse environmental health hazards like malnutrition, poor hygiene, infections, and accidents. It is not affected by perinatal hazards. In the 1-4 year group, second year is the period with the highest risk of death (mainly due to childhood infections like measles, whooping cough, diphtheria, diarrhea, and ARI), after which the rates decline progressively.

Solution to Question 9:

The most common mode of industrial lead poisoning (plumbism) is due to inhalation of fumes and dust of lead. Lesser common modes include ingestion of small quantities of lead trapped in the upper respiratory tract, ingestion due to contaminated hands, and absorption through the skin (only organic lead). Lead poisoning is a notifiable and compensable disease in India.

Lead exerts its toxic action by combining with the SH-group of enzymes involved in porphyrin synthesis and carbohydrate metabolism, disrupting normal membrane permeability, and causing potassium leakage in RBCs. Symptoms of lead toxicity include:

Inorganic lead toxicity:

• Abdominal colic, also known as Painter's colic

• loss of appetite, obstipation, constipation

• Anemia (stippling of red cells)

• Wrist drop or foot drop

• Blue-line on the gums known as Burtonian line

Organic lead toxicity (mainly affects CNS):

• Headache

• Delirium, confusion

• Insomnia

Earliest and consistent sign: Facial pallor.

Clinical symptoms of lead poisoning occur when blood levels are >70μg/dl. Measurement of Coproporphyrin (CPU) in urine is a useful screening test. Periodic examination of workers must be done and atmosphere lead concentration must be kept <2mg/cubic meter at the workspace. Handwashing before eating and prohibition of taking food in workplaces is essential. D-penicillamine is used as a chelating agent for treatment.

Solution to Question 10:

Pentavalent vaccine is a combination of five vaccines that provide protection against 5 life-threatening diseases such as diphtheria, pertussis, tetanus, hepatitis B, and Haemophilus influenzae type b (Hib). Administration of pentavalent vaccine reduces the number of pricks to a child.

It is given at 6, 10, and 14 weeks in the immunization program, and can replace the need for separate administration of Hepatitis B and DPT vaccines. However, the Hepatitis B birth dose and DPT booster doses should be continued. It is given 0.5ml intramuscularly in the anterolateral aspect of the mid-thigh.

Pentavalent vaccine is a freeze-sensitive vaccine and therefore should be stored and transported at 2- 8° C temperature. It is a very safe vaccine with a few mild side effects like pain, redness and swelling at the injection site, fever, vomiting, abnormal crying, and loss of appetite.

Solution to Question 11:

Glove paper cover is a general solid waste that is to be collected in the black bin.

All non-infectious general wastes like food waste, paper waste, and mineral water bottles should be discarded in black bins.

Soiled linen bedsheets contaminated by blood or body fluids should be disposed in the yellow bin. All plastic recyclable wastes are discarded in red bins (gloves and contaminated gloves).

Solution to Question 12:

In a homogenous sample or population, the total coefficient of correlation (r) will be the average of the individual correlation coefficients. Therefore in the given condition, it will be the same as 0.6

The given scattergram shows the correlation coefficient of height and weight of 4 homogenous samples. Averaging methods are appropriate when several different samples are independently drawn from the same or a homogenous population and the interest is in aggregating those values.

Therefore the total correlation coefficient of the whole group will be:

r = (sum of correlation coeffiecient of each sample) ÷ (total no. of samples)

= (0.6 + 0.6 + 0.6 + 0.6) ÷ 4 = 2.4 ÷ 4

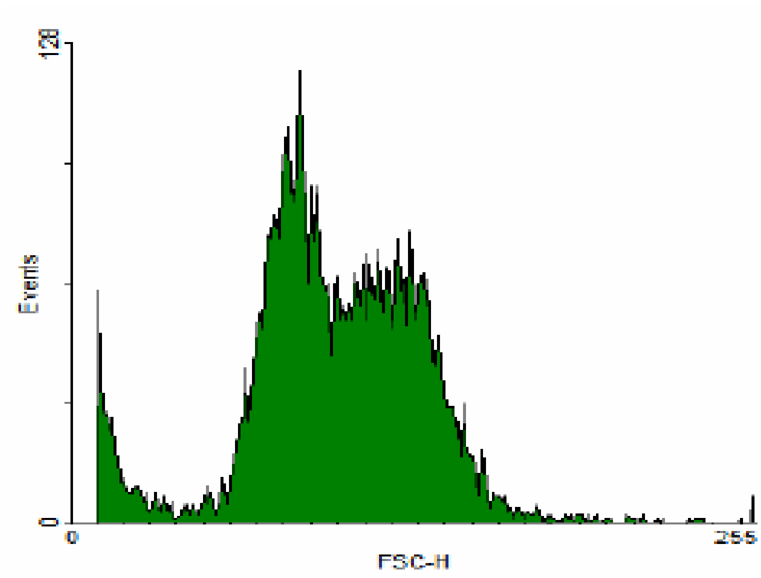
= 0.6

Note: A correlation between 2 variables does not demonstrate any causal relationship between those variables, no matter how strong it is. Correlation is merely a measure of a statistical association between the variables. Therefore inferring a causal relationship between two variables on the basis of a correlation is a common and fundamental error.

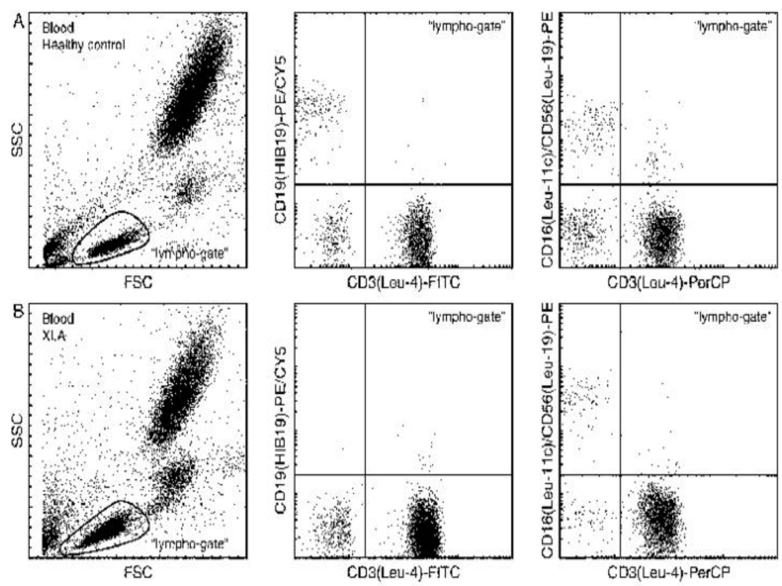
Solution to Question 13:

Flow cytometry results are typically presented by two-dimensional dot plots or one-dimensional histograms.

Histograms compare only a single parameter. A large number of events detected at one particular intensity will be displayed as a spike on the histogram. They are fast to read and easy to understand.



Dot plots can compare 2 or more parameters simultaneously on a 2D scatterplot. Each distinct event is represented as a single dot and events with similar intensities will cluster together in the same region on the scatter plot. They are a more complex, illustrative representation of data.



Pie charts and bar diagrams are used for representing qualitative data. A line diagram is used to show the trend of events or changes over passage of time.

Solution to Question 14:

Rotavirus vaccine is associated with a potential risk of intussusception, especially when the first dose of these vaccines is given to infants aged >12 weeks. Because of this risk, the rotavirus vaccine is not a part of catch-up immunization programs, as the age of the child is difficult to ascertain in such programs.

Rotavirus vaccines are oral, live attenuated vaccines and are of 3 types:

• Rotarix (human monovalent) vaccine is administered in 2 oral doses:

• First dose: At 6 weeks, no later than 12 weeks (due to the risk of intussusception).

• Second dose: With a gap of 4 weeks from the first dose, to be given by 16 weeks and no later than 24 weeks.

• RotaTeq (pentavalent bovine-human reassortant) vaccine is administered in 3 oral doses at 2, 4, and 6 months. The first dose is given between 6-12 weeks and subsequent doses within a 4-10 weeks gap. All the 3 doses should be given before 32 weeks.

• Rotavac (monovalent liquid frozen) vaccine is administered in 3 oral doses, 4 weeks apart at 6,10 and 14 weeks. It should not be administered to children more than 8 months of age.

Solution to Question 15:

The graph shown is of a boxplot, and option B represents the correct answer. 1 is normal, 2 is positively skewed, 3 is negatively skewed, and 4 is normal with an outlier.

A boxplot is a method for graphically depicting groups of numerical data through their quartiles. Box plots may also have lines extending vertically from the boxes (whiskers) indicating variability outside the upper and lower quartiles, hence the terms box-and-whisker plot.

It shows the five-number summary of a set of data including,

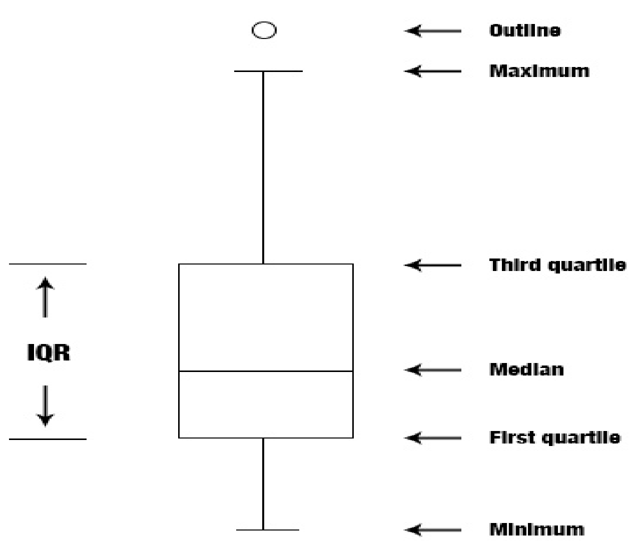
• Minimum score

• First (lower) quartile

• Median

• Third (upper) quartile and

• Maximum score



In the box plot, a simple box extends from the 25th percentile (lower limit of the box) to the 75th percentile (upper limit of the box) and the line through the box indicates the median value (50th percentile).

The top and bottom ends of the whiskers are the highest and lowest value of that group excluding outliers

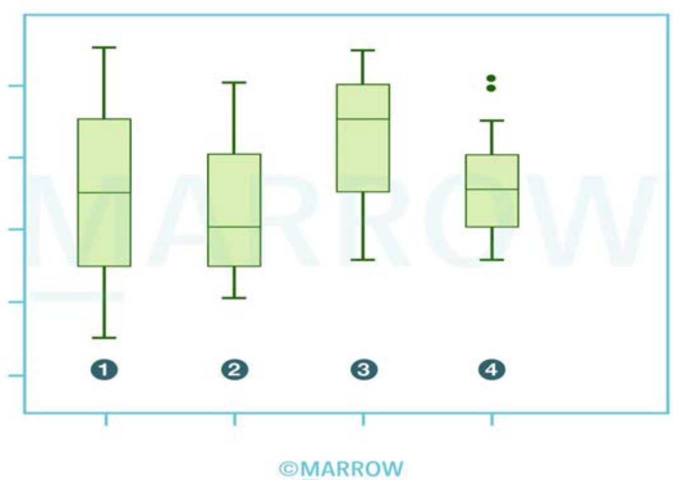
Interpreting the box plot:

• When the median is in the middle of the box, and the whiskers are about the same on both sides of the box i.e when each quartile is of the same length, then the distribution is symmetric or normal (as shown in box plot-1).

• When the median is closer to the bottom of the box, and if the whisker is shorter on the lower end of the box, then the distribution is positively skewed (skewed right as shown in boxplot-2).

• When the median is closer to the top of the box, and if the whisker is shorter on the upper end of the box, then the distribution is negatively skewed (skewed left as shown in boxplot-3).

• Small circles or unfilled dots are drawn on the chart to indicate where suspected outliers lie. Filled circles are used for known outliers (as shown in boxplot-4).



Solution to Question 16:

The age standardized mortality ratio (SMR) is an indicator of the death rate of a population adjusted to a standard age distribution. It is obtained by indirect standardization and cannot be used to compare multiple groups with different age structures.

Standardized death rate (SDR) obtained by direct standardization enables comparisons of mortality rates between populations with different age compositions by eliminating the confounding effect caused by variations in age structures (Options C and D). This adjustment process can also account for factors such as sex, race, and parity.

Standardization can be achieved through two methods: direct or indirect. In both cases, a standard population is selected as a reference, characterized by known numbers in each age and sex group.

The direct method of standardization requires age-specific death rates (ASDR) along with the corresponding subgroup sizes. By applying these rates to the standard population, the expected deaths across all age groups can be calculated assuming that both populations share the same age-specific death rates. Dividing the total expected deaths by the standard population yields the SDR.

The calculated standardized death rate in direct standardization can be compared with different population groups with different age group compositions.

The table below shows the calculation of age-specific death rates per 1000 population for a sub-population:

Crude death rate = 64/7500 = 8.533 per 1000 population.

Applying ASDR to the standard population, we can get the SDR

SDR per 1000 population = (212/23500) x 1000 = 9.021

Indirect standardization allows adjustment when age-specific rates are either unavailable or unstable due to small numbers. Examples of indirect standardization methods include standardized mortality ratio (SMR), life table analysis, survival analysis, regression analysis, and multivariate analysis.

SMR estimates expected deaths, assuming that the study group experiences the same death rates as a standard population.

The table below shows the calculation of the standardized mortality ratio for a study group population.

SMR = Observed deaths / Expected deaths x 100 = (65/59.85) x 100= 108.6

If the SMR exceeds 100, it indicates that the study group population carries a higher mortality risk compared to the overall population. However, it's important to note that the SMR cannot be directly compared to other groups with different age compositions.

Age(years)	Mid-year population	Deaths in the year	ASDR	Standard population	Expected deaths
0-15	2000	10	5	6000	30
15-45	4000	24	6	12000	72
>45	1500	30	20	5500	110
Total	7500	64		23500	212

Age(years)	National population death rates per 1000	Study group population	Observed deaths	Expected deaths
0-15	3.0	1200	*	3.6
15-45	6.5	2500	*	16.25
>45	10	4000	*	40
Total			65	59.85

Solution to Question 17:

The components of the Human Development Index (HDI) include life expectancy at birth, literacy, and standard of living.

Human Development Index (HDI) is a measure of a country's average achievements in three dimensions of human development:

• A long and healthy life, as measured by life expectancy at birth

• Knowledge (literacy), as measured by mean years of schooling and expected years of schooling, and

• A decent standard of living, as measured by GNI (gross national income) per capita.

HDI values range from 0 to 1, with 1 as the maximum value.

Note- Life expectancy at 1 year, literacy and infant mortality rate are components of the Physical quality of life index (PQLI).

Solution to Question 18:

External validity of a screening test is related to generalizability.

Validity (accuracy) refers to what extent the test accurately measures which it sets out to measure. It refers to the closeness with which measured values agree with "true" values.

While assessing the validity of a research study, two kinds of validity are involved:

• External validity: If the study sample is truly representative of the population from which it was drawn, then the study is said to have external validity. Its findings can be applied to the entire population as well and therefore it is said to be generalizable.

• Internal validity: Here the results of the study are valid for the sample of patients who were actually studied.

Note: Precision (reliability or reproducibility) is a measure of the repeatability of a test. That is, the test must give consistent results when repeated more than once on the same individual or material, under the same conditions.

Solution to Question 19:

All the above tests are related to the pasteurization of milk except the iodine test. It is done to test if milk is adulterated with starch or cereal flour.

The iodine test is done by adding 2 drops of iodine solution containing potassium iodide and iodine to 1ml of milk. The appearance of a blue color indicates that the milk is adulterated with starch. A control sample of milk that is not adulterated with starch will develop a slightly yellowish color.

Pasteurization can be done by the following methods,

• Holder/Vat method: 63-66 °C for 30 minutes and then quickly cooled to 5°C.

• HTST method (High-temperature short time): 72°C for at least 15 seconds, and then cooled rapidly to 4°C.

• UHT method (Ultra-high temp): Rapidly heated to 125 °C for a few seconds, then cooled and bottled as quickly as possible.

Tests in pasteurization:

• Phosphatase test: It is based on the fact that raw milk contains phosphatase enzyme which gets destroyed at temperatures corresponding to the standard time and temperature required for pasteurization.

• Standard plate count: It is used to determine the bacteriological quality of pasteurized milk.

• Coliform count: The presence of coliform organisms in pasteurized milk indicates either improper pasteurization or post-pasteurization contamination.

• Methylene blue reduction test is an indirect method for the detection of microorganisms in milk. This test is carried out on the milk accepted for pasteurization. Definite quantities of methylene blue are added to 10 ml of milk at a uniform temperature of 37°C. The milk is considered satisfactory if it fails to reduce the methylene blue dye in 30 minutes under standard conditions.

Solution to Question 20:

An increase in the incidence of non-communicable diseases over a few years is described as a secular trend.

The secular trend describes the occurrence of disease over a prolonged period, usually years. Even though it may have short-term fluctuations, there is a tendency of the disease occurrence to change consistently in a particular direction or a definite movement in one direction.

Examples of the secular trend include coronary heart disease, lung cancer, and diabetes, which have shown a consistent upward trend in incidence as well as prevalence, in developed countries. Similarly, there has been a consistent decline in the diseases such as tuberculosis, typhoid fever, diphtheria, and polio.

Other options:

Option A: Seasonal trends reflect seasonal changes in disease occurrence following changes in environmental conditions that enhance the ability of the agent to replicate or be transmitted. For example, measles (usually occurs in early spring), varicella, etc. Non-infectious conditions like hay fever and snake bites may also show seasonal variations.

Option B: Cyclic trend is shown by some diseases that occur in cycles spread over short periods of time which may be, days, weeks, months, or years. Examples include the influenza pandemic, which is known to occur at intervals of every 7–10 years due to antigenic variations.

Option C: Periodic fluctuations include both seasonal and cyclic trends.

Solution to Question 21:

Among the given options, follow-up is not a part of case-control studies.

Case-Control Studies are also called retrospective studies. This study proceeds from effect to cause and is done after the exposure and outcome have occurred. They focus on diseases or health problems that have already developed. They are less time-consuming and have no risk of attrition because they do not require follow-up.

There are four basic steps in conducting a case-control study:

• Selection of cases and controls

• Matching- This is done to ensure comparability between cases and controls. Controls are selected such that they are similar to cases with respect to certain pertinent selected variables that influence the disease outcome.

• Measurement of exposure - This is obtained by interviews, questionnaires, or study of the past records of cases.

• Analysis in the form of exposure rates, risk ratio, and odds ratio.

Odds Ratio (OR) calculated from a case-control study, is a measure of the strength of the association between risk factor and outcome.

Note: Follow-up is a part of cohort studies. The elements in cohort studies include selecting study subjects, obtaining data on exposure, selecting comparison groups, follow up and analysis.

Solution to Question 22:

Cohort study is the best study design to demonstrate the relationship between cause and effect.

A cohort study is a type of analytical study, which progresses from cause to effect, as exposure has occurred but not disease. It is also known as forward-looking study, incidence study, prospective study, and longitudinal study.

A well-designed cohort study is considered the most reliable means of showing an association between a cause or suspected risk factor and subsequent disease (or effect) because it eliminates many of the problems of the case-control study.

Relative risk (RR) calculated using cohort study demonstrates the relationship between cause and effect. It is a direct measure of the strength of association between suspected cause and effect. The larger the RR, the greater the strength of the association between the suspected factor and disease.

The advantages of cohort study are the following:

• The incidence can be calculated.

• Several possible outcomes related to exposure can be studied simultaneously.

• Provides a direct estimate of relative risk.

• Dose-response ratios can be calculated.

• Bias like misclassification can be minimized.

Other options:

Option A: In a case-control study, the study begins from the outcome and goes retrospectively to find out the exposure among the cases and controls.

Option C: A cross-sectional study is a prevalence study which is more useful for the study of chronic diseases. It is more appropriate for the study of the distribution of a disease in a population than studying its etiology.

Option D: Clinical trials are research studies performed on people that are aimed at evaluating new tests and treatments and their effects on human health outcomes.

Solution to Question 23:

Hepatitis B subunit vaccines are composed of the HBsAg (Hepatitis B virus surface antigen).

Subunit vaccines are produced from macromolecules derived from immunogenic components of pathogens by recombinant DNA technology. They differ from inactivated whole-cell vaccines, by containing only the antigenic parts of the pathogen. These parts are necessary to elicit a protective immune response.

Hepatitis B vaccine is made by inserting the HbsAg gene into the yeast cell. This modified yeast cell produces large amounts of hepatitis B surface antigen, which is purified and harvested and used to produce the vaccine. The recombinant hepatitis B vaccine is identical to the natural hepatitis B surface antigen, but does not contain virus DNA and is unable to produce infection.

Solution to Question 24:

According to verbal autopsy report 2010-2013 data, the correct sequence of causes of neonatal mortality from maximum to least is prematurity and low birth weight> birth injuries and birth asphyxia> sepsis> congenital anomalies. (1>2>4>3)

Neonatal deaths are deaths that occur during the neonatal period, that is between birth and 28 completed days after birth. It is calculated by using the following formula:

Neonatal mortality rate = Number of deaths of children under 28 days of age in a year x 1000/Total live births in the same year

Important causes of neonatal mortality include (from maximum to least)

• Prematurity & low birth weight

• Birth asphyxia & birth trauma

• Neonatal pneumonia

• Other non-communicable diseases

• Sepsis

• Congenital anomalies

• Diarrhoeal diseases

• Injuries

• Tetanus

Note: Recent 2020 SRS (Sample Registration System) report shows the following:

• Neonatal mortality rate to be 20.

• Infant mortality rate to be 28.

• Under 5 mortality rate to be 32.

Solution to Question 25:

All the above steps are involved in meta-analysis except for randomization.

Meta-analysis is a set of statistical techniques for combining quantitative results from multiple studies to produce a summary of empirical knowledge on a given topic. It refers to a method of a systematic review.

Four basic steps involved in a meta-analysis are:

• Identification: This is the first step in a meta-analysis and it includes finding all of the articles or studies related to the topic.

• Selection: After assembling all the related studies, it is important to select the right or necessary studies by using inclusion (also called eligibility) criteria. Examples are minimum sample size (as very small studies may be unrepresentative), study design (controlled trials only vs randomized controlled trials only), etc.

• Abstraction: Once an appropriate group of studies has been identified, the author has to abstract the relevant data from each study.

• Analysis: This is the last step of meta-analysis.

Objectives of meta-analysis include,

• Summarize results from individual studies.

• Analyze differences in results among studies.

• Overcome small sample sizes of individual studies to detect the effects of interest.

• Determine if new studies are needed to investigate an issue.

• Generate new hypotheses for future studies.

Note- Randomization involves randomly assigning patients to different groups, like experimental and control groups. Clinical trials usually involve randomization and are hence called randomized clinical trials (RCTs). It is done to prevent confounding.

Solution to Question 26:

In the given scenario, the new drug was found to decrease mortality but not cure the disease so the duration of the disease is increased thereby increasing the prevalence. There is no change in the incidence of the disease.

Incidence refers to the number of new cases in a defined population in a specified period of time.

Prevalence refers to the number of old and new cases at a given point of time or over a period of time in a given population.

Prevalence depends on two factors, the incidence, and duration of illness. If the population is stable and incidence and duration are unchanging, the relationship between incidence and prevalence can be expressed as:

$P = I \times D$

(I - Incidence and D - Mean duration)

This equation shows longer the duration of the disease, the greater the prevalence.

Note:

• In chronic diseases like TB, prevalence will be higher than incidence.

• If a new treatment for a disease is effective, incidence does not change but prevalence decreases as the duration of disease decreases.

• If the treatment only prevents death but doesn't cure the disease, it may cause a paradoxical increase in prevalence.

Community Medicine NEET 2021

Question 1:

Which of the following thermometers is used to measure the low air velocity rather than the cooling power of the air?

- **a)** Kata thermometer
- **b)** Globe thermometer
- **c)** Wet globe thermometer
- **d)** Dial thermometer

Question 2:

There is an outbreak of buboes in the community. What is the vector responsible for this condition?

- **a)** Xenopsylla cheopis
- **b)** Phlebotomus argentipes
- **c)** Ixodes tick
- **d)** Female Anopheles mosquito

Question 3:

The Pathanamthitta district of Kerala was affected by floods and the government distributed doxycycline tablets for prophylaxis. Which other chemical will be distributed along with it?

- **a)** Zinc phosphide
- **b)** Malathion
- **c)** Lindane
- **d)** Paris green

Question 4:

Following admission of a road traffic accident case, there is spillage of blood on the hospital floor. Which disinfectant will you use to clean the floor?

a) Ethyl alcohol

b) Chlorhexidine

c) Sodium hypochlorite

d) Formaldehyde

Question 5:

The air pollution index chart of 4 consecutive days in Delhi station is given below. The air quality index of November 23 is classified as?

a) Moderately polluted

b) Poor

c) Very poor

d) Severe

Question 6:

There is an outbreak of acute encephalitis in the community and a vaccination drive is launched. Which of the following is true about the vaccine given in this condition?

a) Live and subcutaneous

b) Killed and intramuscular

c) Live and intramuscular

d) Killed and subcutaneous

Question 7:

The evaluation based on the treatment given to the patient and their clinical management in a health care facility measures which of the following?

a) Outcome

b) Process

c) Structure

d) Input

Question 8:

A girl child has had recurrent yeast infections and respiratory virus infections since she was 3 months old. Which of the following vaccine is contraindicated considering her immune status?

a) Killed IPV

b) Measles/MMR

c) DPT

d) TT/Td

Question 9:

Which of the following is a conditioning influence for malnutrition?

a) Child rearing

b) Infectious diseases

c) Food habits

d) Socioeconomic factors

Question 10:

Which of the following is not a component of the global hunger index (GHI)?

a) Infant mortality rate

b) Under 5 mortality

c) Percentage of undernourished population

d) Percentage of undernourished children under 5 years

Question 11:

A researcher wants to know whether there is an association of CRP values with the risk of myocardial infarction/stroke. The study group was divided into quintiles and the following table is formed based on the study results. Which of the following statements is true regarding the relationship between CRP and the risk of myocardial infarction/stroke?

a) CRP has no association with risk of MI/stroke

b) Increase in CRP increases the risk of MI/stroke

c) Increase in CRP decreases the risk of MI/stroke

d) In quintile 1, there is no risk of MI/stroke

Question 12:

A child presented with pharyngitis. A throat swab was obtained and sent for culture. The used swab should be discarded in which color bin?

a) Red

b) Yellow

c) White

d) Blue

Question 13:

Which of the following is the SI unit for the measurement of brightness of light from a point source?

a) Candela

b) Lux

c) Lambert

d) Lumen

Question 14:

A pregnant woman whose niece lives with her in the same house contracted Varicella. She tested negative for serum antibodies against varicella. What would this mean?

a) She is immune to zoster

b) She is susceptible to zoster

c) She is susceptible to chicken pox

d) She is immune to chicken pox

Question 15:

The school health programs are managed by which of the following?

a) Primary health center

b) District hospital

c) Sub-centre

d) Sub-divisional hospital

Question 16:

A man from Chattisgarh presented with progressive muscle weakness and leg spasms. Pure motor paresis was seen on examination. What is the most appropriate history to be elicited in this patient?

a) History of similar illness in the past

b) History of fever

c) History of vaccination

d) History of diet

Question 17:

In a 10-year-old school child, which of the following vaccines is given as a part of the immunization programme?

a) BCG

b) MMR

c) TT/Td

d) DPT

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	c
5	d
6	b
7	b
8	b
9	b

10	a
11	b
12	b
13	a
14	c
15	a
16	d
17	c

Detailed Explanations

Solution to Question 1:

Kata thermometer is used to measure the low air velocity rather than the cooling power of the air.

It was originally used to measure the cooling power of the air (air temperature, humidity, and air movement) but currently, it is used only as an anemometer to measure low air velocities.

A wet kata reading of 20 and above and a dry kata reading of 6 and above were taken as indices of thermal comfort.

Kata thermometer is shown in the image below.

Other options:

Option B: Globe thermometer is used for the direct measurement of mean radiant temperature of the surroundings.

Options C and D: Wet globe thermometer consists of a dial thermometer. It provides a measure of the cooling capacity of the working environment.

Solution to Question 2:

The vector for bubonic plague is Xenopsylla cheopis. It is also called a rat flea. It also transmits murine typhus, chiggerosis, and hymenolepsis diminuta. It can be identified by the presence of a conical-shaped head, the presence of thorax, abdomen, and three pairs of legs.

Plague is a zoonotic disease caused by Yersinia pestis. It is a gram-negative, non-motile, coccobacillus that exhibits bipolar staining with special stains like Wayson's stain. The reservoir of infections is wild rodents like field mice, gerbils, skunks, etc. In India, Tatera indica is the main reservoir. The important vector of plague is rat flea, also called Xenopsylla cheopis.

Humans acquire the disease by the bite of an infected flea, by direct contact with the tissues of the infected animal, or by droplet infection from cases of pneumonic plague.

Human plague mainly presents as:

Bubonic plague- This is the most common type of disease. The patient develops a sudden fever, chills, headache, and painful lymphadenitis. Within a few days, the lymph nodes in the groin, axilla, or neck enlarge and are called buboes. This is followed by the suppuration of the buboes. Bubonic plague cannot spread from person to person as the bacilli are locked up in the buboes and do not find a way or easy exit.

Pneumonic plague- It develops as a complication of bubonic-septicemic plague.

Septicemic plague- It occurs when the infection spreads to blood.

Other options

Option B- Species Phlebotomus argentipes, belongs to the group of sand fly. It transmits kala azar. Diseases caused by other species of sand flies are sandfly fever and oriental sore.

Below image shows sand fly

Option C- Ixodes ticks (hard ticks) transmit rocky mountain spotted fever, Russian spring-summer encephalitis, Colorado tick fever, tularemia, tick paralysis and human babesiosis.

Below image shows Ixodes tick

Option D- Female Anopheles mosquitoes are vectors for malaria and filariasis (not in India). They can be identified by the presence of spotted wings. When resting on a surface, they make an angle with the surface.

The below image shows anopheles mosquito.

Solution to Question 3:

Leptospirosis, followed by acute diarrhoeal diseases and dengue are most commonly prevalent following floods in Kerala. Malathion will be the choice of insecticide used in this scenario.

Leptospirosis is a bacterial zoonotic disease transmitted via water contaminated with rodent urine. Infected rodents excrete large amounts of leptospira in their urine and floods facilitate the spread of the organism.

Prevention of leptospirosis:

• Chemoprophylaxis with Doxycycline 200 mg once a week for 6 weeks is recommended for high-risk populations in endemic areas. In case of fever, Doxycycline 100 mg twice a day for seven days is the recommended dose.

• Rubber shoes and gloves should be used.

• Anti-rodent measures must be taken.

• Mapping of water bodies must be done to establish a proper drainage system.

Other diseases usually transmitted after floods are typhoid fever, malaria, cholera, hepatitis A & E.

General measures to control disease spread during floods are:

• Chlorine tablets are distributed to disinfect the household water supply. One tablet can chlorinate 20 liters of water. Drinking water disinfectants like bleaching powder are also used.

• Malathion is an organophosphorus synthetic insecticide used for vector control (adult mosquitoes). It is a contact poison that is sprayed using ultra-light volume fogging machines.

• Indoor residual sprays with insecticides and larvicides like abate for drinking water are recommended.

• Tetanus booster dose for previously vaccinated people with open injuries may be indicated.

• General hygiene and sanitary measures must be followed.

Other health risks posed by floods are hypothermia, respiratory tract infections, snake bites, and injuries/drowning.

Other options:

Option A: Zinc phosphide is a rodenticide used to kill animals like mice and rats. It is a highly toxic compound and can produce toxicity by oral ingestion, inhalation, and through damaged skin.

Option C: Lindane is used as an insecticide for hardwood logs, lumber, and crops. It also has topical pediculicidal and scabicidal activities and is used in pet treatment.

Option D: Paris green is an insecticide that is not much used due to the availability of lesser phytotoxic agents.

Solution to Question 4:

The disinfectant used in blood and body fluid spillage management is sodium hypochlorite.

Procedure to be followed in the event of blood/body fluid spillage :

• Wear a pair of non-sterile gloves

• Broken glass (if any) of the container is cleaned first using tongs or a pan and brush and discarded in the sharps container. Caution is taken to not use hands.

• Disposable paper towels are used to absorb as much as possible.

• Water and detergent are used to clean the surface until visibly clean.

• The area is saturated with 0.5% Sodium Hypochlorite. This is prepared by diluting 5.25% sodium hypochlorite bleach at a 1:10 ratio.

• The tongs, pan, and brush are washed under running water and left to dry

• The gloves are removed and discarded.

• Hands are washed with soap and water using proper technique and dried with a disposable towel.

• The spillage is recorded in an incident book.

Solution to Question 5:

According to the data presented in the question, the air quality index for the 23rd of November is 407 which is classified as severe.

Air Quality Index (AQI) -

A tool used for effective communication of air quality status on a scale that is easily comprehensible. Complex air quality data of various pollutants are compiled to give a single score (index value), nomenclature and colour that determines the air pollution of that particular location.

Eight pollutants are taken into consideration to calculate the AQI -

• PM10 (Airborne particulate matter with a diameter of 10 microns or less)

• PM2.5 (Airborne particulate matter with a diameter of 2.5 microns or less)

• NO_2 (Nitrogen dioxide)

• SO_2 (Sulfur dioxide)

• CO (Carbon monoxide)

• O_3 (Ozone/Trioxygen)

• NH_3 (Ammonia)

• Pb (Lead)

AQI categories and the range are as follows :

AQI Category	AQI Range
Good	0 - 50
Satisfactory	51 - 100
Moderately polluted	101 - 200

AQI Category	AQI Range
Poor	201 - 300
Very poor	301 - 400
Severe	401 - 500

Solution to Question 6:

The given scenario is suggestive of Japanese encephalitis for which a killed vaccine is administered intramuscularly.

The Japanese encephalitis (JE) virus is transmitted by the bite of Culex mosquitoes, specifically Culex tritaeniorhynchus and Culex vishnui (common in India) species. They present with fever and headache in mild disease form. Still, severe disease presents with rapid onset of high fever, headache, neck stiffness, disorientation, coma, seizures, spastic paralysis, and ultimately death.

Killed JE vaccines (Inactivated SA 14-14-2 cell culture-derived and inactivated Vero cell culture-derived) are gradually replacing live attenuated vaccines. This change is because the killed vaccines are reported to have better immunogenicity than the live attenuated vaccine.

The SA 14-14-2 (live attenuated) vaccine is based on a stable neuro-attenuated strain of the JE virus with a Yellow Fever vaccine acting as a viral vector. Adverse effects of the vaccine include fever and rash.

Dose: 0.5 ml

Site: Live vaccine: Subcutaneous, left upper arm

Killed vaccine: Intramuscular, anterolateral side of left mid-thigh

They can be administered up to 15 years of age with a month gap between the two doses.

The JE virus vaccine has been included in the National Immunization Schedule (NIS) to be administered in endemic areas (Uttar Pradesh, Assam, West Bengal, Karnataka) in two doses. The first dose is administered at 9-12 completed months. The second dose is administered at 16-24 months. The schedule however mentions the use of both live and killed vaccines.

Note: According to the national immunisation schedule, both live attenuated, killed-inactivated vaccines are now used for vaccination against the JE virus.

Solution to Question 7:

The evaluation based on the treatment given to the patient and their clinical management in a health care facility measures the "process" of medical care.

The basic steps of evaluation are as follows :

• Determine what is to be evaluated

• Establish standards and criteria

• Plan the methodology to be applied

• Gather information

• Analyze the results

• Take action

• Re-evaluate

The three types of evaluation are:

• Evaluation of "structure": This is an evaluation of whether facilities, equipment, manpower, and organization meet a standard approved by experts.

• Evaluation of "process": The process of medical care includes the problems of recognition, diagnostic procedures, treatment and clinical management, care, and prevention. They are evaluated by comparing with a predetermined standard. Eg; Evaluation of the physician (or nurse) performance is known as "Medical (or nursing) Audit".

• Evaluation of "outcome": This is concerned with the benefits gained by people using the health services. The traditional outcome components are the "5 Ds" of ill-health, i.e; disease, discomfort, dissatisfaction, disability, and death.

Elements of evaluation are relevance, adequacy, accessibility, acceptability, effectiveness, efficiency and impact.

Solution to Question 8:

The clinical scenario showing an immunocompromised child should not receive the Measles/MMR vaccine from the given options as it is a live attenuated vaccine which is contraindicated in immunocompromised people.

Contraindications for live vaccines:

• Immune deficiency diseases

• Suppressed immune responses as a result of

• Leukaemia

• Lymphoma

• Malignancies

• Corticosteroid therapy

• Treatment with alkylating agents and antimetabolic agents

• Therapeutic radiation

• Pregnancy

Measles vaccine:

Live attenuated vaccine for measles is safe and effective with no evidence of person to person transfer of vaccine strains. It is freeze-dried product requiring reconstitution. 0.5ml is administered subcutaneously in the right upper arm as two doses at the 9th month and between the 16th-24th month. They also contain a small amount of neomycin.

Other options :

Option A - Killed IPV is a form of killed/inactivated vaccine where the absolute contraindication is a severe local or general reaction to a previous dose.

Options C and D - DPT and TT/Td fall under toxoids/protein vaccines that are usually highly efficacious and safe immunizing agents.

Solution to Question 9:

From the given options, infectious diseases are the conditioning influence for malnutrition.

Ecology of Malnutrition :

The ecological factors related to malnutrition as follows

• Conditioning influences: A conditioning factor that plays a huge role in malnutrition is infectious diseases which causes a vicious cycle where infections cause malnutrition which in turn causes minor childhood ailments. Diarrhoeal diseases, intestinal parasites, measles, whooping cough, malaria, tuberculosis contributes significantly to malnutrition. The occurrence of infections are largely based on the environmental conditions that the child is exposed to.

• Cultural influences: Poor diets as a result of cultural and regional influences also play a role in malnutrition. They are as follows with an example for each:

• Food habits, traditions, customs, beliefs and attitudes: Rice and wheat are largely staple in India. Papaya is avoided as it is thought to cause abortion.

• Food fads: Personal likes and dislikes to a certain group of foods.

• Cooking practices: Prolonged boiling of cereals in open pans which leads to a loss of nutritive value

• Child-rearing practices: Premature curtailment of breastfeeding

• Socio-economic factors: Poverty plays a huge role in causing malnutrition. This is usually a result of ignorance, insufficient education, lack of knowledge regarding the nutritive value of foods, inadequate sanitary environment, large family size, etc. which affects the quality of life.

• Food production: The amount of food produced is directly proportional to the amount of food consumed. Developed countries have 5.8 hectares of arable land compared to the 0.6 hectares of land per Indian. Even distribution of food among all sectors of society is also a factor.

• Health and other services: Remedial actions that can be taken by the health sector to reduce malnutrition are as follows -

• Nutritional surveillance - Continuous monitoring of factors that affect nutrition; groups affected are identified.

• Nutritional rehabilitation - Urgent care for individuals suffering from severe malnutrition.

• Nutrition supplementation - Supplementary feeds for mothers and children

• Health education - Health education programmes to increase awareness about malnutrition and nutrition value of food.

Solution to Question 10:

The infant mortality rate is not a component of the global hunger index.

The Global Hunger Index (GHI) has three dimensions. These dimensions are measured by indicators.

GHI score = (1/3 X standardized PUN) + (1/6 X standardised CWA) + (1/6 X standardised CST) + (1/3 X Standardised CM)

Where,

• PUN is the proportion of the population that is undernourished

• CWA is the prevalence of wasting in children under 5 years of age

• CST is the prevalence of stunting in children under 5 years of age

• CM is the proportion of children dying under the age of 5 years of age.

The result lies on a scale from 0 to 100, with 0 meaning no hunger and 100 being the worst score. The GHI is given by the International Food Policy Research Institute. In 2021, India's GHI was 27.5. It ranks 101 among 116 countries. In the 2022,India's GHI stands at 29.1, and India ranks 107th out of the 121 countries.

Dimensions	Indicators
Inadequate food supp ly	Proportion of the population undernourished
Child mortality	Under 5 mortality rate
Child undernutrition	Wasting in children under 5 yearsStunting in children un der 5 years

Solution to Question 11:

From the study results, option B is the true statement i.e; an increase in the CRP increases the risk of myocardial infarction/stroke.

Relative risk (RR)/risk ratio is the ratio of the incidence of the disease (or death) among exposed and the incidence among non-exposed.

$$RR = \frac{\text{Incidence of disease (or death) among exposed}}{\text{Incidence of disease (or death) among non-exposed}}$$

It is a direct measure of the strength of the association between suspected cause and effect. The larger the RR, the greater the strength of the association between the suspected factor and disease. As there is an increase in the RR with increasing CRP levels, the risk of an MI/stroke also increases with elevated CRP levels.

RR = 1 indicates no association.

RR >1 indicates a positive association.

RR of 2 or higher indicates a >/=100% increase in risk.

Relative risk calculation (RR) = [a/(a+b)]/ c/(c+d)] where,

In the question, the table shows an increase in relative risk with an increase in CRP values. This thereby indicates that an increase in the CRP values increases the risk of myocardial infarction/stroke.

Note: Relative risk provides information on the increase or decrease in the likelihood of a condition based on exposure. It does not provide the absolute risk of the event occurring. The last option states that "In quintile 1 there is no risk of MI/stroke". Even though the relative risk is 1, it only indicates no association but does not exclude the occurrence of MI/ Stroke.

	Disease		
Exposure		yes	no
yes	a	b	
no	c	d	

Solution to Question 12:

The used throat swab should be discarded in the yellow color bin as it is a clinical laboratory waste.

Microbiology, biotechnology, and clinical laboratory waste are collected in autoclave-safe bags. They must be pre-treated before disposal by incineration, plasma pyrolysis, or deep burial.

Solution to Question 13:

The SI unit for the measurement of brightness of light from a point source is Candela.

The "power" of light from a given point source which radiates in all directions is also known as it's luminous intensity of the point of source. This is measured in candela or candle light.

The following table shows different units used for the measurement of light as recommended by the International Organization for Standardization:

Quantity Measured	Description	SI Unit	Other Units Used
Luminous intensity	Brightness of a point source	Candela	Candle power
Luminous flux	Flow of light	Lumen	-
Illumination(Illuminance)	Amount of light reaching a surface	Lux (per unit area)	Foot candle, phot, lm/cm^2
Luminance	Amount of light reflected from a surface	Lambert	Foot lambert, candle/cm^2

Solution to Question 14:

The given clinical scenario showing the absence of serum antibodies is indicative of susceptibility to chickenpox.

Varicella zoster virus causes two host responses which present as chickenpox and herpes zoster. They are characterized by a vesicular rash that is accompanied by fever and malaise (chickenpox). The immune system eliminates the infection except for some virions that establish latent infections inside nerve cells which are reactivated when immunity is compromised and present as herpes zoster.

One attack with the virus gives durable immunity while a second attack of the virus is rare. In the absence of immunity, all ages are equally susceptible to the disease. The acquisition of maternal antibodies protects the infant during the first few months of life. The IgG antibodies produced during the first attack with the virus persist for life and their presence is correlated with protection against varicella. The absence of antibodies in the given scenario signifies that the patient is susceptible to chicken pox. Hence, the serologic screening of serum for lgG antibodies is used to assess immunity or susceptibility of a person to varicella, especially in unvaccinated people, e.g. in healthcare workers.

The presence of cell-mediated immunity appears to be important in the recovery from VZ infections and in protection against the reactivation of latent VZ virus which causes Herpes zoster.

Varicella in pregnant women :

The contraction of the virus by a pregnant woman increases the risk of the fetus to congenital varicella syndrome. This is seen in mothers who are affected during the first 20 weeks of gestation. Affected babies present with the following :

• Foetal wastage

• Cutaneous scars

• Atrophied limbs

• Microcephaly

• Low birth weight

• Cataract

• Microphthalmia

• Chorioretinitis

• Deafness

• Cerebra-cortical atrophy

Prevention:

• Varicella-Zoster lmmunoglobulin (VZIG) is given as post-exposure prophylaxis within 72 hours of exposure and has been recommended for prevention of chickenpox in exposed susceptible individuals particularly in immunosuppressed persons such as:

• Persons receiving immunosuppressive therapy

• Persons with congenital cellular immunodeficiency

• Persons with acquired immunodeficiency including HIV/AIDS

• Exposed pregnant women

• Newborns

• Premature infants of low birth weight

• Live attenuated vaccine is administered to children who have not had chicken pox between the age of 12 to 18 months.

Solution to Question 15:

The school health programs are managed by the primary health centre (PHC).

The School Health Committee (1961) in India recommended that the school health services should be a part of the general health services. Since the general health services are administered through PHCs, school health programs are also managed by the PHCs under their jurisdiction.

The School Health Committee advocated the formation of school health committees at the village level, block level, district level, state level, and national level. The National School Health Council will be an advisory and coordinating body.

PHCs provide the following school health services:

• Screening of general health assessment of anemia/nutritional status, visual acuity, hearing problem, dental check-up, physical disabilities, learning disorders, and behavior problems, etc.

• Basic medicines for common ailments

• Immunization

• Micronutrient (Vitamin A, iron, and folic acid) management

• Deworming

• Mid-day meal

Note: National Programme for Mid-Day Meal in Schools has recently been renamed Pradhan Mantri Poshan Shakti Nirman (PM POSHAN).

Solution to Question 16:

The given clinical features are suggestive of neurolathyrism and excessive consumption of the pulse Lathyrus sativus is the main cause. Hence, a diet history should be elicited in this patient to confirm the diagnosis.

Lathyrism is a paralyzing disease of humans and animals. It is due to the consumption of Khesari dal/Lathyrus sativus. When the husk is removed, it looks similar to red gram dal or Bengal gram dal. It is more prevalent in parts of Madhya Pradesh, Uttar Pradesh, Odisha, and Bihar.

In humans, it is referred to as neurolathyrism because it affects the nervous system. It is characterized by gradually developing spastic paralysis of the lower limbs.

In animals, it is referred to as osteolathyrism/odoratism because the pathological changes in the bones lead to skeletal deformities.

It usually affects young males aged 15-45 years and it is manifested in stages:

- Latent stage
- No-stick stage
- One-stick stage
- Two-stick stage
- Crawler stage

BOAA (Beta oxalyl amino alanine) is the toxic compound causing Lathyrism. The water-soluble property of the toxin is used in removing it from the pulses by soaking it in hot water and rejecting the soaked water.

Parboiling can be used for detoxifying the crop on large-scale operations. Overnight soaking in lime water followed by boiling is also effective in destroying the toxin.

Vitamin C in doses of 500-1000mg can be used for the prevention and treatment of lathyrism.

The Prevention of Food Adulteration Act in India has banned Lathyrus sativus in all forms (whole/split/flour).

Solution to Question 17:

In the given scenario, a ten-year-old schoolchild will be administered the TT/Td vaccine according to the National Immunization Schedule.

The Tetanus and adult diphtheria (Td) vaccine is a combination of tetanus and diphtheria with a lower concentration of diphtheria antigen (d) as recommended for older children and adults. The MoHFW recommends the replacement of TT vaccine with Td vaccine for all age groups, including pregnant women. A dose of 0.5ml is administered in the intramuscular route in the upper arm.

Scheduled to be administered at 10 and 16 years of age.

For pregnant women :

- Td-1: early in pregnancy
- Td-2: 4 weeks after Td-1

• Td (Booster): if pregnancy occurs within 3 years of last pregnancy and 2 Td doses were received.

Other options:

Option A - According to the National Immunization Schedule, BCG should be given at birth or as early as possible up to 1 year of age. It provides protection against CNS and disseminated TB. A dose of 0.1 ml is administered intradermally in the left upper arm.

Option B - Measles/MR is given at 9 completed months (1st dose) and a second dose is given at 16-24 months. Measles can be given till 5 years of age. A dose of 0.5ml is administered subcutaneously in the right upper arm.

Option D - DPT vaccine is administered as a part of the pentavalent vaccine at the 6th, 10th and 14th weeks of life. DPT booster doses are administered at 16-24 months (first booster) and 5-6 years (second booster). It can be administered until 7 years of age (with booster doses). A dose of 0.5ml is administered in the anterolateral aspect of the mid-thigh. The 2nd booster dose can be administered in the upper arm.

Community Medicine INI-CET 2022

Question 1:

As per the National Tuberculosis Elimination Program (NTEP), HIV TB prevalence of ______ would deem a district as a high-priority district?

- **a)** >10%
- **b)** >12%
- **c)** >15%
- **d)** >20%

Question 2:

India is a country with different cultures and diverse languages. Which of the following steps should a physician take to address the patient for better outcomes?

- **a)** 1, 2
- **b)** 2, 3
- **c)** 3, 4
- **d)** 1, 4

Question 3:

5-year-old Ram and 3-year-old Sham are Rama Devi's sons. Sham developed a rash, but Rama Devi ignored it, assuming it to be an allergy. After 3 days, Ram also developed a rash along with a fever. Out of fear, Rama Devi took Ram to the primary health center (PHC). The healthcare workers at the PHC suspected it to be measles. Choose all correct options from among the following:

- **a)** 1 and 4 are correct
- **b)** 1 only is correct
- **c)** 1, 2, 3 and 4 are correct
- **d)** 1 and 3 are correct

Question 4:

You are measuring a patient's blood pressure at a primary health center when the sphygmomanometer fell and broke and mercury was spilled onto the ground. How will you clean the mercury?

a) Collect it by using a card onto a sheet of paper and put the mercury into a jar

b) Collect it by sweeping it with a broom and throw it outside

c) Collect it by sweeping it and flush it in the toilet

d) Collect it by hand and then wash hands with water

Question 5:

The true statement about screening tests for genetic diseases is:

a) It is always invasive

b) Screening test has better accuracy than diagnostic test

c) It defines risk of transmission of disease to the child

d) Screening requires genetic mapping

Question 6:

Which of the following is the true statement regarding measures to prevent typhoid transmission in the community?

a) Typhoid vaccine administration is the best method of preventing transmission

b) Hygiene practices and clean sanitation control is more important than typhoid vaccine

c) In 10% cases, person to person transmission occurs

d) Drug resistance in typhoid is not as big a problem as in TB

Question 7:

Which of the following is true about the management of a baby born to an HIV-positive mother?

a) If replacement feeding, the neonate should be tested for HIV, if negative, ART is not required.

b) In exclusive breastfeeding, give nevirapine for 6 months only when baby is positive

c) If replacement feeding, the neonate need not receive nevirapine if HIV negative

d) If exclusive replacement feeding, give nevirapine for 6 weeks, even if the baby is negative

Question 8:

What is the branch of statistics dealing with describing the internal collection of data?

- **a)** Theoretical statistics
- **b)** Descriptive statistics
- **c)** Analytical statistics
- **d)** Inferential statistics

Question 9:

Isolation in a separate room is mandatory for which of the following air-borne infections?

- **a)** Measles
- **b)** Rabies
- **c)** Nipah
- **d)** Aseptic meningitis

Question 10:

All the following cause widening of confidence interval except?

- **a)** Increase in sample size
- **b)** Decrease in sample size
- **c)** Increase in variability
- **d)** Increase in confidence level

Question 11:

False statement regarding rabies immunoglobin is:

- **a)** It is exclusively given in the deltoid
- **b)** The dose is 20 IU/kg
- **c)** Not given after 7 days of the 1st dose of the vaccine
- **d)** If human RIG is unavailable, equine is preferred

Question 12:

A 1-year-old child is brought to the healthcare unit that you are working on. His mother mentions that he had received the first dose of DPT vaccine at 6 weeks of age but has missed vaccination since then. What would you advise the patient?

a) Do not give DPT

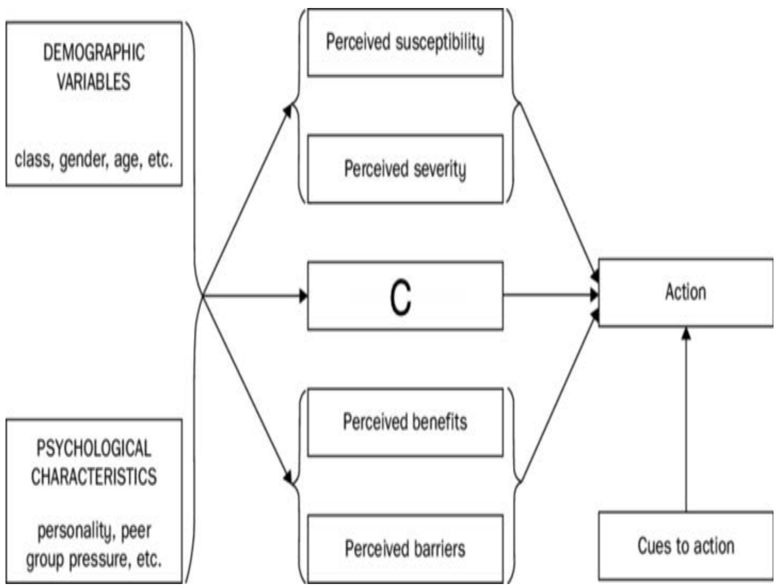
b) Restart DPT

c) Give second dose of DPT

d) Give dT alone

Question 13:

The following image shows a health belief model. What does box C stand for?



a) Perceived control

b) Behavioral modification

c) Health motivation

d) Self-efficacy

Question 14:

A point-of-care diagnostic test with a sensitivity of 98% shows a negative test result in a patient. What does this imply?

a) The patient would need another point-of-care diagnostic test for the same clinical condition.

b) Prevalance of the disease is 2%.

- **c)** Patient is less likely to have the disease.
- **d)** Patient has been wrongly ruled out due to false negativity.

Question 15:

The table given below shows a multivariate analysis of the relation between the risk of developing blindness and age, literacy rate, residence, and gender. Which of the following is true?

- **a)** 2 and 3
- **b)** 1 and 3
- **c)** 1 and 2
- **d)** 3 and 4

Question 16:

A patient with lymphatic filarisis is being managed as shown in the image below. What is the level of prevention done here?

- **a)** Specific protection
- **b)** Disability limitation
- **c)** Primary prevention
- **d)** Health promotion

Question 17:

Which of the following indicators can help in determining whether the health system is giving importance to identifying leprosy in the community?

a) Treatment completion rate

b) Annual new case detection rate per lac

c) Treatment initiation rate

d) Proportion of newly diagnosed patients with grade-2 disability

Question 18:

Which of the following is true regarding the vaccine vial monitor (VVM)?

a) 3,4

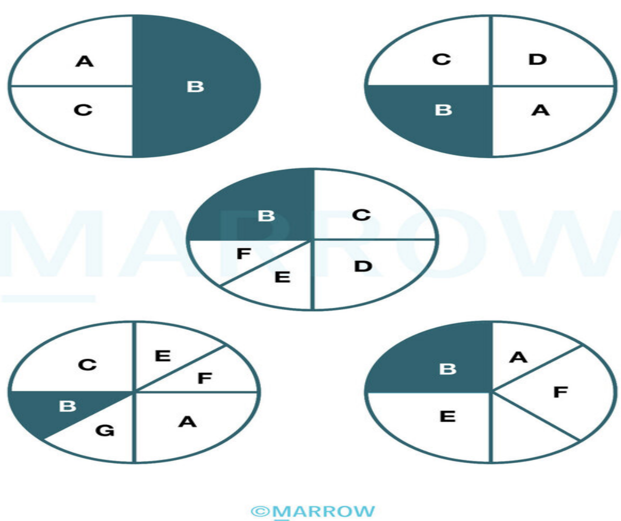
b) 1,2

c) 1, 2, 3, 4, 5

d) 5,6

Question 19:

Which of the following models of disease causation is depicted in the image below?



a) Sufficient component cause model

b) Epidemiological triad

c) PERT model

d) Web of causation model

Question 20:

Attributable risk calculated to understand the relation between smoking and lung cancer is 90%. Which of the following is true?

a) 10% cases of lung cancer can be eliminated in smokers if smoking is eliminated

b) 90% cases of lung cancer can be eliminated in population if smoking is eliminated

c) 90% cases of lung cancer can be eliminated in smokers if smoking is eliminated

d) 10% cases of lung cancer can be eliminated in population if smoking is eliminated

Question 21:

Following an earthquake, the building codes were changed to earthquake-resistant buildings. Which of the following step of disaster management does this fall under?

a) Disaster response

b) Disaster reconstruction

c) Disaster rehabilitation

d) Disaster mitigation

Question 22:

Which of the following statements are true regarding drugs used in anti-tubercular treatment?

a) A, B, C, D

b) B, C

c) A, C

d) A, D

Question 23:

Which of the following criteria should be satisfied to start MDT?

a) 1,4

b) 1,3

c) 1,3,4

d) 1,2,3,4

Question 24:

Match the following stages with their relevant descriptions.

a) 1-c, 2-d, 3-a, 4-d

b) 1-d, 2-b, 3-a, 4-c

c) 1-b, 2-d, 3-c, 4-d

d) 1-a, 2- c, 3-d, 4-b

Question 25:

Contraceptive failure is assessed using which of the following methods?

a) Pharmacological index

b) Pearl index

c) Performance index

d) Efficacy index

Question 26:

Match the following scientists with their discoveries:

a) 1-a, 2-b, 3-a

b) 1-b, 2-a, 3-c

c) 1-c, 2-b, 3-a

d) 1-b, 2-c, 3-a

Question 27:

Identify the formula given below:

$$\sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

a) Standard deviation

b) Standard error

c) Median

d) Variance

Question 28:

A child presents with a dog bite from a street dog. On examination, an abrasion of 1.5 cm is present, with slight oozing. Which of the following statements are correct?

a) 2, 3, 5, 6

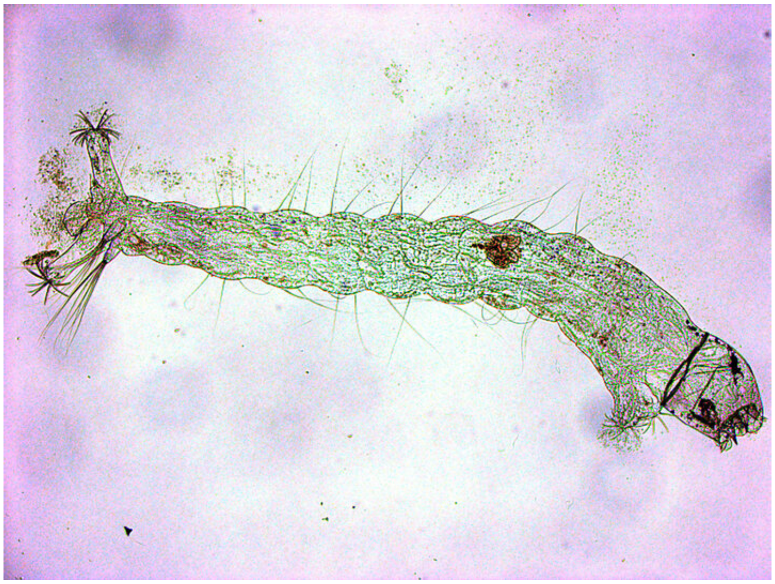
b) 1, 2, 5, 6

c) 1, 2, 3, 5

d) 1, 2, 3, 6

Question 29:

Water from a broken septic tank is collected in a container and breeding mosquitoes are seen. The image of the larva is given below. Identify the species of the mosquito.



a) Anopheles

b) Aedes

c) Mansonia

d) Culex

Question 30:

The drug shown in the image below is administered for:

a) Malaria

b) Dengue

c) Sexually transmitted infections

d) AIDS

Question 31:

Under the Anemia Mukt Bharat initiative, mild to moderate anemia in pregnant women < 34 weeks of gestation is treated using:

a) 3 iron and folic acid (IFA) tablets, taken OD

b) IV iron sucrose for non-compliance with oral tablets

c) 2 iron and folic acid tablets OD + IV iron sucrose

d) IM ferric carboxy maltose (FCM)

Answer Key

Question No.	Correct Option
1	a
2	d
3	c
4	a
5	c
6	b
7	d
8	b
9	a
10	a
11	a
12	c
13	c
14	c
15	d
16	b
17	d
18	b
19	a

20	c
21	d
22	d
23	b
24	b
25	b
26	d
27	a
28	d
29	d
30	a
31	b

Detailed Explanations

Solution to Question 1:

As per the National Tuberculosis Elimination Programme (NTEP), an HIV-TB prevalence of >10% would deem a district as a high-priority district.

RNTCP (Revised National Tuberculosis Control Programme) has been revised by the government of India as NTEP. The vision and goals of the NTEP programme respectively are

• Tuberculosis (TB) free India with zero deaths, diseases and poverty due to tuberculosis

• To achieve a rapid decline in the burden of TB, morbidity and mortality while working towards the elimination of TB in India by 2025.

NTEP has categorized and prioritized the districts in the country based on the following criteria to intensify tuberculosis control.

Categorization	Criteria	Category
High TB	Total TB case notification rate > 180 per lakh	A
High TB-HIV(Tuberculosis and HIV co-infected patients)	>10%proportion of known HIV positives amongst TB patients tested for HIV	B
High DR-TB (Drug-resistant tuberculosis)	> 25% proportion of relapse out of incident smear-positive TB cases	C
Very low case finding effort	Annual TB suspect examination rate of < 400 per lakh population	D

Categorization	Criteria	Category
Average	None of above	E
High case finding but Low T CNR	Annual suspect examination rate >1200 per lakh population and total case notification rate < 80 per lakh population	F

Solution to Question 2:

The physician should have good communication and consider the patient's religious, and cultural perceptions for better outcomes (statements 1 and 4 are correct).

Statement 1 is true since the proper usage of vocabulary that is easy for the patient to understand is essential for the patient. The doctor should clarify the patient's interpretation of the words used. Thus insisting on good communication is essential for better health outcomes.

Statement 2 is not true. An interpreter is needed if the patient is not a native speaker of the language spoken in that region. The interpreter can be a family member above the age of 16 or an independent interpreter. Communication gets affected with open-ended questioning getting lost in the double translation. The answer returned might not be adequate for the doctor and the consultation as such becomes longer than usual. The doctor will have to change to a more direct style of questioning to prevent any misunderstanding.

Statement 3 is not true while statement 4 is true. Personality, culture, and beliefs are important in creating trust between the doctor and patient and are an important component in the medical interview for a better outcome. The doctor must listen to the patients' concerns, and preferences.

Health communication may fail if the message is not adequate. A good message should have the following:

- It should be in line with the objective
- It needs to be meaningful
- Based on the felt needs
- Clear and understandable
- Specific and accurate
- Culturally and socially appropriate.

Solution to Question 3:

All 4 statements are correct from the given clinical scenario.

Statement 1 is correct: Ram is the index case as he was the first case to come to the attention of the investigating authority (PHC)

Statement 2 is correct: From the given clinical scenario, considering the family of Ramadevi as a unit, Sham is the primary case (the first person who gets the communicable disease and introduces it to the community).

Statement 3 is correct:

• Incubation period from the time of exposure to the development of rash is 7 to 18 days with an average of 14 days.

• Ram developed the rash 3 days after Sham. Considering the period of communicability of rash as 4 days prior to or 4 days later than the onset of rash, it is evident that Sham was in an infectious state and Ram has contracted the infection from Sham

Statement 4 is correct: Post-exposure prophylaxis (PEP) can be given in the form of a vaccine or immunoglobulin. The vaccine can be given within 72 hours of exposure, while immunoglobulin (IVIG) can be given up to 6 days following exposure. Hence, Ram could would have been eligible to receive PEP if he were brought to the PHC along with Sham.

Solution to Question 4:

Mercury is cleaned by collecting it by using a card onto a sheet of paper and putting the mercury into a jar.

After a mercury spill, the following steps should be followed:

• Isolation of the spill and ventilation of the area

• Collection of items to clean the spill: zipper-top plastic bags, trash bags, gloves, paper towels, cardboard or squeegee, eye dropper, tape, flashlight, and powdered sulfur.

• Clean up the spill.

• After putting on the gloves, the glass pieces are picked and placed on a towel, rolled and disposed of into a zipper-top plastic bag, and finally sealed.

• The beads of mercury are cleaned by using cardboard or a squeegee. They are rolled onto a sheet of paper, disposed into the zipper bag, and sealed.

• Powdered sulfur may be used to absorb small beads that are not visible

• Looking for any missed spill using a flashlight

• Remove other mercury contaminated items and wipe off any mercury beads with a wet towel into a plastic trash bag

• Proper disposal of contaminated clean-up materials.

• Keep the area well ventilated for 24 hours, keep pets and children away from the area

The use of a vacuum cleaner or broom to clean the spill should be avoided. This will break the mercury into smaller droplets and spread it further. Mercury should not be poured down any drain. Do not use a washing machine to clean the clothes contaminated with mercury. This can contaminate the water and sewage. Mercury-contaminated shoes or clothing should not be worn.

Solution to Question 5:

The true statement about screening tests for genetic diseases is it defines the risk of transmission of disease to the child.

A screening test is an initial examination done in apparently healthy individuals while diagnostic tests are done on individuals with positive test results for further workup and treatment.

The aim of performing a screening test is to favorably change the natural history of the disease by an early diagnosis and treatment in individuals who are positive for the particular screening test. They are less invasive and less accurate than diagnostic tests.

Genetic screening is a type of public health program done in high-risk asymptomatic individuals in which tests are used to detect an inherited predisposition to disease, risk of transmitting the disease, prevention of the disease, early treatment, or family planning. For example, prenatal screening during pregnancy.

Genetic mapping is a tool used to identify new genes and their functions. It also may give information on which chromosome contains the defective gene and where the gene is present on that chromosome. Gene maps have been used to pinpoint the gene responsible for rare single-gene inherited disorders. Screening does not require genetic mapping.

Solution to Question 6:

The true statement regarding typhoid transmission preventive measures is that hygiene practices and clean sanitation control are more important than the typhoid vaccine. It helps in breaking the chain of transmission.

Typhoid fever is caused by Salmonella typhi and is mainly transmitted to humans by the fecal-oral route.

Clinical features include high fever with chills associated with headache, malaise, cough, sore throat, abdominal pain, and constipation or diarrhea are seen in the early phase. The fever occurs in a step-ladder fashion reaching a plateau in 7-10 days. A maculopapular rash may be seen on the trunk and chest known as rose spots. In the late stages, hepatosplenomegaly, abdominal distension, relative bradycardia, and a dicrotic pulse can occur.

Diagnosis is by isolation of the organism from blood, bone marrow, and stools. Blood culture is mainly used for the diagnosis. Other tests such as the Widal tests are useful.

Management is mainly by sanitation. It is the ultimate method to control typhoid. It breaks the transmission of the disease. It is combined with proper health education.

Antibiotic therapy is started early and the choice of antibiotics depends on the susceptibility of the organism to the drug. In the case of drug-susceptible typhoid, ciprofloxacin is useful.

Drug resistance to typhoid has been emerging. Similar to drug resistance in TB, there are multidrug-resistant (MDR) and extensively drug-resistant (XDR) S. Typhi strains. MDR strains are resistant against at least 1 out of 3 or >3 categorically differentiated antimicrobials, such as ampicillin, cotrimoxazole, and chloramphenicol. XDR are those which are resistant to all except 1-2 antimicrobials.

Ceftriaxone, cefotaxime, and (oral) cefixime are effective for the treatment of MDR enteric fever.

Immunization is a complementary approach in management since the protective efficacy can be overcome by a high inoculum in the case of food-borne diseases.

The typhoid vaccines available are the Vi polysaccharide vaccine and the Ty21a vaccine. Ty21a is a live attenuated vaccine recommended after 6 years and is given orally on days 1,3,5 and 7. Vi polysaccharide vaccine can be given after 2 years as a single dose with a booster dose for every 2 years.

Solution to Question 7:

If exclusive replacement feeding, giving nevirapine for 6 weeks, even if the baby is negative is true regarding the management of a baby born to an HIV-positive mother.

The national AIDS control organization (NACO) has recommended the following for an infant born to an HIV-positive mother.

According to the National AIDS Control Organization (NACO), antiretroviral therapy (ART) should be given to all HIV-positive pregnant women including during labor hours and breastfeeding. They are started on a triple-drug ART. ART should be given irrespective of the CD4 count, and clinical stage and should be continued lifelong.

Immediate care of HIV-exposed infants includes wiping of baby's mouth and nostrils, handling of infants with gloves, immediate cord clamping, and initiation of feeding. An infant born to an HIV-positive mother should be breastfed exclusively or by replacement feeds exclusively for 6 months of age.

Infants should receive ARV prophylaxis. A single drug or dual drug regimen is followed depending on the risk of transmission which can be low or high risk.

• Low-risk infants are those born to mothers with a low viral load which is detected after 32 weeks of pregnancy till delivery.

• High-risk infants are those born to HIV-positive mothers, not on ART, with viral load not estimated beyond 32 weeks, viral load not suppressed after 32 weeks, or diagnosed with HIV within 6 weeks of delivery.

HIV Risk	ARV prophylaxis	Duration
Low-risk infants	Syrup nevirapine or zidovudine	Birth to6 weeksof ageirrespective of feeding
High-risk infants	Syrup nevirapine + syrup zidovudine	Exclusive replacement feeding: Birth to 6 weeksExclusive breastfeeding: Birth till 12 weeks

Solution to Question 8:

Descriptive statistics is the branch that deals with describing the internal collection of data.

Statistical methods are mainly of two types

• Descriptive statistics is a type of statistical method which helps to describe, organise or summarise the data. It refers only to the actual data. They are described by the measures of central tendency and the degrees of dispersion.

• Inferential statistics involves going beyond the actual data and making inferences to test the hypothesis.

Other types of statistics:

• Analytical statistics is mainly used to test the hypothesis.

• Theoretical statistics is a mathematical approach which involves describing, predicting the events or analyzing the relationship between things.

Solution to Question 9:

Isolation in a separate room is mandatory for measles among the given options. It is transmitted by droplet infection as well as via the airborne route. The period of isolation is from the onset of symptoms (catarrhal stage) through the fourth day of the rash.

Isolation is defined as the separation of an infected person or animal for the period of communicability in places to prevent direct or indirect transmission to susceptible individuals.

Measles is an infectious disease mainly seen in the childhood age group. It is caused by an RNA paramyxovirus and is transmitted to humans primarily by droplet infection and also through airborne transmission.

Features include

• The prodromal phase-characterized by fever, cough, coryza, and Koplik's spots which are rashes seen in the buccal mucosa opposite the first and second molar.

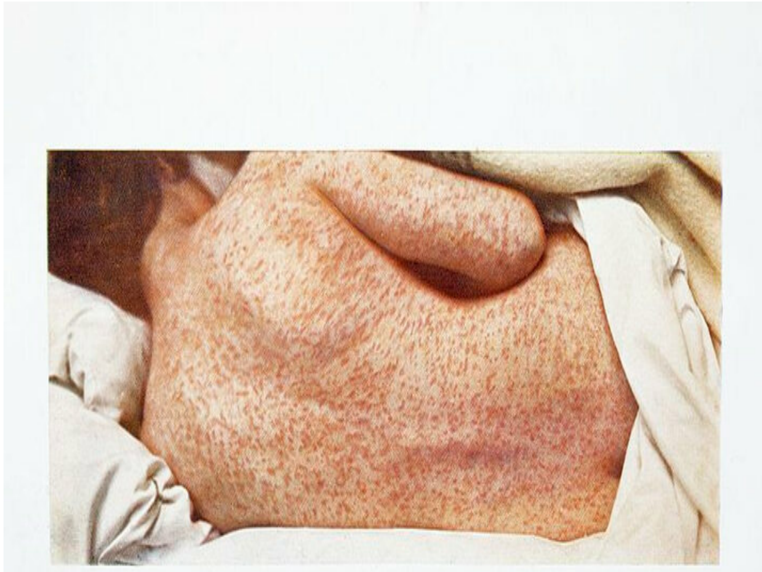
• The eruptive phase- macular or maculopapular rash begins behind the ears and spreads to the face, neck, and extremities.

• Post measles phase is characterized by weight loss.

Some of the complications of measles include otitis media, laryngotracheobronchitis (croup), diarrhea, post-infectious measles, or subacute sclerosing pan encephalitis (SSPE). Subacute sclerosing panencephalitis is a rare but fatal complication.

Treatment is mainly aimed at supportive care such as relieving the symptoms, treating dehydration, and isolating patients.

The below images show a characteristic rash in measles and Koplik's spots on the buccal mucosa.



Other options:

Option B: Rabies is transmitted via a bite or scratch from an infected animal. Contraction of rabies by inhalation of virus-containing aerosols is extremely rare. Human-to-human transmission is possible but has not been confirmed.

Option C: Nipah is transmitted mainly from direct contact with sick pigs or their contaminated tissues. To prevent human-to-human transmission, close unprotected contact with infected people is avoided. Regular hand hygiene should be followed when caring for or visiting infected people.

Option D: If the diagnosis is confirmed as aseptic meningitis, the patient can often be discharged home. Isolation is required if the meningitis is of infectious etiology.

Solution to Question 10:

All the given options cause a widening of confidence interval except for an increase in sample size. An increase in sample size leads to a narrower confidence interval.

Research is usually done with samples drawn from a population, rather than by using the entire population. In order to get a fair idea of the population parameters, sample statistics and estimates of error in the sample are used to get a range of values. This range is the confidence interval (CI). It is estimated based on a desirable confidence level.

The confidence interval is calculated as the difference between upper and lower confidence limits.

The width of the CI is an indication of the utility of our population parameter estimation. Factors affecting the width of the confidence interval include:

• Desired confidence level- width of CI is directly proportionate to the confidence level

• Variability in the sample- width of CI is directly proportionate to the variability in the data

• Sample size- width of the CI is inversely related to the sample size

Therefore, increases in variability and confidence levels and decreases in sample size lead to wider confidence intervals.

Solution to Question 11:

The false statement regarding rabies immunoglobulin is it is exclusively given in the deltoid. Rabies immunoglobulin is given as much as possible into the wound site or around it.

Rabies immunoglobulin (RIG) is passive immunization. It is administered only once, preferably within 24 hours of the exposure, ie, on day 0 of the anti-rabies vaccine (ARV). If RIG was not given with the ARV 1st dose, it can be given up to the 7th day after the first dose of ARV. It is not given after 7 days of receiving the 1st dose of ARV.

The dose is 20 IU/kg body weight for human rabies immunoglobulin (HRIG) and for equine rabies immunoglobulin (ERIG), the dose is 40 IU/kg body weight. When HRIG is not available, ERIG can be used. They are potent, safe, and less expensive.

The entire RIG dose or as much as anatomically possible should be infiltrated into the wound or as close as possible. If it is likely that there are additional small wounds, the remaining RIG should be given IM as close as possible to the site of wound or exposure.

All patients with category III exposures should be given RIG. This includes:

• single/multiple transdermal bites/scratches

• licks on broken skin

• contamination of mucous membranes with saliva

• contact with bats

In immunocompromised patients, RIG should be given to both category II and III exposures.

Solution to Question 12:

The advice for the patient is to give the second dose of DPT (Diphtheria-pertussis-tetanus).

DPT is a combined vaccine given for immunization against the three diseases diphtheria, pertussis, and tetanus. It is given to infants as early as 6 weeks following birth. The pertussis component of the vaccine is not recommended beyond 6 years due to the decrease in the severity of pertussis infection with age. 7 years of age onwards, only Td combinations are used.

It is given in 3 doses with an interval of 4 weeks followed by booster doses at 18-24 months and at 5 years of age. The dosage is 0.5 mL and is given in the anterolateral aspect of the thigh intramuscularly. The immunization can be resumed without repeating the previous dose in case of any delayed or interrupted immunization.

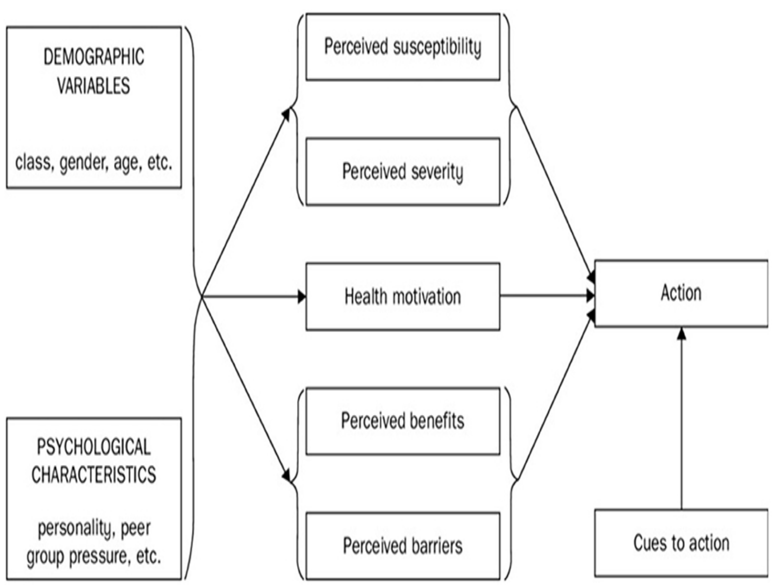
Adverse effects include fever, and neurological complications such as encephalitis, convulsions, infantile spasms and Reye's syndrome.

Anaphylaxis is a contraindication for subsequent vaccination with diphtheria, pertussis, or tetanus vaccine. Encephalopathy following vaccination is a contraindication for pertussis vaccine. DT is given for successive doses.

Events such as inconsolable cry, fever more than 40.5°C, seizures with or without fever are considered as precautions, and not contraindications for DTwP.

Note: IAP guidelines state that "temperature more than >40.5°C is a precaution and not contraindication". Park's 27th Ed states that "temperature above 40°C is a contraindication to DPT". This topic is controversial, however, IAP guidelines are commonly followed by practitioners.

Solution to Question 13:



Box C stands for health motivation.

The health belief model was proposed from the idea that a person's health behavior changes in accordance with certain beliefs.

It can be categorized as follows:

• Individual perceptions.

• Variable factors that modify beliefs.

• Action.

Individual perceptions:

Perceived susceptibility refers to a person's own beliefs about the probability of developing a disease. If the perceived susceptibility is high, an individual is more likely to engage in health-promoting behavior to reduce the chances of contracting the illness.

Perceived severity is an individual's assessment of the severity of a particular health problem. If the disease is perceived to have a high severity, then the person is more likely to adopt a healthier lifestyle.

Perceived threat comprises perceived susceptibility and severity. People with a higher perceived threat tend to take on health-promoting activities.

Action:

Perceived benefits refer to a person's subjective judgment on how a particular health habit is modifying his chances of developing a disease. If the person believes that a habit is likely to reduce his/her chances of susceptibility to a disease or lower the severity of a disease, he/she is more likely to follow that healthy habit.

Perceived barriers to action also influence if a person wants to follow healthy behavior. The benefits of a particular health habit must overweigh the barriers that a person must overcome to continue following that behavior.

Cues to action may be internal or external. Internal cues include pain and clinical symptoms. External cues include awareness through media, physicians, and friends.

Behavioral modification (Option B) is part of the action after assessing all of the above-mentioned concepts.

Self-efficacy (Option D) can be defined as a person's capacity in achieving his health goals. A person's confidence in his/her ability to overcome limitations to attain his/her health goals was found to be a major factor in determining the outcome of his behavioral changes.

The health belief system is thought to be useful in analyzing health behavior patterns and formulating disease prevention plans based on these beliefs. All internal and external factors, including demographic and psychological characteristics, aim at health motivation to attain the action phase of a health belief system.

Other options:

Option A: Perceived control refers to a person's internal belief in the control they would have over things that are part of the past or the future or over a process, behavior, and outcome. Perceived control can influence all of the categories of the health belief system.

Solution to Question 14:

A negative result in a diagnostic test with a sensitivity of 98% implies that the patient is less likely to have the disease.

Sensitivity is the proportion of all people with the disease who test positive. It is the probability of the test being positive when the disease is present. The sensitivity of a test is given by those who have the disease and test positive (true positives) out of the total diseased population.

Other statements:

Option A: The patient would not need another point-of-care diagnostic test for the same clinical condition as the sensitivity of the diagnostic test is 98%.

Option B: Prevalence refers to the number of old and new cases at a given point in time or over a period of time in a given population. It cannot be derived from the given data.

Option D: The higher the sensitivity of a diagnostic test, the lower will be the number of false negatives. Since the test has a 98% sensitivity, the patient most likely does not have the disease.

Solution to Question 15:

The correct statements drawn from the values given in the table are 3 and 4.

An odds ratio is a measure of the association between an exposure and an outcome. It represents the odds that an outcome will occur given a particular exposure in comparison to the odds of the outcome occurring in the absence of that exposure.

An odds ratio of 1 indicates that the risk factor is not related to the disease.

An odds ratio of less than 1 indicates that a person with the disease is less likely to have been exposed to the factor, implying that the factor may be a protective factor against the disease.

An odds ratio greater than 1 indicates that the odds ratio is significant, and the factor can be ascertained as a risk factor.

The p-value is the probability of obtaining a result at least as extreme as the one that was actually observed in the biological or clinical experiment. A p-value of less than 0.05 indicates that the test result is statistically significant.

Other options:

Statement 1: Female population is at greater risk of developing blindness is a false statement; since the odds ratio is <1, the female population will be at lesser risk of developing blindness than the male population.

Statement 2: People dwelling in rural areas are at increased risk of developing blindness is also false; a p-value of >0.05 means that this association is not statistically significant.

Solution to Question 16:

The image shows an established case of chronic lymphatic filariasis, where hygiene is being maintained to prevent further disability by secondary bacterial infection. Disability limitation is a mode of intervention in tertiary prevention.

Other options:

Option A: Specific protection is a mode of intervention in primary prevention which aims at limiting the number of cases of a particular disease. This includes immunization, use of specific nutrients, chemoprophylaxis, accident protection, and avoidance of allergens.

Option C: Primary prevention is a level of prevention which is defined as action taken prior to the onset of a disease, which removes the possibility that a disease will ever occur. It is accomplished by measures designed to promote general health and well-being.

Option D: Health promotion is a mode of intervention in primary prevention, and is the process of enabling people to increase control over, and improve health. Health promotion may be achieved through health education, environmental modification, nutritional interventions or lifestyle changes.

Solution to Question 17:

The proportion of newly diagnosed patients with grade 2 disability is useful in determining whether the health system is giving importance to identifying leprosy in the community.

The National Leprosy Eradication Programme (NLEP) was introduced in 1983. The strategy of NLEP is to control the disease through a reduction in the infection and the infective source, thus breaking the chain of disease transmission. It functions under National Health Mission (NHM).

The vision of the NLEP is 'Leprosy-free India'. Disability prevention and medical rehabilitation (DPMR) is the priority of the programme. The mission of NLEP is to provide quality services free of cost to all people through the integrated healthcare system.

Main indicators for evaluating case detection:

• Proportion of grade-2 disabilities among new cases

• Proportion of females among new cases

• Proportion of MB among new cases

• Proportion of child (0-14 years) among new cases

• Child rate per 1,00,000 population

• Scheduled caste new case detection rate

• Scheduled tribe new case detection rate

Solution to Question 18:

The correct statements are 1 and 2.

Vaccine vial monitors (VVM) are heat-sensitive indicators routinely used to monitor heat exposure of the vaccine and determine vaccine viability as the vaccine vial moves across the supply chain. It records cumulative heat exposure through a gradual change in color.

If the color of the inner square is the same or is darker than the outer circle, it indicates that the vaccine has been exposed to high temperatures and should be discarded.

A VVM can be used to assess vaccine potency and not vaccine efficacy. Vaccine potency is the ability of the vaccine to provide adequate immunity to the patient. Vaccine potency is irreversibly lost when the vaccine is exposed to unfavorable temperatures; hence, VVM indicating heat exposure can be used to assess whether the vaccine potency is likely to be intact or not. Vaccine efficacy, on the other hand, is a measure of the immunity conferred by the vaccine or the ability of the vaccine to prevent the disease in a vaccinated person under ideal conditions, such as clinical trials. The term vaccine efficacy is used in relation to findings of clinical trials, whereas the term vaccine potency is used for regular usage of the vaccine. Vaccine efficacy is determined only during clinical trials, whereas vaccine potency is tested for every batch of the vaccine that is manufactured.

There are currently 4 types of VVMs; they are chosen to match the heat sensitivity of the vaccines based on the VVM number:

• VVM 30: High stability

• VVM 14: Medium stability

• VVM 7: Moderate stability

• VVM 2: Least stable

Purpose of VVMs:

• Ensures that vaccines damaged by heat are not administered.

• Decision on which vaccines to be kept after a cold chain break occurs, minimizing unnecessary vaccine wastage.

• Decision on which vaccine is to be used first. Vaccines showing significant heat exposure should be used before those that show lower heat exposure, even if their expiry date is longer.

The expiry date of a vaccine cannot be relaxed even if VVM is in an acceptable range. Expired vaccines must be discarded. The expiry date of a vaccine cannot be determined using a VVM as it does not monitor the vaccine components. It only ensures that the vaccines have been exposed to the required temperatures and are viable for use.

There are two different locations for VVMs:

• On the label of the vaccine: The vaccine vial, once opened, can be kept for subsequent immunization sessions for up to 28 days, regardless of the formulation of the product.

• In a location other than on the label (e.g., cap or neck of ampoule): The vaccine vial, once opened should be discarded at the end of the immunization session or within six hours of opening, whichever comes first, regardless of the formulation of the product.

Solution to Question 19:

The model of disease causation depicted below is a sufficient component cause model.

It was developed by Rothman based on the idea that an outcome can only happen when a sufficient cause is present. A disease can have multiple sufficient causes. But, each sufficient cause can consist of a pie chart of component causes, a causal pie. One component cause can be a part of more than one sufficient cause. A component cause can be considered necessary if it is a part of a disease's sufficient causes.

In the given model, each pie chart represents a sufficient cause made of component causes. Since B is present in all the charts, it can be considered a necessary cause. All the others are contributing causes.

Other options:

Option B: An epidemiological triad depicts the interrelationship between the causative agent, the host, and the environment. All of them contribute to disease development. The aim of an epidemiological triad is to identify how to break this relationship and prevent the outbreak of the disease.

Option C: PERT is used for network analysis. It is a type of quantitative management technique that helps in detailed planning and comprehensive supervision.

Option D: The web of causation considers all the predisposing factors and their complex interrelationship with each other. This model is ideally suited to study chronic disease. The web of causation does not imply that the disease is controlled only when all the multiple chains of causation are removed. Sometimes removal of one significant link is sufficient to control the disease.

Solution to Question 20:

Attributable risk (AR) describes the extent of outcome (lung cancer) that can be attributed to exposure (smoking). In the given scenario, an AR of 90% means that 90% of lung cancer cases among smokers are attributable to smoking. Therefore, 90% cases of lung cancer can be eliminated in smokers if smoking is eliminated.

AR is the difference in incidence rates of disease between an exposed group and an unexposed group. AR is indicative of the extent of the disease attributed to the exposure. It also refers to the amount of disease that could be eliminated in the exposed persons if the exposure is controlled or eliminated. It also helps to determine the impact a successful preventive or public health program might have in case the exposure to cause is reduced or eliminated.

Population-attributable risk (PAR) is the difference in the incidence of disease in the total population and the incidence of disease among those not exposed to the supposed cause. PAR is useful in providing an estimate of the amount by which the disease could be reduced in that whole population if the suspected factor was eliminated or modified.

For example, a PAR of 90% in the context of smoking and lung cancer means that 90% cases of lung cancer in the population can be attributed to smoking.

Relative risk is defined as the ratio of incidence of the disease (outcome) among exposed and unexposed groups. It helps in establishing the association between a cause and a disease. However, it does not help in assessing the extent of the cause bringing about the disease (outcome).

Solution to Question 21:

The measures that are taken to prevent disasters from causing emergencies or to lessen the likely effects of emergencies are part of disaster mitigation.

Disaster mitigation measures aim to reduce the vulnerability of the system. It complements disaster preparedness and response activities and includes measures taken to lessen the likely effects of emergencies. Some of these measures are:

• Flood mitigation

• Land-use planning

• Improved building codes

• Reduction and protection of vulnerable populations or structures.

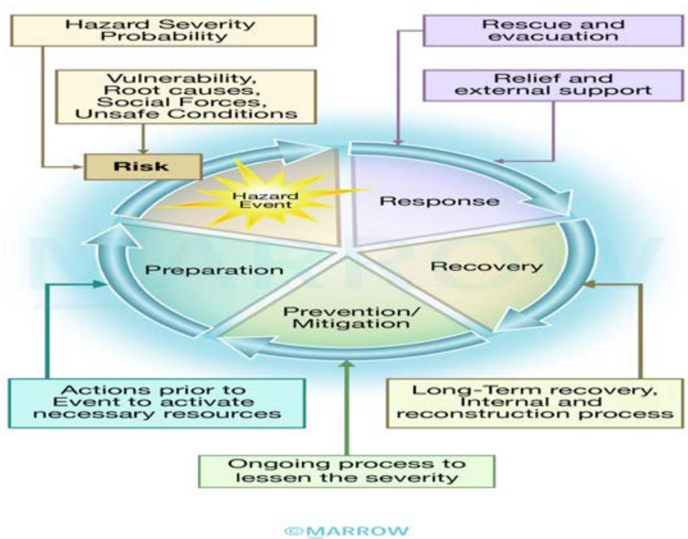
Other options:

Option A - Disaster response includes activities such as rescue and first aid, field care, triage and tagging that form a part of the emergency care needed in the first few hours after a disaster.

Option B - Disaster reconstruction is a part of the recovery phase after a disaster.

Option C - Disaster rehabilitation refers to measures that lead to the restoration of pre-disaster conditions. These measures include water supply restoration, food safety, basic sanitation, personal hygiene, and vector control.

The disaster management cycle is given below:



Solution to Question 22:

Isoniazid inhibits mycolic acid synthesis and Bedaquiline inhibits ATP synthase.

Other options:

Pyrazinamide is not an enzyme inducer. Pyrazinamide is converted into its active form pyrazinoic acid. Pyrazinoic acid disrupts mycobacterial cell membrane metabolism and transport functions.

Amongst the anti-tubercular drugs, rifampicin is an enzyme inducer.

Rifampicin can be used in treatment of meningococcal carriers, prophylaxis in contacts with Haemophilus influezae type b, and serious staphylococcal infections.

Solution to Question 23:

To start multidrug therapy (MDT), the criteria that should be satisfied are a hypopigmented patch with sensory loss and peripheral nerve thickening with sensory loss.

Leprosy (Hansen's disease) is a chronic infectious disease caused by Mycobacterium leprae. It mainly affects the skin and peripheral nerves. The disease has two polar forms, lepromatous leprosy, and tuberculoid leprosy, lying at the two ends of a long spectrum of the disease.

Leprosy is diagnosed by the detection of at least one of the following cardinal signs:

• Definite loss of sensation in a pale hypopigmented patch

• Thickened or enlarged peripheral nerve, with loss of sensation and/or weakness of the muscles supplied by that nerve (statement 2)

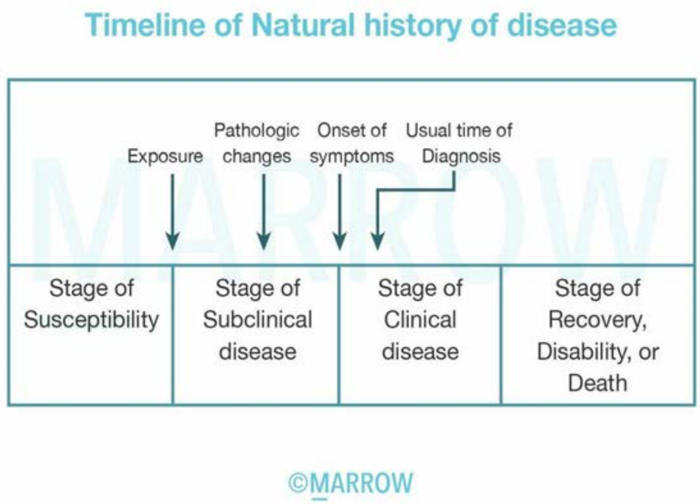
• Presence of acid-fast bacilli in a slit-skin smear.

However, a skin smear is not a prerequisite for implementing MDT. Since MDT has been introduced, many procedures have been simplified, so that leprosy patients can be detected by health workers in the field and MDT can be immediately started.

Solution to Question 24:

The correct match is 1-d, 2-b, 3-a, and 4-c.

Natural history of disease refers to the progression of a disease in an individual over time, in the absence of treatment. It has two phases, a period of pre-pathogenesis and a period of pathogenesis.



The period of pre-pathogenesis is before the onset of disease in humans. An interaction between the agent, host and the environment occurs in this phase before the pathogenesis. It starts with appropriate exposure to or accumulation of factors sufficient for the disease process to begin in a susceptible host. The exposure is a microorganism in an infectious disease. For lung cancer, it is any factor that initiates the process, such as asbestos fibres or components in tobacco smoke. Primary prevention (health promotion and specific protection) is applicable in this period.

The period of pathogenesis starts after the disease process has been triggered and pathological changes occur without the individual being aware of it. It begins with the entry of the organism till recovery, death, or disability. The subclinical or asymptomatic period between exposure and onset of symptoms is called the incubation period for infectious diseases, and the latency period for chronic diseases. The onset of clinical symptoms is the beginning of the clinical stage of the disease. Secondary (halting of the disease process) and tertiary prevention (measures to limit disability) are applicable in this period.

Rehabilitation has been defined as the use of medical, social, educational, and vocational measures for training the individual to the highest possible level of functional ability. It is a part of tertiary prevention like providing callipers for walking patients and arranging schooling after being affected by polio.

Solution to Question 25:

Pearl index is the common method used to measure contraceptive failure rate.

It is defined as the number of failures per 100 woman-years of exposure (HWY). This rate is given by the formula:

Pearl index = Total accidental pregnancies x 1200/Total months of exposure

Solution to Question 26:

The correctly matched pairs are as follows:

• Walter Reed showed that yellow fever is transmitted by the Aedes mosquito.

• Edward Jenner - smallpox vaccination

• James Lind - prevention of scurvy by intake of citrus fruits

Scientist/ Doctor	Achievements
Susruta	Father of Indian SurgeryFather of Plastic and Cosmetic SurgeryAuthor of “Sushruta Samhita”
Hippocrates	First True EpidemiologistFather of MedicineAuthor of “Airs, Water and Place”
Charaka	Father of Indian MedicineAuthor of “Charaka Samhita”
Avicenna	Author of “Canon of Medicine” and “The Book on Healing”
Andreas Vesalius	Father of Modern AnatomyAuthor of “Fabrica”
Hammurabi	Author of “Babylonian Code of Hammurabi”
John Snow	Father of Epidemiology
Louis Pasteur Robert Koch / Ferdinand Cohn	Father of Bacteriology
Samuel Hahnemann	Father of Homeopathy

Solution to Question 27:

The formula given above is used to calculate the standard deviation.

Standard deviation (or σ) is a measure of how dispersed the data is in relation to the mean. Data clustered around the mean have a low standard deviation, while data that are more spread out have a high standard deviation. A standard deviation close to zero indicates that data points are close to the mean, whereas a high or low standard deviation indicates data points are respectively above or below the mean.

Standard error = SD/√n, where SD is the standard deviation and n is the sample size

Standard error will reduce when the sample size is increased.

The median divides the frequency distribution in half when all the scores are listed in order. When a distribution has an odd number of elements, the median is the middle value. When the distribution has an even number of elements, the median is the average of the two middle scores.

Variance is the mean of the squares of all the deviation scores of the distribution.

Solution to Question 28:

A child presenting with abrasion with slight oozing after a dog bite comes under category III. The correct statements are 1, 2, 3, and 6.

The following measures should be taken immediately to stop the contraction of rabies:

• Cleansing: The most important step in the prevention of rabies is washing the affected area with plenty of soap and water under a running tap for 15 minutes. This is a standard procedure that minimizes the contraction of rabies.

• Chemical treatment: After washing and drying the wound, disinfectants like povidone-iodine and spirit can be applied.

• Suturing: Bite wounds are not sutured generally. If suturing is unavoidable, rabies immunoglobulin should be infiltrated in the depth and around the wound, and suturing should be delayed by a few hours.

• In unavoidable situations, minimal loose sutures may be applied to arrest bleeding.

• Other measures: Local antibiotics and tetanus toxoids should be given.

Different wound categories for post-exposure prophylaxis:

Post-exposure prophylaxis regimen for rabies:

Intramuscular regimen: In this regimen, 1 ml or 0.5 ml of vaccine is injected into the deltoid in patients with category II and III wounds.

• Essen regimen: A 5-dose regimen in which 1 dose of vaccine is given on days 0, 3, 7, 14, and 28.

• Zagreb regimen: A 4-dose multisite regimen in which 2 doses of vaccine are given on day 0, followed by 1 dose on days 7 and 21.

Intradermal regimen: In this regimen, 0.1 ml of injection is given at 2 sites on days 0, 3, 7, and 28.

Equine Rabies Immunoglobulin at a dose of 40 IU/kg (a maximum of 3000 IU) body weight is given once. It is given to all Category III bites and also Category II bites in the case of immune-compromised persons.

Categories of wound	Post-exposure prophylaxis
Category I- touching or feeding animals, licks on intact skin	None
Category II- nibbling of uncovered skin, minor scratches, or abrasionswithout bleeding	Immediate vaccination andlocal treatment of the wound
Category III- single or multiple transdermal bites or scratches, licks on broken skin contacts with bats	Immediate vaccination andadministration of rabiesimmunoglobulin; localtreatment of the wound

Solution to Question 29:

The species of mosquito that breeds in dirty water is Culex.

Culex fatigans are found everywhere in India and they transmit Bancroftian filariasis. Culex mosquitoes are also vectors for Japanese encephalitis, West Nile fever, and viral arthritis.

It can be identified by the presence of unspotted wings and hunched back. The larva has a head that is suspended downwards at an angle to the water surface and a siphon tube is present.

It breeds in dirty water collections like stagnant drains, septic tanks, and burrow pits. The peak biting time is during midnight. The preferred biting site is the legs, mainly below the knee.

Solution to Question 30:

The image given below shows artemisinin-based combination therapy (ACT), which is used in the treatment of falciparum malaria.

ACT consists of artesunate given for 3 days and sulphadoxine-pyrimethamine given for 1 day. It is supplied by the government of India under the national vector-borne disease control program (NVDCP).

It is given to all confirmed cases of Plasmodium falciparum infection, found positive by microscopy or rapid diagnostic tests. Primaquine with dose of 0.75 mg/kg of body weight is given on day 2 along with it. Artemether-lumefantrine is used for the treatment of falciparum malaria in the northeastern states of India.

This drug is not given in pregnancy and in children less than 5 kg of body weight.

Solution to Question 31:

According to the Anemia Mukt Bharat initiative, pregnant women with mild and moderate anemia is treated with

• Two iron and folic acid tablets (60 mg elemental iron and 500 mcg folic acid) daily, orally given by the health provider during the ANC contact

• Parenteral iron (IV iron sucrose or ferric carboxy maltose [FCM]) may be considered as the first line of management in pregnant women who are detected to be anemic late in pregnancy or in whom compliance is likely to be low (high chance of loss to follow-up)

Anemia Mukt Bharat 2018 scheme:

This is an initiative that aims to strengthen the existing mechanisms and foster newer strategies for tackling anemia. It focuses on six target beneficiary groups, through six interventions and six institutional mechanisms to achieve their target by 2022.

Interventions:

• Prophylactic iron and folic acid supplementation.

• Deworming.

• Intensified year-round Behaviour Change Communication Campaign (Solid Body, Smart Mind) including ensuring delayed cord clamping in newborns.

• Testing of anemia using digital methods and point-of-care treatment.

• Mandatory provision of Iron and Folic Acid fortified foods in government-funded health programs.

• Addressing non-nutritional causes of anemia in endemic pockets, with a special focus on malaria, hemoglobinopathies, and fluorosis.

Beneficiaries:

• Children, 6-59 months of age

• Children, 5-9 years of age

• Adolescent girls and boys, 10-19 years of age

• Women of reproductive age, 20-49 years of age

• Pregnant women

• Lactating mothers, 0-6 months

Institutions through which activities are carried out:

• Intra-Ministerial Coordination

• National Anemia Mukt Bharat Unit

• National Centre of Excellence and Advanced Research on Anemia Control (NCEAR-A)

• Convergence with other Ministries

• Strengthening Supply Chain and Logistics

• Anemia Mukt Bharat Dashboard and Digital Portal

Community medicine NEET 2022

Question 1:

The average life expectancy for a woman in Japan is 87 years. Due to recent advances in testing for cervical cancer, there is an increase in life expectancy by 15 years. The healthcare utility value is 0.8. Which of the following can be calculated from the parameters given?

- **a)** HALE
- **b)** DALY
- **c)** DFLE
- **d)** QALY

Question 2:

You are working in a primary health center (PHC) situated in a high seismic zone. Which of the following will you do as part of preparedness for an emergency?

- **a)** 1, 2, 3, 4
- **b)** 2, 3, 4
- **c)** 1, 2, 3
- **d)** 1, 2, 4

Question 3:

Although many animals are implicated in the spread of rabies, dogs are the most common ones. Also, it usually affects children in developing countries. Knowing this, what is the most cost-effective and logical way to reduce the incidence of rabies?

- **a)** Testing all the dogs for rabies
- **b)** Reduce stray dog population and vaccinate all dogs
- **c)** Increase the laboratory facilities
- **d)** Increase capacity of healthcare workers for surveillance

Question 4:

Which of the following steps is not included in STEP approach of WHO?

- **a)** Therapeutic assessment
- **b)** Physical assessment
- **c)** Psychological assessment
- **d)** Behavioral assessment

Question 5:

A male patient diagnosed with tuberculosis took complete treatment. Sputum examination was done after the completion of the intensive and the continuation phases. It was found to be negative. What is the status of the patient?

- **a)** Cured
- **b)** Treatment completed
- **c)** Lost to follow up
- **d)** Treatment failed

Question 6:

A cohort study was conducted with drinkers and non-drinkers of green tea to study its effect on diabetes mellitus. The risk ratio was found to be 0.84. Which of the following statements is correct?

- **a)** Green tea reduces the risk of diabetes
- **b)** Green tea increases the risk of diabetes
- **c)** Data insufficient to establish causal association
- **d)** The value (0.85) tends to be close to 1, hence there is no effect

Question 7:

Which statement refers best to the criteria for starting an urban community health center?

- **a)** Caters to a population of 1-1.5 lakh
- **b)** Referral centre for 2- 3 primary health centers
- **c)** No sub-district and district hospitals present in the area
- **d)** Should have a 100-bed facility in metro cities

Question 8:

How is a broken vaccine vial disposed, according to biomedical waste management?

a) Puncture proof blue bin

b) White container

c) Yellow container

d) Red container

Question 9:

In an urban area in the state of Madhya Pradesh, a primigravida goes for institutional delivery after being motivated by an ASHA worker. What are the benefits they will receive (in terms of money in rupees) under the Janani Suraksha Yojana?

a) 1000 for mother and 400 for ASHA

b) 1400 for mother and 600 for ASHA

c) 600 for mother and 400 for ASHA

d) 400 for mother and 600 for ASHA

Question 10:

Which of the following agency provides seed and manure in applied nutrition programs in schools?

a) CARE

b) UNDP

c) UNICEF

d) WHO

Question 11:

Many children from a particular community coming to a hospital were detected to have acute lymphoblastic leukemia (ALL). It was assumed that it is due to the presence of cytotoxic waste in the water of that community. If a case-control study has to be done to find whether the chemical and ALL are associated, what will be taken as the control?

a) Children from the area exposed, but unaffected with the disease

b) Children from the area not exposed and affected with the disease

c) Children coming to your OPD, who do not have the disease

d) All children with ALL irrespective of exposure status

Question 12:

An auxiliary nurse midwife has to conduct a vaccination camp in a village. She received 2 open vials, one of which is a pentavalent vaccine and the other is an MR vaccine. What can she do regarding the utilization of these vials?

a) Use MR vaccine and discard pentavalent vaccine

b) Use pentavalent vaccine and discard MR vaccine

c) Use both

d) Discard both

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	a
5	a
6	a
7	d
8	a
9	a
10	c
11	c
12	b

Detailed Explanations

Solution to Question 1:

QALY (quality-adjusted life year) can be calculated by the utility value.

QALY is a disability rate indicator. It is based on the number of years that would be added to life due to medical intervention. A utility value of 1.0 is given for each year of perfect health and 0 for death. It is the measure of the disease burden which includes both quality and quantity of life.

Disability rates that are related to illness and injury are used as supplementation to mortality and morbidity indicators. It is based on the concept that a full range of daily activities implies good health. It is divided into:

• Event-type indicators: It counts the number of days of restricted activity, bed disability days, and work loss days.

• Person-type indicators: It counts limitation of mobility and limitation of activity.

Other options:

Option A: HALE (health adjusted life expectancy) is an indicator that estimates the number of years a newborn is expected to live in full health based on the current rates of ill health and mortality. HALE is calculated using the life expectancy at birth with an adjustment for time spent in poor health. It was known as DALE (disability-adjusted life expectancy) previously.

Option B: DALY (disability adjusted life years) is an indicator that estimates the number of years lost due to ill-health, disability, or early death. It is calculated by:

DALY= Years of life lost (YLL) + Years lived with disability (YLD)

Option C: DFLE (disability-free life expectancy) is an indicator that estimates the average number of years an individual is expected to live free of disability if the current pattern of mortality and disability continues to apply. It is also known as active life expectancy and is calculated by:

DFLE = Life expectancy - Duration of disability

Solution to Question 2:

Steps 1, 2, and 3 are done as part of preparedness for an emergency by someone working in a PHC situated in a high seismic zone.

Emergency preparedness refers to long-term development activities that strengthen the capacity of a country to effectively manage all kinds of emergencies. It involves the following steps:

• Disaster preparedness by making sure all financial and other resources are available

• Increase public awareness through campaigns and loudspeakers

• Conduct a simulation for the disaster and assess the response

• Evaluate the risk of disaster

• Information sessions coordination with news media

• Adopt standards and regulations

Disaster risk management is a systematic process using administrative decisions, policies, and strategies to lessen the impacts of natural hazards and disasters. The steps are:

• Response - This includes activities such as:

• Search, rescue, and first aid: This step is important since the organized relief services will be able to address only a small fraction of demand after a major disaster.

• Field care: This includes making health care available regardless of facilities and operating status.

• Triage: Classifying the injured based on the severity of the injuries and chances of survival with medical intervention.

• Tagging: Tags with name, age, triage category, diagnosis and treatment are put for identification purposes.

• Identification of the dead: The dead bodies should be shifted from the disaster site to the mortuary.

• Rehabilitation - This includes measures that lead to the restoration of pre-disaster conditions. These measures include water supply restoration, food safety, basic sanitation, personal hygiene, and vector control.

• Mitigation - This includes measures to prevent disasters from causing emergencies or to lessen the likely effects of emergencies. Some of them are flood mitigation, land use planning, improved building codes, and the reduction and protection of vulnerable populations or structures.

• Preparedness - This is a program that aims at strengthening the capacity of a country to manage all types of emergencies efficiently. Its objective is to ensure that appropriate measurements are taken for the relief and rehabilitation of disaster victims.

Some measures that should be followed by all in case of an emergency include:

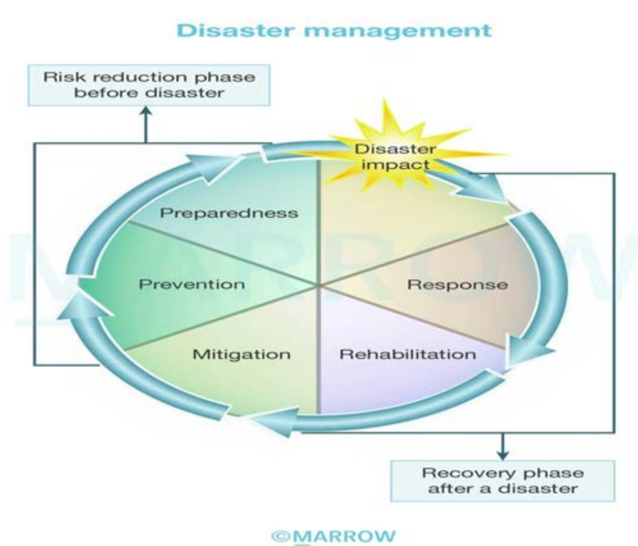
• Avoiding telephone use, unless it is to call for help

• Listen to and carry out the radio or loudspeaker broadcast messages

• Carry out the official instructions given over the radio or by loudspeaker

• Keep a family emergency kit ready

Note: Vaccination of the general population post-disaster against typhoid, cholera, and tetanus is not recommended.



Solution to Question 3:

Reducing the stray dog population and vaccinating all dogs is the most cost-effective and logical way to reduce the incidence of rabies. The Animal Birth Control initiative from the Government of India is directed towards this goal.

Rabies is a highly fatal viral disease caused by Lyssavirus type 1, affecting the central nervous system. Dogs are considered the major source of rabies infection. Thus immunization of dogs at the age of 3-4 months and booster doses at regular intervals is recommended to control rabies. The type of vaccines used are:

• BPL inactivated nervous tissue vaccine - it is a single-dose vaccine. Booster is advised after 6 months and subsequently every year.

• Modified live virus vaccine - it is also a single-dose vaccine and a booster is given every 3 years.

Other methods to control rabies include:

• Registration and licensing of all domestic dogs

• The immediate killing of dogs and cats bitten by rabid animals

• 6 months quarantine for imported dogs

• Educating people regarding dog care and prevention of rabies.

Note: Strengthening laboratory facilities and surveillance are also the methods used to reduce the incidence according to the National Rabies Control Program (NRCP)

Humans get infected after a deep bite or scratch by an infected animal. The transmission also occurs when the saliva of the infected dog comes into direct contact with human mucosa or fresh skin wounds.

Rabies in man is also known as hydrophobia. The clinical features usually begin as prodromal symptoms like headache, and fever followed by neurological symptoms. The patients usually present with confusion, hallucination, irritability, hydrophobia, aerophobia, and muscle spasms. They also present with symptoms of autonomic dysfunction such as hypersalivation, lacrimation, cardiac arrhythmias, etc.

Laboratory tests for rabies include:

• Fluorescent antibody test is a highly reliable and best test for rapid detection of rabies antigen in the infected specimens

• Microscopic examination of the brain tissue to demonstrate Negri bodies.

Solution to Question 4:

Therapeutic assessment is not included in the STEP approach of WHO.

STEP approach of WHO is a survey methodology developed to assess risk factors of non-communicable diseases(NCD). It helps the countries to establish an NCD surveillance system. The steps include the following:

• Questionnaires contain questions about behavioral risk factors like tobacco use, alcohol use, diet, and physical activity.

• Physical measurements contain blood pressure, height, weight, and waist size

• Biochemical measurements contain blood glucose, blood lipids, urinary sodium, and creatinine.

Note: Mental health and suicide are included in optional behavioral risk factors (step 1). Hence, psychological assessment is also included in the STEP approach.

Surveillance is defined as systematic and continuous data analysis and taking appropriate preventive actions according to the results obtained. Identification of cases that are missed and not notified is known as sentinel surveillance. It is used in communicable diseases such as HIV, malaria, etc. The study population for HIV sentinel surveillance are sex workers, IV drug abusers, transgender people, migrant workers, truck drivers, etc.

Solution to Question 5:

The negative sputum examination at the end of the treatment implies the status of the patient is cured.

Tuberculosis (TB) is a communicable disease caused by Mycobacterium tuberculosis. The disease mainly affects the lungs and causes pulmonary tuberculosis.

A bacteriologically confirmed TB case is one in which the biological specimen tests positive on smear microscopy and culture. A clinically diagnosed TB case is one that is diagnosed as an active TB case by a clinician, who has decided to give the patient a full course of TB treatment.

There are two phases in the treatment of TB. The intensive phase lasts for 2 months and four drugs are used. They are isoniazid, rifampicin, pyrazinamide, and ethambutol. The continuation phase lasts for 4 months, where isoniazid is given along with rifampicin and Ethambutol.

After the completion of anti-tubercular treatment, the sputum/culture is tested and one of the outcomes given below is assigned to the patient:

• Cured - A TB patient whose smear or culture turned negative in the last month of treatment and on at least one previous occasion

• Treatment completed - A TB patient who completed treatment without any failure, but has no record to show that sputum smear or culture results in the last month of treatment and on at least one previous occasion were negative; it may be either because tests were not done or because results are unavailable

• Treatment failed - A TB patient whose sputum smear or culture is still positive after treatment for 5 months or more

• Lost to follow-up - A TB patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more

• Died - A TB patient who dies before starting or during the course of treatment for any reason

• Not evaluated - A TB patient for whom no treatment outcome is assigned; this includes cases "transferred out" to another treatment unit and cases for whom the treatment outcome is unknown to the reporting unit whose treatment I interrupted

Solution to Question 6:

Green tea reduces the risk of diabetes mellitus is the correct statement since the risk ratio is 0.84.

Relative risk (RR) or risk ratio is the ratio of the incidence of the disease (or death) among exposed and the incidence among non-exposed. It can be determined from a cohort study.

It is a direct measure of the strength of the association between suspected cause and effect. The larger the RR, the greater the strength of the association between the suspected factor and disease.

• RR < 1 indicates inverse association(protective effect).

• RR = 1 indicates no association

• RR > 1 indicates a positive association.

• RR of 2 indicates a 100% increase in risk.

In the question, the risk ratio is less than 1. Hence it implies the incidence of diabetes mellitus is reduced when exposed to green tea.

Solution to Question 7:

An urban community health center (U-CHC) has a 100-bed facility in metro cities.

U-CHC acts as a referral center for 4–5 urban primary health centers (U-PHC). It is usually a 30–50 bedded facility that caters to a population of 2,50,000.

When it comes to metro cities, it caters to a population of 5,00,000 with a 100-bed facility. These hospitals will have surgeons, gynecologists, physicians, and pediatricians, along with x-ray and laboratory facilities.

The specialists at the community health center can refer a patient to the state-level hospital or the nearest medical college hospital directly. The subdistrict and district hospitals operate in addition to CHC to cater to the population.

Services of CHC:

• Care of routine and emergency cases in medicine and surgery

• Maternal, newborn, and child health

• Family planning

• All the national health program functions

• Physical medicine and rehabilitation services

Solution to Question 8:

A broken vaccine vial is disposed of in a puncture-proof blue bin, according to biomedical waste management (BMW).

According to BMW, broken or discarded contaminated glass materials such as vials, ampules, etc are disposed of in boxes marked with blue color. It is then disinfected either by washing with detergent, autoclaving, or microwaving and sent for recycling.

The methods used for treatment and disposal of healthcare waste are:

• Incineration: This is a dry oxidation process done using high temperatures.

• Chemical disinfection: Chemicals are added to waste to kill or inactivate the microorganisms.

• Wet and dry thermal treatment: Shredded infectious waste is exposed to high temperature, high-pressure steam and it resembles an autoclave sterilization process.

• Microwave irradiation: Microorganisms are killed using high-frequency microwave.

• Land disposal.

• Inertization: Waste is mixed with cement and other substances before disposal to reduce toxicity.

Solution to Question 9:

According to Janani Suraksha Yojana (JSY) scheme, the mother will receive 1000 rupees and the ASHA worker will receive 400 rupees.

Janani Suraksha Yojana (JSY) is a program under the National Rural Health Mission (NRHM), whose aim is to reduce maternal mortality by encouraging delivery in government health facilities. Under the scheme, a financial incentive is also provided for giving birth in a government health facility.

The Accredited Social Health Activist (ASHA) is the link between pregnant women and health institutions. She is responsible for making institutional antenatal and postnatal care available.

States are divided into low-performing and high-performing on the basis of institutional delivery rate. The 10 low-performing states (LPS) with a low institutional delivery rate are Uttar Pradesh, Uttarakhand, Madhya Pradesh, Jharkhand, Bihar, Rajasthan, Chattisgarh, Odisha, Assam, and Jammu & Kashmir. All the other states are high-performing states (HPS).

The incentives under JSY are as follows:

Under this scheme, the cost of cesarean section and management of obstetric complications is also subsidized, up to Rs 1500 per delivery to the government institutions, where government specialists are not present.

Category	Rural area	Urban area
LPS	Mother package: 1400/-ASHA's package:600/-Total:2000/-	Mother package: 1000/-ASHA's package:400/-Total:1400/-
HPS	Mother package: 700/-ASHA's package:600/-Total:1300/-	Mother package: 600/-ASHA's package:400/-Total:1000/-

Solution to Question 10:

UNICEF provides seed and manure in applied nutrition programs in schools.

Applied Nutrition Program is implemented with the aim of producing various types of productive food by the community for the community itself.

UNICEF assists the program by providing implements, seeds, manure, and water supply equipment. If free land is available in schools, the facilities provided by UNICEF should be used in developing school gardens.

The produce obtained from the gardens can be utilized for school feeding programs and for nutrition education.

Another strategy called GOBI was also implemented by UNICEF with the aim of a child health revolution.

Other options:

Option A- CARE- Cooperative for Assistance and Relief Everywhere aims at providing food for children in the age group of 6-11 years. It is also helping in projects like integrated nutrition and health project, better health and nutrition projects, anemia control projects, etc.

Option B- UNDP- The United Nations Development Program provides funds for technical assistance. Its main objective is to help poorer nations in developing their human and natural resources.

Option D- WHO- World Health Organization aims at the attainment of the highest level of health by all people, which helps them to lead a socially and economically productive life.

Solution to Question 11:

Children coming to the OPD, who do not have the disease, will be taken as the control for the case-control study.

The prerequisite for selecting a control group is that they must be free from the disease under study. They must be as similar as possible to the cases, except for the fact that they should not have the disease. Usually, the control group consists of persons who have not been exposed to the disease or factor whose influence is under study. Therefore, among the options, the best answer would be the children coming to the OPD, who do not have the disease as controls for the study.

Case-control studies are also called retrospective studies. This study proceeds from effect to cause and is done after the exposure and outcome have occurred.

They focus on diseases or health problems that have already developed. They are less time-consuming and have no risk of attrition because they do not require follow-up.

There are four basic steps in conducting a case-control study:

• Selection of cases and controls.

• Matching- This is done to ensure comparability between cases and controls. Controls are selected such that they are similar to cases with respect to certain factors, like age, that influence the disease outcome.

• Measurement of exposure - This is obtained by interviews, questionnaires, or study of the past records of cases.

• Analysis in the form of exposure rates, risk ratio, and odds ratio.

The Odds Ratio (OR) calculated from a case-control study, is a measure of the strength of the association between risk factor and outcome.

Solution to Question 12:

She can use the pentavalent vaccine and discard the MR vaccine. According to the open vial policy, the pentavalent vaccine can be reused, whereas MR can not be reused.

Open vial policy was implemented to allow the reuse of partially used multidose vials under the Universal Immunization Program (UIP) for up to 28 days if certain conditions are met. This was implemented to reduce vaccine wastage.

It applies only to DPT, TT, hepatitis B, oral polio vaccine (OPV), liquid pentavalent, PCV, injectable IPV, inactivated Japanese encephalitis (JE) vaccine, and rotavirus vaccine. It does not apply to Measles/MR and BCG vaccines.

Note: Open vial policy does not apply to rotavirus vaccine with VVM on cap, it applies only to new rotavirus vaccine with VVM on label.

Community Medicine NEET 2023

Question 1:

In a village, it is observed that several farmers have crossed gait and use a stick for support to stand up and walk. Due to poor yield from farms, they consume meals containing rice and pulses only. Supplementing their diet with which of the following vitamins could have prevented this?

- **a)** Vitamin A
- **b)** Vitamin D
- **c)** Vitamin C
- **d)** Vitamin B

Question 2:

A 35-year-old homeless man presented with a 1-month history of fever, cough, and weight loss. Both sputum smears turned out to be negative, but the chest x-ray ordered was suggestive of tuberculosis. According to the recent NTEP guidelines, which is the next best line of management?

- **a)** Repeat sputum smears
- **b)** Ask for CBNAAT
- **c)** Ask for line probe assay
- **d)** Wait until TB culture results to start ATT

Question 3:

Research is being conducted to find the association between aniline dye exposure and bladder cancer in workers who have worked in the industry for >20 years. Two groups were formed: one directly involved with dye handling and the other group consisting of office clerks not directly exposed to the dye. Years of occupation were noted from records. What type of study is being performed?

- **a)** Retrospective cohort study
- **b)** Prospective cohort study
- **c)** Case-control study

d) Intervention and response

Question 4:

The blood pressure of a population was tracked from childhood to adulthood. It was observed that those who had lower BP in childhood had low BP in adulthood, while those who had higher BP in childhood had high BP in adulthood. This can be best described as____

a) Rule of halves

b) Tracking of blood pressure

c) STEPwise approach

d) Primordial approach

Question 5:

A 30-week primigravida c/o reduced vision at night. She has been avoiding papaya, mango, and other fruits throughout her pregnancy as she thinks they could be abortifacients. It is the primary duty of which of the following workers to provide counseling and information to the patient?

a) ANM

b) AWW

c) Trained birth attendant

d) ASHA

Question 6:

A 22-year-old female comes to the STI clinic with minimal vaginal discharge. On speculum examination, erosions are seen on the cervix. Which of the following kit should be given to this patient?

a) Green

b) Red

c) Gray

d) Yellow

Question 7:

The years of potential life lost could be attributed to______.

a) Years lost to morbidity

b) Years lost due to premature death

c) Years lost to disability

d) Years lost to poor quality of life

Question 8:

In a 10-year-old school child, which of the following vaccines is given as a part of the school immunization program?

a) Measles vaccine

b) Rotavirus vaccine

c) TT/Td vaccine

d) Hepatitis B vaccine

Question 9:

Which of the following statements is true about cancer treatment according to the Colombo plan?

a) Help with PET scan units for diagnosis of cancer

b) Human resource strengthening

c) Setting up chemotherapy units

d) Setting up cobalt therapy units

Question 10:

A poor farmer with a history of successive crop failure develops progressive spastic paraparesis, signs of upper motor neuron paralysis, and gait instability. Name the toxin responsible for this condition.

a) Aflatoxin

b) Beta-oxalyl-amino-alanine

c) Ergot alkaloids

d) Fusarium toxin

Question 11:

A young male came to the hospital with a clean-cut wound without any bleeding. The patient received a full course of tetanus vaccination 10 years ago. What is the best management for this patient?

a) Human tetanus immunoglobulin and full course of vaccine

b) Human tetanus immunoglobulin only

c) Single-dose tetanus toxoid

d) Full course tetanus toxoid

Question 12:

How is water collected for bacteriological examination during a disease outbreak?

a) Collect water from already leaking taps

b) Before collecting, let water flow for at least 1 minute

c) Water sample container is kept close to the tap avoid spillage

d) Collect from a gentle stream of water to avoid splashing

Question 13:

The average daily dietary nutrient intake level sufficient to meet the nutrient requirements of nearly all (97-98%) healthy individuals in a particular life stage and gender group is known as _____.

a) Adequate intake

b) Dietary goal

c) Estimated average requirement

d) Recommended dietary allowance

Question 14:

A boys' hostel has an outbreak of fever cases with headache, followed by the development of pleomorphic rashes sparing palms and soles. What is the next best step in the management of suspected cases?

a) Isolate for 6 days after giving acyclovir, followed by VZIG within 72 hours of exposure

b) Isolate for 12 days after giving acyclovir, followed by VZIG within 48 hours of exposure

c) Isolate for 6 days

d) Only give VZIG

Question 15:

Which of the following is the sensitive indicator to assess the availability, utilization, and effectiveness of healthcare in a community?

a) Infant mortality rate

b) Maternal mortality rate

c) Immunization coverage

d) Disability-adjusted life years

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	b
5	d
6	c
7	b
8	c
9	d
10	b
11	c
12	d
13	d
14	c
15	a

Detailed Explanations

Solution to Question 1:

The likely diagnosis based on the given clinical scenario is neurolathyrism. Administration of vitamin C at doses of 500-1000 mg is beneficial for this.

Lathyrism is a paralyzing disease of humans and animals. It is due to the consumption of khesari dal or Lathyrus sativus. When the husk is removed, it looks similar to red gram or Bengal gram. It is mostly consumed by poor agricultural laborers because it is relatively cheap. Lathyrism is more prevalent in parts of Madhya Pradesh, Uttar Pradesh, Odisha, and Bihar. The Prevention of Food Adulteration Act in India has banned Lathyrus sativus in all forms (whole/ split/ flour).

In humans, it is referred to as neurolathyrism because it affects the nervous system. It is characterized by gradually developing spastic paralysis of the lower limbs. Beta-oxalyl amino alanine (BOAA) is the toxic compound causing this disease. It is water-soluble and can be removed by the following methods:

• Soaking pulses in hot water and discarding the water

• Overnight soaking in lime water, followed by boiling

• Parboiling - for large-scale detoxification

Solution to Question 2:

The given case with the classical symptoms of evening rise of temperature and productive cough of more than 2 weeks in duration prompts a diagnosis of pulmonary tuberculosis. According to the NTEP guidelines, the next best line of management is CBNAAT.

The initial diagnosis of pulmonary TB is done with sputum smear examination and chest radiography. Chest radiography features suggestive of TB include upper lobe infiltrates and cavities. However, these are not very specific to TB. The presence of these features should prompt microbiological evaluation for Mycobacterium tuberculosis.

A positive smear examination, irrespective of CXR findings, is considered as microbiologically positive with sensitivity status unknown. Drug sensitivity status via NAAT is to be done following a sputum smear before initiation of treatment. This helps in the early detection and treatment of drug-resistant TB. This has been integrated into the regular TB services to detect rifampicin resistance at the time of diagnosis. Depending on the sensitivity of the tubercle bacilli, appropriate ATT is started.

In cases of high clinical suspicion, positive CXR findings, or in people with HIV, a CB-NAAT is done. Two sputum samples are sent for cartridge-based nucleic acid amplification (CBNAAT), which detects the presence of Mycobacterium DNA along with resistance against rifampicin within 2 hours. It analyses sputum or any other biological specimen (except blood and blood-contaminated specimens) using real-time PCR technology.

The interpretation of CBNAAT is as follows:

• If the CBNAAT turns out to be positive/MTB detected, the 2nd sample is sent for culture and drug sensitivity testing (C&DST) immediately.

• If the CBNAAT turns out to be negative for rifampicin resistance, a first-line line probe assay is ordered, to look for isoniazid resistance.

• If FL-LPA shows no resistance to isoniazid, the patient is treated as drug-sensitive, and a conventional daily regime of 2HRZE and 4HRE is given.

• If FL-LPA shows resistance to isoniazid, reflex testing is done using second-line line probe assay (SL-LPA).

• If SL-LPA turns out to be negative, treat as isoniazid or H-mono-/poly-resistance

• If SL-LPA turns out to be resistant to some drugs, modify the treatment regime accordingly.

• If CBNAAT is positive for rifampicin resistance, the second sample is sent for first-line line probe assay (FL-LPA), second-line line probe assay (SL-LPA), and BACTEC MGIT 960 to look for isoniazid (H) mono-/poly-resistance

• If there is no resistance/H-mono-/poly- resistance, treat as shorter oral bedaquiline-containing MDR regime.

• If there is resistance, then treat as longer oral bedaquiline-containing MDR regime.

• If the 2nd sample also turns out to be indeterminate, a fresh sample is taken for LPA or liquid culture, to ascertain drug resistance. Meanwhile, treatment initiation is done and modified subsequently, as per results.

Other options:

Option A: Repeat sputum smears are not the next best line of management in this case, as the patient already had two negative sputum smears and a chest x-ray suggestive of tuberculosis.

Option C: Line probe assay would not be the immediate next step in management. First, CBNAAT should be performed to detect Mycobacterium tuberculosis and identify rifampicin resistance.

Option D: Waiting for TB culture results to start ATT is not the best management strategy, as it can take several weeks for the results to come back, and the patient may deteriorate in the meantime.

Solution to Question 3:

The study given here is a retrospective cohort study.

A retrospective cohort study is when both the exposure and outcome have occurred before the investigations and study is undertaken.

In this scenario, the workers in the industry were divided into two groups, viz those who are directly exposed and the clerical staff who are not directly exposed to the suspected carcinogen. The incidence of carcinogenesis among these two groups is studied going backward tracking their records.

An increased risk of bladder cancer was observed among long-term workers directly exposed to the chemical in aniline dye industries when compared with the other study group. It is also called a historical cohort study, a prospective study in retrospect, or a non-concurrent prospective study. The cohort is identified from past records and is assessed to date for the outcome. The same cohort is followed up prospectively in the future for further assessment of the outcome.

Other options:

Option B - Prospective cohort study: This type of study involves following a group of participants over time and monitoring their exposure to a risk factor and the development of an outcome. In this case, the outcome (bladder cancer) and exposure (aniline dye) have already occurred, making this a retrospective cohort study.

Option C - Case-control study: A case-control study involves comparing individuals with a specific outcome (cases) to those without the outcome (controls) and assessing their exposure history. This study design is not being used in the given scenario, as the workers are divided into two groups based on their exposure status and followed for the development of bladder cancer.

Option D - Intervention and response: This option is not a type of epidemiological study design. It could refer to an intervention study, in which an intervention is applied to a group of participants, and the response is measured. However, this is not applicable to the current scenario.

Solution to Question 4:

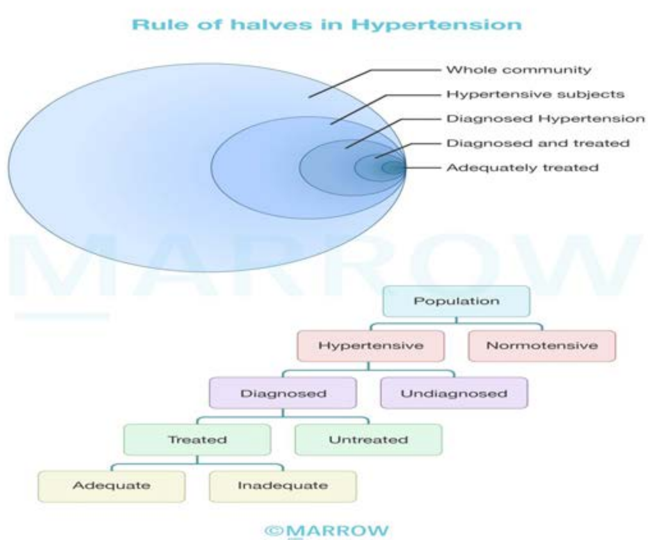
The above phenomenon is best described as the "tracking" of blood pressure.

In this phenomenon, the blood pressure of individuals is followed up from early childhood into adulthood. The BP in individuals with low BP levels tend to remain low, and in those with high BP levels, they tend to remain high. This phenomenon of persistence of rank order of blood pressure has been described as "tracking".

Other options:

Option A: The rule of halves depicts the disease burden of hypertension in the community. It states that only about half of the hypertensive subjects in the general population are aware of the condition. Only half of those aware are being treated and only half of those being treated are considered to be adequately treated.

A pictorial representation of the rule of halves in hypertension is shown below:



Option B: The WHO STEPwise approach to non-communicable disease (NCD) risk factor surveillance (STEPS) is a simple, standardized method for collecting, analyzing, and disseminating data on key NCD risk factors such as tobacco use, alcohol use, physical inactivity, unhealthy diet, overweight and obesity, raised blood pressure, raised blood glucose, and abnormal blood lipids.

Solution to Question 5:

Amongst the given options, it is the duty of the ASHA to provide counseling and information to this pregnant patient

An ASHA is a trained female community health activist and a resident woman of the village with a formal education of at least up to 10th grade. She should preferably be in the age group of 25 to 45 years (married/divorced/widow) and have qualities such as good communication skills and leadership skills. The general norm of selection recommends having one ASHA for 1000 population, but in tribal, hilly, and desert areas, the norm could be relaxed to one ASHA per habitation.

Roles of ASHA:

• To create awareness and provide information to the community on determinants of health such as nutrition, basic sanitation, and hygienic practices

• To counsel women on birth preparedness, the importance of safe delivery, breast-feeding and complementary feeding, immunization, and contraception

• To counsel women on the prevention of common infections including reproductive tract infection/sexually transmitted infection

• To mobilize the community and help them in accessing health and health-related services.

• To arrange and escort pregnant women and children requiring treatment/admission to the nearest pre-identified health facility

• To inform sub-center/primary health center about the births and deaths in her village and any unusual health problems/disease outbreaks in the community

• To promote the construction of household toilets under the total sanitation campaign

Other Options:

Option A (ANM): An Auxiliary Nurse Midwife (ANM) is a trained health worker who provides basic healthcare services, including antenatal care, delivery, and postnatal care. Although an ANM can provide counseling and information to pregnant women, the primary responsibility for this role lies with the ASHA worker.

Option B (AWW): An Anganwadi Worker (AWW) is responsible for providing basic health care, nutrition, and preschool education to children under the age of six, pregnant women, and lactating mothers. While AWWs play a vital role in maternal and child health, counseling pregnant women on health-related issues is not their primary responsibility.

Option C (Trained birth attendant): A trained birth attendant is responsible for assisting with childbirth and providing postpartum care. Although they may provide some counseling and information, this role is primarily focused on the delivery process rather than ongoing counseling throughout pregnancy.

Solution to Question 6:

This patient with vaginal discharge and erosions on the cervix has cervicitis from a sexually transmitted infection. She should be given the gray kit which contains cefixime and azithromycin.

Under the STD control program, pre-packed color-coded kits have been provided for free supply to treat sexually transmitted (STI) or reproductive tract infections (RTI). Management of STDs under the syndromic approach is based on specific signs and symptoms and is not dependent on laboratory investigations.

A patient with just vaginal discharge is presumed to have vaginitis and is given a green kit that contains secnidazole and fluconazole. But this patient also has cervical erosion, which is a sign of cervicitis. She should be treated with the grey kit. This kit contains a single tablet of 1 gram azithromycin for chlamydia and a single tablet of 400 mg cefixime for gonorrhea.

Solution to Question 7:

The years of potential life lost could be attributed to the years lost due to premature death.

It is a mortality indicator that shows the number of years that a dying person could have expected to survive (usually taken as 75 years). Mortality indicators are used as the starting point in the evaluation of health status. They are used as indirect indicators of health.

The concept of years of potential life lost (YPLL) is used to estimate the societal impact of premature death. It helps prioritize public health interventions and allocate resources to address significant health problems. YPLL, along with other metrics like DALYs and QALYs, aids in assessing the burden of disease and the effectiveness of healthcare interventions. These measurements can help inform policymakers and healthcare providers in making informed decisions about healthcare delivery and resource allocation.

Strategies to reduce YPLL include prevention and early intervention, improving access to healthcare, and addressing social determinants of health that contribute to health disparities and premature death.

Other options:

Option A: Years lost to morbidity refers to the time spent living with a disease or disability, which is not directly related to the concept of years of potential life lost.

Option C: Years lost to disability is a component of disability-adjusted life years (DALY). DALY indicates the number of years lost due to ill health, disability, or early death.

DALY = YLL +YLD

• Years lost to disability (YLD): Here, the number of cases due to injury and illness is multiplied by the average duration of the disease and a weighting factor. (Severity of the disease on a scale of 0 (perfect health) to 1 (dead))

• Years of lost life (YLL) - Here, the number of deaths at each age is multiplied by the expected remaining years of life. (according to a global standard of life expectancy)

Option D: Quality-adjusted life year (QALY) indicates how the health status of an individual impacts his quality and quantity of life. It shows the years of life lost due to injury or illness. It is commonly used in evaluating the outcomes of medical interventions.

Solution to Question 8:

A ten-year-old school child will be administered the TT/Td vaccine, as part of the school immunization program, as per the National Immunization Schedule.

Tetanus and adult diphtheria (Td) vaccine is a combination of tetanus and diphtheria antigens, with a lower concentration of diphtheria antigen (d), as recommended for older children and adults. The MoHFW recommends the replacement of the TT vaccine with Td vaccine for all age groups, including pregnant women. A dose of 0.5 mL is administered in the intramuscular route in the upper arm.

Scheduled to be administered at 10 and 16 years of age.

For pregnant women:

• Td-1: early in pregnancy

• Td-2: 4 weeks after Td-1

• Td (Booster): if pregnancy occurs within 3 years of the last pregnancy and 2 Td doses were received.

Other options:

Option A - According to the National Immunization Schedule, Measles/MR vaccine is given at 9 completed months (1st dose), and a second dose is given at 16-24 months. Measles vaccine can be given till 5 years of age. A dose of 0.5 mL is administered subcutaneously in the right upper arm.

Option B - Rotavirus vaccines are oral, live attenuated vaccines and are of 3 types: Rotavac, RotaTeq, and Rotarix. Rotavac (monovalent liquid frozen) vaccine is administered in 3 oral doses, 4 weeks apart at 6, 10, and 14 weeks. Though CDC guidelines recommend that it should not be administered to children more than 8 months of age, the National Immunisation Guidelines in India mention that the first dose of rotavirus vaccine can be administered up to the age of 1 year.

Option D - Hepatitis B vaccine is an example of a subunit vaccine made using HBsAg. The first dose (0.5 mL, intramuscular) is given at birth or as early as possible within 24 hours on the anterolateral side of mid-thigh. Subsequent doses are given as a part of the pentavalent vaccine at 6 weeks, 10 weeks, and 14 weeks (can be given until one year of age).

Solution to Question 9:

Aid to set up cobalt therapy units for cancer treatment at medical institutions in India was a part of the Colombo plan.

The Colombo Plan is a regional organization that upholds the concept of a collective intergovernmental effort to strengthen the economic and social development of member countries in the Asia-Pacific region.

It was formed at a meeting of the commonwealth foreign ministers with the headquarters at Colombo in January 1950. It comprises 20 developing countries within the region and 6 non-regional members - Australia, Canada, Japan, New Zealand, the UK, and USA. It involves foreign aid and technical assistance for the economic and social development of the region.

The contribution of Canada in supplying cobalt therapy units to medical institutions in India was an important item of aid under the Colombo plan. AIIMS New Delhi was established with financial assistance from New Zealand under the Colombo plan. Cobalt therapy is a form of radiotherapy that uses cobalt-60, a radioactive isotope, to generate high-energy gamma rays for treating cancer. Cobalt therapy units were an important tool for cancer treatment in the mid-20th century, and their provision under the Colombo plan significantly improved cancer treatment capabilities in India.

The Colombo Plan has evolved over the years and now focuses on a broader range of development goals, including education, public administration, agriculture, and industry. This reflects the changing needs of the Asia-Pacific region and the importance of a comprehensive approach to development.

Other Options:

Option A (Help with PET scan units for diagnosis of cancer): While PET scan units are important for cancer diagnosis, they were not part of the Colombo plan's cancer treatment initiatives in India.

Option B (Human resource strengthening): Although the Colombo plan aims to improve the economic and social development of member countries, human resource strengthening in the context of cancer treatment is not explicitly mentioned as part of the plan.

Option C (Setting up chemotherapy units): The Colombo plan's focus in cancer treatment was on providing aid for cobalt therapy units, not chemotherapy units.

Solution to Question 10:

The diagnosis based on the given clinical scenario is neurolathyrism, and the toxin responsible for the condition is beta-oxalyl-amino-alanine.

Lathyrism is a paralyzing disease of humans and animals. It is due to the consumption of Khesari dal/Lathyrus sativus. It is mostly consumed by agricultural laborers of poor economic status because it is relatively cheap. It is more prevalent in parts of Madhya Pradesh, Uttar Pradesh, Odisha, and Bihar.

In humans, it is referred to as neurolathyrism because it affects the nervous system. It is characterized by gradually developing spastic paralysis of the lower limbs. In animals, it is referred to as osteolathyrism as it leads to skeletal deformities. It usually affects young males aged 15-45 years and it is manifested in stages:

- Latent stage
- No-stick stage
- One-stick stage

- Two-stick stage
- Crawler stage

Beta-oxalyl-amino-alanine (BOAA) is the toxic compound causing lathyrism. The water-soluble property of the toxin is used in removing it from the pulses by soaking it in hot water and rejecting the soaked water. Vitamin C in doses of 500-1000 mg can be used for the prevention and treatment of lathyrism.

Other options:

Option A: Aflatoxins are produced by certain fungi, such as Aspergillus flavus and Aspergillus parasiticus, are the most potent hepatotoxins. These fungi can contaminate food grains such as groundnut, maize, parboiled rice, sorghum, wheat, and tapioca. It is mainly due to improper storage. It can be prevented by proper storage after drying. Moisture content should be kept below 10%. They do not cause the neurological symptoms described in the question.

Option B: Ergot alkaloids are produced by ergot or Claviceps purpurea, a field fungus. Food grains like bajra, rye, sorghum, and wheat have a tendency to get contaminated with this ergot fungus. These get harvested along with the food grains and consumption of this ergot-infested grain leads to ergotism.

Acute cases present with nausea, vomiting, giddiness, and drowsiness for a period of up to 24 to 48 hours after the ingestion of ergot grain. In cases of chronic consumption, they present with painful cramps in limbs, and peripheral gangrene due to vasoconstriction of capillaries. However, they do not cause the progressive spastic paraparesis and upper motor neuron paralysis characteristic of neurolathyrism. Ergot-infested grains are removed by floating them in 20% saltwater. Handpicking can also be done.

Option D: Fusarium toxin is produced by the fungus Fusarium incarnatum. They are known to contaminate sorghum and rice.

Solution to Question 11:

The patient with a clean-cut wound without any bleeding, who received a full course of tetanus vaccination 10 years ago, should be given a single dose of tetanus toxoid vaccine and appropriate wound care.

Wound management involves cleaning the wound thoroughly to remove soil, foreign bodies, and necrotic tissue.

Note: In cases where combined immunization is necessary, the patient is given 250 units of human tetanus immunoglobulin in one arm and 0.5 mL of adsorbed tetanus toxoid (PTAP or APT) in the other arm or gluteal region. The antitoxin is given for immediate temporary protection, and the tetanus toxoid is given for long-lasting protection.

Solution to Question 12:

During a disease outbreak, for collection of a water sample for bacteriological examination from a tap in regular use, water is collected from a gentle stream of water to avoid splashing.

If water samples are to be collected from a tap that is not regularly used, they should be first sterilized by heating. Then, the tap is cooled and opened fully and the initial water stream is let out to flow before the sample is collected.

The sample bottle must be held near the base with a hand and the stopper and paper cover over it are removed together and held in the fingers.

Other options:

Option A (Collect water from already leaking taps): Collecting water from leaking taps is not recommended as it increases the risk of contamination and may not provide an accurate representation of water quality.

Option B (Before collecting, let water flow for at least 1 minute): This option is close to the correct answer but falls short, as it is recommended to let the water flow for at least two minutes to flush out stagnant water in the pipe.

Option C (Water sample container is kept close to the tap to avoid spillage): While keeping the sample container close to the tap is a good practice to avoid spillage, it is not the main concern during the collection of water samples for bacteriological examination.

Solution to Question 13:

The average daily dietary nutrient intake level sufficient to meet the nutrient requirements of nearly all healthy individuals in a particular life stage and gender group is known as recommended dietary allowance (RDA).

Recommended dietary allowances (RDAs) are developed by the Food and Nutrition Board of the Institute of Medicine, National Academy of Sciences, and are part of the Dietary Reference Intakes (DRIs). DRIs include not only RDAs but also adequate intakes (AIs), tolerable upper intake levels (ULs), and estimated average requirements (EARs).

RDAs are essential for planning and assessing diets for healthy individuals and serve as the basis for food labeling, nutrition policy, and dietary guidelines. Meeting the RDA for a specific nutrient helps ensure that an individual's nutrient intake is sufficient to prevent deficiency and maintain optimal health. However, individual nutrient needs may vary depending on factors such as age, gender, health status, and activity level.

Other options

Option A: Adequate intake is the recommended average daily intake level which is based on experimentally found estimates of nutrient intake by a group of apparently healthy population (assumed to be adequate). It is used when RDA cannot be determined. It is referred to as acceptable intake in the indian context.

Option B: Dietary goals are country-specific quantitative targets for selected nutrients aimed at preventing long-term chronic diseases. They are not directly focused on meeting the nutrient requirements of individuals but rather focus on population-level nutritional health.

Option C: The estimated average requirement (EAR) is the daily nutrient intake level estimated to meet the requirements of half the healthy individuals in a specific life stage and gender group. It is used to calculate the RDA but is not the same as the RDA.

The RDA is calculated as the estimated average requirement (EAR) plus twice the standard deviation (SD).

RDA = EAR + (2 × SD)

Solution to Question 14:

The scenario describes an outbreak of fever cases with headache followed by the development of pleomorphic rashes sparing palms and soles, which points to the diagnosis of chickenpox. In a boys' hostel, it is reasonable to assume that the majority of the affected individuals are not in the high-risk groups for severe varicella infection. For the management of suspected cases in this situation, isolation for 6 days from the onset of the rash is the appropriate measure to prevent further spread of the infection. Acyclovir and VZIG are not necessary for healthy individuals.

Varicella zoster immunoglobulin (VZIG) is not indicated for everyone. VZIG is a passive immunization used as post-exposure prophylaxis for individuals at high risk of developing severe varicella (chickenpox) infection after being exposed to the varicella-zoster virus. It should be administered within 72 hours of exposure to provide the best protection, although it may still be effective if given up to 10 days after exposure.

High-risk individuals who may benefit from VZIG include

• Immunocompromised individuals, such as those receiving immunosuppressive therapy, those with congenital cellular immunodeficiency, and those with acquired immunodeficiency, including HIV/AIDS.

• Pregnant women who have been exposed to the virus and have no history of chickenpox or vaccination, as varicella infection during pregnancy can lead to congenital varicella syndrome or severe neonatal varicella infection.

• Newborns whose mothers developed chickenpox within five days before delivery or within 48 hours after delivery, as they are at risk of severe neonatal varicella infection.

For healthy individuals who have been exposed to the varicella-zoster virus, the varicella vaccine is usually recommended as post-exposure prophylaxis, rather than VZIG. The vaccine should be administered within 3 to 5 days of exposure to help prevent or reduce the severity of the disease.

Live attenuated vaccine is administered to children who have not had chickenpox between the age of 12 to 18 months.

The below image shows multiple vesicular lesions on the trunk is suggestive of chickenpox.

Solution to Question 15:

The infant mortality rate (IMR) is a sensitive indicator of the availability, utilization, and effectiveness of health care in a community.

Infant mortality rate (IMR) = Number of deaths in the first year of life/Total number of live births x 1000.

It is one of the universally accepted healthcare indicators for the whole population and the socioeconomic conditions under which they live.

Other options:

Option B - Maternal mortality rate: While maternal mortality rate is an important indicator of maternal healthcare services, it is not as sensitive as the infant mortality rate in reflecting the availability, utilization, and effectiveness of healthcare in a community.

Option C - Immunization coverage: Immunization coverage is an essential factor in preventing diseases and promoting public health. However, it is a narrower indicator than infant mortality rate, focusing primarily on preventive healthcare services for children. Full immunization coverage is defined as the proportion of infants 9-11 months old who have received the 3 doses of DPT vaccine, BCG vaccine, 3 doses of OPV, and one dose of measles vaccine against the estimated number of infants during a specific year.

Option D - Disability-adjusted life years: DALYs are a useful measure of the overall burden of disease in a population, but they are less sensitive in assessing the availability, utilization, and effectiveness of healthcare services in a community compared to infant mortality rate.

Community Medicine INI-CET 2023

Question 1:

All of the following statements are true for descriptors of central tendency except:

- **a)** Mode is affected by extreme values
- **b)** Mean is affected by extreme values
- **c)** Median is calculated by overall observation
- **d)** Mode can be used as a qualitative data descriptor

Question 2:

The formula given below corresponds to:

$$\sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

- **a)** Standard deviation
- **b)** Variance
- **c)** Mean
- **d)** Median

Question 3:

All of the following may be used to find the normalcy of data except:

a) Plot histogram

b) Mantel-Haenszel test

c) Box whisker

d) Shapiro Wilk test

Question 4:

In Kaplan Meier survival curve, the bands used for the confidence interval is?

a) Unequal precision band

b) Hirshberg band

c) Holmes band

d) Hall-Wellner Band

Question 5:

30,000 women were followed up for 10 years for the development of breast cancer. 1200 women developed cancer and were given a questionnaire to assess possible risk factors. Additionally, 2000 women from the same study were used as controls and were also given the questionnaire. What is this type of study called?

a) Case cohort

b) Retrospective cohort

c) Nested case control

d) Cross control cohort

Question 6:

A person had a dog bite. Vaccination is up to date for both the dog and the man. The person's history is significant for having undergone splenectomy five years ago. What is the best management in this case?

a) Observation

b) Amoxiclav

c) Metronidazole

d) Ciprofloxacin

Question 7:

A child came for DPT vaccination at 10 weeks. He has a history of fever of more than 40°C and an inconsolable cry at 6 weeks of age following vaccination. What should be done next?

a) Avoid DPT vaccination

b) Give DT vaccine

c) Give DPT vaccine

d) Defer by 4 weeks

Question 8:

High fat intake according to FDA is?

a) Diet should contain 400 kcal from carbohydrates

b) Diet should contain total 1000 kcal

c) Diet should contain 800 kcal from fat

d) Diet should contain 80% fats

Question 9:

The slogan coined by WHO 'Do it Right' in the first aid of a snake bite includes all of the following except?

a) Reassurance

b) Tell sequence of events in detail after the bite

c) Go to hospital

d) Incisions

Question 10:

Calculate the kappa statistics for the following data set comparing IGRA and Mantoux test.

a) 0.4

b) 0.6

c) 0.8

d) 0.5

Question 11:

Which of the following is not correctly matched?

a) Observational study — MOOSE

b) Systematic review — PRISMA

c) Diagnostic Study - CONSORT

d) Case report — CARE

Question 12:

Which of the following best represents the definition of culture?

a) It is static and not related to health and disease

b) A set of beliefs which are static

c) Socially acquired learned behaviour

d) Rules set according to religion which are dynamic

Question 13:

According to National Programme for Control of Blindness and Visual Impairment which of the following is incorrect about the reach in approach?

a) People with cataract are treated in a makeshift operation centre

b) IOL placement by small incision cataract surgery is preferred

c) Reach in approach has lesser post operative complications

d) People with cataract are identified and sent to higher centre

Question 14:

A 68-year-old lady was brought to the hospital by her daughter, with episodes of falls, confusion, chronic diarrhea, and cough. The intern decided to conduct a geriatric assessment. Which of the following factors indicated such an examination?

a) Age, sex, falls

b) Age, cough, chronic diarrhoea

c) Age, fall, confusion

d) Age, sex, cough, chronic diarrhea

Question 15:

Which of the following best helps assess the quality of healthcare provided?

a) Multiple logistic regression model

b) Bar graph

c) Scattergram

d) Ishikawa diagram/ Fish bone diagram

Question 16:

A mother brought her 6 month old infant baby to an Anganwadi worker Lata for advice regarding the vaccination of her child. The last vaccines he had received were OPV3, PENTA 3, IPV2, and ROTA3 at 14 weeks of age. BCG and Hep B were given at birth. When is the next vaccination due?

a) MR vaccine at left arm intramuscularly and Vitamin A drops at the end of 9 months

b) MR vaccine at right arm subcutaneously and Vitamin A drops at the end of 8 months

c) MR vaccine at left arm intramuscularly and Vitamin A drops at the end of 8 months

d) MR vaccine at right arm subcutaneously and Vitamin A drops at the end of 9 months

Question 17:

Which of the following statements is true about palliative care?

a) A,B,C,D

b) C,E,F

c) B,D

d) A,B,E,F

Question 18:

Which of the following is the highest level of community participation?

a) Considering their ideas

b) Felt needs of community considered

c) Contribution of resources

d) Involving in planning of health related campaigns

Question 19:

A patient hailing from an area infested with mosquitoes presents with fever, chills, and myalgias. WBC count was 4000 cells/cu.mm, RBC was normal, and platelets were 90000 cells/cu.mm. He also had a maculopapular rash. What is the likely diagnosis?

a) Dengue

b) Typhoid

c) Ebola haemorrhagic fever

d) Rift valley fever

Question 20:

Which of the following statements are correctly matched?

a) Statement 1,2

b) Statement 3,4

c) Statement 2,3

d) Statement 1,2,3

Question 21:

Two doctors examined 100 chest X - Rays of patients. The kappa statistic was found to be 0.71. What does it mean?

a) Good agreement

b) Perfect agreement

c) Poor agreement

d) No agreement

Question 22:

Which of the following is an example of a longitudinal observational and analytical study?

a) Case control Study

b) Cohort Study

c) Randomized Controlled Trial

d) Ecological Study

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	d
5	c
6	b
7	c
8	b
9	d
10	a
11	c
12	c
13	a
14	c
15	d
16	d
17	c
18	d
19	a
20	c
21	a
22	b

Detailed Explanations

Solution to Question 1:

Mode is not affected by extreme values in a given set of data.

The mean of a given set of data is responsive to each and every value in the distribution. Thus, the mean is the measure of central tendency which is maximally affected by extreme values (option B). It is usually an inappropriate measure for describing skewed distributions. However, repeated samples from the same population will always tend to have similar means. Mean is able to resist fluctuations between different samples drawn from the same population better than mode or median.

To obtain the median of a given set of data, the values are first arranged in ascending or descending order of magnitude, and then the value of the middle observation is noted. Since the data has to be first arranged in order before the median can be determined, an overall observation is necessary for its calculation (option C).

Mode is the observed value in the frequency distribution that occurs with the greatest frequency. It is completely uninfluenced by extreme values in the distribution. The mode can be used to describe qualitative data (option D).

Solution to Question 2:

The formula given above is used to calculate the standard deviation.

Standard deviation (or σ) is a measure of how dispersed the data is in relation to the mean. Data clustered around the mean have a low standard deviation, while data that are more spread out have a high standard deviation. A standard deviation close to zero indicates that data points are close to the mean, whereas a high or low standard deviation indicates data points are respectively above or below the mean.

Variance (option B) is the mean of the squares of all the deviation scores of the distribution. It is numerically equal to the square of the standard deviation.

Mean (option C) is obtained by adding together the individual observations and then dividing by the total number of observations.

Median (option D) divides the frequency distribution in half when all the scores are listed in order. When a distribution has an odd number of elements, the median is the middle value. When the distribution has an even number of elements, the median is the average of the two middle scores.

Solution to Question 3:

Mantel-Haenszel method is used to calculate adjusted odds ratio during stratified analysis. It is not a test to detect normalcy of data.

A plot histogram and a box whisker plot are visual methods to assess if a given set of data is distributed normally. A normal distribution curve is an example of a plot histogram.

Sometimes just the visual representation of the data might not be enough to tell whether the data is normally distributed. This is where Shapiro-Wilk test comes in. It examines how closely a given set of data fits a normal distribution. The sample data is first ordered and standardized. An assessment is then made how the sample quantiles fit the standard normal quantiles.

Shapiro-Wilk test is commonly used on small samples.

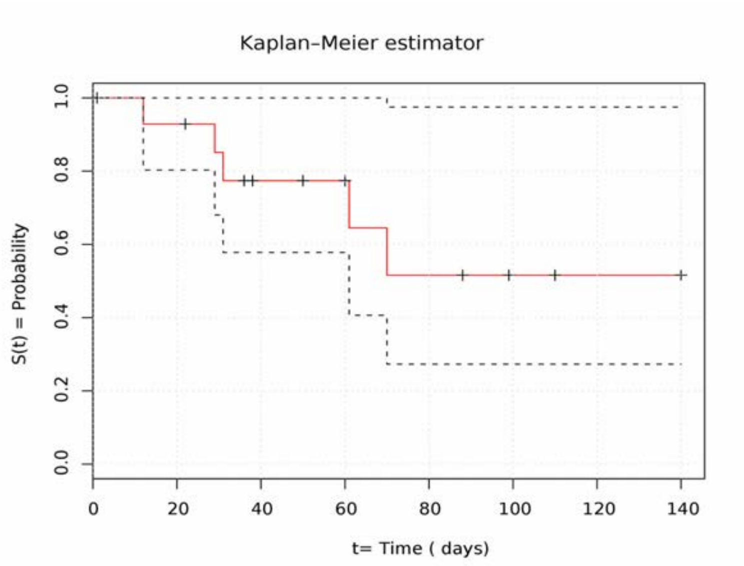
Stratified analysis of the data is done to reduce confounding. In such a scenario, multiple odds ratios are obtained — one for each stratum of the data. These have to be combined to give an adjusted odds ratio. This represents the effect of the exposure adjusted for the confounding factor. Mantel-Haenszel method is used for this purpose.

Solution to Question 4:

The bands used for confidence interval in the Kaplan-Meier survival curve are Hall-Wellner bands.

When using standard statistical software, at each point in time, the upper and lower limits of the confidence interval are also obtained. Hall-Wellner bands, Gill bands, and Nair's Equal Precision confidence bands are used for this purpose.

Kaplan-Meier plot is a graphical representation of the number of people surviving during a given length of time. Often there are two or more groups represented in the plot, such as those treated versus those not treated. The graph often has a number of steps as the number of people surviving is a discrete variable. The image given below shows a Kaplan-Meier plot. The dotted lines represent the upper and lower limits of the 95% confidence interval.



Solution to Question 5:

The type of study described in the question is a nested case-control study.

A nested case-control study is essentially a case-control study that is present inside a cohort study. The cases are all the cohort members who developed the disease being studied at a given point in time during the follow-up period. However, each case is matched to one or more controls — individuals who did not develop the disease but are otherwise comparable to cases in terms of age and gender. Cases and controls are then compared in terms of their respective exposures to the risk factor. Nested case-control studies thus represent an interim analysis of the exposure and outcome data of a cohort study.

A case-cohort study (option A) involves selecting a subset of the entire cohort at the beginning of the study. This sub-cohort acts as the control group for making comparisons with multiple case groups. For example, if a cohort study is being done to assess smoking as a risk factor for ischemic heart disease and lung cancer, the same sub-cohort can be used as the control group for both outcomes. If it was a nested case-control study, it would require two separate control groups. Case-cohort studies are more efficient than nested case-control studies when multiple outcomes are being studied.

In a retrospective cohort study (option B), the outcomes have already occurred before the study begins. The investigator has to analyze the history of prior exposures and the duration of such exposures. Even though the exposure and the outcome have already occurred, the investigator goes from the exposure $\rightarrow$ outcome in a retrospective cohort study. This is in contrast to case-control studies, where the investigator goes from outcome $\rightarrow$ exposure.

Cross control cohort (option D) is not a type of study design.

Solution to Question 6:

Amoxicillin-clavulanate (Amoxiclav) is the drug of choice in this patient.

Antibiotic prophylaxis is indicated in:

• High-risk animal bites: cat bites in any location and hand bites by any animals, including humans. Cat and human bites have a higher risk of infection versus dog bites. Moreover, hand bites have a risk of developing an infection in the closed spaces of the hand.

• High-risk patients: comorbidities such as diabetes, liver disease, and absence of functional spleen. Patients without a functional spleen are at an increased risk for overwhelming sepsis primarily due to Capnocytophaga species.

Pasteurella species is the most common isolate from cat and dog bites. Streptococci, S. aureus and Eikenella corrodens are found in human bites. Amoxicillin-clavulanate is the first-line drug of choice for prophylaxis. In patients with penicillin allergy, clindamycin plus doxycycline/co-trimoxazole/fluoroquinolone can be used.

Solution to Question 7:

The next step for this patient should be to administer DPT vaccine.

Events such as persistent inconsolable crying (more than 3 hours in duration), hyperpyrexia (fever > 40°C), hypotonic-hyporesponsive episodes within 48 hours of DTwP administration, and seizures, with or without fever, within 72 hours DTwP administration are considered precautions but not contraindications to future doses of DTwP. These events don't recur with the next dose and have been shown to have no permanent sequelae.

Absolute contraindications to any pertussis vaccination are a history of anaphylaxis or development of encephalopathy within 7 days following previous DTwP vaccination. In case of anaphylaxis, further immunization with diphtheria, tetanus, and pertussis vaccines are contraindicated as it cannot be said with certainty which component caused anaphylaxis. For patients with a history of encephalopathy following vaccination, any pertussis vaccine is contraindicated, and only diphtheria and tetanus vaccines may be used.

Note: This answer might be controversial. The above answer is as per the IAP Guidebook on Immunization.

Solution to Question 8:

High fat intake according to FDA is a diet that should contain a total of 1000 kcal.

High fat diets have low carbohydrates, <50 grams/day. This equates to <250 kcal/day in the form of carbohydrates, not 400 kcal (option A). The percentage of calories from fat is >50% (option D), which would equate to about 500-600 kcal from fats (option C). The diet should contain a total of 800 - 1000 kcal.

Ketogenic diets mainly incorporate low carbohydrates (<50 grams/day), low protein, and high fats. Gluconeogenesis from the glucogenic amino acids in the diet and fat oxidation are utilized to meet the metabolic demands of the body. Ketogenic diets are especially useful for the treatment of intractable epilepsy.

Solution to Question 9:

Incisions are not a part of the management of a snake bite.

The first aid following a snake bite consists of the mnemonic: carry no right.

Carry: The victim is not allowed to walk, even for a short distance. Walking generates muscle-pumping activity which disperses the venom into the systemic circulation.

No: No tourniquet, no electrotherapy, no cutting, no pressure immobilization, no suction to the bite site

R: Reassure the patient since 70% of the bites are from non-venomous species.

I: Immobilize in the same way as a fractured limb. A splint can be applied to the bitten extremity to limit the movement and assist with positioning. The extremity should be maintained in a neutral position at approximately heart level.

GH: Get to a hospital immediately. The most important step is to rapidly transport the patient to a medical facility equipped to provide supportive management.

T: Tell the doctor of any systemic symptoms which may have appeared on the way to the hospital.

Note that elapid envenomations that are primarily neurotoxic and have no significant effects on local tissue may be managed by pressure immobilization. In this technique, the entire limb is immediately wrapped with a bandage and then immobilized. This technique restricts lymphatic drainage and has been shown to delay the absorption of venom into the systemic circulation. However, the offending snake should have been reliably identified as neurotoxic and the rescuer should be trained in applying this pressure bandage.

Solution to Question 10:

Kappa statistic is used to evaluate the agreement between IGRA and TST test results.

Kappa is calculated as:

$$\kappa = \frac{p_O - p_E}{1 - p_E}$$

where, pO is observed agreement and pE is expected agreement.

The table in the question has been redrawn with alphabets in the cells to better illustrate the formulae:

	IGRA (+)	IGRA (–)
Mantoux (+)	80(a)	40(b)
Mantoux (–)	20(c)	60(d)

In this question, the pO will be all the situations in which both IGRA and Mantoux tests are positive or negative — that is, both of them are in agreement with one another.

$$p_0 = \frac{a+d}{a+b+c+d} = \frac{140}{200} = 0.7$$

pE represents the probability of random agreement between the tests. It is calculated as a sum of both the tests being positive together at random and negative together at random.

$$p_E = p_{pos} + p_{neg}$$

$$p_{pos} = \frac{a+b}{a+b+c+d} \times \frac{a+c}{a+b+c+d}$$

$$p_{pos} = \frac{120}{200} \times \frac{100}{200} = 0.3$$

$$p_{neg} = \frac{c+d}{a+b+c+d} \times \frac{b+d}{a+b+c+d}$$

$$p_{neg} = \frac{80}{200} \times \frac{100}{200} = 0.2$$

$$p_E = 0.3 + 0.2 = 0.5$$

Substituting the calculated values in the formula for kappa:

$$K = \frac{0.7 - 0.5}{1 - 0.5} = \frac{0.2}{0.5} = 0.4$$

Strength of agreement varies with kappa coefficient as follows:

- 0.81 to 1: perfect agreement between the two tests/methods
- 0.61 to 0.80: substantial agreement
- 0.41 to 0.60: moderate agreement
- 0.21 to 0.40: fair agreement
- 0.10 to 0.20: slight agreement

Solution to Question 11:

The guidelines used for reporting the results of RCT are CONSORT (consolidated standards of reporting trials).

To improve the quality of reporting of health research, certain standards have been developed. These standards vary according to the type of study in question. Many journals now require the authors to complete the relevant checklist according to these standards prior to submitting their paper for publication. These standards include:

Many of these guidelines are available via the EQUATOR network.

Guideline	Expansion	Area
CONSORT	Consolidated standards of reporting trials	Randomized controlled trials
TREND	Transparent reporting of non-randomized designs	Behavioural and public health interventions with non-randomized designs

Guideline	Expansion	Area
STROBE	Strengthening the reporting of observational studies in epidemiology	Cohort, case-control, and cross-sectional studies
STRAEGA	Strengthening the reporting of genetic association studies	Modification of STROBE for genetic studies
STARD	Standards for the reporting of diagnostic accuracy studies	Diagnostic tests
CARE	Case reporting	Clinical case reports and reports of patient encounters
SAMPL	Statistical analyses and methods in published literature	Basic statistical reporting for articles published in biomedical journals
PRISMA	Preferred reporting items for systematic reviews and meta-analyses	Meta-analyses of randomized controlled trials
MOOSE	Meta-analysis of observational studies in epidemiology	Meta-analyses of observational studies

Solution to Question 12:

Culture is the socially acquired learned behaviour. It is passed onto generations through formal and informal learning. It is the representation of customs, beliefs, laws, arts, and moral values of a society. Cultural factors have a large part to play in health.

Other closely related terms include:

Customs are the traditionally accepted ways of behaving that are specific to a society. They can be folkways, which are referred to as the right ways of doing things in less vital areas of human conduct. Mores are stringent customs followed by a society.

Socialization is the process by which a person in a society imbibes its beliefs, customs, etc. through daily interaction with its members.

A community is a social group determined by geographical boundaries and/or common values and interests.

Acculturation is the process by which there is the diffusion of cultures between two culturally different groups of people. Acculturation does not take place by folklore.

Folklore is the passage of customs and beliefs of a community through generations, merely by word of mouth.

Solution to Question 13:

Patients with cataract are not operated in makeshift operation centres but are rather shifted to base hospitals (reach-in approach) as the visual acuity results indicate that the Base Hospital had significantly higher final corrected visual acuity compared to the Peripheral Eye Camps.

The Base Hospital approach, also known as the reach-in approach, and the Peripheral Eye Camp approach, also known as the reach-out approach, are two community-oriented strategies implemented within the framework of the National Programme for the Control of Blindness. These techniques aim to address the existing backlog of cataract-related blindness. Both methods have demonstrated efficacy and possess distinct advantages and disadvantages. Both community participation and intersectoral coordination, together with the utilization of suitable technology at an affordable price, are essential components for both endeavours.

Based on recent developments, there is a shift in emphasis from the 'Reach-out' approach to the 'Reach-in' approach. The Reach-in Approach involves the organization of screening camps in rural and remote locations that lack access to eye-care institutions. Comprehensive eye care treatments, including refraction, are offered to patients. Subsequently, those requiring cataract surgery are conveyed to the nearest hospitals that possess adequate resources and facilities, following the base hospital strategy. To conduct ocular surgery, it is imperative to comply with the guidelines established by the Government of India.

• It is imperative to have at least one anesthesiologist who is well prepared to address typical crises to ensure patient safety and optimal care.

• It is recommended that a minimum of one, ideally two, operating surgeons be present, with each surgeon limiting their daily caseload to no more than 30 procedures.

• Currently, the preferred approach for cataract extraction is extracapsular surgery, with a preference for manual small incision cataract surgery (SICS) or phacoemulsification, together with the implantation of a posterior chamber intraocular lens.

• It is ideal to limit the number of surgical procedures conducted in a day to no more than 50.

• Simultaneous surgery on both eyes should be avoided.

• Patients who have a high surgical risk due to conditions such as severe diabetes, severe hypertension, and cardiac issues should not undergo surgery in camp settings. Cases related to such issues should be appropriately referred to the base hospitals.

National Programme for the Control of Blindness and Visual Impairment (NPCBVI). The levels of eye care, according to NPCBVI are:

Primary level: Sub-district level hospitals, community health centres, mobile ophthalmic units, upgraded PHCs, link workers, and Panchayats.

Secondary level: District hospitals, NGO eye hospitals

Tertiary levels: Regional institutes of Ophthalmology, medical colleges.

The goal is to reduce the prevalence of blindness to 0.3% by 2020.

Solution to Question 14:

This patient's age, history of falls and episodes of confusion necessitate a comprehensive geriatric assessment.

Various comprehensive geriatric assessment programs use the following before performing a full-fledged assessment:

• Age

• The presence of concurrent medical conditions, such as heart failure or cancer.

• The presence of psychosocial illnesses like depression or isolation.

• Diseases unique to the elderly, include cognitive decline, mobility issues, and falls

• Prior or anticipated high health-care utilization

• Changes in living situation (for example, from independent living to assisted living, nursing facility, or in-home caretakers)

The inclusion of geriatric evaluation is imperative within the broader framework of the patient's health trajectory, taking into account their future care objectives and personal preferences. Functional status refers to the comprehensive assessment of an individual's health effect within the framework of their surroundings and social support network. It encompasses the individual's capacity to do the physical and social tasks required for their typical activities and duties. The constituents of functional status encompass activities of daily living (ADLs), instrumental activities of daily living (IADLs), and advanced activities of daily living (AADLs). Two often employed measures in the field are the Katz index, which assesses activities of daily living (ADLs), and the Lawton scale, which evaluates instrumental activities of daily living (IADLs). The most effective approach for ascertaining AADLS involves the utilization of open-ended inquiries, such as "In what manner do you allocate your time throughout the day?" or "What activities do you find enjoyable?"

There is a growing trend in geriatric assessment to integrate perspectives from non-clinical office workers, patients, and their relatives or loved ones. This collaborative approach may encompass the use of standing orders, forms, and questionnaires. An advisable strategy involves doing an annual evaluation of functional status, commencing at the age of 65 years, using the Medicare Annual Wellness Visit. For individuals below the age of 75 years with multiple comorbidities or those aged 75 years and above, it is recommended to augment this assessment with supplementary components. Furthermore, it is recommended that some components of geriatric assessment, such as evaluating activities of daily living (ADLs) and instrumental activities of daily living (IADLs), measuring gait, balance, and the risk of falls, evaluating mood and affect, as well as assessing cognition, be conducted following significant illnesses, particularly those necessitating hospitalization.

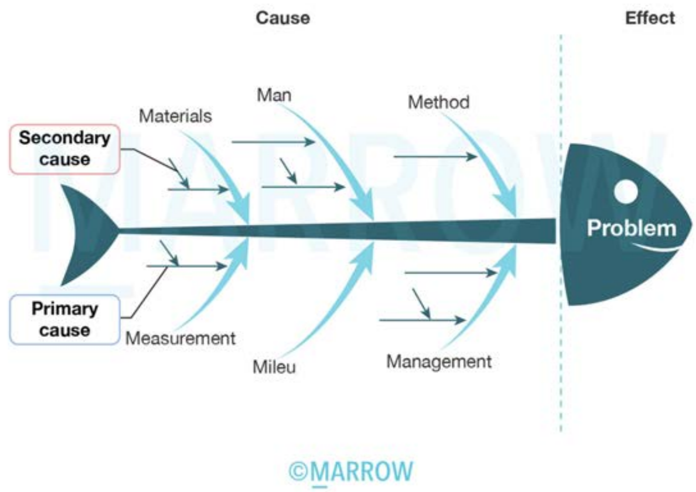
Solution to Question 15:

Ishikawa diagram or else better known as the fishbone diagram helps sketch out problems that affect healthcare in a community and thus uses those as pointers to better address them in the community.

The Ishikawa diagram, often known as the fishbone diagram, is employed by teams aiming to comprehend the numerous potential factors that contribute to a quality of care issue. This tool enables teams to concentrate their development endeavours on these identified causes. The utilization of this tool is applicable across several phases of the quality improvement framework, with its primary application commonly observed within the context of root cause analysis.

The fishbone graphic bears a resemblance to the skeletal structure of a fish. To depict the diagram about a quality of care issue, the problem (effect) is encapsulated within a box positioned on the extreme right of the diagram. A vertical line, referred to as the centre line or spine, is then positioned to the left of the box where the problem is documented. Subsequently, diagonal lines resembling fish bones are sketched emanating from the central line, also known as the spine. The diagonal lines depicted in the diagram signify distinct categorizations of the underlying factors contributing to the issue. Supplementary branches may also be delineated from the primary skeletal structure to symbolize underlying factors leading to the causes.

The groupings are typically structured into categories, commonly known as the five Ps (patients/clients, providers, policies, processes and procedures, and place/equipment), the six Ms (machine, method, materials, measurement, man, and Mother Nature), and the four Ss (surroundings, suppliers, systems, and skills). Nevertheless, the groupings need to be tailored to the organization and problem being investigated. The responsibility for ensuring this specificity lies with the quality improvement team. Once the categories have been delineated, the quality improvement team proceeds to partake in a brainstorming session to identify causes that are specific to each respective grouping. The aforementioned procedure persists until the team reaches a point where they are unable to ascertain any further underlying factors.

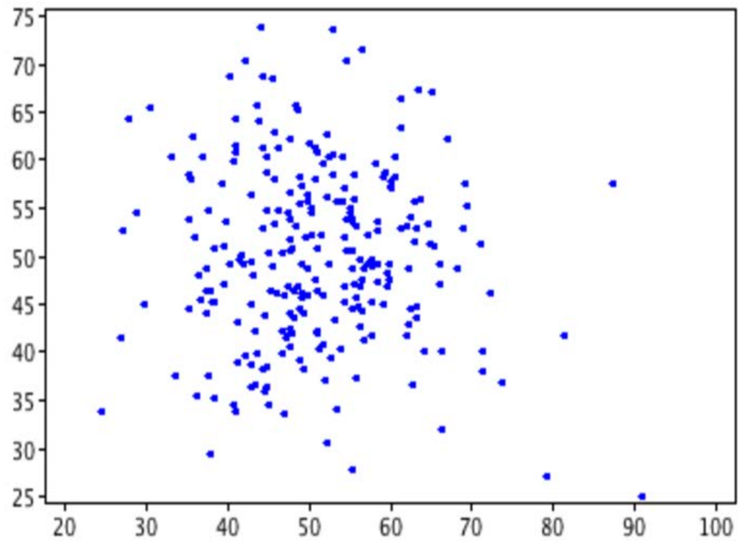


Other options:

Option A: Logistic regression may be used in some situations where we are not interested in a numerical outcome (like risk, etc) but we want a nominal or categorical one (like death/survival, developing a disease/not developing a disease). Here, the relationship between the variables will not be a linear one as the outcome from survival to death in relationship with the risk factor can change suddenly. Thus these outcomes are represented by 0 and 1.

Option B: A bar graph is a pictorial way of presenting a set of numbers, the magnitudes of which are proportional to the length of the bars in the graph. It can be simple, multiple, or component bar graphs and enable values to be compared visually. They are used to compare a single variable value between several groups.

Option C: A scatter plot is a statistical map used to study the correlation between two given variables. In scatter plots or scatter diagrams, the pattern made by plotted points along the two axes indicates the possible relationship. If the dots cluster along a straight line, then the relationship is linear. If there is no cluster, then there might be no relationship. Below is an example of a scatter diagram without a linear relation.



Solution to Question 16:

According to the National Immunization Schedule, the Measles/MR vaccine is given at 9 completed months (1st dose), and a second dose is given at 16-24 months. Measles vaccine can be given till 5 years of age. A dose of 0.5 mL is administered subcutaneously in the right upper arm.

Vitamin A schedule in the National Immunisation Schedule(NIS)

• 1 lakh IU at 9 completed months along with the Measles-Rubella vaccine

• 2 lakh IU is administered bi-annually for children between 1-5 years of age

Total number of doses administered- 9 doses,17 lakhs IU

Solution to Question 17:

True statements are:

• The patient is sovereign

• The aim is to improve the quality of life

Palliative care helps patients and their families cope with physical, psychological, social, and spiritual obstacles of life-threatening disease. It is interesting to note that even the quality of life of the caregiver also improves. Palliative care is needed by 56.8 million individuals annually, including 25.7 million in their final year according to WHO, while only 14% of palliative care patients worldwide receive it.

Palliative care improves the quality of life of life-threatening illness patients and their families. Early detection, assessment, and treatment of physical, psychosocial, and spiritual suffering and additional concerns diminish discomfort. Addressing suffering goes beyond physical symptoms. Palliative care supports patients and caregivers collaboratively. This includes practical assistance and bereavement counselling. It supports patients to live as fully as possible till death.

The human right to health includes palliative care. Patient-centred and integrated health services should focus on individual needs and preferences.

Many diseases require palliative care. Most palliative care patients have chronic conditions like cardiovascular disease (38.5%), cancer (34%), respiratory disease (10.3%), AIDS (5.7%), and diabetes (4.6%). Kidney failure, chronic liver illness, multiple sclerosis, Parkinson's, rheumatoid arthritis, neurological disease, dementia, congenital defects, and drug-resistant TB may require palliative care.

Most common and serious symptoms that patients undergoing palliative care experience are pain and respiratory difficulties. Almost 80% of AIDS, cancer, and 67% of cardiovascular and COPD patients would experience moderate to severe agony at death.

Opioids also relieve dyspnea and diminish the increased pain experienced by patients. Early symptom control is ethical to reduce suffering and respect the dignity of the patient.

Solution to Question 18:

The highest level of community participation is for the community to be involved in the planning of health-related campaigns.

Engaging in the strategic planning of health-related campaigns signifies a substantial degree of community involvement, surpassing the mere contemplation of ideas or the provision of resources. The process of planning campaigns necessitates proactive participation and cooperation, signifying a heightened level of engagement in decision-making procedures that have a direct influence on the welfare of the community. This level of participation also includes engaging with the community's perceived needs through actively contributing to the strategic design and implementation of initiatives that aim to enhance health outcomes.

Earlier socialisation was envisaged as an idea for the utilisation of services but recently WHO and UNICEF deemed community participation would be of even greater help. They have defined it as "the process by which individuals and families assume responsibility for their health and welfare for those of the community and develop the capacity to contribute to their and the community's development." It also signifies the involvement of the community in the process of planning, organizing, and managing their healthcare services. This concept is commonly referred to as "Health by the People."

There are three ways in which a community would be able to participate:

1) The community can offer various resources such as facilities, human resources, logistical assistance, and maybe financial help.

2) Active community engagement is crucial in the processes of planning, management, and assessment.

3) Another significant contribution that individuals may make is by actively participating in and utilizing healthcare services.

Other options:

Options A, B and C are also methods by which a community will be able to participate but planning of health-related campaigns is considered the highest form of community participation.

Solution to Question 19:

The given clinical scenario with a patient hailing from a mosquito-infested area with fever and decreased platelet counts is diagnostic of dengue.

Symptoms of severe dengue are headache, myalgias, retroorbital pain, end-organ damage, pleural & ascitic effusions and thrombocytopenia (<100,000/ uL). Complications include mild hepatic injury, seizures, arrythmias and acute kidney injury.

Diagnosis is made by NS1 antigen detection and IgM antibody assays. Treatment includes crystalloid infusion in mild cases and colloid infusion in severe cases, oxygen administration and close monitoring.

Dengue is classified into dengue with or without warning signs and severe dengue. Dengue is to be suspected when a high fever (40°C/104°F) is accompanied by any two of the following symptoms during the febrile phase (2-7 days) which include severe headache, retro-orbital pain, arthralgia or myalgia, nausea, vomiting and maculopapular rash.

Warning signs in dengue are:

• severe abdominal pain

• persistent vomiting

• tachypnea

• bleeding gums or nose

• fatigue

• restlessness

• hepatomegaly

• malena or hematemesis

Solution to Question 20:

True statements are:

• Contemplation phase - A smoker is considering giving up his habit of smoking. (Recognizes his problem and thinks about to bring about a change)

• Maintenance phase - A person who quit smoking one year back and has abstained from smoking ever since. (He has sustained his abstinence from smoking)

Motivation enhancement therapy (MET) is a systematic intervention approach for bringing about change in the addictive behaviour seen in people with substance use disorders. It is based on the principle of motivational psychology and aims to enable the patient to deal with his addiction by producing internally motivated change.

Addictive behaviors do not change abruptly, and go through gradual stages. The stages of motivation enhancement therapy (MET) include:

Precontemplation - individuals in this stage are often unaware that their behaviour (eg. alcohol abuse) is problematic or is producing negative consequences.

Contemplation - individuals in this stage recognize their problems and begin to think about changing their behaviour. They weigh the pros and cons of change. They may still feel ambivalent about changing their behaviour.

Preparation or determination - individuals in this stage make decisions to change their habits and are ready to take action. They start to take small steps toward behaviour change.

Action - individuals in this stage change their behaviour and intend to keep moving forward with this change in behaviour

Maintenance - individuals in this stage sustain their change in behaviour for a while and intend to maintain this change going forward. People work to prevent relapse to earlier stages.

Other statements:

Statement 1: This statement is false as people in the pre-contemplation stage are unaware that their behaviour is problematic.

Statement 4: This statement is also false as people in the action stage make strides to change their behaviour.

Solution to Question 21:

A kappa statistic of 0.71 indicates a good agreement between the two doctors who collected the data.

The strength of agreement varies with the kappa coefficient as follows:

0.81 to 1: perfect agreement between the two tests/methods

0.61 to 0.80: substantial agreement/ good agreement

0.41 to 0.60: moderate agreement

0.21 to 0.40: fair agreement

0.10 to 0.20: slight agreement

Most of the studies test the association between an exposure variable and the disease. In certain cases, it is anticipated that there will be a correlation between the variables, and the challenge is to determine the extent of this correlation or association. This assertion holds special significance in the realm of reliability studies, wherein the researcher seeks to quantitatively assess the extent to which a given variable (such as a particular finding on a chest X-ray) can be reproduced through several measurements which is the utility of the kappa statistic. It is also important to note that a chi-square test can also be used but would only qualitatively assess the data and not quantitatively which of course is accomplished by the use of kappa statistic.

Kappa statistic can be calculated as:

$$\kappa = \frac{p_O - p_E}{1 - p_E}$$

where pO is observed agreement and pE is expected agreement.

Solution to Question 22:

Cohort study is an example of a longitudinal observational and analytical study.

A cohort study is a type of analytical study, which progresses from cause to effect, as exposure has occurred but not disease. It is also known as a forward-looking study, incidence study, prospective study, and longitudinal study. A well-designed cohort study is considered the most reliable means of showing an association between a cause or suspected risk factor and subsequent disease (or effect).

Types of epidemiological studies:

• Observational studies - allow nature to take its course, the investigator measures but does not intervene.

• Descriptive studies - limited to the description of the occurrence of a disease in a population- (hypothesis formulation).

• Analytical studies - go further by analyzing the relationship between health status and other variables- (hypothesis testing).

• Ecological studies

• Cross-sectional studies

• Case-control studies

• Cohort studies

• Experimental studies or intervention studies involve an active attempt to change a disease determinant or the progress of a disease and are similar in design to experiments in other sciences- (hypothesis confirmation).

• Randomized controlled trials

• Field trials

• Community trials

Other options:

Option A: Although case-control studies are inherently analytical, they are not longitudinal. Longitudinal studies involve the collection of variables about a subject over an extended duration.

Option C: A randomized control trial is an experimental study in which an intervention is given to a group of patients participating in the study and compared to another group of patients with similar characteristics who are not given that intervention to be studied. It uses patients as the unit of study.

Option D: An ecological study is a type of analytical study that analyzes the health status and its relation with other variables. It is also called a correlational study, and it uses the population as a unit of study. Longitudinal studies involve the collection of variables about a subject over an extended duration.

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Question 1:

Which of the following statements is incorrect about rotavirus vaccine?

a) Available as monovalent and pentavalent vaccines

b) It is a subunit vaccine

c) Not recommended after 6 months of age

d) Oral route of administration

Question 2:

Which of the following is not a polysaccharide vaccine?

a) Pneumococcal

b) Meningococcal

c) Hepatitis B

d) Haemophilus influenza type B

Question 3:

Which of the following vaccines is not recommended after 65 years?

a) Influenza

b) Herpes Zoster

c) Pneumococcal

d) Human papilloma virus

Question 4:

Arrange the steps of the experimental study :

a) C - B - A - D

b) C - A - B - D

c) A - B - C - D

d) D - C - B - A

Question 5:

Which of the following groups is not included in the high-risk core population for HIV transmission?

a) IV drug users

b) Female sex workers

c) Men who have sex with men

d) Long-distance truck drivers

Question 6:

A student went on a trip to Jaisalmer, where the temp is 40 degrees, and the relative humidity is 20%. What will the mechanism of heat loss be?

a) Evaporation

b) Radiation

c) Conduction with air

d) Conduction with objects

Question 7:

Malaria detection on the field is done by which of the following?

a) Thick smear

b) Thin smear

c) HRP2 antigen

d) Quantitative buffy coat

Question 8:

Which of the following is the best study design as per the hierarchy of evidence?

a) Meta analysis

b) Case control study

c) Cohort study

d) Large RCT

Question 9:

Dr. Sarah made a presentation on Infectious diseases. She spoke about the prevention and treatment of Zika virus. Which of the following is true with respect to the prevention and treatment of Zika virus?

a) Sexual abstinence, using condoms, and delaying pregnancy helps control the disease

b) Purified, inactivated FDA-approved vaccine is found to be helpful in prevention

c) Symptomatic treatment with NSAIDs

d) Immunoglobulin and plasmapheresis are used in the treatment of Zika

Question 10:

ICER is a unit for which of the following economic analysis?

a) Cost effectiveness

b) Cost benefit

c) Cost utility

d) Cost margin

Question 11:

You have gone from your clinic to a village for screening where people have diabetes and hypertension. What type of screening is done for family members of these patients who show up in large numbers at your clinic?

a) Multiphasic screening

b) Mass screening

c) High-risk screening

d) Anonymous screening

Question 12:

All are live vaccines except

a) Rota virus

b) Hepatitis B

c) Measles

d) Varicella

Question 13:

A pus swab has been sent for microbiological examination to your lab. After testing the sample, how will you discard it?

a) Red bin

b) Yellow bin

c) Blue bin

d) Black bin

Question 14:

Universal health coverage (UHC) means that all people and communities receive the quality health services they need, without financial hardship. Which of the following are the challenges to implementing UHC?

a) Inequalities in the distribution, adequate support, and access to quality-assured products only in low-income settings

b) The purpose of UHC is to find a balance between countries' income, health provision, and financial burden

c) To provide healthcare to everyone regardless of the country's income

d) UHC only deals with the financial aspect of health and cannot cover health provisions for all

Question 15:

Which is not a part of the Anemia Mukt Bharat Programme?

a) 6 beneficiaries

b) 6 interventions

c) 6 institutional mechanisms

d) 6 health functionaries

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	a
5	d
6	a
7	c
8	a
9	a
10	a
11	c
12	b
13	b
14	c
15	d

Detailed Explanations

Solution to Question 1:

Rotavirus vaccine is not a subunit vaccine.

Rotavirus vaccines are oral, live attenuated vaccines and are of 3 types:

• Rotarix (human monovalent) vaccine is administered in 2 oral doses:

• First dose: At 6 weeks, no later than 12 weeks (due to the risk of intussusception).

• Second dose: With a gap of 4 weeks from the first dose, to be given by 16 weeks and no later than 24 weeks.

• RotaTeq (pentavalent bovine-human reassortant) vaccine is administered in 3 oral doses at 2, 4, and 6 months. The first dose is given between 6-12 weeks and subsequent doses within a 4-10 weeks gap. All the 3 doses should be given before 32 weeks.

• Rotavac (monovalent liquid frozen) vaccine is administered in 3 oral doses, 4 weeks apart at 6,10 and 14 weeks. It should not be administered to children more than 8 months of age.

Option C: According to the MoHFW, the rotavirus virus vaccine can be given till the age of 1 year and the potential benefit from vaccination outweighs the risk of intussusception. However, per the CDC guidelines, the maximum age for all rotavirus vaccines is 8 months, 0 days. Since this is a recall question, the options appeared as such in the exam. Additionally, rotavirus is a live vaccine, making option 2 (which states that it is a subunit vaccine) the distinctly wrong statement and the better answer.

Solution to Question 2:

Hepatitis B vaccine is a recombinant protein vaccine.

Polysaccharide vaccines are a type of subunit vaccine containing the capsular polysaccharide antigen. They generate a T-cell independent response.

Examples include:

• Pneumococcus

• Meningococcus

• Haemophilus influenzae type b (Hib)

Recombinant protein vaccines: With the help of recombinant DNA technology, the purified recombinant proteins from the cultured microorganisms result in cleaner vaccine preparations with a better safety profile.

Examples include:

• HBV

• HPV

• Lyme disease

• Cholera toxin-B

• COVID-19

Solution to Question 3:

Human papillomavirus vaccine is not recommended after 65 years.

In otherwise healthy elderly patients, the recommended vaccines are:

• Inactivated or recombinant influenza vaccine, 1 dose annually

• Td or Tdap booster, every 10 years

• Zoster recombinant vaccine

• Pneumococcal vaccine

• COVID-19 vaccine

The zoster recombinant vaccine is not indicated for the prevention of varicella. This vaccine is given to prevent episodes of herpes zoster and post-herpetic neuralgia. It is indicated regardless of a previous episode of Zoster.

All adults ≥65 years of age should also receive pneumococcal vaccination. This is in the form of pneumococcal conjugate vaccine (PCV) first, followed by pneumococcal polysaccharide vaccine

(PPSV23) after an interval.

Solution to Question 4:

C - B - A - D is the correct order of steps in an experimental study.

Defining inclusion and exclusion criteria → Randomization → Blinding → Statistical analysis

Phases of experimental study:

1. Planning phase

• Literature review

• Trial design

• Protocol

• Budget, funding

• Approvals - ethics, collaborating institutes, funding agencies, sponsors

2. Preparatory phase

• Organization, staffing

• Delegation, responsibility, and accountability

• Standardization of protocols, training, data management, privacy

• Pilot testing, quality control

3. Operational phase

• Selection of target population

• Choosing a sample study population

• Eligibility screen: Exclusion and inclusion criteria

• Informed consent

• Subject registration, baseline evaluations

• Randomization

• Blinding

• Conduct a study on the groups allocated

• Follow-ups, evaluation protocol

• Exit plan, termination point

• Trial monitoring

4. Analysis phase

• Preliminary clarifications

• Initial Data Analysis, definitive analysis

• Statistical inference

5. Conclusion

• Validity, interpretation

• Writing reports, manuscripts, dissemination

Randomization is the random allocation of the participants into the study and control groups in the experimental studies.

Blinding is done after randomization to reduce the possibility of bias, such as subject variation bias (due to participants), observer bias, and evaluation bias (by the investigator).

Solution to Question 5:

Long-distance truck drivers are not included in the high-risk core population for HIV transmission. They are part of the bridge population.

Target interventions for the control of AIDS transmission are done for high-risk core populations like:

• sex workers

• Men who have sex with men

• Transgender

• Injecting drug users

Bridge populations like migrants and long-distance truckers also receive targeted interventions for the control of AIDS transmission.

The services offered through targeted interventions include detection and treatment for sexually transmitted diseases and condom distribution.

Solution to Question 6:

Evaporation is the mechanism of heat loss in this student.

Evaporation happens when perspiration evaporates from the skin. This is a significant source of heat loss during activity but not at rest in normal temperature settings.

When the temperature of the surroundings is more than that of the skin, the body gains heat by both radiation and conduction. Under these conditions, the heat can be lost from the body only by evaporation or sweating.

Different methods of heat loss:

• Radiation is heat loss in the form of infrared rays. (Option B)

• Conduction is heat loss that occurs by direct contact with an object. (Option C and D)

• Convection is heat loss resulting from air movement

Solution to Question 7:

In field settings, rapid diagnostic tests, such as those detecting Histidine-Rich Protein 2 (HRP2) antigen, are commonly used for malaria detection.

Rapid diagnostic tests are simple, quick, and require minimal resources, making them ideal for field use where laboratory facilities may be unavailable. HRP-2 antigen detection used in rapid diagnostic tests for malaria is specific for P. falciparum.

Rapid diagnostic tests are based on the detection of antigens using immunochromatographic methods. They are less sensitive than microscopy and cannot differentiate between species except P. falciparum. The 2 antigens detected are:

• pLDH (parasite lactate dehydrogenase)- pan malarial antigen

• HRP-2 (histidine-rich protein 2)- specific for P.falciparum.

Other options

Option A: Thick smears are more sensitive for detecting malaria parasites, they require laboratory facilities and skilled personnel, making them less practical for field settings.

Option B: Thin smears are used primarily for species identification and quantification of parasites, but like thick smears, they are unsuitable for field settings due to resource requirements.

Option D: Quantitative buffy coat (QBC): This method is highly sensitive but requires special equipment (a fluorescent microscope and centrifuge), making it unsuitable for field use.

Solution to Question 8:

Meta-analysis is the best study design as per the hierarchy of evidence.

The correct order of ranking of the hierarchy of evidence is:

Meta-analysis and systemic review > Randomised control trial > Cohort study > Case-control study > Case reports.

Solution to Question 9:

Sexual abstinence, using condoms, and delaying pregnancy help control Zika virus infections.

Zika virus can be transmitted by the bite of infected Aedes aegypti and Aedes albopictus. Zika virus belongs to the Flaviviridae family. It is an arbovirus that is transmitted by Aedes mosquitoes, primarily Aedes aegypti. The reservoir of infection is unknown and has not yet been reported in India. Zika virus can also be transmitted via sexual contact, blood transfusion, and also from mother to fetus during pregnancy.

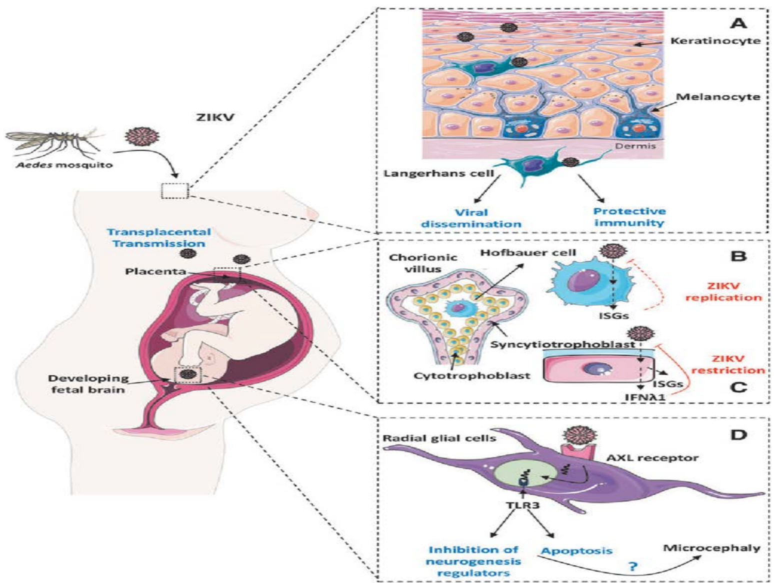
Infection with Zika virus is usually asymptomatic. The symptoms are usually mild and include fever, skin rashes, conjunctivitis, and muscle and joint pain. These symptoms usually resolve in 2-7 days. Complications include Guillain-Barre■ syndrome, neuropathy, and myelitis, and are particularly seen in adults and older children.

Zika virus infection during pregnancy is associated with complications such as fetal loss, preterm birth, and stillbirth. The transmission of the infection to the baby can lead to congenital Zika syndrome which is characterized by microcephaly. Other malformations include limb contractures, high muscle tone, eye abnormalities, and hearing loss. Congenital malformations can follow both symptomatic and asymptomatic infections.

RT-PCR is the standard test for the diagnosis of Zika virus infection. In India, NCDC Delhi and the National Institute of Virology Pune, are the only institutes that can provide the diagnosis in an acute febrile stage.

There is no treatment available for Zika virus infection(options C & D). Plenty of rest and fluids along with antipyretics and/or analgesics are indicated for symptom relief. Medical attention should be sought if symptoms worsen. There is no vaccine available for the prevention of Zika virus infection(option B). The key measure for prevention is protection against mosquito bites. WHO also recommends contraceptive counselling for sexually active men and women, especially in regions with active Zika virus transmission.

The image below depicts the transplacental transmission of Zika virus infection.



Solution to Question 10:

ICER is a unit for Cost-effectiveness.

Incremental cost-effectiveness analysis (ICER) is a difference in cost/difference in effectiveness. Effectiveness is usually measured in Quality Adjusted Life Years (QALYs), which reflect both the quality and quantity of life.

The Cost-effectiveness analysis measures the costs and outcomes of different interventions and calculates the cost-effectiveness ratio. The benefits of the health care system are expressed in terms of results achieved (Eg, the number of lives saved or the number of days free from disease). This is a more promising tool in the health field than cost-benefit analysis. However, in many cases, it is not possible to use cost-effective analysis.

Other options:

Option B: Cost-benefit analysis: Here, the economic benefits of any program are compared with the cost of the program. The benefits are expressed in monetary terms (whereas in cost-effectiveness analysis, it is expressed in terms of results achieved)to identify whether it is economically sound and to select the best one from several alternate programs. The main drawback of this technique is that the benefits of a particular program in the health field cannot always be expressed in monetary terms. Usually, the benefits are expressed in terms of births or deaths prevented or illnesses avoided or overcome.

Option C: Cost-utility: compares the cost of a program to the health improvement it provides.

Option D: Cost margin: the difference between a product's revenues and how much it costs to make.

Solution to Question 11:

The given scenario of screening the family members of diabetic and hypertensive patients is an example of high-risk screening.

High-risk or selective screening is done on high-risk groups like in this case. Another example: MRI screening of the breast is done annually in 25-year-old women with BRCA 1, and BRCA 2 mutations in the first-degree relative.

Other options

Option A: Multiphasic screening is the application of two or more screening tests in combination at one time to a large number of people instead of screening separately for individual diseases. This procedure includes a health questionnaire, clinical examination, and investigations that can be performed rapidly with the appropriate staffing organization and equipment.

Option B: Mass screening is a screening of a whole or large population like using chest X-rays for TB screening in large populations. It is done for diseases with high prevalence.

Option D: Anonymous screening is a screening that allows people to take tests without using their name, commonly used for HIV testing. They will receive a number to access the test results instead of providing their name.

Solution to Question 12:

Hepatitis B is a recombinant vaccine, not a live vaccine.

Live attenuated vaccines are prepared from organisms that have lost their capacity to induce full-blown disease but retain their immunogenicity and are contraindicated in pregnancy unless the risk of infection exceeds the risk of harm to the fetus. Varicella, Mumps, BCG, Measles, and Rubella vaccines are examples of live attenuated vaccines.

Recombinant protein vaccines are prepared by recombinant DNA technology by expressing protective protein antigens in yeast cells like the hepatitis B vaccine. They are generally better purified from cultured microorganisms resulting in cleaner vaccine preparations with a better safety profile.

Inactivated or killed vaccines are prepared by growing the organism in culture media and inactivating them with heat and chemicals (usually formalin). They are usually safe in pregnancy. Examples include cholera, rabies, and polio (IPV) vaccines.

Toxoid vaccines are a type of subunit vaccine prepared by detoxifying the toxins produced by organisms like tetanus bacilli. In general, they are highly efficacious and safe immunising agents.

Solution to Question 13:

The used pus swab should be discarded in the yellow colour bin as it is clinical laboratory waste.

Microbiology, biotechnology, and clinical laboratory waste are collected in autoclave-safe bags. They must be pre-treated before disposal by incineration, plasma pyrolysis, or deep burial.

Solution to Question 14:

Providing healthcare to everyone, regardless of the country's income, is one of the challenges to implementing Universal Health Coverage. A challenge related to this is mobilising sufficient resources in low-income countries to provide equitable care.

UHC aims at ensuring that all people have access to quality health services without suffering financial hardship.

Components:

- Equity in access: Everyone receives the healthcare they need.
- Quality of services: Delivered in a manner that improves health outcomes.
- Financial protection: Avoiding financial hardship for those accessing services.

Challenges:

- Health inequities.
- Resource constraints in low- and middle-income countries.
- Balancing financial protection and service delivery.

Strategies to Achieve UHC:

- Strengthening health systems.
- Universal health insurance.
- Public-private partnerships.

Other options

Option A: Inequalities in the distribution, adequate support, and access to quality-assured products are a challenge, but they occur across income settings, not only in low-income ones.

Option B: The purpose of UHC to find a balance between countries' income, health provision, and financial burden describes a goal of UHC and not a challenge.

Option D: UHC encompasses both financial protection and access to quality health services.

Solution to Question 15:

Among the listed options, 6 health functionaries are not a part of the 6x6x6 strategy employed by the Anemia Mukt Bharat Program.

The Anemia Mukt Bharat scheme is an initiative that aims to strengthen the existing mechanisms and foster newer strategies for tackling anaemia. The 6 x 6 x 6 strategy represents:

• 6 beneficiaries (option A)

• 6 interventions (option B)

• 6 institutional mechanisms (option C)

The 6 beneficiaries include:

• Preschool children (6-59 months of age)

• Children (5-9 years of age)

• Adolescents (10-19 years of age)

• Pregnant women

• Lactating mothers

• Young women in the reproductive age group (20-24 years of age)

The 6 interventions include:

• Prophylactic iron and folic acid supplementation

• Deworming

• Intensified year-round "behaviour change communication campaign"

• Testing of anaemia using digital methods and point-of-care treatment

• Mandatory provision of iron and folic acid-fortified foods in government-funded health programs

• Addressing non-nutritional causes of anaemia in endemic pockets, with a special focus on malaria, hemoglobinopathies, and fluorosis

The 6 institutional mechanisms include:

• Intra-ministerial coordination

• National Anemia Mukt Bharat unit

• National Center of Excellence and Advanced Research on Anemia Control

• Convergence with other ministries

• Strengthening supply chain and logistics

- Digital Anemia Mukt Bharat dashboard and online portal

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Question 1:

Identify the image representing the National Leprosy eradication program:

a) 1

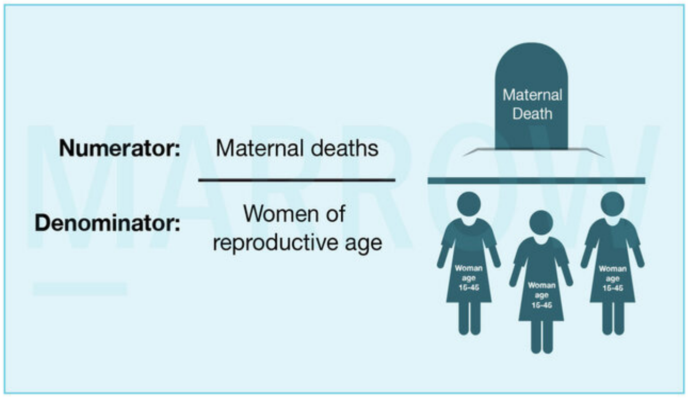
b) 2

c) 3

d) 4

Question 2:

What is calculated with the following information?



©MARROW

a) Maternal Mortality Rate

b) Maternal Mortality Ratio

c) Death rate in reproductive women

d) Total fertility rate

Question 3:

Every pregnant woman receives a free antenatal check-up on the 9th of each month, along with free medicines and delivery services, including cesarean sections, under the JSSK program. What does JSSK stand for?

a) Janani Shishu Swasthya Karyakram

b) Janani Shishu Suraksha Kendra

c) Janani Shishu Suraksha Karyakram

d) Janani Shishu Swasthya kendra

Question 4:

What is the communication barrier if interpretation of a words are understood in a different manner?

a) Linguistic barrier

b) Semantic barrier

c) Psychological barrier

d) Cultural barrier

Question 5:

30 young volunteers with previous lifestyle modifications were contributing to drug trial for cholesterol levels based on statin therapy for 9 months. Which of the following tests is performed?

a) Chi-square test

b) Paired t-test

c) McNemar's test

d) Unpaired T-test

Question 6:

A case-control study is conducted on the association between intake of Vitamin C and the occurrence of respiratory illness. What is the order in which the following steps are conducted?

a) 3-2-1-4

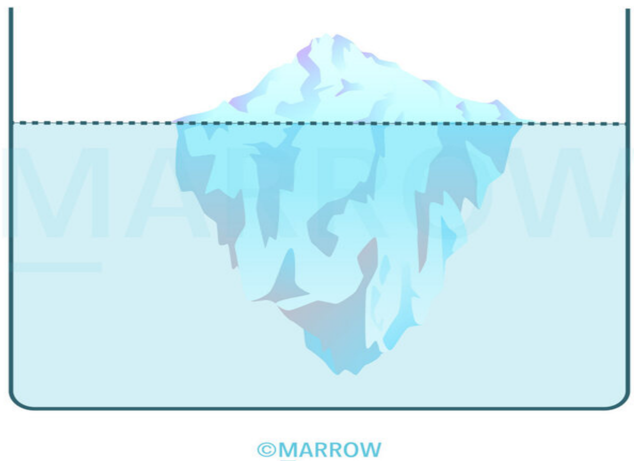
b) 1-2-3-4

c) 4-3-2-1

d) 3-1-2-4

Question 7:

Which among the following diseases will be the best representation of the submerged portion?



a) Tetanus

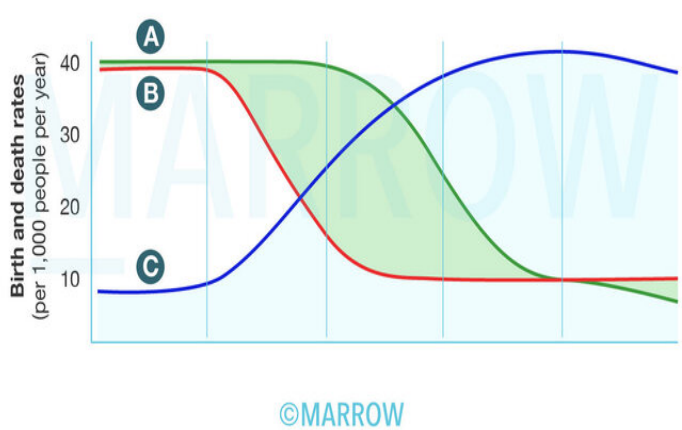
b) Influenza

c) Chicken Pox

d) Rabies

Question 8:

What does line B represent in the graph given?



a) Death rate

b) Birth rate

c) Total population

d) Population growth rate

Question 9:

In a sudden onset disaster, the 'recovery' phase consists of which of the following?

a) i, ii, iii and iv

b) ii, iii and iv only

c) ii and iii only

d) i, ii and iv only

Question 10:

A pharmacist randomly selects from three groups of medicines: plant-based, animal-derived, and synthetic. What type of sampling is this?

a) Simple random sampling

b) Stratified sampling

c) Systemic sampling

d) Cluster sampling

Question 11:

The children in a school underwent a health check-up whose ages were 2,3,4,4,5,6,7,8. What is their median age?

a) 4

b) 4.5

c) 4.8

d) 5

Question 12:

Identify the use of this image

a) Cooling power of air

b) Air temperature

c) Air humidity

d) Radiant heat

Question 13:

A student was making a presentation with charts but lost titles while editing. What could be the heading for the following chart?

a) Physical Quality of Life Index

b) Multidimensional Poverty Index

c) Disability-adjusted life years

d) Human Development Index

Question 14:

A mother delivers a baby in a rural area with the assistance of a skilled birth attendant. The attendant cuts the umbilical cord cleanly, ties the ends with new threads, bathes the baby with warm water, wipes the baby's eyes with a sterile wet cloth from the medial to lateral corner, and then hands the baby to the mother for skin-to-skin contact. Which of these steps is incorrect?

a) Using new threads to tie the ends of the umbilical cord

b) Wiping the baby's eyes from medial to lateral corner

c) Skin-to-skin contact with the mother

d) Bathing the child with warm water

Question 15:

A homeless man comes with diarrhea, dementia, and dermatitis. He consumes alcohol occasionally. He admits to having 'X' as his staple diet. Which of the following is X most likely to be?

a) Maize

b) Wheat

c) Milk

d) Peanuts

Question 16:

In the Motivation model of health education, there are three stages for the adoption of new ideas. What is the third one other than awareness & motivation?

a) Contemplation

b) Reflection

c) Action

d) Dedication

Question 17:

An unvaccinated 5-year-old child presents with a sore throat and difficulty in swallowing. On examination, she has a greyish membrane over her tonsils. She has a fully vaccinated 2-year-old brother who was asymptomatic. What is the appropriate management of this contact?

a) No intervention

b) Booster dose of toxoid

c) Single dose of immunoglobulin

d) Prophylactic dose of penicillin

Question 18:

A disease has an incubation period of 3-14 days. It has a median incubation of 10 days. A person had to travel abroad and he was not vaccinated. As the disease is of international significance, how long will he have to be quarantined?

- **a)** 20 days
- **b)** 14 days
- **c)** 28 days
- **d)** 10 days

Question 19:

The image shown corresponds to which health program and disease?

- **a)** AROHI for HIV
- **b)** AMITHA for polio
- **c)** SAPANA for leprosy
- **d)** GAURI for cataract

Question 20:

A study is investigating the relationship between diet and the incidence of a particular disease. Among 100 vegetarians, 30 individuals are found to have the disease, while among 100 non-vegetarians, 50 individuals are affected. Which biostatistical method is most appropriate to determine if there is no significant association between diet and disease incidence?

a) Mann-Whitney test

b) Correlation coefficient

c) T-squared test

d) Chi-square test

Question 21:

Which of the following options best defines the correct sequence of the phases of the demographic cycle?

a) High expanding $\rightarrow$ Early stationary $\rightarrow$ Late expanding $\rightarrow$ Low stationary $\rightarrow$ Declining

b) High stationary $\rightarrow$ Early expanding $\rightarrow$ Late expanding $\rightarrow$ Low stationary $\rightarrow$ Declining

c) Low stationary $\rightarrow$ Early expanding $\rightarrow$ Declining

d) Late expanding $\rightarrow$ High stationary $\rightarrow$ Declining

Question 22:

A man presents with sudden onset bilateral leg swelling, diarrhea, and difficulty breathing. He reports consuming mustard oil purchased from a local vendor prior to the onset of symptoms. A health inspector is investigating the cause of his condition. Which test is the most appropriate to detect the adulterant in the mustard oil?

a) Paper Chromatography Test

b) Methylene Blue Test

c) Phosphatase Test

d) Nitric Acid Test

Question 23:

In a population of 30,000, there were 500 live births & 14 still births in the year 2023. There were 18 children that died before the age of 1 year, 12 before 28 days of life & among that, 9 died in the first 7 days of life. Calculate the early neonatal mortality rate?

a) 24

b) 18

c) 32

d) 36

Question 24:

A 37-year-old woman from Bihar presents with fever and unexplained weight loss over the past 6 months. Examination reveals massive splenomegaly and blood investigations show anemia, thrombocytopenia, and hypergammaglobulinemia. What is the vector of the organism causing this condition?

- **a)** Phlebotomus species
- **b)** Aedes mosquito
- **c)** Mite
- **d)** Female anopheles mosquito

Question 25:

A hospital collected nasopharyngeal swabs from 1000 patients every day from the onset of the disease. The organism count was found to be maximum on the 5th day, which also showed the maximum infectivity (infecting 825 patients). What is this period of 5 days from the onset of the disease known as?

- **a)** Generation time
- **b)** Median Incubation period
- **c)** Serial interval
- **d)** Lead time

Question 26:

Which of the following are not included in the School Health Programme under Ayushman Bharat?

- **a)** Monthly IFA tablets for all children
- **b)** Health screening
- **c)** Deworming for school going children
- **d)** Sanitary napkins for adolescent girls

Question 27:

Which of the following is true about longevity dividend?

- **a)** The intervention can delay the onset of the disease

b) Small amounts of damage is accumulated over time

c) The older you are the more prone you are to disease

d) Economic and health benefit for the society

Question 28:

For standards of ventilation, the widely quoted standard is that of De Chaumont who advocated a fresh air supply of 3000 cubic feet per person per hour. It is calculated based on which parameter?

a) Humidity of air

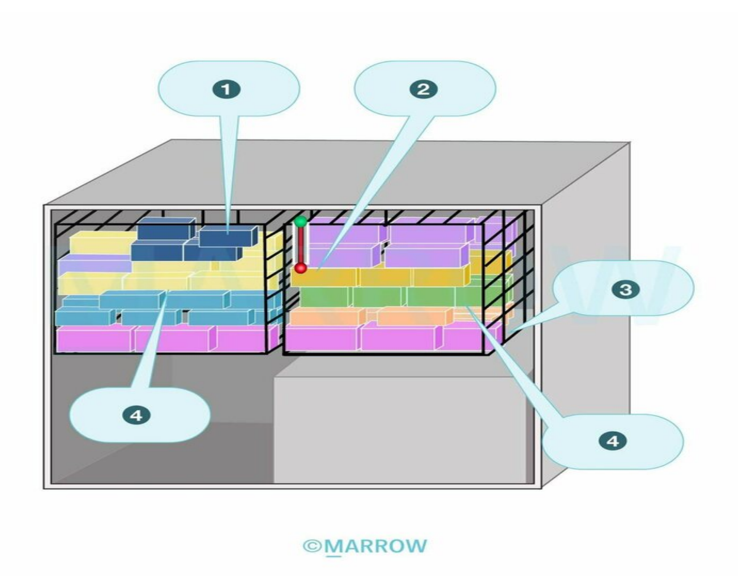
b) O2 content of air

c) Temperature of air

d) CO2 content of air

Question 29:

Which vaccine will be placed at zone 3 in the device shown below?



a) Pentavalent

b) Hep B

c) OPV

d) DPT

Question 30:

Pick the correct statement regarding the Sanchol cholera vaccine

- **a)** Bivalent vaccine given in 2 liquid doses orally to children >=1 year of age
- **b)** Bivalent given in single dose orally to children >1 years of age
- **c)** Monovalent given in 2 liquid doses orally with a bicarbonate buffer
- **d)** Monovalent given in a single dose orally with a bicarbonate buffer

Question 31:

In a town, there were 1000 live births in a period of 6 months. During the same period, 12 women died due to the following causes. What is the maternal mortality ratio?

- **a)** 800 per 100000 live births
- **b)** 20 per 1000 live births
- **c)** 125 per 1000 live births
- **d)** 400 per 100000 live births

Question 32:

Which of the following steps of disaster management is in correct order?

- **a)** Disaster impact – Mitigation – Rehabilitation – Response
- **b)** Rehabilitation – Response – Disaster Impact – Mitigation
- **c)** Disaster impact – Response – Rehabilitation – Mitigation
- **d)** Response – Disaster impact – Rehabilitation – Mitigation

Question 33:

Which of the following are components of the physical quality of life index (PQLI)?

- **a)** 1,3 and 5
- **b)** 2 and 5
- **c)** 1,2 and 4
- **d)** 1,3 and 4

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	b
5	b
6	a
7	b
8	a
9	c
10	b
11	b
12	a
13	d
14	d
15	a
16	c
17	d
18	b
19	c
20	d
21	b
22	a
23	b
24	a
25	a
26	a
27	a
28	d
29	c
30	a
31	a
32	c
33	d

Detailed Explanations

Solution to Question 1:

The given image 2 represents the National Leprosy Eradication Programme (NLEP).

Characteristics of the NLEP emblem:

• The lotus in the NLEP emblem symbolizes beauty and purity.

• A partially affected thumb, a normal forefinger, and the shape of a house symbolize that leprosy can be cured and that a patient with leprosy can be a useful member of society.

• The rising sun symbolizes hope and optimism.

• The emblem captures the spirit of hope and positivity in the eradication of leprosy.

NLEP is a centrally sponsored health scheme of the Ministry of Health and Family Welfare, Government of India.

Other options:

Option A: Poshan Abhiyaan: It aims to enhance key nutritional indicators for both children and women.

Option C: National Health Mission: It is a public health initiative focused on ensuring universal access to affordable, equitable, and high-quality healthcare services.

Option D: Ayushman Bharat: It provides nationwide medical coverage and cashless hospitalization benefits for enrolled families in both public and private hospitals.

The given image shows the correct logos for these programs

Solution to Question 2:

The given expression is used for calculating the maternal mortality rate. The maternal mortality rate is expressed as the "number of maternal deaths in a given period per 100,000 women in reproductive age during the same time period".

Other options:

Option B: The maternal mortality ratio is the "number of maternal deaths during a given period per 100,000 live births during the same time".

Option C: The death rate in reproductive women is expressed as the "number of deaths among women aged 15-49 years during a given period per 1,000 women of the same age group during the same time period.

Option D: The Total Fertility Rate (TFR) is expressed as the "average number of children a woman would have over her lifetime if she were to experience the current age-specific fertility rates throughout her reproductive years (typically ages 15-49).

Solution to Question 3:

JSSK stands for Janani Shishu Suraksha Karyakram. It is an initiative by the Indian government aimed at reducing maternal and infant mortality.

The provisions of JSSK include:

• Free transport

• Free treatment

• Free drugs and consumables

• Free diagnostics and diet

• Free and cashless delivery

• Free C-Section

• Free provision of blood

Note: Previously under JSSK, the free entitlements were provided only for sick newborns till 30 days after birth. Now it is expanded to cover infants also.

Solution to Question 4:

The communication barrier that occurs when the interpretation of words is understood differently is called a semantic Barrier.

Semantic barriers arise from misunderstandings due to differences in the meaning of words, phrases, or symbols. These barriers can cause confusion and misinterpretation, leading to ineffective communication.

example: In a public health campaign, "herd immunity" may be misunderstood by some as meaning they don't need vaccination or as related to animals. This confusion can reduce vaccination rates and is a perfect example of a semantic barrier.

other options:

Option A: Linguistic Barrier: This barrier occurs when the language or dialect used in communication is not understood by one or more parties. It can include differences in language, vocabulary, or grammar, making it difficult to convey the intended message.

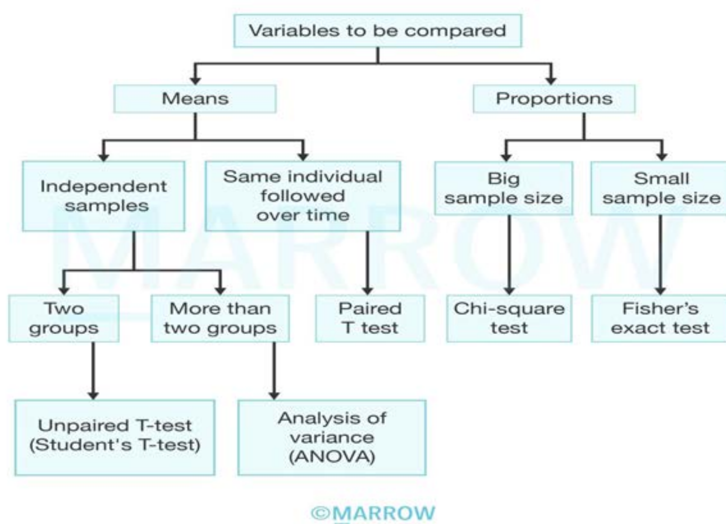
Option C: Psychological Barrier: Psychological barriers involve emotions, attitudes, or mental states that affect communication. These could include stress, fear, anger, or low self-esteem, which can hinder the ability to send, receive, or interpret messages accurately.

Option D: Cultural Barrier: Cultural barriers arise when differences in cultural backgrounds, beliefs, values, or practices lead to misunderstandings. These barriers can affect how messages are interpreted and can lead to miscommunication if cultural contexts are not considered.

Solution to Question 5:

This study involves a quantitative variable, i.e., blood cholesterol, and compares the before and after results of the same group. Hence, a paired T-test is used.

Tests of significance are used to check for significant association between two or more variables. Examples are the Student t-test, Paired t-test, Chi-square test, ANOVA, and Fischer-exact test.



McNemar's test is a non-parametric test for analysing paired nominal data. This test examines the marginal homogeneity of two dichotomous variables using a 2 x 2 contingency table. One independent variable with two dependent groups and one nominal variable with two categories (dichotomous) are needed for the test.

Solution to Question 6:

The correct order of steps in this case-control study is: Selecting case and control groups (3) -> Matching (2) -> Measurement of exposure (1) -> Analysis (4)

Case-Control Studies are also called retrospective studies. This study proceeds from effect to cause and is done after the exposure and outcome have occurred. They focus on diseases or health problems that have already developed. They are less time-consuming and have no risk of attrition because they do not require follow-up.

There are four basic steps in conducting a case-control study:

• Selection of cases and controls

• Matching- This is done to ensure comparability between cases and controls. Controls are selected such that they are similar to cases with respect to certain pertinent selected variables that influence the disease outcome.

• Measurement of exposure - This is obtained by interviews, questionnaires, or study of the past records of cases.

• Analysis in the form of exposure rates, risk ratio, and odds ratio.

Solution to Question 7:

The best representation of the submerged portion of the iceberg phenomenon would be influenza, as it often has a large number of asymptomatic or mild cases that go undetected, especially in milder seasons.

Tetanus (option A), chickenpox(option C), rabies (option D) and measles are typically acute and severe, with most cases being symptomatic and well-recognized when they occur. They do not show the iceberg phenomenon.

Iceberg phenomenon is shown by:

• Polio

• Influenza

• Rubella

• Mumps

• Japanese Encephalitis

• Diphtheria

• Hepatitis A and B

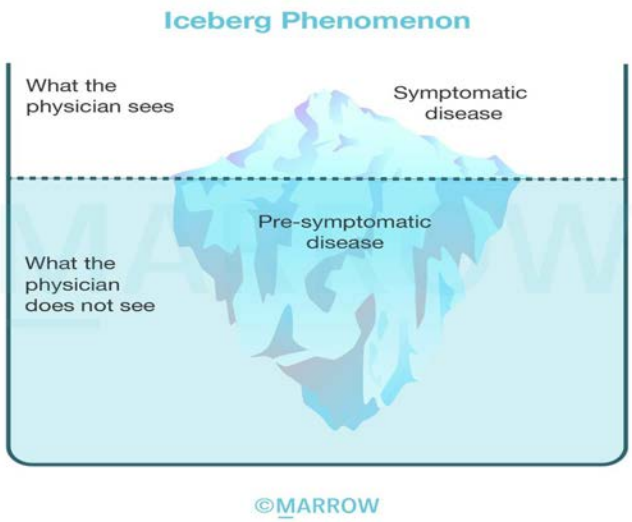
• Tuberculosis

• Diabetes

• Malnutrition

• Hypertension

Note: If a disease has more number of subclinical infections, it shows the iceberg phenomenon of disease depicted in the image below.



Solution to Question 8:

The line B here represents the death rate.

Line A represents the Birth rate

Line C represents the Total population

Solution to Question 9:

In disaster management, the recovery phase consists of response and rehabilitation.

Disaster preparedness refers to long-term development activities that strengthen the capacity of a country to manage all kinds of emergencies effectively.

• Response - This includes activities such as:

• Triage: Classifying the injured based on the severity of the injuries and chances of survival with medical intervention.

• Tagging: Tags with name, age, triage category, diagnosis and treatment are put for identification purposes.

• Identification of the dead: The dead bodies should be shifted from the disaster site to the mortuary.

• Rehabilitation - This includes measures that lead to the restoration of pre-disaster conditions. These measures include water supply restoration, food safety, basic sanitation, personal hygiene, and vector control.

• Mitigation - This includes measures to prevent disasters from causing emergencies or to lessen the likely effects of emergencies. Some of them are flood mitigation, land use planning, improved building codes, and the reduction and protection of vulnerable populations or structures.

• Preparedness - This is a program that aims at strengthening the capacity of a country to manage all types of emergencies efficiently. Its objective is to ensure that appropriate measurements are taken for the relief and rehabilitation of disaster victims.

Some measures that should be followed by all in case of an emergency include:

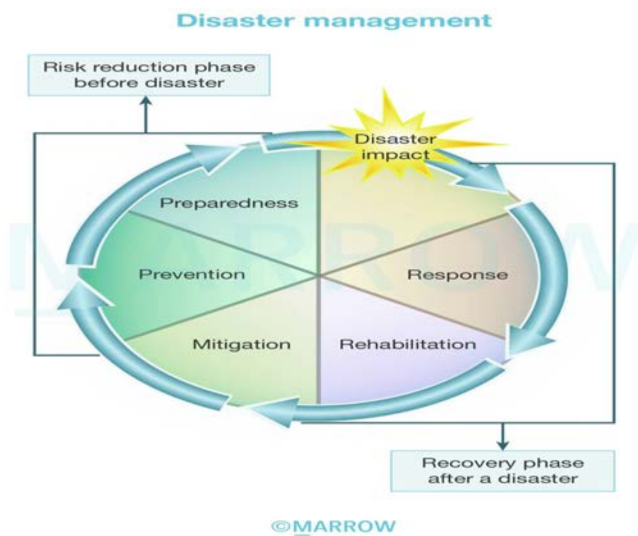
• Avoiding telephone use, unless it is to call for help

• Listen to and carry out the radio or loudspeaker broadcast messages

• Carry out the official instructions given over the radio or by loudspeaker

• Keep a family emergency kit ready

Note: Vaccination of the general population post-disaster against typhoid, cholera, and tetanus is not recommended.



Solution to Question 10:

The type of sampling described is stratified random sampling.

In stratified random sampling, the population is divided into distinct subgroups or "strata" (in this case, plant-based, animal-derived, and synthetic medicines). A random sample is then taken from each subgroup.

Other options:

Option A: In simple random sampling, every member of the population has an equal chance of being selected,

Option C: In systemic sampling, elements from a larger population are selected at regular intervals, known as sampling intervals.

Option D: In cluster sampling, the population is divided into distinct groups, or "clusters", and then a random sample of these clusters is selected

Solution to Question 11:

The median age here is 4.5 years.

In a distribution that has an odd number of elements, the median is the middle value. In a distribution that has an even number of elements, the median is the average of the middle values, i.e., the arithmetic mean of the two middle values.

Here: elements: 2,3,4,4,5,6,7,8, therefore median = (4+5)/2= 4.5

Solution to Question 12:

The image shows a Kata thermometer, it is a specialized device used to measure the cooling power of air.

Solution to Question 13:

The index that is calculated using the given indicators such as life expectancy at birth, mean and expected years of schooling, and the per capita income is the Human Developmental Index (HDI).

Human Development Index (HDI) is defined as a composite index focusing on three basic dimensions of human development. It assesses abilities

• To lead a healthy long life - measured by life expectancy at birth

• To acquire knowledge- measured by mean years of schooling

• To achieve a decent standard of living- measured by gross national income per capita

Hence the 3 dimension indices used to calculate the human development index are as follows life expectancy index, education index, and gross national index.

Other options:

Option A- Physical Quality of Life Index (PQLI) uses three indicators which include infant mortality, life expectancy at age one, and literacy rate.

Option B- Multidimensional Poverty Index (MPI) - It is an international measure of acute multidimensional poverty by capturing the acute deprivations in health, education, and living standards that a person faces simultaneously. It is assessed using 3 dimensions and 10 indicators which are as follows:

• Health - Nutrition and child mortality

• Education- Years of schooling and school attendance

• Standard of living -

• Cooking fuel

• Sanitation

• Drinking water

• Electricity

• Housing

• Assets

Option C- Disability-adjusted life years (DALY): It is the amount of time lost due to ill health. It is expressed as the number of years lost due to ill-health, disability, or early death. The components of DALY are:

• Years of life lost due to premature death

• Years lived with disability (YLD)

• Years lived in a state less than full health

Solution to Question 14:

The incorrect step is bathing the child with warm water immediately after birth. It is generally recommended to delay bathing in the immediate newborn period to help maintain the baby's body temperature and prevent hypothermia.

Other options:

Option A: Using new threads to tie the ends of the umbilical cord is correct and helps prevent infection and ensure proper cord care.

Option B: Wiping the baby's eyes from the medial to lateral corner is appropriate to prevent the spread of any potential infection from the eye's inner to outer corners.

Option C: Skin-to-skin contact with the mother is beneficial as it helps stabilize the baby's temperature, promotes bonding, and supports early breastfeeding.

Solution to Question 15:

This patient with symptoms of dementia, diarrhea and dermatitis most likely has niacin/vitamin B3 deficiency. X is likely to be maize in his diet.

Niacin deficiency causes pellagra, found mostly among people eating corn/maize-based diets, which is low in niacin and tryptophan. Excess leucine in jowar and maize disrupts the conversion of tryptophan to niacin.

Niacin can be synthesized in the body from tryptophan. Two compounds- nicotinic acid, and nicotinamide have the biological activity of niacin. The nicotinamide ring of the coenzymes NAD (nicotinamide adenine dinucleotide) and NADP (nicotinamide adenine dinucleotide phosphate) participate in oxidation/reduction reactions.

The initial symptoms of pellagra include loss of appetite, generalized weakness, irritability, abdominal pain, and vomiting. This is followed by bright red glossitis and a characteristic pigmented, scaly skin rash in sun-exposed areas. This rash is known as Casal's necklace when it forms a ring around the neck.

The 4Ds of pellagra:

• Diarrhea

• Dermatitis

• Dementia, leading to

• Death

It is diagnosed by measuring urinary levels of 2-methyl nicotinamide and 2-pyridone.

It is treated by oral supplementation of 100–200 mg of nicotinamide or nicotinic acid three times daily for 5 days.

Casal's necklace finding in niacin deficiency

Solution to Question 16:

The third stage in the Motivation model of health education, following awareness and motivation, is action.

In the process of adopting new ideas or behaviours, individuals progress through distinct stages. The first stage, awareness, involves gaining general information about health needs and issues, which serves as the foundation for further exploration. Without awareness, individuals may not recognize the importance of making changes to their health behaviours.

The second stage, motivation, encompasses interest, evaluation, and decision-making. Here, individuals express curiosity about the information presented, seek out additional details, and weigh the pros and cons of adopting the new behaviour. This stage often requires interpersonal communication and support from peers and family, as individuals contemplate the implications of their choices.

Finally, in the action stage, individuals actively adopt and implement the new behaviour or idea into their lives. This involves not just the decision to change but also taking concrete steps to integrate the new practice into their daily routine. It is during this stage that the new behaviour becomes part of an individual's values and lifestyle, leading to sustained change. Effective communication strategies are essential to support individuals as they navigate through these stages, ensuring a successful transition to healthier behaviours.

Note: The three models of health education:

• Medical model: Transfer of proven medical information into education

• Motivational model: Awareness → Motivation → Action or Internalization

• Social intervention model: Changing of a social environment

Solution to Question 17:

The given clinical features of sore throat and difficulty in swallowing with a greyish-white membrane over the tonsils in an unvaccinated child is suggestive of pharyngeal diphtheria. The appropriate management for the asymptomatic 2-year-old brother of the child with diphtheria is to initiate prophylactic antibiotic treatment with penicillin, regardless of his immunization status.

Diphtheria is caused by the bacteria Corynebacterium diphtheriae which spreads via droplets through the respiratory tract (most important), cutaneous lesions, and fomites. Management includes antitoxin therapy as a single empirical dose of 20,000-100,000 units and antibiotics. Penicillin G/ Benzylpenicillin is the drug of choice for diphtheria. However, in case of an allergy to the same, Erythromycin is used. Other antibiotics used are clindamycin, rifampin, and tetracycline.

All household contacts of diphtheria should be closely monitored for 7 days. Culture should be taken from both the throat and nose. After culture, all contacts should receive the following:

• Antibiotic prophylaxis (benzathine penicillin G or erythromycin), regardless of immunization status.

• If fewer than three doses of diphtheria toxoid have been given, or vaccination history is unknown, an immediate dose of diphtheria toxoid should be given and the primary series completed according to the current schedule.

• If more than 5 years have elapsed since the administration of the diphtheria toxoid-containing vaccine, a booster dose should be given.

• If the most recent dose was within 5 years, no booster is required.

At least two successive negative cultures obtained 24 hr apart after completion of therapy are needed to confirm the elimination of the organism.

Solution to Question 18:

A person who is not vaccinated and has been exposed to a disease of international significance (like COVID) with an incubation period of 3-14 days should be quarantined for 14 days(longest incubation period).

Quarantine is defined as the limitation of freedom of movement of well-persons or animals, exposed to a communicable disease for a period of time not longer than the longest usual incubation period, to prevent contact with those not exposed. This is done to prevent the potential spread of the disease during its incubation period, which is the time between exposure and the onset of symptoms.

The quarantined person:

- Should stay in a well-ventilated single room and stay away from high-risk individuals
- Should not attend any religious or social gatherings
- Must wash hands often
- Their surroundings must be disinfected with 1% sodium hypochlorite solution

In case the person being quarantined becomes symptomatic, all his contacts should be home quarantined for 14 days and followed up for another 14 days or till the report turns out negative on lab testing.

Note: The median incubation period refers to the time between exposure to a pathogen and the onset of symptoms in 50% of cases. And isolation is the separation and restriction of movements or activities of persons who are sick, to prevent the transmission of disease.

Solution to Question 19:

The given image is of SAPANA for leprosy. Sapana refers to a local school-going girl who will help to spread awareness in the community.

The other developments under NLEP are:

- Mycobacterium Indicus Prani (MiP) vaccine for multibacillary leprosy patients and their contacts (it is in a project mode).
- Nikusth: A web-based system for the data management of leprosy cases.

Note: Meena is a nine-year-old girl, who is the symbol of UNICEF's Girl Empowerment programme to impart important messages on gender, child rights, education, health and development in the South Asian region.

The posters below depict the role play by Sapna:

Solution to Question 20:

To determine whether there is an association between diet (vegetarians vs. non-vegetarians) and the incidence of disease (present or absent), the best biostatistical method to use is the Chi-square test.

Other options:

Option A: Mann-Whitney test is used for comparing two independent groups on a continuous or ordinal outcome, but in this case, both variables (diet and disease) are categorical.

Option B: The correlation coefficient measures the strength of a linear relationship between two continuous variables, but it is not appropriate for categorical data.

Option C: T-squared test is typically used for multivariate hypothesis testing and is not suited for testing relationships between categorical variables like diet and disease incidence.

Solution to Question 21:

The correct sequence of the phases of the demographic cycle is High stationary → Early expanding → Late expanding → Low stationary → Declining.

India is currently in the "late expanding" stage which is the third stage in the demographic cycle where declining death rates and declining birth rates are seen.

Another important concept of the demographic cycle is the demographic gap which is the difference between birth rate and death rate. The demographic gap is maximum in the early-expanding phase, characterized by a high birth rate and low death rate. As the demography of the country transitions to the late-expanding phase, both birth rate and death rate falls causing a decreasing demographic gap. Although the birth rate is falling in the late-expanding phase, it is still higher than the death rate, which causes the population to increase.

Solution to Question 22:

The clinical scenario is suggestive of epidemic dropsy. Paper chromatography test is used to identify the presence of toxic alkaloids from Argemone mexicana seeds that adulterate mustard oil.

Epidemic dropsy is caused by the toxic compound sanguinarine, which is present in argemone oil extracted from argemone seeds. It is commonly used as an adulterant in mustard oil.

Epidemic dropsy results in sudden onset of swelling of lower limbs, diarrhea, and breathlessness. Some patients may develop hypersecretory glaucoma. It can lead to death due to cardiac failure.

Adulteration with argemone oil can be detected by the following tests:

• Paper chromatography - most sensitive test

• Nitric acid test

Solution to Question 23:

The early neonatal mortality rate here would be 18 per 1,000 live births.

It can be calculated as shown:

Test description :

Early neonatal mortality rate $= \dfrac{\text{Number of deaths within 7 days}}{\text{Total number of live births}} \times 1000$

Given data :

Total number of live births : 500

Number of deaths within the first 7 days : 9

Calculation :

Early neonatal mortality rate $= \dfrac{9}{500} \times 1000 = 18$

The neonatal mortality rate (NMR) refers to the rate of deaths within the first 28 days of life. It is calculated using the following parameters:

• Early neonatal deaths- death in the first 7 days of life

• Late neonatal deaths - after 7th day but before 28th completed day of life

Solution to Question 24:

From the given clinical features, the most likely disease is visceral leishmaniasis or Kala azar caused by Leishmania donovani, and the vector is a female Phlebotomus sandfly. A patient from Bihar presenting with fever, weight loss, and hepatosplenomegaly with a smear examination showing anemia, thrombocytopenia, and hypergammaglobulinemia point towards the diagnosis.

Leishmaniasis is transmitted by the bite of an infected female Phlebotomus sandfly. In India, it is endemic to Bihar, West Bengal, Jharkhand, and Uttar Pradesh. Patients present with high fever, hepatosplenomegaly, weight loss, and emaciation. The skin becomes dry, rough, and darkly pigmented. There is anemia, leukopenia, and thrombocytopenia.

The parasite exists in 2 forms:

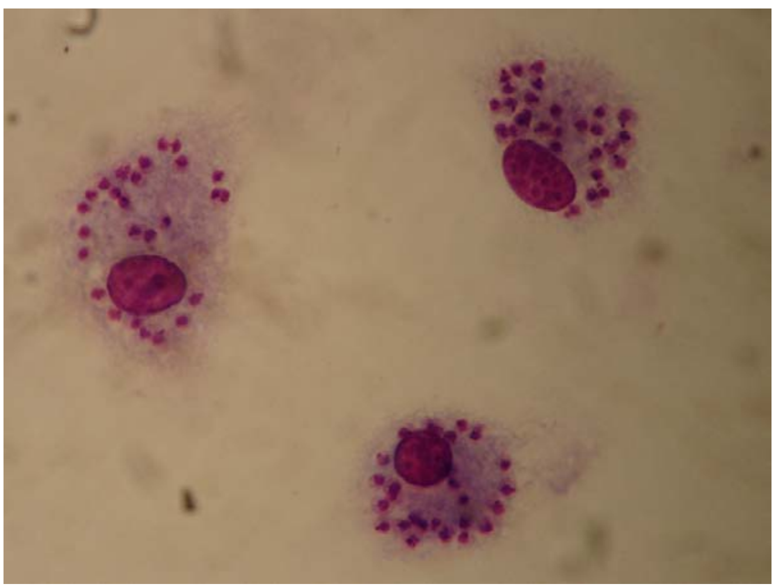
• Amstigote form (also called LD bodies)- It is seen within macrophages and contains a kinetoplast.

• Promastigote form- It is the infective form in the sandfly. The kinetoplast is seen at the anterior end.

Kinetoplast are non-nuclear DNA present in addition to nucleus. They consist of blepharoplast and parabasal body. Flagellum originates near the kinetoplast.

Diagnosis of kala-azar is mainly by demonstration of amastigotes within macrophages in tissue aspirates such as blood smears, bone marrow, splenic, or lymph node aspirate. It is the gold standard for diagnosis.

The image below shows amastigotes located intracellularly within macrophages.



Treatment is with a pentavalent antimonial compound. In Bihar, there is resistance to antimony so amphotericin-B-deoxycholate or miltefosine is preferred.

In human beings, different species of Leishmania can cause the following diseases:

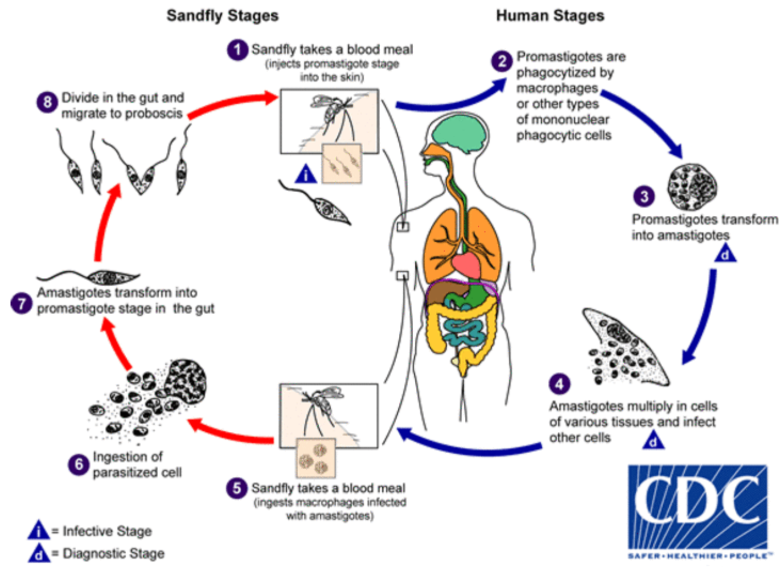
• Visceral leishmaniasis, or kala-azar,

• Cutaneous leishmaniasis,

• Mucocutaneous leishmaniasis, and

• Diffuse cutaneous leishmaniasis.

The life cycle of Lesihmania is shown below.



Other options:

Option B: The Aedes mosquito is the vector for diseases such as dengue, chikungunya, and Zika virus, none of which fit the clinical profile of visceral leishmaniasis.

Option C: Mites are vectors for diseases such as scrub typhus, which typically presents with fever and an eschar, but it does not explain the massive splenomegaly or hematologic findings seen in this case.

Option D: Ixodid ticks are vectors for tick-borne diseases such as Lyme disease and babesiosis. These conditions do not present with the clinical features observed in this patient.

Solution to Question 25:

The time interval between the receipt of infection by a host and the maximal infectivity of that host is known as the generation time.

The time of maximum communicability may precede or follow the incubation period.

Other Options:

Option B: The median incubation period refers to the time between exposure to a pathogen and the onset of symptoms in 50% of cases.

Option C: Serial interval is the gap in time between the onset of the primary case and the second case. The term serial interval is used in an outbreak of disease.

Option D: Lead time is the period between the first possible time of detection and the usual time of diagnosis.

Solution to Question 26:

Weekly Iron Folic Acid Supplementation is included in School Heath Programme not monthly IFA tablets for all children.

The Weekly Iron Folic Acid Supplementation (WIFS) program is a school health program that provides iron and folic acid (IFA) tablets which aims to reduce the prevalence of anemia in school going children.

The school health programs are managed by the primary health center (PHC).

The School Health Committee (1961) in India recommended that the school health services should be a part of the general health services administered through PHCs. Formation of school health committees at the village, block, district, state, and national levels was recommended. The National School Health Council was created as an advisory and coordinating body.

PHCs provide the following school health services:

• Screening and general assessments for

• Anemia/nutritional status

• Visual acuity and hearing problems

• Dental check-up

• Physical disabilities

• Learning disorders and behavior problems

• Basic medicines for common ailments

• Immunization

• Micronutrient management (vitamin A, iron, and folic acid)

• Deworming

• Mid-day meal

• Provison of sanitary napkins

Note: National Programme for Mid-Day Meal in Schools has recently been renamed Pradhan Mantri Poshan Shakti Nirman (PM POSHAN).

Solution to Question 27:

The intervention can delay the onset of the disease (Statement 1) is true about longevity dividend.

The longevity dividend is a horizontal approach to public health which is based on a broader strategy which means to foster health for all generations. This can be done by developing a new model for health promotion and disease prevention. The longevity-dividend model aims to prevent or delay the root causes of disease and disability by decimating the one major risk factor which is biological aging.

Solution to Question 28:

For standards of ventilation, CO_2 content of air is the paramater used to calculate fresh air supply of 3000 cubic feet per person per hour which was advocated by De Chaumont.

Ventilation involves the quality control of incoming air but not quality control of outgoing air. It also involves the replacement of vitiated air(air with decreased oxygen content) by a supply of fresh outdoor air. It aims to provide a comfortable thermal environment and reduce the risk of infections.

Fresh air supply of f 3000 cubic feet per person per hour is a good standard for air requirement. It is observed that the amount of CO_2 due to respiration is not more than 2 parts in 1000 parts of air, then the air in the room is considered fresh.

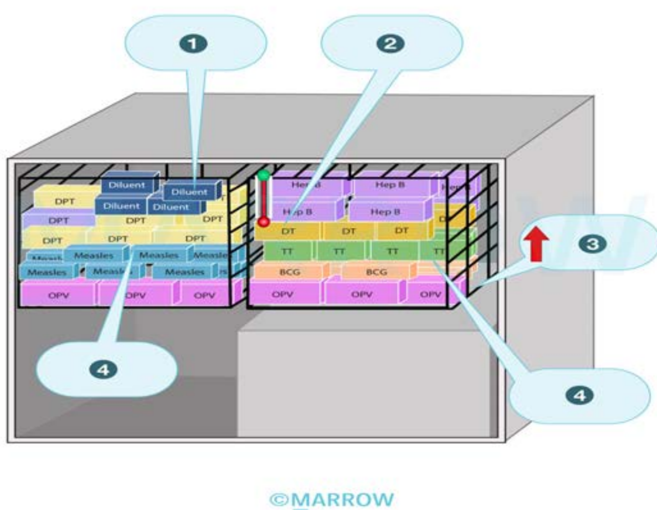
Solution to Question 29:

The image shown in the question is of an ice-lined refrigerator (ILR). The OPV vaccines are stored in the zone 3 that corresponds to the lowest

ILRs are the main method of vaccine storage in primary health centers. They are an important part of the cold chain and they maintain a temperature of 2 to 8 °C for the storage of vaccines.

They have different temperature zones for different vaccines. The upper zone is for freeze-sensitive vaccines like cholera, hep B, rotavirus(liquid), etc while the lower zone is for heat-sensitive vaccines like OPV, measles, etc.

All vaccines are kept in the basket provided with ILR. Vaccines like OPV, BCG, measles, and JE can be kept at the bottom of the basket while, DPT, TT, Hep B, IPV, and pentavalent vaccines and diluents are kept in the upper part of the basket.



Solution to Question 30:

The Sanchol cholera vaccine is a bivalent vaccine given in 2 liquid doses orally to children >=1 year of age.

The Sanchol cholera vaccine is a bivalent oral cholera vaccine (OCV) that targets both Vibrio cholerae O1 and Vibrio cholerae O139 serogroups, which are the primary causes of cholera outbreaks. It is designed to provide immunity by stimulating the production of antibodies in the intestines, the site of infection.

Key Features of the Sanchol Vaccine:

Bivalent Nature: The vaccine protects against two strains of the cholera bacterium: O1, which is responsible for most global cholera cases, and O139 which is seen in South Asia.

Dosing Schedule: The Sanchol vaccine is administered in two oral doses. The doses are given 14 days apart for children aged 1 year and above as well as adults. This two-dose regimen is essential for achieving effective immunity. The vaccine is provided in liquid form, making it easy to administer. It does not require any special preparation, such as mixing with a bicarbonate buffer, which is needed for some older cholera vaccines.

Age Recommendation: The vaccine is recommended for individuals aged 1 year and older. It is particularly useful in endemic regions or during outbreaks to prevent the spread of the disease.

Efficacy and Duration: Sanchol has been shown to provide significant protection against cholera for up to 3-5 years after vaccination, although booster doses may be required in areas with ongoing transmission.

The vaccine is widely used in mass vaccination campaigns, especially in areas prone to cholera outbreaks, such as refugee camps or regions with poor sanitation. It has been a critical tool in reducing the incidence of cholera in many endemic areas.

Solution to Question 31:

This town's maternal mortality ratio (MMR) is 800 per 100,000 live births.

Maternal mortality refers to deaths during pregnancy or within 42 days of termination of pregnancy, from causes related to pregnancy. Excluding the deaths from traffic accidents and snake bite, the total number of maternal deaths is 8.

MMR = (Number of maternal deaths in a given year / Number of live births in the same area and year) x 1,00,000 (or 1000).

Hence, MMR = 8 x 100000/1000 = 800 per 100,000 live births.

Solution to Question 32:

The correct order is: Disaster impact – Response – Rehabilitation – Mitigation

• Disaster impact and response includes:

• Search, rescue and first aid must be done first.

• Field care

• Triage

• Lagging

• Identification of dead

• Rehabilitation or recovery includes:

• Water supply

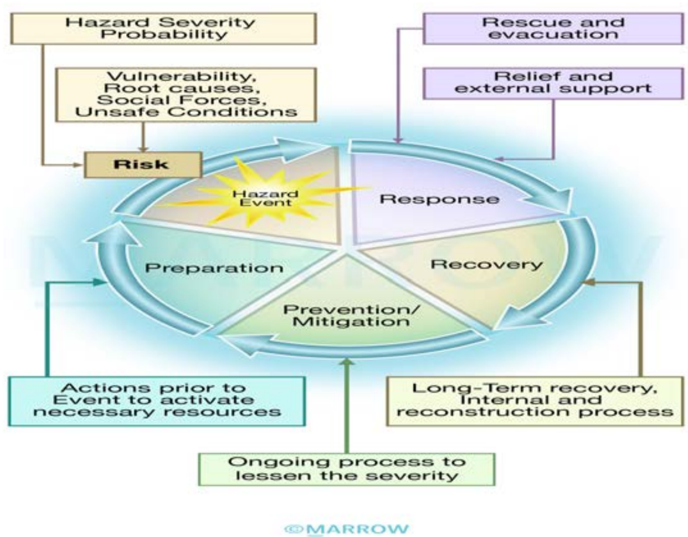
• Sanitation and personal hygiene

• Food safety

• Vector control

• Mitigation:

• Measures designed to either prevent hazards or reduce the effects of the disaster leading to disaster preparedness.



Solution to Question 33:

Infant mortality rate, literacy rate, and life expectancy at 1 year of age are the components of the physical quality of life index (PQLI).